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(54) **HETERODIMERIC ANTIBODIES THAT BIND
PROSTATE SPECIFIC MEMBRANE
ANTIGEN (PSMA) AND CD3**

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None

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(57) **ABSTRACT**

Provided herein are novel antigen binding domains and
antibodies (e.g., heterodimeric antibodies) that bind Prostate
Specific Membrane Antigen (PSMA). In exemplary embod-
iments, the anti-PSMA antibodies also bind CD3. Such
antibodies that bind PSMA and CD3 are useful, for example
in the treatment of PSMA-related cancer.

6 Claims, 122 Drawing Sheets

Specification includes a Sequence Listing.

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Figure 1A

Monomer 1	Monomer 2
F405A	T394F
S364D	Y349K
S364E	L368K
S364E	Y349K
S364F	K370G
S364H	Y349K
S364H	Y349T
S364Y	K370G
T411K	K370E
V397S/F405A	T394F
K370R/T411K	K370E/T411E
L351E/S364D	Y349K/L351K
L351E/S364E	Y349K/L351K
L351E/T366D	L351K/T366K
P395T/V397S/F405A	T394F
S364D/K370G	S364Y/K370R
S364D/T394F	Y349K/F405A
S364E/F405A	Y349K/T394F
S364E/F405S	Y349K/T394Y
S364E/T411E	Y349K/D401K
S364H/D401K	Y349T/T411E
S364H/F405A	Y349T/T394F
S364H/T394F	Y349T/F405A
Y349C/S364E	Y349K/S354C
L351E/S364D/F405A	Y349K/L351K/T394F
L351K/S364H/D401K	Y349T/L351E/T411E
S364E/T411E/F405A	Y349K/T394F/D401K
S364H/D401K/F405A	Y349T/T394F/T411E
S364H/F405A/T411E	Y349T/T394F/D401K

Figure 1B

Monomer 1	Monomer 2
K370E/T411D	T411K
L368E/K409E	L368K
Y349T/T394F/S354C	S364H/F405A/Y349C
T411E	D401K
T411E	D401R/T411R
Q347E/K360E	Q347R
L368E	S364K
L368E/K370S	S364K
L368E/K370T	S364K
L368E/D401R	S364K
L368E/D401N	S364K
L368E	E357S/S364K
L368E	S364K/K409E
L368E	S364K/K409V
L368D	S364K
L368D/K370S	S364K
L368D/K370S	S364K/E357L
L368D/K370S	S364K/E357Q
T411E/K360E/Q362E	D401K
K370S	S364K
L368E/K370S	S364K/E357Q
K370S	S364K/E357Q
T411E/K360D	D401K
T411E/K360E	D401K
T411E/Q362E	D401K
T411E/N390D	D401K
T411E	D401K/Q347K
T411E	D401K/Q347R
T411E/K360D/Q362E	D401K

Figure 1C

Monomer 1	Monomer 2
T411E/K360E/N390D	D401K
T411E/Q362E/N390D	D401K
T411E/Q347R	D401K/K360D
T411E/Q347R	D401K/K360E
T411E/K360	D401K/Q347K
T411E/K360D	D401K/Q347R
T411E/K360E	D401K/Q347K
T411E/K360E	D401K/Q347R
T411E/S364K	D401K/K370S
T411E/K370S	D401K/S364K
Q347E	E357Q
Q347E	E357Q/Q362K
K360D/Q362E	Q347R
K360D/Q362E	D401K
K360D/Q362E	Q347R/D401K
K360E/Q362E	Q347R
K360E/Q362E	D401K
K360E/Q362E	Q347R/D401K
Q362E/N390D	D401K
Q347E/K360D	D401N
K360D	Q347R/N390K
K360D	N390K/D401N
K360E	Y349H
K370S/Q347E	S364K
K370S/E357L	S364K
K370S/E357Q	S364K
K370S/Q347E/E357L	S364K
K370S/Q347E/E357Q	S364K

Figure 1D

Monomer 1	Monomer 2
L368D/K370S/Q347E	S364K
L368D/K370S/E357L	S364K
L368D/K370S/E357Q	S364K
L368D/K370S/Q347E/E357L	S364K
L368D/K370S/Q347E/E357Q	S364K
L368E/K370S/Q347E	S364K
L368E/K370S/E357L	S364K
L368E/K370S/E357Q	S364K
L368E/K370S/Q347E/E357L	S364K
L368E/K370S/Q347E/E357Q	S364K
L368D/K370T/Q347E	S364K
L368D/K370T/E357L	S364K
L368D/K370T/E357Q	S364K
L368D/K370T/Q347E/E357L	S364K
L368D/K370T/Q347E/E357Q	S364K
L368E/K370T/Q347E	S364K
L368E/K370T/E357L	S364K
L368E/K370T/E357Q	S364K
L368E/K370T/Q347E/E357L	S364K
L368E/K370T/Q347E/E357Q	S364K
T411E/Q362E	D401K/T411K
T411E/N390D	D401K/T411K
T411E/Q362E	D401R/T411R
T411E/N390D	D401R/T411R
Y407T	T366Y
F405A	T394W
T366Y/F405A	T394W/Y407T
Y407A	T366W
T366S/L368A/Y407V	T366W
T366S/L368A/Y407V/Y349C	T366W/S354C

Figure 1E

Monomer 1	Monomer 2
K392D/K409D	E356K/D399K
K370D/K392D/K409D	E356K/E357K/D399K
I199T/N203D/K247Q/R355Q/N384S/K392N/V397M/Q419E/K447_	Q196K/I199T/P217R/P228R/N276K
I199T/N203D/K247Q/R355Q/N384S/K392N/V397M/Q419E/K447_	Q196K/I199T/N276K
N384S/K392N/V397M/Q419E	N276K
D221E/P228E/L368E	D221R/P228R/K409R
C220E/P228E/L368E	C220R/E224R/P228R/K409R
F405L	K409R
T366I/K392M/T394W	F405A/Y407V
T366V/K409F	L351Y/Y407A
T366A/K392E/K409F/T411E	D399R/S400R/Y407A
L351K	L351E
I199T/N203D/K247Q/R355Q/Q419E/K447_	Q196K/I199T/P217R/P228R/N276K
I199T/N203D/K247Q/R355Q/Q419E/K447_	Q196K/I199T/N276K
I199T/N203D/K274Q/R355Q/N384S/K392N/V397M/Q419E/K447_	
N208D/Q295E/N384D/Q418E/N421D	
N208D/Q295E/Q418E/N421D	
Q196K/I199T/P217R/P228R/N276K	
Q196K/I199T/N276K	
E269Q/E272Q/E283Q/E357Q	
E269Q/E272Q/E283Q	
E269Q/E272Q	
E269Q/E283Q	
E272Q/E283Q	
E269Q	

Figure 2

<u>Variant constant region</u>	<u>Substitutions</u>
pl-ISO(-)	I199T/N203D/K274Q/R355Q/N384S/K392N/N397M/Q419E/K447_
pl_ISO(-)-Fc only	K274Q/R355Q/N384S/K392N/V397M/Q419E/K447_
pl_(-)_isosteric_A	N208D/Q295E/N384D/Q418E/N421D
pl_(-)_isosteric A-Fc only	Q295E/N384D/Q418E/N421D
pl_(-)_isosteric_B	N208D/Q295E/Q418E/N421D
pl_(-)_isosteric_B-Fc only	Q295E/Q418E/N421D
pl_ISO(+RR)	Q196K/I199T/P217R/P228R/N276K
pl_ISO(+RR)-Fc only	P217R/P228R/N276K
pl_ISO(+)	Q196K/I199T/N276K
pl_ISO(+)-Fc only	N276K
pl_(+)_isosteric_A	E269Q/E272Q/E283Q/E357Q
pl_(+)_isosteric_B	E269Q/E272Q/E283Q
pl_(+)_isosteric_E269Q/E272Q	E269Q/E272Q
pl_(+)_isosteric_E269Q/E283Q	E269Q/E283Q
pl_(+)_isosteric_E272Q/E283Q	E272Q/E283Q
pl_(+)_isosteric_E269Q	E269Q

Figure 3

Ablation Variants

G236R

S239G

S239K

S239Q

S239R

V266D

S267K

S267R

H268K

E269R

299R

299K

K322A

A327G

A327L

A327N

A327Q

L328E

L328R

P329A

P329H

P329K

A330L

A330S/P331S

I332K

I332R

V266D/A327Q

V266D/P329K

S267R/A327Q

S267R/P329K

G236R/L328R

E233P/L234V/L235A/G236_/S239K

E233P/L234V/L235A/G236_/S267K

E233P/L234V/L235A/G236_/S239K/A327G

E233P/L234V/L235A/G236_/S267K/A327G

E233P/L234V/L235A/G236_

S239K/S267K

267K/P329K

Figure 4

Heavy Chain 1 (-) e.g. Fab-Fc	Heavy Chain 2 (+) e.g. scFv-Fc or Fab-scFv-Fc
	C220S
Heterodimeric skew variants L368D/K370S	Heterodimeric skew variants S364K/E357Q
Isosteric pl substitutions N208D/Q295E/N384D/Q418E/N421D	
FcKO E233P/L234V/L235A/G236_/S267K	FcKO E233P/L234V/L235A/G236_/S267K
±M428L/N434S	±M428L/N434S

Figure 5

Positive Charged scFv Linkers

Name	Sequence	Length	Charge	SEQ ID NO:
Gly-Ser 15	GGGGSGGGSGGGGS	15	0	5
Whitlow linker	GSTSGSGKPGSGEGSTKG	18	+1	6
6paxA_1 (+A)	IRPRAIGGSKPRVA	14	+4	7
+B	GKGGSGKGGSGKGGGS	15	+3	8
+C	GGKSGGKGGSGGKGS	15	+3	9
+D	GGGKSGGKSGGGKS	15	+3	10
+E	GKGKSGKSGKSGKGS	15	+6	11
+F	GGGKSGGKSGGKGGGS	15	+3	12
+G	GKPGSGKPGSGKPGS	15	+3	13
+H	GKPGSGKPGSGKPGSGKPGS	20	+4	1
+I	GKGKSGKSGKSGKSGKGS	20	+8	14

Negative Charged scFv Linkers

Name	Sequence	Length	Charge	SEQ ID NO:
Gly-Ser 20	GGGGSGGGSGGGGSGGGGS	20	0	15
3hsc_2 (-A)	STAGDTHLGGEDFD	14	-4	16
-B	GEGSGEGGSGEGGS	15	-3	17
-C	GGEGSGEGSGEGGS	15	-3	18
-D	GGGESGGGESGGGES	15	-3	19
-E	GEGESGEGESGEGES	15	-6	20
-F	GGGESGGEGSGEGGS	15	-3	21
-G	GEGESGEGESGEGESGEGES	20	-8	22

Additional scFv Linkers

GGGGSGGGSGGGGS	SEQ ID NO:5
GGGGSGGGSGGGGSGGGGS	SEQ ID NO:15
GSTSGSGKPGSGEGSTKG	SEQ ID NO:6
PRGASKSGSASQTGSAPGS	SEQ ID NO:23
GTAAAGAGAAAGAAAGAAG	SEQ ID NO:24
GTSGSSGSGSGSGSGGGG	SEQ ID NO:25
GKPGSGKPGSGKPGSGKPGS	SEQ ID NO:1

Figure 6

Useful domain linkers

Name	Sequence	SEQ ID NO:
(GGGGS) ₁ or GGGGS	GGGGS	2
(GGGGS) ₂	GGGGSGGGGS	26
(GGGGS) ₃	GGGGSGGGGSGGGGS	5
(GGGGS) ₄	GGGGSGGGGSGGGGSGGGGS	15
(GGGGS) ₅	GGGGSGGGGSGGGGSGGGGSGGGGS	27
(GGGGS) ₆	GGGGSGGGGSGGGGSGGGGSGGGGSGGGGS	28
(GGGGS) ₇	GGGGSGGGGSGGGGSGGGGSGGGGSGGGGSGGGGS	29
(GGGGA) ₁ or GGGGA	GGGGA	30
(GGGGA) ₂	GGGGAGGGGA	31
(GGGGA) ₃	GGGGAGGGGAGGGGA	32
(GGGGA) ₄	GGGGAGGGGAGGGGAGGGGA	33
(GGGGA) ₅	GGGGAGGGGAGGGGAGGGGAGGGGA	34
(GGGGA) ₆	GGGGAGGGGAGGGGAGGGGAGGGGAGGGGA	35
(GGGGA) ₇	GGGGAGGGGAGGGGAGGGGAGGGGAGGGGAGGGGA	36
30AA-linker	DPALVHQRPAAPPGGGSGGGSGGGSGGG	37
(GKPGS) ₁ or GKPGS	GKPGS	38
(GKPGS) ₅	GKPGSGKPGSGKPGSGKPGSGKPGS	39
(GKPGS) ₆	GKPGSGKPGSGKPGSGKPGSGKPGSGKPGS	40
(GGGES) ₁ or GGGES	GGGES	41
"half hinge"	KTHTCPPCP	42
"full hinge C220S variant"	EPKSSDKTHTCPPCP	43
"flex half hinge"	GGGSGGGGSKTHTCPPCP	44
"charged half hinge1"	GKPGSGKPGSKTHTCPPCP	45
"charged half hinge2"	GKPGSKTHTCPPCP	46

Figure 7A

1 + 1 Fab-scFv-Fc Backbone 1**>Fab-Fc Side (SEQ ID NO: 47)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGDVFS
VMHEALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 48)

EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSVMHEALHNHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 2**>Fab-Fc Side (SEQ ID NO: 49)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGDVFS
VMHEALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 50)

EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSVMHEALHNHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 3**>Fab-Fc Side (SEQ ID NO: 51)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCEVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGDVFS
VMHEALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 52)

EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSVMHEALHNHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 4**>Fab-Fc Side (SEQ ID NO: 53)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSFFLYSKLEVDKSRWEQGDVFS
VMHEALHNHYTQKSLSLSPGK

Figure 7B

>scFv-Fc Side (SEQ ID NO: 54)

EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTPPVLDSDGSGFFLYSKLTVDKSRWQQGNVFSVMSHEALHNHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 5**>Fab-Fc Side (SEQ ID NO: 55)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSRDELTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSGFFLYSKLTVDKSRWEQGDVFS
VMHEALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 56)

EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDQLTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTPPVLDSDGSGFFLYSKLTVDKSRWQQGNVFSVMSHEALHNHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 6**>Fab-Fc Side (SEQ ID NO: 57)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYASTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSGFFLYSKLTVDKSRWEQGDVFS
VMHEALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 58)

EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYASTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREQMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTPPVLDSDGSGFFLYSKLTVDKSRWQQGNVFSVMSHEALHNHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 7**>Fab-Fc Side (SEQ ID NO: 59)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYSSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSGFFLYSKLTVDKSRWEQGDVFS
VMHEALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 60)

EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREQMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTPPVLDSDGSGFFLYSKLTVDKSRWQQGNVFSVMSHEALHNHYTQKSLSLSPGK

Figure 7C

1 + 1 Fab-scFv-Fc Backbone 8**>Fab-Fc Side (SEQ ID NO: 61)**

ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLG
TKTYTCNVDPKPSDTKVDKRVESKYGPPCPCPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVVSQEDPEVQ
FNWYVDGVEVHNAKTKPREEEFNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTISKAKGQPREPQVYTL
PPSQEEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSFFLYSRLTVDKSRWEEGDFVSCSV
MHEALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 62)

ESKYGPPCPCPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVVSQEDPEVQFNWYVDGVEVHNAKTKPREEQ
FNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTISKAKGQPREPQVYTLPPSQEQMTKNQVKLTCLVKGF
YPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMEALHNHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 9**>Fab-Fc Side (SEQ ID NO: 63)**

ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSNFG
TQTYTCNVDPKPSDTKVDKTVKCCVECPPCAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVQF
NWYVDGVEVHNAKTKPREEEFNSTFRVSVLTVVHQDWLNGKEYKCKVSNKGLPAPIEKTISKTKGQPREPQVYTL
PSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPMLDSDGSFFLYSKLTVDKSRWEQGDVFC
SVMH EALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 64)

ERKCCVECPPCAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVQFNWYVDGVEVHNAKTKPREEQF
NSTFRVSVLTVVHQDWLNGKEYKCKVSNKGLPAPIEKTISKTKGQPREPQVYTLPPSREQMTKNQVKLTCLVKGFY
PSDIAVEWESNGQPENNYKTPPMLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMEALHNHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 10**>Fab-Fc Side (SEQ ID NO: 65)**

ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSNFG
TQTYTCNVDPKPSDTKVDKTVKCCVECPPCAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVQF
NWYVDGVEVHNAKTKPREEEFNSTFRVSVLTVVHQDWLNGKEYKCKVSNKGLPAPIEKTISKTKGQPREPQVYTL
PSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPMLDSDGSFFLYSKLTVDKSRWEQGDVFC
SVMH EALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 66)

ERKCCVECPPCAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVQFNWYVDGVEVHNAKTKPREEQF
NSTFRVSVLTVVHQDWLNGKEYKCKVSNKGLPAPIEKTISKTKGQPREPQVYTLPPSREQMTKNQVKLTCLVKGFY
PSDIAVEWESNGQPENNYKTPPMLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMEALHNHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 11**>Fab-Fc Side (SEQ ID NO: 67)**

ASTKGPSVFPLAPSSKSTSGGTAAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLG
TQTYICNVNHKPSDTKVDKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGDVFC
SVMLHEALHSHYTQKSLSLSPGK

Figure 7D

>scFv-Fc Side (SEQ ID NO: 68)

EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREQMTKNQVKLTCLVK
GFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVVFSCSVLHEALHSHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 12**>Fab-Fc Side (SEQ ID NO: 69)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLG
TQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVAGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVVFSCS
VMHEALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 70)

ERKSSDKTHTCPRPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFKWYVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREQMTKNQVKLTCLVK
GFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVVFSCVMHEALHNHYTQKSLSLSPGK

Figure 8A

2 + 1 Fab2-scFv-Fc Backbone 1**>Fab-Fc Side (SEQ ID NO: 71)**

ASTKGPSVFPLAPSSKSTSGGTAAAGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVDFSCS
VMHEALHNHYTQKSLSLSPGK

>Fab-scFv-Fc Side (SEQ ID NO: 72)

APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVDFSCVMHEALHNHYTQKSLSLSPGK

2 + 1 Fab2-scFv-Fc Backbone 2**>Fab-Fc Side (SEQ ID NO: 73)**

ASTKGPSVFPLAPSSKSTSGGTAAAGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVDFSCS
VMHEALHNHYTQKSLSLSPGK

>Fab-scFv-Fc Side (SEQ ID NO: 74)

APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVDFSCVMHEALHNHYTQKSLSLSPGK

2 + 1 Fab2-scFv-Fc Backbone 3**>Fab-Fc Side (SEQ ID NO: 75)**

ASTKGPSVFPLAPSSKSTSGGTAAAGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCEVSGFYPSDIAVEWESDGPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVDFSCS
VMHEALHNHYTQKSLSLSPGK

>Fab-scFv-Fc Side (SEQ ID NO: 76)

APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVDFSCVMHEALHNHYTQKSLSLSPGK

2 + 1 Fab2-scFv-Fc Backbone 4**>Fab-Fc Side (SEQ ID NO: 77)**

ASTKGPSVFPLAPSSKSTSGGTAAAGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESDGPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVDFSCS
VMHEALHNHYTQKSLSLSPGK

Figure 8B

>Fab-scFv-Fc Side (SEQ ID NO: 78)

APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSDGSGFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK

2 + 1 Fab2-scFv-Fc Backbone 5**>Fab-Fc Side (SEQ ID NO: 79)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSRDELTKNQVSLTCDVSGFYPSDIAVEWESDGGQENNYKTTTPVLDSDGSGFFLYSKLTVDKSRWEQGDVFCFS
VMHEALHNHYTQKSLSLSPGK

>Fab-scFv-Fc Side (SEQ ID NO: 80)

APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDQLTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSDGSGFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK

2 + 1 Fab2-scFv-Fc Backbone 6**>Fab-Fc Side (SEQ ID NO: 81)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYASTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGGQENNYKTTTPVLDSDGSGFFLYSKLTVDKSRWEQGDVFCFS
VMHEALHNHYTQKSLSLSPGK

>Fab-scFv-Fc Side (SEQ ID NO: 82)

APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYASTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREQMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSDGSGFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK

2 + 1 Fab2-scFv-Fc Backbone 7**>Fab-Fc Side (SEQ ID NO: 83)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYSSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGGQENNYKTTTPVLDSDGSGFFLYSKLTVDKSRWEQGDVFCFS
VMHEALHNHYTQKSLSLSPGK

>Fab-scFv-Fc Side (SEQ ID NO: 84)

APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYSSTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREQMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSDGSGFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK

Figure 8C

2 + 1 Fab2-scFv-Fc Backbone 8**>Fab-Fc Side (SEQ ID NO: 85)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDQGPNENNYKTTPPVLDSDGSFFLYSKLTVDKSRWEQGDFVSCS
VLHEALHSHYTQKSLSLSPGK

>Fab-scFv-Fc Side (SEQ ID NO: 86)

APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPNENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVVFSCSVLHEALHSHYTQKSLSLSPGK

2 + 1 Fab2-scFv-Fc Backbone 9**>Fab-Fc Side (SEQ ID NO: 87)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLG
TQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVAGFYPSDIAVEWESNGQPNENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVVFSCS
VMHEALHNHYTQKSLSLSPGK

>Fab-scFv-Fc Side (SEQ ID NO: 88)

APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFKWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPNENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVVFSCSVLHEALHNHYTQKSLSLSPGK

Figure 9**Constant Light Domain – Kappa (SEQ ID NO: 89)**

RTVAAPSVFIFFPPSDEQLKSGTASVCLLNIFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYLSSTLTLSKADYEKHKVYACEVTHQG
LSSPVTKSFNRGEC

Constant Light Domain – Lambda (SEQ ID NO: 90)

GQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADSSSPVKAGVETTTPSKQSNNKYAASSYLSLTPE
QWKSHRSYSQCQVTHEGSTVEKTVAPTECS

Figure 10A

CD3 High – [anti-CD3] H1.30 L1.47 scFv		
	Sequence	SEQ ID NO:
scFv	<u>EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSK</u> <u>YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNGFGDSY</u> <u>VSWFAYWGQGTLLTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQEPSLTVSPG</u> <u>GTVTLTGSSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSL</u> <u>LGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTCLTVL</u>	91
Variable Heavy (vh) Domain	<u>EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSK</u> <u>YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNGFGDSY</u> <u>VSWFAYWGQGTLLTVSS</u>	92
vhCDR1	<u>TYAMN</u>	93
vhCDR2	<u>RIRSKYNNYATYYADSVKG</u>	94
vhCDR3	<u>HGNGFGDSYVSWFAY</u>	95
Variable Light (vl) Domain	<u>QAVVTQEPSLTVSPGGTTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGVPARFSGSLLGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTCLTV</u> <u>L</u>	96
vlCDR1	<u>GSSTGAVTTSNYAN</u>	97
vlCDR2	<u>GTNKRAP</u>	98
vlCDR3	<u>ALWYSNHWV</u>	99
Linker	<u>GKPGSGKPGSGKPGSGKPGS</u>	100

CD3 High-Int #1 – [anti-CD3] H1.32 L1.47 scFv		
	Sequence	SEQ ID NO:
scFv	<u>EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSK</u> <u>ANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNGFGDSY</u> <u>VSWFAYWGQGTLLTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQEPSLTVSPG</u> <u>GTVTLTGSSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSL</u> <u>LGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTCLTVL</u>	101
Variable Heavy (vh) Domain	<u>EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSK</u> <u>ANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNGFGDSY</u> <u>VSWFAYWGQGTLLTVSS</u>	102
vhCDR1	<u>TYAMN</u>	103
vhCDR2	<u>RIRSKANNYATYYADSVKG</u>	104
vhCDR3	<u>HGNGFGDSYVSWFAY</u>	105
Variable Light (vl) Domain	<u>QAVVTQEPSLTVSPGGTTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGVPARFSGSLLGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTCLTV</u> <u>L</u>	106
vlCDR1	<u>GSSTGAVTTSNYAN</u>	107
vlCDR2	<u>GTNKRAP</u>	108
vlCDR3	<u>ALWYSNHWV</u>	109
Linker	<u>GKPGSGKPGSGKPGSGKPGS</u>	110

Figure 10B

CD3 High-Int #2 – [anti-CD3] H1.89 L1.47 scFv		
	Sequence	SEQ ID NO:
scFv	<u>EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSK</u> <u>YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNGFGDEY</u> <u>VSWFAYWGQGTLLTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQEPSLTVSPG</u> <u>GTVTLTGSSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSL</u> <u>LGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTCLTVL</u>	111
Variable Heavy (vh) Domain	<u>EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSK</u> <u>YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNGFGDEY</u> <u>VSWFAYWGQGTLLTVSS</u>	112
vhCDR1	<u>TYAMN</u>	113
vhCDR2	<u>RIRSKYNNYATYYADSVKG</u>	114
vhCDR3	<u>HGNGFGDEYVSWFAY</u>	115
Variable Light (vl) Domain	<u>QAVVTQEPSLTVSPGGTTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGVPARFSGSLLGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTCLTV</u> <u>L</u>	116
vlCDR1	<u>GSSTGAVTTSNYAN</u>	117
vlCDR2	<u>GTNKRAP</u>	118
vlCDR3	<u>ALWYSNHWV</u>	119
Linker	<u>GKPGSGKPGSGKPGSGKPGS</u>	120

CD3 High-Int – [anti-CD3] H1.90 L1.47 scFv		
	Sequence	SEQ ID NO:
scFv	<u>EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSK</u> <u>YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNGFGDPY</u> <u>VSWFAYWGQGTLLTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQEPSLTVSPG</u> <u>GTVTLTGSSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSL</u> <u>LGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTCLTVL</u>	121
Variable Heavy (vh) Domain	<u>EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSK</u> <u>YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNGFGDPY</u> <u>VSWFAYWGQGTLLTVSS</u>	122
vhCDR1	<u>TYAMN</u>	123
vhCDR2	<u>RIRSKYNNYATYYADSVKG</u>	124
vhCDR3	<u>HGNGFGDPYVSWFAY</u>	125
Variable Light (vl) Domain	<u>QAVVTQEPSLTVSPGGTTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGVPARFSGSLLGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTCLTV</u> <u>L</u>	126
vlCDR1	<u>GSSTGAVTTSNYAN</u>	127
vlCDR2	<u>GTNKRAP</u>	128
vlCDR3	<u>ALWYSNHWV</u>	129
Linker	<u>GKPGSGKPGSGKPGSGKPGS</u>	130

Figure 10C

Anti-CD3-Intermediate – [anti-CD3] H1.33 L1.47 scFv		
	Sequence	SEQ ID NO:
scFv	<u>EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSK</u> <u>YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNGFGDSY</u> <u>VSWFDYWGQGTLVTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQEPSLTVSPG</u> <u>GTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSL</u> <u>LGKAALTISGAQPEDEADYYCALWYSNHWVFSGGTGKLT</u>	131
Variable Heavy (vh) Domain	<u>EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSK</u> <u>YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNGFGDSY</u> <u>VSWFDYWGQGTLVTVSS</u>	132
vhCDR1	<u>TYAMN</u>	133
vhCDR2	<u>RIRSKYNNYATYYADSVKG</u>	134
vhCDR3	<u>HGNGFGDSYVSWFDY</u>	135
Variable Light (vl) Domain	<u>QAVVTQEPSLTVSPGGTGTTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGVPARFSGSLLGKAALTISGAQPEDEADYYCALWYSNHWVFSGGTGKLT</u> <u>L</u>	136
vlCDR1	<u>GSSTGAVTTSNYAN</u>	137
vlCDR2	<u>GTNKRAP</u>	138
vlCDR3	<u>ALWYSNHWV</u>	139
Linker	<u>GKPGSGKPGSGKPGSGKPGS</u>	140

CD3 High-Int – [anti-CD3] H1.31 L1.47 scFv		
	Sequence	SEQ ID NO:
scFv	<u>EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMSWVRQAPGKGLEWVGRIRSK</u> <u>YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNGFGDSY</u> <u>VSWFAYWGQGTLVTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQEPSLTVSPG</u> <u>GTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSL</u> <u>LGKAALTISGAQPEDEADYYCALWYSNHWVFSGGTGKLT</u>	141
Variable Heavy (vh) Domain	<u>EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMSWVRQAPGKGLEWVGRIRSK</u> <u>YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNGFGDSY</u> <u>VSWFAYWGQGTLVTVSS</u>	142
vhCDR1	<u>TYAMS</u>	143
vhCDR2	<u>RIRSKYNNYATYYADSVKG</u>	144
vhCDR3	<u>HGNGFGDSYVSWFAY</u>	145
Variable Light (vl) Domain	<u>QAVVTQEPSLTVSPGGTGTTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGVPARFSGSLLGKAALTISGAQPEDEADYYCALWYSNHWVFSGGTGKLT</u> <u>L</u>	146
vlCDR1	<u>GSSTGAVTTSNYAN</u>	147
vlCDR2	<u>GTNKRAP</u>	148
vlCDR3	<u>ALWYSNHWV</u>	149
Linker	<u>GKPGSGKPGSGKPGSGKPGS</u>	150

Figure 10D

CD3 High[VL-VH] – [anti-CD3] L1.47 H1.30 scFv		
	Sequence	SEQ ID NO:
scFv	QAVVTQEPSTLVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN KRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTV L/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCAASGFTFST YAMNWVRQAPGKGLEWVGRIIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQM NSLRAEDTAVYYCVRHGNFGDSYVSWFAYWGQGLTVTVSS	151
Variable Light (vl) Domain	QAVVTQEPSTLVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN KRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTV L	152
vICDR1	GSSTGAVTTSNYAN	153
vICDR2	GTNKRAP	154
vICDR3	ALWYSNHWV	155
Variable Heavy (vh) Domain	EVQLVESGGGLVQPGGSLRLSCAASGFTFSTYAMNWVRQAPGKGLEWVGRIIRSK YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNFGDSY VSWFAYWGQGLTVTVSS	156
vhCDR1	TYAMN	157
vhCDR2	RIRSKYNNYATYYADSVKG	158
vhCDR3	HGNFGDSYVSWFAY	159
Linker	GKPGSGKPGSGKPGSGKPGS	160

CD3 High-Int #1[VL-VH] – [anti-CD3] L1.47 H1.32 scFv		
	Sequence	SEQ ID NO:
scFv	QAVVTQEPSTLVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN KRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTV L/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCAASGFTFST YAMNWVRQAPGKGLEWVGRIIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQM NSLRAEDTAVYYCVRHGNFGDSYVSWFAYWGQGLTVTVSS	161
Variable Light (vl) Domain	QAVVTQEPSTLVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN KRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTV L	162
vICDR1	GSSTGAVTTSNYAN	163
vICDR2	GTNKRAP	164
vICDR3	ALWYSNHWV	165
Variable Heavy (vh) Domain	EVQLVESGGGLVQPGGSLRLSCAASGFTFSTYAMNWVRQAPGKGLEWVGRIIRSK ANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNFGDSY VSWFAYWGQGLTVTVSS	166
vhCDR1	TYAMN	167
vhCDR2	RIRSKANNYATYYADSVKG	168
vhCDR3	HGNFGDSYVSWFAY	169
Linker	GKPGSGKPGSGKPGSGKPGS	170

Figure 10E

CD3 High-Int #2[VL-VH] – [anti-CD3] L1.47 H1.89 scFv		
	Sequence	SEQ ID NO:
scFv	QAVVTQEPSTLVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN KRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTV L/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCAASGFTFST YAMNWVRQAPGKGLEWVGRIIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQM NSLRAEDTAVYYCVRHGNFGDEYVSWFAYWGQGLTVTVSS	171
Variable Light (vl) Domain	QAVVTQEPSTLVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN KRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTV L	172
vICDR1	GSSTGAVTTSNYAN	173
vICDR2	GTNKRAP	174
vICDR3	ALWYSNHWV	175
Variable Heavy (vh) Domain	EVQLVESGGGLVQPGGSLRLSCAASGFTFSTYAMNWVRQAPGKGLEWVGRIIRSK YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNFGDEY VSWFAYWGQGLTVTVSS	176
vhCDR1	TYAMN	177
vhCDR2	RIRSKYNNYATYYADSVKG	178
vhCDR3	HGNFGDEYVSWFAY	179
Linker	GKPGSGKPGSGKPGSGKPGS	180

CD3 High-Int[VL-VH] – [anti-CD3] L1.47 H1.90 scFv		
	Sequence	SEQ ID NO:
scFv	QAVVTQEPSTLVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN KRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTV L/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCAASGFTFST YAMNWVRQAPGKGLEWVGRIIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQM NSLRAEDTAVYYCVRHGNFGDPYVSWFAYWGQGLTVTVSS	181
Variable Light (vl) Domain	QAVVTQEPSTLVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN KRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTV L	182
vICDR1	GSSTGAVTTSNYAN	183
vICDR2	GTNKRAP	184
vICDR3	ALWYSNHWV	185
Variable Heavy (vh) Domain	EVQLVESGGGLVQPGGSLRLSCAASGFTFSTYAMNWVRQAPGKGLEWVGRIIRSK YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNFGDPY VSWFAYWGQGLTVTVSS	186
vhCDR1	TYAMN	187
vhCDR2	RIRSKYNNYATYYADSVKG	188
vhCDR3	HGNFGDPYVSWFAY	189
Linker	GKPGSGKPGSGKPGSGKPGS	190

Figure 10F

Anti-CD3-Intermediate[VL-VH] – [anti-CD3] L1.47 H1.33 scFv		
	Sequence	SEQ ID NO:
scFv	QAVVTQEPSTLVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN KRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTV L/GKPGSGKPGSGKPGSGKPGS/ EVQLVESGGGLVQPGGSLRLSCAASGFTFST YAMNWVRQAPGKGLEWVGRIIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQM NSLRAEDTAVYYCVRHGNFGDSYVSWFDYWGQGTLLTVSS	191
Variable Light (vl) Domain	QAVVTQEPSTLVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN KRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTV L	192
vICDR1	GSSTGAVTTSNYAN	193
vICDR2	GTNKRAP	194
vICDR3	ALWYSNHWV	195
Variable Heavy (vh) Domain	EVQLVESGGGLVQPGGSLRLSCAASGFTFSTYAMNWVRQAPGKGLEWVGRIIRSK YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNFGDSY VSWFDYWGQGTLLTVSS	196
vhCDR1	TYAMN	197
vhCDR2	RIRSKYNNYATYYADSVKG	198
vhCDR3	HGNFGDSYVSWFDY	199
Linker	GKPGSGKPGSGKPGSGKPGS	200

CD3 High-Int[VL-VH] – [anti-CD3] L1.47 H1.31 scFv		
	Sequence	SEQ ID NO:
scFv	QAVVTQEPSTLVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN KRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTV L/GKPGSGKPGSGKPGSGKPGS/ EVQLVESGGGLVQPGGSLRLSCAASGFTFST YAMSWVRQAPGKGLEWVGRIIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQM NSLRAEDTAVYYCVRHGNFGDSYVSWFAYWGQGTLLTVSS	201
Variable Light (vl) Domain	QAVVTQEPSTLVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN KRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTV L	202
vICDR1	GSSTGAVTTSNYAN	203
vICDR2	GTNKRAP	204
vICDR3	ALWYSNHWV	205
Variable Heavy (vh) Domain	EVQLVESGGGLVQPGGSLRLSCAASGFTFSTYAMSWVRQAPGKGLEWVGRIIRSK YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNFGDSY VSWFAYWGQGTLLTVSS	206
vhCDR1	TYAMS	207
vhCDR2	RIRSKYNNYATYYADSVKG	208
vhCDR3	HGNFGDSYVSWFAY	209
Linker	GKPGSGKPGSGKPGSGKPGS	210

Figure 11A

Human PSMA sequence

>sp|Q04609

MWNLLHETDSAVATARRPRWLCAGALVLAGGFFLLGFLFGWFIKSSNEATNITPKHNMKAFLDELKAENIKKFLYNF
TQIPHLAGTEQNFQLAKQIQSQWKEFGLDSVELAHYDVLLSYPNKTHPNYISIIINEDGNEIFNTSLFEPFPPPGYENV
SDIVPPFSAFSPQGMPEGDLVYVNYARTEDFFKLERDMKINCSCGKIVARYGKVFRGNKVKNQAQLAGAKGVILYSDP
ADYFAPGVKSYPDGWNLPGGGVQQRGNILNLNGAGDPLTPGYPA NEYAYRRGIAEAVGLPSIPVHPIGYYDAQKLEK
MGGSAPPDSSWRGSLKVPYNVGPFGFTGNFSTQKVKMHIHSTNEVTRIYNVIGTLRGAVEPDRYVILGGHRDSWVFGG
IDPQSGAAVVHEIVRSFGTLKKEGWRPRRTILFASWDAEEFGLLGSTEWAEENSRLQLQERGVAYINADSSIEGNYTL
RVDCTPLMYSLVHNLTKELKSPDEGFEGKSLYESWTKKSPSPFSGMPRISKLGSGNDFEVFFQRLGIASGRARYTK
NWETNKFSGYPLYHSVYETYELVEKFYDPMFKYHILTVAQVRGGMVFELANSIVLPFDCRDYAVVLRKYADKIYSISM
KHPQEMKTYSVSFDSLFSAVKNFTEIASKFSERLQDFDKSNPIVLRMMNDQLMFLERAFIDPLGLPDRPFYRHVIYA
PSSH NKYAGESFPFGIYDALFDIESKVDPSKAWGEVKRQIYVAAFTVQAAAETLSEVA (SEQ ID NO: 211)

Human PSMA sequence, extracellular domain

>sp|Q04609|44-750

KSSNEATNITPKHNMKAFLDELKAENIKKFLYNFTQIPHLAGTEQNFQLAKQIQSQWKEFGLDSVELAHYDVLLSY
NKTHPNYISIIINEDGNEIFNTSLFEPFPPPGYENVSDIVPPFSAFSPQGMPEGDLVYVNYARTEDFFKLERDMKINC
GKIVARYGKVFRGNKVKNQAQLAGAKGVILYSDPADYFAPGVKSYPDGWNLPGGGVQQRGNILNLNGAGDPLTPGYPA
NEYAYRRGIAEAVGLPSIPVHPIGYYDAQKLEKMGGSAPPDSSWRGSLKVPYNVGPFGFTGNFSTQKVKMHIHSTNE
VTRIYNVIGTLRGAVEPDRYVILGGHRDSWVFGGIDPQSGAAVVHEIVRSFGTLKKEGWRPRRTILFASWDAEEFGL
LGSTEWAEENSRLQLQERGVAYINADSSIEGNYTLRVDCTPLMYSLVHNLTKELKSPDEGFEGKSLYESWTKKSPSP
FSGMPRISKLGSGNDFEVFFQRLGIASGRARYTKNWETNKFSGYPLYHSVYETYELVEKFYDPMFKYHILTVAQVRG
MVFELANSIVLPFDCRDYAVVLRKYADKIYSISMKHPQEMKTYSVSFDSLFSAVKNFTEIASKFSERLQDFDKSNPI
VLRMMNDQLMFLERAFIDPLGLPDRPFYRHVIYAPSSH NKYAGESFPFGIYDALFDIESKVDPSKAWGEVKRQIYVAA
FTVQAAAETLSEVA (SEQ ID NO: 212)

Mouse PSMA sequence

>sp|O35409

MWNALQDRDSAEVLGHRQRWLRVGTLVLALTGTFLIGFLFGWFIKPSNEATGNVSHSGMKKEFLHELKAENIKKFLY
NFTTRPHLAGTQNNFELAKQIHDQWKEFGLDLVELSHYDVLLSYPNKTHPNYISIIINEDGNEIFKTSLSSEQPPPGYE
NISDVVPPYSAFSPQGTPEGDLVYVNYARTEDFFKLEREMKISCSCGKIVARYGKVFRGNMVKNAQLAGAKGMILYS
DPADYFVPAVKSYPDGWNLPGGGVQQRGNVNLNLNGAGDPLTPGYPA NEHAYRHELTNAVGLPSIPVHPIGYDDAQKLL
EHMGGPAPPDSSWKGGGLKVPYNVGPFGAGNFSTQKVKMHIHSYTKVTRIYNVIGTLKGALEPDRYVILGGHRDAWVF
GGIDPQSGAAVVHEIVRSFGTLKKKGRPRRTILFASWDAEEFGLLGSTEWAEHSRLQLQERGVAYINADSSIEGNY
TLRVDTPLMYSLVYNLTKEQLSPDEGFEGKSLYDSWKEKSPSPEFIGMPRISKLGSGNDFEVFFQRLGIASGRARY
TKNWKTNKVSSYPLYHSVYETYELVVKFYDPTFKYHILTVAQVRGAMVFELANSIVLPFDCQSYAVALKKYADTIYNI
SMKHPQEMKAYMISFDSLFSAVNNFTDVASKFNQRLQELDKSNPILLRIMNDQLMYLERAFIDPLGLPGRPFYRHII
YAPSSH NKYAGESFPFGIYDALFDISSKVNASKAWNEVKRQISIAFTTVQAAAETLREVA (SEQ ID NO: 213)

Mouse PSMA sequence, extracellular domain

>sp|O35409|45-752

KPSNEATGNVSHSGMKKEFLHELKAENIKKFLYNFTTRPHLAGTQNNFELAKQIHDQWKEFGLDLVELSHYDVLLSY
PNKTHPNYISIIINEDGNEIFKTSLSSEQPPPGYENISDVVPPYSAFSPQGTPEGDLVYVNYARTEDFFKLEREMKISC
SGKIVARYGKVFRGNMVKNAQLAGAKGMILYSDPADYFVPAVKSYPDGWNLPGGGVQQRGNVNLNLNGAGDPLTPGYPA
ANEHAYRHELTNAVGLPSIPVHPIGYDDAQKLEHMGGPAPPDSSWKGGGLKVPYNVGPFGAGNFSTQKVKMHIHSY
KVTRIYNVIGTLKGALEPDRYVILGGHRDAWVFGGIDPQSGAAVVHEIVRSFGTLKKKGRPRRTILFASWDAEEFGL
LLGSTEWAEHSRLQLQERGVAYINADSSIEGNYTLRVDCTPLMYSLVYNLTKEQLSPDEGFEGKSLYDSWKEKSPSP
EFIGMPRISKLGSGNDFEVFFQRLGIASGRARYTKNWKTNKVSSYPLYHSVYETYELVVKFYDPTFKYHILTVAQVRG
AMVFELANSIVLPFDCQSYAVALKKYADTIYNI SMKHPQEMKAYMISFDSLFSAVNNFTDVASKFNQRLQELDKSNP
ILLRIMNDQLMYLERAFIDPLGLPGRPFYRHIIYAPSSH NKYAGESFPFGIYDALFDISSKVNASKAWNEVKRQISIA
FTTVQAAAETLREVA (SEQ ID NO: 214)

Figure 11B

Macaca fascicularis PSMA sequence

>tr|G7PNF

MWNLLHETDSAVATARRPRWLCAGALVLAGGFFLLGFLFGWFIKSSSEATNITPKHNMKAFLDELKAENIKKFLHNF
TQIPHLAGTEQNFQLAKQIQSQWKEFGLDSVELTHYDVLLSYPNKTHPNYISIIINEDGNEIFNTSLFEPPPAGYENV
SDIVPPFSAFSPQGMPEGDLVYVNYARTEDFFKLERDMKINCSGKIVARIYGVFRGNKVKNQLAGATGVILYSDP
DDYFAPGVKSYPDGWNLPGGGVQRCNINLNLGAGDPLTPGYPAHEYAYRRGMAEAVGLPSIPVHPIGYYDAQKLEK
MGGSASPDSSWRGSLKVPYNVGPFGFTGNFSTQKVKMHIHSTSEVTRIYNVIGTLRGAVEPDRYVILGHRDSWVFGG
IDPQSGAAVVHEIVRSFGMLKKEGWRPRRTILFASWDAEEFGLLGSTEWAEENSRLQERGVAIYNADSSIEGNYTL
RVDCTPLMYSLVYNLTKELESPDEGFEGKSLYESWTKKSPSPFSGMPRISKLGSGNDFEVFFQRLGIASGRARYTK
NWETNKFSSYPLYHSVYETYELVEKFYDPMFKYHLTVAQVRGGMVFELANSVVLPFDCRDYAVVLRKYADKIYNISM
KHPQEMKTYSVSFDLSLFAVKNFTEIASKFSERLRDFFDKSNPIILLRMMNDQLMFLERAFIDPLGLPDRPFYRHVIYA
PSSHNKYAGESFPFIYDALFDIESKVDPSQAWGEVVKRQISIAFTTVQAAAETLSEVA (SEQ ID NO: 215)

Macaca fascicularis PSMA sequence, extracellular domain (predicted)

>tr|G7PNF|44-750

KSSSEATNITPKHNMKAFLDELKAENIKKFLHNF TQIPHLAGTEQNFQLAKQIQSQWKEFGLDSVELTHYDVLLSY
PNKTHPNYISIIINEDGNEIFNTSLFEPPPAGYENVSDIVPPFSAFSPQGMPEGDLVYVNYARTEDFFKLERDMKINCS
GKIVARIYGVFRGNKVKNQLAGATGVILYSDPDDYFAPGVKSYPDGWNLPGGGVQRCNINLNLGAGDPLTPGYPA
HEYAYRRGMAEAVGLPSIPVHPIGYYDAQKLEKMGGSASPDSSWRGSLKVPYNVGPFGFTGNFSTQKVKMHIHSTSE
VTRIYNVIGTLRGAVEPDRYVILGHRDSWVFGGIDPQSGAAVVHEIVRSFGMLKKEGWRPRRTILFASWDAEEFGL
LGSTEWAEENSRLQERGVAIYNADSSIEGNYTLRVDCTPLMYSLVYNLTKELESPDEGFEGKSLYESWTKKSPSP
FSGMPRISKLGSGNDFEVFFQRLGIASGRARYTKNWETNKFSSYPLYHSVYETYELVEKFYDPMFKYHLTVAQVRG
GMVFELANSVVLPFDCRDYAVVLRKYADKIYNISMKHPQEMKTYSVSFDLSLFAVKNFTEIASKFSERLRDFFDKSNPI
ILLRMMNDQLMFLERAFIDPLGLPDRPFYRHVIYAPSSHNKYAGESFPFIYDALFDIESKVDPSQAWGEVVKRQISIA
FTTVQAAAETLSEVA (SEQ ID NO: 216)

Figure 12

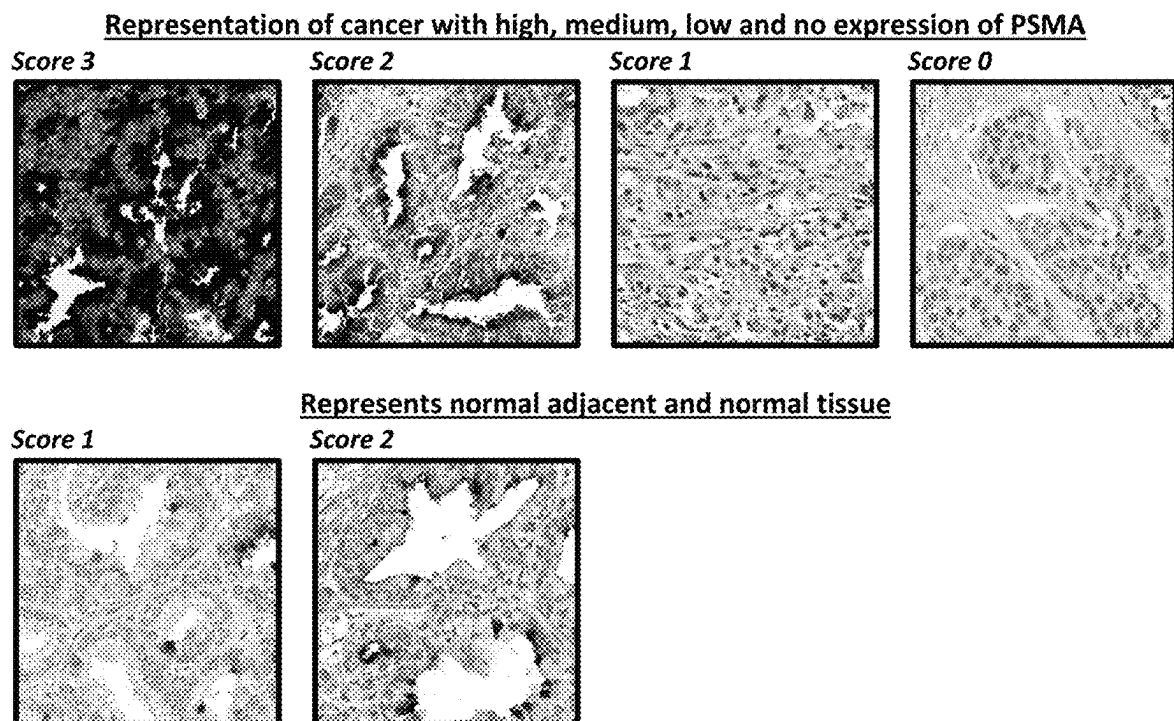


Figure 13

	(n)	PREVALENCE			
		Score 3	Score 2	Score 1	Score 0
		% (n)	% (n)	% (n)	% (n)
PROSTATE ADENOCARCINOMA	160	55% (88)	28% (45)	14% (23)	3% (4)
NORMAL ADJACENT PROSTATE TISSUE	16	0% (0)	56% (9)	44% (7)	0% (0)
NORMAL PROSTATE TISSUE	16	13% (2)	50% (8)	38% (6)	0% (0)
TOTAL CORES	192				

Figure 14

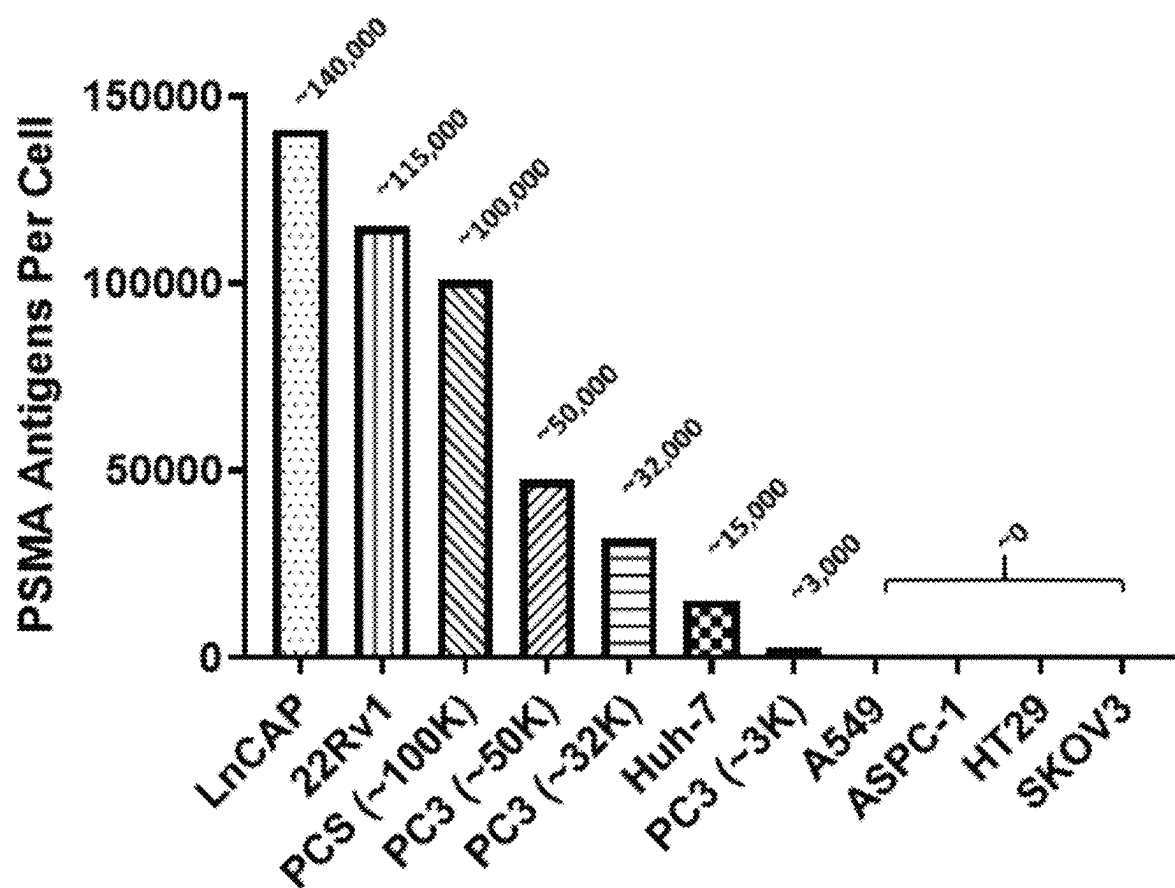


Figure 15

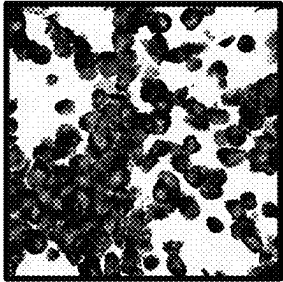
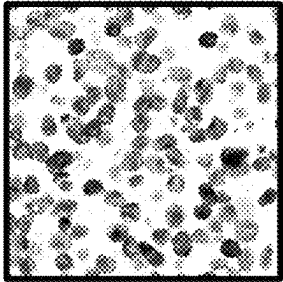
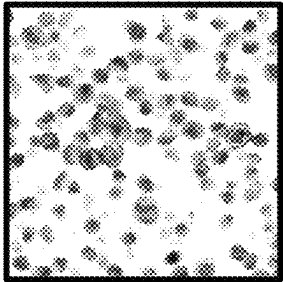
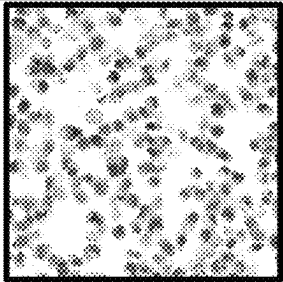
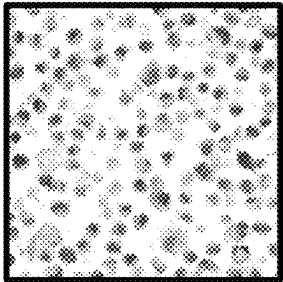
Score 3	LNCaP (~140K PSMA)	
Score 2	PC3 (~100K PSMA)	
Score 1	PC3 (~50K/~32K PSMA)	
	Huh-7 (~15K PSMA)	
Score 0	PC3 (~3K PSMA)	

Figure 16

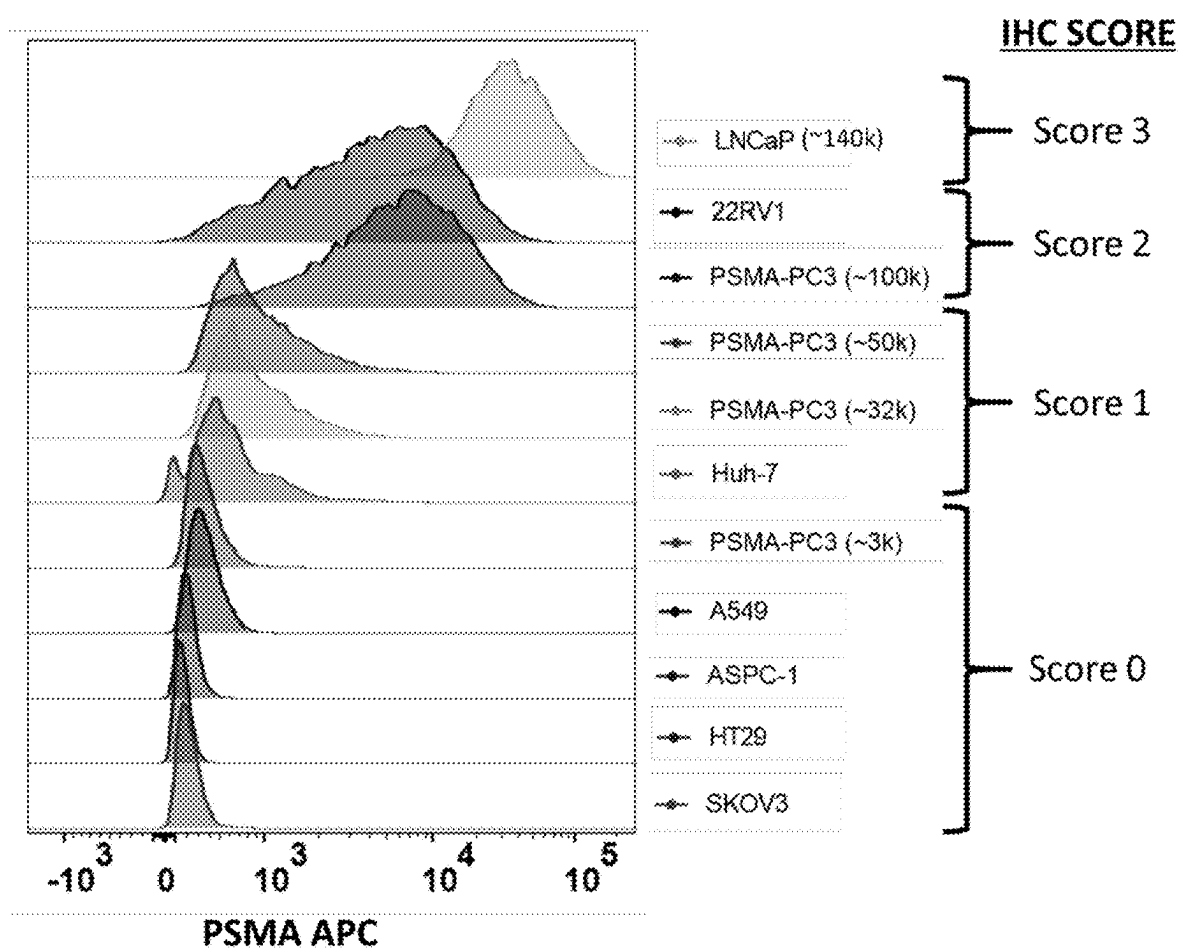


Figure 17

PSMA-H H1L1

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYMELSSLSRSEDVAVYCAAGWNFDYWGQGTLLVTVSS	217
vhCDR1	EYTIH	218
vhCDR2	NINPNNGGTTYNQKFQ	219
vhCDR3	GWNFDY	220
Variable light (vl) domain	DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISSLQAEDVAVYFCQQYNSYPLTFGAGTKVEIK	221
vlCDR1	RASQDVGTAVD	222
vlCDR2	WASTRHT	223
vlCDR3	QQYNSYPLT	224

>XENP31858 PSMA-H H1L1 IgG1 PVA /S267K**Chain 1 - PSMA-H_H1_IgG1_PVA_/S267K Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYMELSSLSRSEDVAVYCAAGWNFDYWGQGTLLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCTPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 225)

Chain 2 - PSMA-H_L1 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISSLQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 226)

>XENP31604 PSMA H H1L1 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYMELSSLSRSEDVAVYCAAGWNFDYWGQGTLLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCTPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 227)

Chain 2 - PSMA-H_L1 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISSLQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 228)

Figure 18A

PSMA-H_L1.1 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASNDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 229)

PSMA-H_L1.2 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASEDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 230)

PSMA-H_L1.3 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 231)

PSMA-H_L1.4 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASSEDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 232)

PSMA-H_L1.5 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASIDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 233)

PSMA-H_L1.6 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGSADVYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 234)

PSMA-H_L1.7 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGNADVYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 235)

PSMA-H_L1.8 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGEADVYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 236)

PSMA-H_L1.9 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGIADVYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 237)

PSMA-H_L1.10 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTIVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 238)

PSMA-H_L1.11 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTSDVYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 239)

PSMA-H_L1.12 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTIVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 240)

PSMA-H_L1.13 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTIVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 241)

PSMA-H_L1.14 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTGVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 242)

PSMA-H_L1.15 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTQVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 243)

PSMA-H_L1.16 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTQVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 244)

PSMA-H_L1.17 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTALDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 245)

Figure 18B

PSMA-H_ L1.18 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 246)

PSMA-H_ L1.19 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVAWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 247)

PSMA-H_ L1.20 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIFWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 248)

PSMA-H_ L1.21 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIQWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 249)

PSMA-H_ L1.22 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIEWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 250)

PSMA-H_ L1.23 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIHWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 251)

PSMA-H_ L1.24 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYYASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 252)

PSMA-H_ L1.25 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYFASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 253)

PSMA-H_ L1.26 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYHASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 254)

PSMA-H_ L1.27 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYQASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 255)

PSMA-H_ L1.28 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYEASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 256)

PSMA-H_ L1.29 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYTASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 257)

PSMA-H_ L1.30 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWISTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 258)

PSMA-H_ L1.31 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWLSTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 259)

PSMA-H_ L1.32 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWTSTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 260)

PSMA-H_ L1.33 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASSRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 261)

PSMA-H_ L1.34 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASQRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 262)

Figure 18C

PSMA-H_L1.35 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASERHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 263)

PSMA-H_L1.36 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASNRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 264)

PSMA-H_L1.37 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASGRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 265)

PSMA-H_L1.38 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRETGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 266)

PSMA-H_L1.39 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHSVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 267)

PSMA-H_L1.40 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHEGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 268)

PSMA-H_L1.41 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHYGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 269)

PSMA-H_L1.42 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCNQYNSYPLTFGAGTKVEIK (SEQ ID NO: 270)

PSMA-H_L1.43 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCEQYNSYPLTFGAGTKVEIK (SEQ ID NO: 271)

PSMA-H_L1.44 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQYNSYPLTFGAGTKVEIK (SEQ ID NO: 272)

PSMA-H_L1.45 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCRQYNSYPLTFGAGTKVEIK (SEQ ID NO: 273)

PSMA-H_L1.46 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCLQYNSYPLTFGAGTKVEIK (SEQ ID NO: 274)

PSMA-H_L1.47 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCHQYNSYPLTFGAGTKVEIK (SEQ ID NO: 275)

PSMA-H_L1.48 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQFNSYPLTFGAGTKVEIK (SEQ ID NO: 276)

PSMA-H_L1.49 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQWNSYPLTFGAGTKVEIK (SEQ ID NO: 277)

PSMA-H_L1.50 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQKNSYPLTFGAGTKVEIK (SEQ ID NO: 278)

PSMA-H_L1.51 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQRNSYPLTFGAGTKVEIK (SEQ ID NO: 279)

Figure 18D

PSMA-H_ L1.52 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQHNSYPLTFGAGTKVEIK (SEQ ID NO: 280)

PSMA-H_ L1.53 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQNSYPLTFGAGTKVEIK (SEQ ID NO: 281)

PSMA-H_ L1.54 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQNSYPLTFGAGTKVEIK (SEQ ID NO: 282)

PSMA-H_ L1.55 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQINSYPLTFGAGTKVEIK (SEQ ID NO: 283)

PSMA-H_ L1.56 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQNSYPLTFGAGTKVEIK (SEQ ID NO: 284)

PSMA-H_ L1.57 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYYSYPLTFGAGTKVEIK (SEQ ID NO: 285)

PSMA-H_ L1.58 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYQSYPLTFGAGTKVEIK (SEQ ID NO: 286)

PSMA-H_ L1.59 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYESYPLTFGAGTKVEIK (SEQ ID NO: 287)

PSMA-H_ L1.60 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYGSYPLTFGAGTKVEIK (SEQ ID NO: 288)

PSMA-H_ L1.61 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYTSYPLTFGAGTKVEIK (SEQ ID NO: 289)

PSMA-H_ L1.62 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSTPLTFGAGTKVEIK (SEQ ID NO: 290)

PSMA-H_ L1.63 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSFPLTFGAGTKVEIK (SEQ ID NO: 291)

PSMA-H_ L1.64 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSWPLTFGAGTKVEIK (SEQ ID NO: 292)

PSMA-H_ L1.65 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSHPLTFGAGTKVEIK (SEQ ID NO: 293)

PSMA-H_ L1.66 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSQPLTFGAGTKVEIK (SEQ ID NO: 294)

PSMA-H_ L1.67 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSEPLTFGAGTKVEIK (SEQ ID NO: 295)

PSMA-H_ L1.68 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSKPLTFGAGTKVEIK (SEQ ID NO: 296)

Figure 18E

PSMA-H_L1.69 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSRPLTFGAGTKVEIK (SEQ ID NO: 297)

PSMA-H_L1.70 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSGPLTFGAGTKVEIK (SEQ ID NO: 298)

PSMA-H_L1.71 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPITFGAGTKVEIK (SEQ ID NO: 299)

PSMA-H_L1.72 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPFTFGAGTKVEIK (SEQ ID NO: 300)

PSMA-H_L1.73 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPKTFGAGTKVEIK (SEQ ID NO: 301)

PSMA-H_L1.74 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPTFGAGTKVEIK (SEQ ID NO: 302)

PSMA-H_L1.75 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPQTFGAGTKVEIK (SEQ ID NO: 303)

PSMA-H_L1.76 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYYASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPQTFGAGTKVEIK (SEQ ID NO: 304)

PSMA-H_L1.77 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTSVDWYQQKPDQSPKLLIYYASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 305)

PSMA-H_L1.78 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYYASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQHNSYPLTFGAGTKVEIK (SEQ ID NO: 306)

PSMA-H_L1.79 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYHASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPQTFGAGTKVEIK (SEQ ID NO: 307)

PSMA-H_L1.80 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTSVDWYQQKPDQSPKLLIYHASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 308)

PSMA-H_L1.81 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYHASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQHNSYPLTFGAGTKVEIK (SEQ ID NO: 309)

PSMA-H_L1.82 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYTASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPQTFGAGTKVEIK (SEQ ID NO: 310)

PSMA-H_L1.83 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTSVDWYQQKPDQSPKLLIYTASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK (SEQ ID NO: 311)

PSMA-H_L1.84 Variable Light

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYTASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQHNSYPLTFGAGTKVEIK (SEQ ID NO: 312)

Figure 19A

>XENP31618 PSMA-H H1 L1.1 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 313)

Chain 2 - PSMA-H_L1.1 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASNDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
314)

>XENP31619 PSMA-H H1 L1.2 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 315)

Chain 2 - PSMA-H_L1.2 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASEDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
316)

>XENP31620 PSMA-H H1 L1.3 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 317)

Chain 2 - PSMA-H_L1.3 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASNDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
318)

Figure 19B

>XENP31621 PSMA-H H1 L1.4 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 319)

Chain 2 - PSMA-H_L1.4 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASSDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
320)

>XENP31622 PSMA-H H1 L1.5 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 321)

Chain 2 - PSMA-H_L1.5 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASIDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
322)

>XENP31623 PSMA-H H1 L1.6 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 323)

Chain 2 - PSMA-H_L1.6 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGSADVWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
324)

Figure 19C

>XENP31624 PSMA-H H1 L1.7 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 325)

Chain 2 - PSMA-H_L1.7 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGNVDWYQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
326)

>XENP31625 PSMA-H H1 L1.8 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 327)

Chain 2 - PSMA-H_L1.8 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGEAVDWYQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
328)

>XENP31626 PSMA-H H1 L1.9 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 329)

Chain 2 - PSMA-H_L1.9 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGIADVWYQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
330)

Figure 19D

>XENP31627 PSMA-H H1 L1.10 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 331)

Chain 2 - PSMA-H_L1.10 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTVDWYQOKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
332)

>XENP31628 PSMA-H H1 L1.11 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 333)

Chain 2 - PSMA-H_L1.11 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTSDWYQOKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
334)

>XENP31629 PSMA-H H1 L1.12 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 335)

Chain 2 - PSMA-H_L1.12 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTVDWYQOKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
336)

Figure 19E

>XENP31630 PSMA-H H1 L1.13 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 337)

Chain 2 - PSMA-H_L1.13 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTIVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
338)

>XENP31631 PSMA-H H1 L1.14 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 339)

Chain 2 - PSMA-H_L1.14 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTGVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
340)

>XENP31632 PSMA-H H1 L1.15 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 341)

Chain 2 - PSMA-H_L1.15 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
342)

Figure 19F

>XENP31633 PSMA-H H1 L1.16 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGLTVTVSS /ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTPPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 343)

Chain 2 - PSMA-H_L1.16 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTQVDWYQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
344)

>XENP31634 PSMA-H H1 L1.17 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGLTVTVSS /ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTPPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 345)

Chain 2 - PSMA-H_L1.17 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTALDWYQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
346)

>XENP31635 PSMA-H H1 L1.18 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGLTVTVSS /ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTPPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 347)

Chain 2 - PSMA-H_L1.18 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTALDWYQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
348)

Figure 19G

>XENP31636 PSMA-H H1 L1.19 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 349)

Chain 2 - PSMA-H_L1.19 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVAWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
350)

>XENP31637 PSMA-H H1 L1.20 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 351)

Chain 2 - PSMA-H_L1.20 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIFWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
352)

>XENP31638 PSMA-H H1 L1.21 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 353)

Chain 2 - PSMA-H_L1.21 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIQWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
354)

Figure 19H

>XENP31639 PSMA-H H1 L1.22 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 355)

Chain 2 - PSMA-H_L1.22 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIEWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
356)

>XENP31640 PSMA-H H1 L1.23 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 357)

Chain 2 - PSMA-H_L1.23 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIHWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
358)

>XENP31641 PSMA-H H1 L1.24 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 359)

Chain 2 - PSMA-H_L1.24 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYYASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
360)

Figure 19I

>XENP31642 PSMA-H H1 L1.25 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 361)

Chain 2 - PSMA-H_L1.25 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYFASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
362)

>XENP31643 PSMA-H H1 L1.26 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 363)

Chain 2 - PSMA-H_L1.26 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYHASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
364)

>XENP31644 PSMA-H H1 L1.27 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 365)

Chain 2 - PSMA-H_L1.27 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYQASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
366)

Figure 19J

>XENP31645 PSMA-H H1 L1.28 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 367)

Chain 2 - PSMA-H_L1.28 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYEASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
368)

>XENP31646 PSMA-H H1 L1.29 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 369)

Chain 2 - PSMA-H_L1.29 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYTASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
370)

>XENP31647 PSMA-H H1 L1.30 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 371)

Chain 2 - PSMA-H_L1.30 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWISTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
372)

Figure 19K

>XENP31648 PSMA-H H1 L1.31 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 373)

Chain 2 - PSMA-H_L1.31 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWLSTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
374)

>XENP31649 PSMA-H H1 L1.32 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 375)

Chain 2 - PSMA-H_L1.32 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWTSTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
376)

>XENP31650 PSMA-H H1 L1.33 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 377)

Chain 2 - PSMA-H_L1.33 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASSRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
378)

Figure 19L

>XENP31651 PSMA-H H1 L1.34 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 379)

Chain 2 - PSMA-H_L1.34 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASQRHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
380)

>XENP31652 PSMA-H H1 L1.35 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 381)

Chain 2 - PSMA-H_L1.35 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASERHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
382)

>XENP31653 PSMA-H H1 L1.36 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 383)

Chain 2 - PSMA-H_L1.36 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASNRHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
384)

Figure 19M

>XENP31654 PSMA-H H1 L1.37 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 385)

Chain 2 - PSMA-H_L1.37 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASGRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
386)

>XENP31655 PSMA-H H1 L1.38 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 387)

Chain 2 - PSMA-H_L1.38 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRETGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
388)

>XENP31656 PSMA-H H1 L1.39 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 389)

Chain 2 - PSMA-H_L1.39 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHSGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
390)

Figure 19N

>XENP31657 PSMA-H H1 L1.40 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 391)

Chain 2 - PSMA-H_L1.40 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHEGVDPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
392)

>XENP31658 PSMA-H H1 L1.41 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 393)

Chain 2 - PSMA-H_L1.41 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHYGVDPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
394)

>XENP31659 PSMA-H H1 L1.42 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 395)

Chain 2 - PSMA-H_L1.42 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVDPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCNQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
396)

Figure 190

>XENP31660 PSMA-H H1 L1.43 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 397)

Chain 2 - PSMA-H_L1.43 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCEQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
398)

>XENP31661 PSMA-H H1 L1.44 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 399)

Chain 2 - PSMA-H_L1.44 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCKQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
400)

>XENP31662 PSMA-H H1 L1.45 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 401)

Chain 2 - PSMA-H_L1.45 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCRQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
402)

Figure 19P

>XENP31663 PSMA-H H1 L1.46 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGLTVTVSS /ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNATKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTPPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 403)

Chain 2 - PSMA-H_L1.46 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCIQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
404)

>XENP31664 PSMA-H H1 L1.47 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGLTVTVSS /ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNATKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTPPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 405)

Chain 2 - PSMA-H_L1.47 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCHQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
406)

>XENP31665 PSMA-H H1 L1.48 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGLTVTVSS /ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNATKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTPPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 407)

Chain 2 - PSMA-H_L1.48 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQFNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
408)

Figure 19Q

>XENP31666 PSMA-H H1 L1.49 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED~~TAVYYCAAGWNFDYWGQGT~~LVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEK~~TISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP~~
ENNYKTTPPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 409)

Chain 2 - PSMA-H_L1.49 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQWNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
410)

>XENP31667 PSMA-H H1 L1.50 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED~~TAVYYCAAGWNFDYWGQGT~~LVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEK~~TISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP~~
ENNYKTTPPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 411)

Chain 2 - PSMA-H_L1.50 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQKNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
412)

>XENP31668 PSMA-H H1 L1.51 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED~~TAVYYCAAGWNFDYWGQGT~~LVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEK~~TISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP~~
ENNYKTTPPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 413)

Chain 2 - PSMA-H_L1.51 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQRNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
414)

Figure 19R

>XENP31669 PSMA-H H1 L1.52 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 415)

Chain 2 - PSMA-H_L1.52 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQHNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
416)

>XENP31670 PSMA-H H1 L1.53 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 417)

Chain 2 - PSMA-H_L1.53 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
418)

>XENP31671 PSMA-H H1 L1.54 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 419)

Chain 2 - PSMA-H_L1.54 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
420)

Figure 19S

>XENP31672 PSMA-H H1 L1.55 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGLTVTVSS /ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 421)

Chain 2 - PSMA-H_L1.55 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQINSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
422)

>XENP31673 PSMA-H H1 L1.56 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGLTVTVSS /ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 423)

Chain 2 - PSMA-H_L1.56 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQINSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
424)

>XENP31674 PSMA-H H1 L1.57 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGLTVTVSS /ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 425)

Chain 2 - PSMA-H_L1.57 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYYSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
426)

Figure 19T

>XENP31675 PSMA-H H1 L1.58 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 427)

Chain 2 - PSMA-H_L1.58 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYQSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
428)

>XENP31676 PSMA-H H1 L1.59 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 429)

Chain 2 - PSMA-H_L1.59 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYESYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
430)

>XENP31677 PSMA-H H1 L1.60 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 431)

Chain 2 - PSMA-H_L1.60 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYQSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
432)

Figure 19U

>XENP31678 PSMA-H H1 L1.61 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGLTVTVSS /ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 433)

Chain 2 - PSMA-H_L1.61 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYTSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
434)

>XENP31679 PSMA-H H1 L1.62 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGLTVTVSS /ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 435)

Chain 2 - PSMA-H_L1.62 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSTPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
436)

>XENP31680 PSMA-H H1 L1.63 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGLTVTVSS /ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 437)

Chain 2 - PSMA-H_L1.63 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSTPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
438)

Figure 19V

>XENP31681 PSMA-H H1 L1.64 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 439)

Chain 2 - PSMA-H_L1.64 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNWPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
440)

>XENP31682 PSMA-H H1 L1.65 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 441)

Chain 2 - PSMA-H_L1.65 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNHPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
442)

>XENP31683 PSMA-H H1 L1.66 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 443)

Chain 2 - PSMA-H_L1.66 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSQPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
444)

Figure 19W

>XENP31684 PSMA-H H1 L1.67 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 445)

Chain 2 - PSMA-H_L1.67 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSEPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
446)

>XENP31685 PSMA-H H1 L1.68 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 447)

Chain 2 - PSMA-H_L1.68 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSKPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
448)

>XENP31686 PSMA-H H1 L1.69 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 449)

Chain 2 - PSMA-H_L1.69 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSRPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
450)

Figure 19X

>XENP31687 PSMA-H H1 L1.70 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 451)

Chain 2 - PSMA-H_L1.70 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSGPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
452)

>XENP31688 PSMA-H H1 L1.71 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 453)

Chain 2 - PSMA-H_L1.71 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPIITFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
454)

>XENP31689 PSMA-H H1 L1.72 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 455)

Chain 2 - PSMA-H_L1.72 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYFFTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
456)

Figure 19Y

>XENP31690 PSMA-H H1 L1.73 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 457)

Chain 2 - PSMA-H_L1.73 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPKTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
458)

>XENP31691 PSMA-H H1 L1.74 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 459)

Chain 2 - PSMA-H_L1.74 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPRTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
460)

>XENP31692 PSMA-H H1 L1.75 IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S Heavy Chain**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 461)

Chain 2 - PSMA-H_L1.75 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYFQTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
462)

Figure 20A

	VL Variant	Alias	Response	K _{Dapp} (M)	k _a (1/Ms)	k _d (1/s)
XENP31604	WT		0.3098	1.07E-09	6.52E+04	6.97E-05
XENP31618	Q27N	PSMA-H_L1.1	0.2072	2.47E-09	6.25E+04	1.54E-04
XENP31619	Q27E	PSMA-H_L1.2	0.3441	<1.0E-12	5.40E+04	<1.0E-07
XENP31620	Q27T	PSMA-H_L1.3	0.2884	5.76E-10	6.19E+04	3.57E-05
XENP31621	Q27S	PSMA-H_L1.4	0.3648	2.73E-10	6.20E+04	1.69E-05
XENP31622	Q27I	PSMA-H_L1.5	0.2117	1.68E-09	6.74E+04	1.13E-04
XENP31623	T31S	PSMA-H_L1.6	0.228	2.24E-10	5.70E+04	1.28E-05
XENP31624	T31N	PSMA-H_L1.7	0.3866	1.73E-09	6.80E+04	1.17E-04
XENP31625	T31E	PSMA-H_L1.8	0.2067	1.67E-09	5.18E+04	8.66E-05
XENP31626	T31I	PSMA-H_L1.9	0.2802	<1.0E-12	7.91E+04	<1.0E-07
XENP31627	A32Y	PSMA-H_L1.10	0.0159	4.43E-09	3.74E+05	1.66E-03
XENP31628	A32S	PSMA-H_L1.11	0.341	4.83E-09	6.06E+04	2.93E-04
XENP31629	A32T	PSMA-H_L1.12	0.2621	2.43E-11	6.73E+04	1.64E-06
XENP31630	A32I	PSMA-H_L1.13	0.0414	2.14E-08	7.96E+04	1.70E-03
XENP31631	A32G	PSMA-H_L1.14	0.1416	6.60E-09	6.53E+04	4.31E-04
XENP31632	A32E	PSMA-H_L1.15	0.0548	1.40E-08	7.84E+04	1.10E-03
XENP31633	A32Q	PSMA-H_L1.16	0.0922	5.62E-09	1.05E+05	5.92E-04
XENP31634	V33L	PSMA-H_L1.17	0.1006	<1.0E-12	6.81E+04	<1.0E-07
XENP31635	V33I	PSMA-H_L1.18	0.076	<1.0E-12	7.65E+04	<1.0E-07
XENP31636	D44A	PSMA-H_L1.19	0.2206	1.40E-09	7.17E+04	1.01E-04
XENP31637	Y49F	PSMA-H_L1.20	0.1277	<1.0E-12	6.61E+04	<1.0E-07
XENP31638	Y49Q	PSMA-H_L1.21	0.1606	5.40E-09	4.04E+04	2.18E-04
XENP31639	Y49E	PSMA-H_L1.22	0.2036	1.30E-08	5.97E+04	7.78E-04
XENP31640	Y49H	PSMA-H_L1.23	0.2286	<1.0E-12	4.77E+04	<1.0E-07
XENP31641	W50Y	PSMA-H_L1.24	0.2198	1.38E-08	6.49E+04	8.97E-04
XENP31642	W50F	PSMA-H_L1.25	0.1664	1.06E-08	5.30E+04	5.62E-04
XENP31643	W50H	PSMA-H_L1.26	0.343	1.38E-08	7.21E+04	9.94E-04
XENP31644	W50Q	PSMA-H_L1.27	0.252	6.78E-09	6.62E+04	4.49E-04
XENP31645	W50E	PSMA-H_L1.28	0.0984	1.29E-08	6.77E+04	8.75E-04
XENP31646	W50T	PSMA-H_L1.29	0.2325	1.15E-08	1.07E+05	1.23E-03
XENP31647	A51I	PSMA-H_L1.30	0.0365	<1.0E-12	6.07E+04	<1.0E-07
XENP31648	A51L	PSMA-H_L1.31	0.0404	3.11E-10	1.13E+05	3.51E-05
XENP31649	A51T	PSMA-H_L1.32	0.1163	<1.0E-12	9.36E+04	<1.0E-07
XENP31650	T53S	PSMA-H_L1.33	0.3225	1.27E-09	9.28E+04	1.18E-04
XENP31651	T53Q	PSMA-H_L1.34	0.3587	1.23E-09	8.38E+04	1.03E-04
XENP31652	T53E	PSMA-H_L1.35	0.2914	<1.0E-12	7.30E+04	<1.0E-07
XENP31653	T53N	PSMA-H_L1.36	0.4066	1.36E-09	8.02E+04	1.09E-04
XENP31654	T53G	PSMA-H_L1.37	0.2885	<1.0E-12	9.41E+04	<1.0E-07

Figure 20B

	<u>VL Variant</u>	<u>Alias</u>	<u>Response</u>	<u>K_{Dapp} (M)</u>	<u>k_a (1/Ms)</u>	<u>k_d (1/s)</u>
XENP31655	H55E	PSMA-H_L1.38	0.3526	<1.0E-12	8.63E+04	<1.0E-07
XENP31656	T56S	PSMA-H_L1.39	0.2726	<1.0E-12	7.92E+04	<1.0E-07
XENP31657	T56E	PSMA-H_L1.40	0.3795	<1.0E-12	7.56E+04	<1.0E-07
XENP31658	T56Y	PSMA-H_L1.41	0.2865	<1.0E-12	7.64E+04	<1.0E-07
XENP31659	Q89N	PSMA-H_L1.42	0.2547	2.60E-09	6.37E+04	1.66E-04
XENP31660	Q89E	PSMA-H_L1.43	0.1808	3.80E-09	7.84E+04	2.98E-04
XENP31661	Q89K	PSMA-H_L1.44	0.3067	7.08E-09	6.15E+04	4.35E-04
XENP31662	Q89R	PSMA-H_L1.45	0.3409	2.13E-08	9.54E+04	2.03E-03
XENP31663	Q89L	PSMA-H_L1.46	0.2259	1.41E-08	5.15E+04	7.25E-04
XENP31664	Q89H	PSMA-H_L1.47	0.2491	2.44E-09	6.91E+04	1.69E-04
XENP31665	Y91F	PSMA-H_L1.48	0.2539	8.34E-11	8.32E+04	6.94E-06
XENP31666	Y91W	PSMA-H_L1.49	0.0325	<1.0E-12	1.87E+05	<1.0E-07
XENP31667	Y91K	PSMA-H_L1.50	0.0782	1.23E-08	8.59E+04	1.06E-03
XENP31668	Y91R	PSMA-H_L1.51	0.0173	1.35E-09	6.10E+05	8.21E-04
XENP31669	Y91H	PSMA-H_L1.52	0.1363	1.75E-08	1.00E+05	1.75E-03
XENP31670	Y91Q	PSMA-H_L1.53	0.0877	3.58E-09	9.64E+04	3.45E-04
XENP31671	Y91E	PSMA-H_L1.54	0.0909	<1.0E-12	6.33E+04	<1.0E-07
XENP31672	Y91I	PSMA-H_L1.55	0.027	1.29E-09	2.30E+05	2.96E-04
XENP31673	Y91L	PSMA-H_L1.56	0.0729	1.15E-09	1.11E+05	1.27E-04
XENP31674	N92Y	PSMA-H_L1.57	0.3016	2.21E-09	8.61E+04	1.90E-04
XENP31675	N92Q	PSMA-H_L1.58	0.3372	5.48E-09	8.04E+04	4.40E-04
XENP31676	N92E	PSMA-H_L1.59	0.1993	1.17E-08	6.53E+04	7.65E-04
XENP31677	N92G	PSMA-H_L1.60	0.3819	2.30E-09	8.33E+04	1.91E-04
XENP31678	N92T	PSMA-H_L1.61	0.3012	1.20E-09	8.73E+04	1.05E-04
XENP31679	Y94T	PSMA-H_L1.62	0.2466	4.66E-09	6.57E+04	3.06E-04
XENP31680	Y94F	PSMA-H_L1.63	0.2898	7.33E-10	1.10E+05	8.06E-05
XENP31681	Y94W	PSMA-H_L1.64	0.3617	4.47E-09	1.11E+05	4.96E-04
XENP31682	Y94H	PSMA-H_L1.65	0.3498	<1.0E-12	8.57E+04	<1.0E-07
XENP31683	Y94Q	PSMA-H_L1.66	0.3608	3.59E-09	8.39E+04	3.01E-04
XENP31684	Y94E	PSMA-H_L1.67	0.2424	1.87E-08	8.27E+04	1.55E-03
XENP31685	Y94K	PSMA-H_L1.68	0.2431	1.96E-08	1.29E+05	2.54E-03
XENP31686	Y94R	PSMA-H_L1.69	0.296	1.77E-08	8.90E+04	1.57E-03
XENP31687	Y94G	PSMA-H_L1.70	0.1983	2.28E-08	6.59E+04	1.50E-03
XENP31688	L96I	PSMA-H_L1.71	0.3765	5.53E-09	7.34E+04	4.05E-04
XENP31689	L96F	PSMA-H_L1.72	0.2938	1.83E-08	7.26E+04	1.33E-03
XENP31690	L96K	PSMA-H_L1.73	0.2085	2.26E-08	1.05E+05	2.36E-03
XENP31691	L96R	PSMA-H_L1.74	0.0456	2.64E-09	1.67E+05	4.40E-04
XENP31692	L96Q	PSMA-H_L1.75	0.2063	3.32E-08	6.81E+04	2.26E-03

Figure 21A

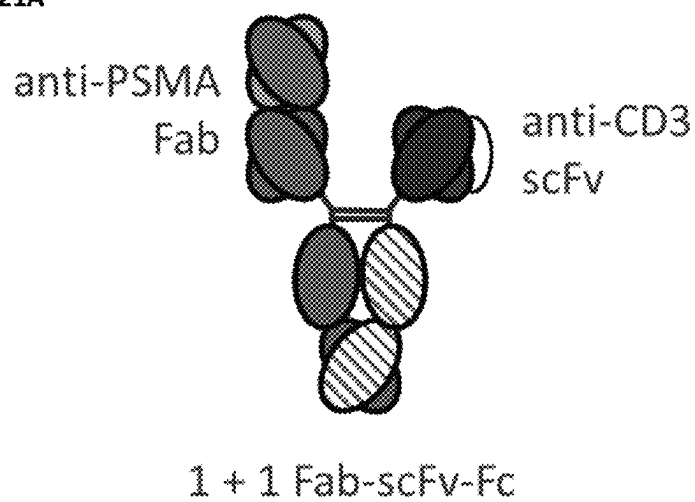


Figure 21B

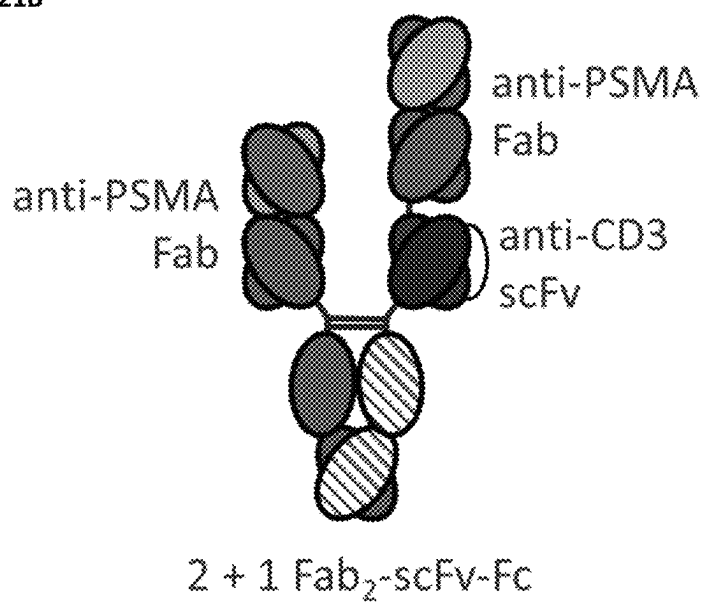


Figure 22A

>XENP14484 PSMA-H_H1L1_Fab-[CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)
_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTILVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGGP
ENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 463)

Chain 2 - [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_C220S/S364K/E357Q

EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKLEWVGRIRSKYNNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRRAEDTAVYYCVRHGNEFGDSYVSWFAYWGQGTILVTVSS/_GKPGSGKPGSGKPGSGKPGS/_QAVVTQE
PSLTVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTISGAQPED
EADYYCALWYSNHWFVGGGTKLTVL/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 464)

Chain 3 - PSMA-H_L1_Light Chain

DIVMTQSPDLSAVSLGERATLSCRASQDVGTAVDWYQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
465)

>XENP33755 PSMA-H_H1_L1.58_Fab-[CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)
_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTILVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGGP
ENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 466)

**Chain 2 - [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S-
IgG1_C220S/S364K/E357Q**

EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKLEWVGRIRSKYNNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRRAEDTAVYYCVRHGNEFGDSYVSWFAYWGQGTILVTVSS/_GKPGSGKPGSGKPGSGKPGS/_QAVVTQE
PSLTVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTISGAQPED
EADYYCALWYSNHWFVGGGTKLTVL/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 467)

Chain 3 - PSMA-H_L1.58_Light Chain

DIVMTQSPDLSAVSLGERATLSCRASQDVGTAVDWYQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYQSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
468)

Figure 22B

>XENP33756 PSMA-H_H1_L1.11 Fab-[CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDQGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 469)

Chain 2 - [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRAEDTAVYYCVRHGNGFGDSYVSWFAYWGQGLTVTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQE
PSLTVSPGGTTLTLCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGKKAALTISGAQPED
EADYYCALWYSNHWVFGGGTKLTVL/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVKLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 470)

Chain 3 - PSMA-H_L1.11 Light Chain

DIVMTQSPDLSAVSLGERATLSCRASQDVGTSDVDWYQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
471)

>XENP33757 PSMA-H_H1_L1.24 Fab-[CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDQGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 472)

Chain 2 - [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRAEDTAVYYCVRHGNGFGDSYVSWFAYWGQGLTVTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQE
PSLTVSPGGTTLTLCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGKKAALTISGAQPED
EADYYCALWYSNHWVFGGGTKLTVL/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVKLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 473)

Chain 3 - PSMA-H_L1.24 Light Chain

DIVMTQSPDLSAVSLGERATLSCRASQDVGTAVDWYQKPDQSPKLLIYASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
474)

Figure 22C

>XENP33758 PSMA-H H1 L1.26 Fab-[CD3] H1.30 L1.47 scFv(GKPGS)4 Fc(216) IgG1 PVA /S267K pl(-)
Isosteric A L368D/K370S-IgG1 C220S/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGSTTYNQKFQGRVTITVDKSTSTAYME
LSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPK
DTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPA
PIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDQGPNYKTTTPVLDSDGSFFLYSKLTV
DKSRWEQGDVFSQSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 475)

Chain 2 - [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S-
IgG1_C220S/S364K/E357Q

EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDSKNTLY
LQMNLSRAEDTAVYYCVRHGNFGDSYVSWFAYWGQGLTVTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQEPSTLVSPGGT
VTLTCCSSTGAVTTSNYANWVQKPKGKSPRGLIGGTNKRAPGVPARFSGSLGGAALTI SGAQPEDEADYYCALWYSNHWV
FGGGLTKLTVL/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNA
KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV
KGFYPSDIAVEWESNGQPNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSQSVMEALHNHYTQKSLSLSPGK
(SEQ ID NO: 476)

Chain 3 - PSMA-H_L1.26 Light Chain

DIVMTQSPDLSAVSLGERATLSRASQDVGTAVDWYQQKPDQSPKLLIYHASTRHTGVPDRFTGSGSGTDFTLTISLSQAED
VAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNALQSGNSQESV
TEQDSKDSSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 477)

>XENP33759 PSMA-H H1 L1.75 Fab-[CD3] H1.30 L1.47 scFv(GKPGS)4 Fc(216) IgG1 PVA /S267K pl(-)
Isosteric A L368D/K370S-IgG1 C220S/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGSTTYNQKFQGRVTITVDKSTSTAYME
LSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPK
DTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPA
PIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDQGPNYKTTTPVLDSDGSFFLYSKLTV
DKSRWEQGDVFSQSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 478)

Chain 2 - [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S-
IgG1_C220S/S364K/E357Q

EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDSKNTLY
LQMNLSRAEDTAVYYCVRHGNFGDSYVSWFAYWGQGLTVTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQEPSTLVSPGGT
VTLTCCSSTGAVTTSNYANWVQKPKGKSPRGLIGGTNKRAPGVPARFSGSLGGAALTI SGAQPEDEADYYCALWYSNHWV
FGGGLTKLTVL/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNA
KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV
KGFYPSDIAVEWESNGQPNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSQSVMEALHNHYTQKSLSLSPGK
(SEQ ID NO: 479)

Chain 3 - PSMA-H_L1.75 Light Chain

DIVMTQSPDLSAVSLGERATLSRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISLSQAED
VAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNALQSGNSQESV
TEQDSKDSSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 480)

Figure 22D

>XENP33760 PSMA-H_H1_L1.68 Fab-[CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S

EVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTIHWVRQAPGQSLWGMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDQGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 481)

Chain 2 - [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRAEDTAVYYCVRHGNGFGDSYVSWFAYWGQGLTVTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQE
PSLTVSPGGTTLTLCGSSTGAVTTSNYANWVQOKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGAALTISGAQPED
EADYYCALWYSNHWVFGGGTKLTVL/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVKLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 482)

Chain 3 - PSMA-H_L1.68 Light Chain

DIVMTQSPDLSAVSLGERATLSCRASQDVGTAVDWYQOKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSKPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
483)

>XENP33761 PSMA-H_H1_L1.29 Fab-[CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S

EVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTIHWVRQAPGQSLWGMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDQGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 484)

Chain 2 - [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRAEDTAVYYCVRHGNGFGDSYVSWFAYWGQGLTVTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQE
PSLTVSPGGTTLTLCGSSTGAVTTSNYANWVQOKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGAALTISGAQPED
EADYYCALWYSNHWVFGGGTKLTVL/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVKLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 485)

Chain 3 - PSMA-H_L1.29 Light Chain

DIVMTQSPDLSAVSLGERATLSCRASQDVGTAVDWYQOKPDQSPKLLIYTASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
486)

Figure 22E

>XENP33762 PSMA-H_H1_L1.52 Fab-[CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)
_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPE
NNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 487)

**Chain 2 - [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S-
IgG1_C220S/S364K/E357Q**

EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRAEDTAVYYCVRHGNFQDSYVSWFAYWGQGTLLVTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQE
PSLTVSPGGTTLTLCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGKAALTISGAQPED
EADYYCALWYSNHWVFGGGTKLTVL/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 488)

Chain 3 - PSMA-H_L1.52 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQHNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
489)

>XENP34234 PSMA-H_H1_L1.78 Fab-[CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)
_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPE
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 490)

**Chain 2 - [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S-
IgG1_C220S/S364K/E357Q**

EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRAEDTAVYYCVRHGNFQDSYVSWFAYWGQGTLLVTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQE
PSLTVSPGGTTLTLCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGKAALTISGAQPED
EADYYCALWYSNHWVFGGGTKLTVL/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 491)

Chain 3 - PSMA-H_L1.78 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYYASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQHNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
492)

Figure 22F

>XENP34235 PSMA-H_H1_L1.81_Fab-[CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGT LVT VSS /ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDQGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 493)

Chain 2 - [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRAEDTAVYYCVRHGNFGDSYVSWFAYWGQGT LVT VSS /GKPGSGKPGSGKPGSGKPGS/QAVVTQE
PSLT VSPGGT VTLT CGSSTGAVTTSNYANWVQOKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTISGAQPED
EADYYCALWYSNHWVFGGGTKLTVL/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVKLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 494)

Chain 3 - PSMA-H_L1.81 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTVAVDWYQOKPDQSPKLLIYHASTRHTGVPDRFTSGSGSDTFTLTIS
LQAEDVAVYFCQQHNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
495)

>XENP34236 PSMA-H_H1_L1.84_Fab-[CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGT LVT VSS /ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDQGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 496)

Chain 2 - [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRAEDTAVYYCVRHGNFGDSYVSWFAYWGQGT LVT VSS /GKPGSGKPGSGKPGSGKPGS/QAVVTQE
PSLT VSPGGT VTLT CGSSTGAVTTSNYANWVQOKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTISGAQPED
EADYYCALWYSNHWVFGGGTKLTVL/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVKLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 497)

Chain 3 - PSMA-H_L1.84 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTVAVDWYQOKPDQSPKLLIYTASTRHTGVPDRFTSGSGSDTFTLTIS
LQAEDVAVYFCQQHNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
498)

Figure 23

>XENP16873 PSMA-H_H1L1_Fab-[CD3]_H1.33_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)
_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSEDVAVYYCAAGWNFDYWGGTTLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 499)

Chain 2 - [CD3]_H1.33_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_C220S/S364K/E357Q

EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRAEDTAVYYCVRHGNFGDSYVSWFDYWGGTTLTVTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQE
PSLTVSPGGTVTLTCGSSTGAVTTSNYANWVQOKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGAALTISGAQPED
EADYYCALWYSNHWVFGGGTKLTVL/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 500)

Chain 3 - PSMA-H_L1 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSPVTKSFNRGEC (SEQ ID NO:
501)

Figure 24

>XENP16874 PSMA-H_H1L1_Fab-[CD3]_H1.31_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pl(-)
_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSEDVAVYYCAAGWNFDYWGGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 502)

Chain 2 - [CD3]_H1.31_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_C220S/S364K/E357Q

EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMSWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRAEDTAVYYCVRHGNFGDSYVSWFAYWGGTLVTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQE
PSLTVSPGGTVTLTCGSSTGAVTTSNYANWVQOKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGAALTISGAQPED
EADYYCALWYSNHWVFGGGTKLTVL/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 503)

Chain 3 - PSMA-H_L1 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEHKQVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
504)

Figure 25

>XENP19722 PSMA-H H1L1 Fab-[CD3] H1.32 L1.47 scFv(GKPGS)4 Fc(216) IgG1 PVA /S267K pl(-)
) Isosteric A L368D/K370S-IgG1 C220S/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_PVA_/S267K_pl(-)_Isosteric_A_L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDVAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 505)

Chain 2 - [CD3]_H1.32_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_C220S/S364K/E357Q

EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRAEDTAVYYCVRHGNFGDSYVSWFAYWGQGTLLTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQE
PSLTVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTISGAQPED
EADYYCALWYSNHVVFVGGGKLTVL/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVKLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 506)

Chain 3 - PSMA-H_L1 Light Chain

DIVMTQSPDLSAVSLGERATLSRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTSGSGSDTFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCILNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSSTLTLSKADYEKKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
507)

Figure 26

>XENP31602 PSMA-H H1L1 Fab-PSMA-H H1L1 CH1 (G4S)2 [CD3] H1.30 L1.47 scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A /PVA /S267K/L368D/K370S-Fc(222) IgG1 PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSGDSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 508)

Chain 2 - PSMA-

H_H1_CH1_(G4S)2_[CD3]_H1.30_L1.47_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/
EVQLVESGGGLVQPGGSLRLSCAASGFTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRAEDTAVYYCVRHGNFGDSYVSWFAYWGQGLTVTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQE
PSLTVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTISGAQPED
EADYYCALWYSNHWVFGGGTKLTVL/GGGGSGGGGS/KTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPSREQMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSGDSFFLYSKLTVDKSR
WQQGNVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 509)

Chain 3 - PSMA-H_L1 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
510)

Figure 27

>XENP31603 PSMA-H H1L1 Fab-PSMA-H H1L1 CH1 (G4S)2 [CD3] H1.32 L1.47 scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A /PVA /S267K/L368D/K370S-Fc(222) IgG1 PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSGDSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 511)

Chain 2 - PSMA-

H_H1_CH1_(G4S)2_[CD3]_H1.32_L1.47_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/
EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRAEDTAVYYCVRHGNFGDSYVSWFAYWGQGLTVTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQE
PSLTVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTISGAQPED
EADYYCALWYSNHWVFGGGTKLTVL/GGGGSGGGGS/KTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPSREQMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSGDSFFLYSKLTVDKSR
WQQGNVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 512)

Chain 3 - PSMA-H_L1 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
513)

Figure 28A

>XENP31855 PSMA-H H1L1 Fab-PSMA-H H1L1 CH1 (G4S)2 [CD3] L1.47 H1.32 scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A /PVA /S267K/L368D/K370S-Fc(222) IgG1 PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCRTSGYTFTEYTIHWVRQAPGQSLWGMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEEQGVDFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 514)

Chain 2 - PSMA-H_H1_CH1_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q

EVQLVQSGAEVKKPGASVKVSCRTSGYTFTEYTIHWVRQAPGQSLWGMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/SGGGSGGGGS/
QAVVTQEPSTLTVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGDSYVSWFAYWGQGTLLTVSS/SGGGSGGGGS/KTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFIYPSDIAVEWESNGQFENNYKTTTPVLDSDGSFFLYSKLTVDKSR
WQQGNVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 515)

Chain 3 - PSMA-H_L1 Light Chain

DIVMTQSPDLSAVSLGERATLSCRASQDVGTAVDWYQKPKDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKKHYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
516)

>XENP32218 PSMA-H H1 L1.58 Fab-PSMA-H H1 L1.58 Fab (G4S) [CD3] L1.47 H1.32 scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A /PVA /S267K/L368D/K370S-Fc(222) IgG1 C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCRTSGYTFTEYTIHWVRQAPGQSLWGMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEEQGVDFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 517)

Chain 2 - PSMA-

H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv(GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q

EVQLVQSGAEVKKPGASVKVSCRTSGYTFTEYTIHWVRQAPGQSLWGMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/SGGGSGGGGS/
QAVVTQEPSTLTVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGDSYVSWFAYWGQGTLLTVSS/SGGGSGGGGS/KTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFIYPSDIAVEWESNGQFENNYKTTTPVLDSDGSFFLYSKLTVDKSR
WQQGNVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 518)

Figure 28B

Chain 3 - PSMA-H_L1.58 Light Chain

DIVMTQSPDLSAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYQSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
519)

>XENP32219 PSMA-H_H1_L1.11 Fab-PSMA-H_H1_L1.11 Fab (G4S) [CD3] L1.47_H1.32_scFv(GKPGS)4 (G4S)2-
IgG1_pl(-) Isosteric A /PVA /S267K/L368D/K370S-Fc(222) IgG1_C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTIKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 520)

Chain 2 - PSMA-**H_H1_(G4S)_ [CD3]_L1.47_H1.32_scFv(GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q**

EVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKEPKSC/SGGGSGGGGS/
QAVVTQEPESLTVPSPGGTVTLTCSSTGAVTTSNYANWVQQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/ EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGDSYVSWFAYWGQGLTLTVSS/SGGGSGGGGS/KTHTCTPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTIKAK
KGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSR
WQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 521)

Chain 3 - PSMA-H_L1.11 Light Chain

DIVMTQSPDLSAVSLGERATLSCRASQDVGTSVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
522)

>XENP32220 PSMA-H_H1_L1.24 Fab-PSMA-H_H1_L1.24 Fab (G4S) [CD3] L1.47_H1.32_scFv(GKPGS)4 (G4S)2-
IgG1_pl(-) Isosteric A /PVA /S267K/L368D/K370S-Fc(222) IgG1_C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTIKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 523)

Figure 28C

Chain 2 - PSMA-

H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv(GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/
QAVVTQEPSTLVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNEFGDSYVSWFAYWGQTLVTVSS/GGGGSGGGGS/KTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPSREOMTKNQVCLTCLVKGFPYSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSR
WQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 524)

Chain 3 - PSMA-H_L1.24 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYYASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKKHYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
525)

>XENP32221 PSMA-H_H1_L1.26 Fab-PSMA-H_H1_L1.26 Fab_(G4S)_[CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2-
IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S-Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDFVSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 526)

Chain 2 - PSMA-

H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv(GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/
QAVVTQEPSTLVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNEFGDSYVSWFAYWGQTLVTVSS/GGGGSGGGGS/KTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPSREOMTKNQVCLTCLVKGFPYSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSR
WQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 527)

Chain 3 - PSMA-H_L1.26 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYYASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKKHYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
528)

Figure 28D

>XENP32222 PSMA-H H1 L1.75 Fab-PSMA-H H1 L1.75 Fab (G4S) [CD3] L1.47 H1.32_scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A /PVA /S267K/L368D/K370S-Fc(222) IgG1 C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A/PVA/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYME
LSSLRSEDVAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVTVFPSSSLGTQTYICNVNHKPSDTKVDKKEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPK
DTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPA
PIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGGQPENNYKTTTPVLDSDGSFFLYSKLTV
DKSRWEQGDVDFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 529)

Chain 2 - PSMA-H_H1_(G4S)_ [CD3]_L1.47_H1.32_scFv(GKPGS)4_Fc(222)_IgG1_C220S/PVA/S267K/S364K/E357Q

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYME
LSSLRSEDVAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVTVFPSSSLGTQTYICNVNHKPSNTKVDKKEPKSC/SGGGSGGGGS/QAVVTQEPSTLVSPG
GTVTLTSGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNH
WVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWV
GRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNFGDSYVSWFAIWGQGLTVTVSS/SGGG
SGGGGS/KTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYN
STYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAV
EWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQGGNVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
530)

Chain 3 - PSMA-H_L1.75 Light Chain

DIVMTQSPDLSAVSLGERATLSRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISSLQAED
VAVYFCQQYNSYPQTTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNALQSGNSQESV
TEQDSKDSSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 531)

>XENP32223 PSMA-H H1 L1.68 Fab-PSMA-H H1 L1.68 Fab (G4S) [CD3] L1.47 H1.32_scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A /PVA /S267K/L368D/K370S-Fc(222) IgG1 C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A/PVA/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYME
LSSLRSEDVAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVTVFPSSSLGTQTYICNVNHKPSDTKVDKKEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPK
DTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPA
PIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGGQPENNYKTTTPVLDSDGSFFLYSKLTV
DKSRWEQGDVDFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 532)

Chain 2 - PSMA-H_H1_(G4S)_ [CD3]_L1.47_H1.32_scFv(GKPGS)4_Fc(222)_IgG1_C220S/PVA/S267K/S364K/E357Q

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYME
LSSLRSEDVAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVTVFPSSSLGTQTYICNVNHKPSNTKVDKKEPKSC/SGGGSGGGGS/QAVVTQEPSTLVSPG
GTVTLTSGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNH
WVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWV
GRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNFGDSYVSWFAIWGQGLTVTVSS/SGGG
SGGGGS/KTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYN
STYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAV
EWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQGGNVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
533)

Figure 28E

Chain 3 - PSMA-H_L1.68 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSKPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
534)

>XENP32224 PSMA-H_H1_L1.29 Fab-PSMA-H_H1_L1.29 Fab (G4S) [CD3] L1.47_H1.32_scFv(GKPGS)4 (G4S)2-
IgG1_pl(-) Isosteric A /PVA /S267K/L368D/K370S-Fc(222) IgG1_C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVVKHEDPEVKFNWYVDGVEVHNAAKTPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTIKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 535)

Chain 2 - PSMA-**H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv(GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/SGGGSGGGGS/
QAVVTQEPESLTVPSPGGTVTLTCSSTGAVTTSNYANWVQQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLS
CAASGFTFTSTYAMNWVRQAPGKGLVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRRAEDTAVYYCVR
HGNFEGDSYVSWFAYWGQGLTVTVSS/SGGGSGGGGS/KTHTCTPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVKHEDPEVKFNWYVDGVEVHNAAKTPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTIKAK
KGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSR
WQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 536)

Chain 3 - PSMA-H_L1.29 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYTASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
537)

>XENP32225 PSMA-H_H1_L1.52 Fab-PSMA-H_H1_L1.52 Fab (G4S) [CD3] L1.47_H1.32_scFv(GKPGS)4 (G4S)2-
IgG1_pl(-) Isosteric A /PVA /S267K/L368D/K370S-Fc(222) IgG1_C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVVKHEDPEVKFNWYVDGVEVHNAAKTPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTIKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 538)

Figure 28F

Chain 2 - PSMA-**H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv(GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/
QAVVTQEPSTLVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNEFGDSYVSWFAYWGGTLVTVSS/GGGGSGGGGS/KTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPSREOMTKNQVCLTCLVKGFPYSDIAVEWESNGQFENNYKTTTPVLDSDGSFFLYSKLTVDKSR
WQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 539)

Chain 3 - PSMA-H_L1.52 Light Chain

DIVMTQSPDLSAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQHNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKKHYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
540)

**>XENP32226 PSMA-H_H1_L1.13 Fab-PSMA-H_H1_L1.13 Fab_(G4S)_[CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2-
IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S-Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q**

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREOMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDFVSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 541)

Chain 2 - PSMA-**H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv(GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/
QAVVTQEPSTLVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNEFGDSYVSWFAYWGGTLVTVSS/GGGGSGGGGS/KTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPSREOMTKNQVCLTCLVKGFPYSDIAVEWESNGQFENNYKTTTPVLDSDGSFFLYSKLTVDKSR
WQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 542)

Chain 3 - PSMA-H_L1.13 Light Chain

DIVMTQSPDLSAVSLGERATLSCRASQDVGTIVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDEVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKKHYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
543)

Figure 28G

>XENP34237 PSMA-H H1 L1.78 Fab-PSMA-H H1 L1.78 Fab (G4S) [CD3] L1.47 H1.32 scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A /PVA /S267K/L368D/K370S-Fc(222) IgG1 C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYME
LSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPK
DTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPA
PIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGPENNYKTTTPVLDSGDSFFLYSKLTV
DKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 544)

Chain 2 - PSMA-

H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYME
LSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKEPKSC/SGGGSGGGGS/QAVVTQEPSTLVSPG
GTVTLTCSGSTGAVTTSNYANWVQKPKSPRGLIGGTNRRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNH
WVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKLEWV
GRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNGFSDSYVSWFAYWGQGTLLTVSS/SGGG
SGGGGS/KTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYN
STYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREQMTKNQVLTCLVKGFYPSDIAV
EWESNGQPENNYKTTTPVLDSGDSFFLYSKLTVDKSRWQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
545)

Chain 3 - PSMA-H_L1.78 Light Chain

DIVMTQSPDLSAVSLGERATLSRASQDVGTAVDWYQQKPDQSPKLLIYYASTRHTGVPDRFTGSGSGTDFTLTITSSLQAED
VAVYFCQHNSYPLTFAGATKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESV
TEQDSKDSYSLSSLTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 546)

>XENP34238 PSMA-H H1 L1.81 Fab-PSMA-H H1 L1.81 Fab (G4S) [CD3] L1.47 H1.32 scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A /PVA /S267K/L368D/K370S-Fc(222) IgG1 C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYME
LSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPK
DTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPA
PIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGPENNYKTTTPVLDSGDSFFLYSKLTV
DKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 547)

Chain 2 - PSMA-

H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYME
LSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKEPKSC/SGGGSGGGGS/QAVVTQEPSTLVSPG
GTVTLTCSGSTGAVTTSNYANWVQKPKSPRGLIGGTNRRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNH
WVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKLEWV
GRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNGFSDSYVSWFAYWGQGTLLTVSS/SGGG
SGGGGS/KTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYN
STYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREQMTKNQVLTCLVKGFYPSDIAV
EWESNGQPENNYKTTTPVLDSGDSFFLYSKLTVDKSRWQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
548)

Figure 28H

Chain 3 - PSMA-H_L1.81 Light Chain

DIVMTQSPDLSAVSLGERATLSRASQDVGTAVDWYQQKPDQSPKLLIYHASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQHNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
549)

>XENP34239 PSMA-H H1 L1.84 Fab-PSMA-H H1 L1.84 Fab (G4S) [CD3] L1.47 H1.32 scFv(GKPGS)4 (G4S)2-
IgG1 pl(-) Isosteric A /PVA /S267K/L368D/K370S-Fc(222) IgG1 C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNATKPREEEYNSTYRVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSGDSFFLYSKLTVDKSRWEQGDVFCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 550)

Chain 2 - PSMA-

H_H1_(G4S)_ [CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q

EVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/
QAVVTQEPSTLTVSPGGTVTLTCCGSSTGAVTTSNYANWVQQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/ EVQLVESGGGLVQPGGSLRLSCA
ASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYADSVKGRFTISRDDSKNTLYLQMNSLRADTAVYYCVR
HGNFGDSYVSWFAYWGQGTLLTVSS/GGGGSGGGGS/KTHTCPPCPAPFVAGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVKHEDPEVKFNWYVDGVEVHNATKPREEQYNSTYRVSVLTVLHQLDNLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSGDSFFLYSKLTVDKSR
WQQGNVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 551)

Chain 3 - PSMA-H_L1.84 Light Chain

DIVMTQSPDLSAVSLGERATLSRASQDVGTAVDWYQQKPDQSPKLLIYTASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQHNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
552)

>XENP34625 PSMA-H H1L1 Fab-PSMA-H H1L1 Fab (G4S) [CD3] L1.47 H1.32 scFv(GKPGS)4 (G4S)2-IgG1 pl(-)
Isosteric A /PVA /S267K/L368D/K370S/M428L/N434S-
Fc(222) IgG1 C220S/PVA /S267K/S364K/E357Q/M428L/N434S

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S/M428L/N434S

EVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNATKPREEEYNSTYRVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSGDSFFLYSKLTVDKSRWEQGDVFCSSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 553)

Figure 28i

Chain 2 - PSMA-

H_H1L1_Fab_(G4S)_[CD3]_L1.47_H1.32_scFv(GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q/M428L/N434S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/SGGGSGGGGS/
QAVVTQEPSTLVSPGGTVTLTCSSTGAVTTSNYANWVQKPKGSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLS
CAASGFTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYC
VVRHGNFGDSYVSWFAYWGQGLTVTVSS/SGGGSGGGGS/KTHCTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEV
TCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVSVLTVHLQDNLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPSREQMTKNQVLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSR
WQQGNVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 554)

Chain 3 - PSMA-H_L1 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
SLQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
555)

**>XENP34626 PSMA-H_H1_L1.58_Fab-PSMA-H_H1_L1.58_Fab_(G4S)_[CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2-
IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S/M428L/N434S-
Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q/M428L/N434S**

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S/M428L/N434S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVSVLTVHL
QDNLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDQGP
ENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 556)

Chain 2 - PSMA-

H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv(GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q/M428L/N434S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/SGGGSGGGGS/
QAVVTQEPSTLVSPGGTVTLTCSSTGAVTTSNYANWVQKPKGSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLS
CAASGFTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYC
VVRHGNFGDSYVSWFAYWGQGLTVTVSS/SGGGSGGGGS/KTHCTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEV
TCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVSVLTVHLQDNLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPSREQMTKNQVLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSR
WQQGNVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 557)

Chain 3 - PSMA-H_L1.58 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
SLQAEDVAVYFCQQYQSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
558)

Figure 28J

>XENP34627 PSMA-H H1 L1.24 Fab-PSMA-H H1 L1.24 Fab (G4S) [CD3] L1.47 H1.32 scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A /PVA /S267K/L368D/K370S/M428L/N434S-Fc(222) IgG1 C220S/PVA /S267K/S364K/E357Q/M428L/N434S

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S/M428L/N434S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVVKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 559)

Chain 2 - PSMA-

H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv(GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q/M428L/N434S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVVKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/CGGGSGGGGS/
QAVVTQEPSTLTVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKGLEWVGRIKSKANNYATYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFCDISYVSWFAFYWCQCTLTVSS/CGGGSGGGGS/KTHTCTPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPSREMQTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSR
WQQGNVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 560)

Chain 3 - PSMA-H_L1.24 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYYASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
561)

>XENP34628 PSMA-H H1 L1.29 Fab-PSMA-H H1 L1.29 Fab (G4S) [CD3] L1.47 H1.32 scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A /PVA /S267K/L368D/K370S/M428L/N434S-Fc(222) IgG1 C220S/PVA /S267K/S364K/E357Q/M428L/N434S

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S/M428L/N434S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVVKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCTPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 562)

Figure 28K

Chain 2 - PSMA-

H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv(GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q/M428L/N434S
EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED~~TAVYYCAAGWNFDYWGQGLTVTVSS~~/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/SGGGSGGGGS/
QAVVTQEPSTLVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTI
SGAQPEDEADYYCALWYSNHWVF~~GGG~~TKLTVL/~~GKPGSGKPGSGKPGSGKPGS~~/EVQLVESGGGLVQPGGSLRLSCA
ASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGDSYVSWFAYWGQGLTVTVSS/SGGGSGGGGS/KTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYITLPPSREQMTKNQVKLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSR
WQQGNVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 563)

Chain 3 - PSMA-H_L1.29 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYTASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
564)

Figure 29

>XENP31853 PSMA-H H1L1 Fab-PSMA-H H1L1 CH1 (G4S)2 [CD3] H1.89 L1.47 scFv(GKPGS)4 (G4S)2-IgG1_pl(-) Isosteric A /PVA /S267K/L368D/K370S-Fc(222) IgG1 PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTPPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 565)

Chain 2 - PSMA-H_H1_CH1_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/
EVQLVESGGGLVQPGGSLRLSCAASGFTFTSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRAEDTAVYYCVRHGNFGDEYVSWFAYWGQGTLLTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQE
PSLTVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGKAALTISGAQPED
EADYYCALWYSNHWVFGGGTKLTVL/GGGGSGGGGS/KTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSR
WQQGNVFSVCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 566)

Chain 3 - PSMA-H_L1 Light Chain

DIVMTQSPDLSAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
567)

Figure 30A

>XENP31856 PSMA-H H1L1 Fab-PSMA-H H1L1 CH1 (G4S)2 [CD3] L1.47 H1.89 scFv(GKPGS)4-IgG1 pl(-)
)_Isosteric A /PVA /S267K/L368D/K370S-Fc(216) IgG1 PVA /S267K/S364K/E357Q

**Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S-
Fc(216)_IgG1_PVA_/S267K/S364K/E357Q**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGGP
ENNYKTPPVLDSDGSFFLYSKLTVDKSRWEEQGVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 568)

**Chain 2 - PSMA-H_H1_CH1_(G4S)2_[CD3]_L1.47_H1.89_scFv(GKPGS)4-IgG1_pl(-)
)_Isosteric_A_/PVA_/S267K/L368D/K370S-Fc(216)_IgG1_PVA_/S267K/S364K/E357Q**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/
QAVVTQEPSTLTVSPGGTITLTCGSSTGAVTTSNYANWVQKPKGSPRGLIGGTNERAPGVPARFSGSLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATFYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGEDEYVSWFAYWGQGTLLTVSS/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 569)

Chain 3 - PSMA-H_L1 Light Chain

DIIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSPVTKSFNRGEC (SEQ ID NO:
570)

>XENP33063 PSMA-H H1 L1.58 Fab-PSMA-H H1 L1.58 Fab (G4S) [CD3] L1.47 H1.89 scFv(GKPGS)4-IgG1 pl(-)
)_Isosteric A /PVA /S267K/L368D/K370S-Fc(216) IgG1 C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGGP
ENNYKTPPVLDSDGSFFLYSKLTVDKSRWEEQGVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 571)

Figure 30B

Chain 2 - PSMA-**H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/
QAVVTQEPSTLTVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGDEYVSWFAYWGQGTLLTVSS/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVKLTCLVKGFPYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 572)

Chain 3 - PSMA-H_L1.58 Light Chain

DIVMTQSPDLSAVSLGERATLSRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYQSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
573)

**>XENP33064 PSMA-H_H1_L1.11_Fab-PSMA-H_H1_L1.11_Fab (G4S) [CD3]_L1.47_H1.89_scFv(GKPGS)4-IgG1_pl(-)
_Isosteric_A_/PVA_/S267K/L368D/K370S-Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q**

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEYNYSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 574)

Chain 2 - PSMA-**H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/
QAVVTQEPSTLTVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGDEYVSWFAYWGQGTLLTVSS/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVKLTCLVKGFPYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 575)

Chain 3 - PSMA-H_L1.11 Light Chain

DIVMTQSPDLSAVSLGERATLSRASQDVGTSDVWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTISS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
576)

Figure 30C

>XENP33065 PSMA-H H1 L1.24 Fab-PSMA-H H1 L1.24 Fab (G4S) [CD3] L1.47 H1.89 scFv(GKPGS)4-IgG1 pl(-)
_Isosteric A /PVA /S267K/L368D/K370S-Fc(216) IgG1 C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A/PVA/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYME
LSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPK
DTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPA
PIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGGQPENNYKTTTPVLDSDGSFFLYSKLTV
DKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 577)

Chain 2 - PSMA-H_H1_(G4S)_ [CD3]_L1.47_H1.89_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA/S267K/S364K/E357Q

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYME
LSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/QAVVTQEPSTLVSPG
GTVTLTCGSSGTAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNH
WVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/ EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWV
GRIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNFGDEYVSWFAYWGQGLTVTVSS/EPKS
SDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRV
VSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 578)

Chain 3 - PSMA-H_L1.24 Light Chain

DIIVMTQSPDLSAVSLGERATLSRASQDVCTAVDWYQQKPDQSPKLLIYYASTRHTGVPDRFTCGSGCTDFTLTISSLQAED
VAVYFCQQYNSYELTFAGATKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESV
TEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 579)

>XENP33066 PSMA-H H1 L1.26 Fab-PSMA-H H1 L1.26 Fab (G4S) [CD3] L1.47 H1.89 scFv(GKPGS)4-IgG1 pl(-)
_Isosteric A /PVA /S267K/L368D/K370S-Fc(216) IgG1 C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A/PVA/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYME
LSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPK
DTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPA
PIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGGQPENNYKTTTPVLDSDGSFFLYSKLTV
DKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 580)

Chain 2 - PSMA-H_H1_(G4S)_ [CD3]_L1.47_H1.89_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA/S267K/S364K/E357Q

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYME
LSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/QAVVTQEPSTLVSPG
GTVTLTCGSSGTAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNH
WVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/ EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWV
GRIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNFGDEYVSWFAYWGQGLTVTVSS/EPKS
SDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRV
VSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 581)

Figure 30D

Chain 3 - PSMA-H₁L1.26 Light Chain

DIVMTQSPDLSAVSLGERATLSRASQDVGTAVDWYQQKPDQSPKLLIYHASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
582)

>XENP33067 PSMA-H H1 L1.75 Fab-PSMA-H H1 L1.75 Fab (G4S) [CD3] L1.47 H1.89 scFv(GKPGS)4-IgG1 pl(-)
Isosteric A /PVA /S267K/L368D/K370S-Fc(216) IgG1 C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H₁ IgG1_{pl(-)} Isosteric A /PVA /S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSEDVAVYCAAGWNFDYWGQGTILVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSPDIAVEWESDGGP
ENNYKTTTPVLDSGDSFFLYSKLTVDKSRWEQGDVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 583)

Chain 2 - PSMA-**H₁ (G4S) [CD3] L1.47 H1.89 scFv(GKPGS)4 Fc(216) IgG1 C220S/PVA /S267K/S364K/E357Q**

EVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSEDVAVYCAAGWNFDYWGQGTILVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/
QAVVTQEPSTLTVSPGGTVTLTQGSSTGAVTTSNYANWVQQKPKGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGDEYVSWFAYWGQGTILVTVSS/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMKNQVLTCLVKGFYPSDIAVEWESNQPENNYKTTTPVLDSGDSFFLYSKLTVDKSRWQQGN
VFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 584)

Chain 3 - PSMA-H₁L1.75 Light Chain

DIVMTQSPDLSAVSLGERATLSRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
585)

>XENP33068 PSMA-H H1 L1.68 Fab-PSMA-H H1 L1.68 Fab (G4S) [CD3] L1.47 H1.89 scFv(GKPGS)4-IgG1 pl(-)
Isosteric A /PVA /S267K/L368D/K370S-Fc(216) IgG1 C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H₁ IgG1_{pl(-)} Isosteric A /PVA /S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSEDVAVYCAAGWNFDYWGQGTILVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSPDIAVEWESDGGP
ENNYKTTTPVLDSGDSFFLYSKLTVDKSRWEQGDVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 586)

Figure 30E

Chain 2 - PSMA-**H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTILTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/
QAVVTQEPSTLTVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGDEYVSWFAYWGQGTILTVSS/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVKLTCLVKGFPYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 587)

Chain 3 - PSMA-H_L1.68 Light Chain

DIIVMTQSPDSLAVSLGERATLSRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSKPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
588)

**>XENP33069 PSMA-H_H1_L1.29 Fab-PSMA-H_H1_L1.29 Fab (G4S) [CD3]_L1.47_H1.89_scFv(GKPGS)4-IgG1_pl(-)
) Isosteric A/PVA_/S267K/L368D/K370S-Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q**

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTILTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 589)

Chain 2 - PSMA-**H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTILTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/
QAVVTQEPSTLTVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGDEYVSWFAYWGQGTILTVSS/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVKLTCLVKGFPYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 590)

Chain 3 - PSMA-H_L1.29 Light Chain

DIIVMTQSPDSLAVSLGERATLSRASQDVGTAVDWYQQKPDQSPKLLIYASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
591)

Figure 30F

>XENP33070 PSMA-H H1 L1.52 Fab-PSMA-H H1 L1.52 Fab (G4S) [CD3] L1.47 H1.89 scFv(GKPGS)4-IgG1 pl(-)
_Isosteric A /PVA /S267K/L368D/K370S-Fc(216) IgG1 C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A/PVA/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYME
LSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPK
DTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPA
PIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGGQPENNYKTTTPVLDSDGSFFLYSKLTV
DKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 592)

Chain 2 - PSMA-H_H1_(G4S)_CD3_L1.47_H1.89_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA/S267K/S364K/E357Q

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYME
LSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/QAVVTQEPSTLVSPG
GTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNH
WVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWV
GRIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNFGDEYVSWFAYWGQGLTVTVSS/EPKS
SDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRV
VSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 593)

Chain 3 - PSMA-H_L1.52 Light Chain

DIVMTQSPDLSAVSLGERATLSRASQDVCTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTCGSGCTDFTLTISSLQAED
VAVYFCQQHNSYFLTFAGATKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESV
TEQDSKDSSTYLSLTSLTSLKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 594)

>XENP33071 PSMA-H H1 L1.13 Fab-PSMA-H H1 L1.13 Fab (G4S) [CD3] L1.47 H1.89 scFv(GKPGS)4-IgG1 pl(-)
_Isosteric A /PVA /S267K/L368D/K370S-Fc(216) IgG1 C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A/PVA/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYME
LSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPK
DTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPA
PIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGGQPENNYKTTTPVLDSDGSFFLYSKLTV
DKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 595)

Chain 2 - PSMA-H_H1_(G4S)_CD3_L1.47_H1.89_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA/S267K/S364K/E357Q

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTSTAYME
LSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/QAVVTQEPSTLVSPG
GTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNH
WVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWV
GRIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNFGDEYVSWFAYWGQGLTVTVSS/EPKS
SDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRV
VSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 596)

Figure 30G

Chain 3 - PSMA-H_L1.13 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTTVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
597)

>XENP34240 PSMA-H H1 L1.78 Fab-PSMA-H H1 L1.78 Fab (G4S) [CD3] L1.47 H1.89 scFv(GKPGS)4-IgG1 pl(-)
_Isosteric A /PVA /S267K/L368D/K370S-Fc(216) IgG1 C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 598)

Chain 2 - PSMA-**H_H1_(G4S)_ [CD3] L1.47_H1.89_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/ GGGGSGGGGS/
QAVVTQEPSTLTVSPGGTTLTCSSTGAVTTSNYANWVQQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGLKLTVL/GKPGSGKPGSGKPGSGKPGS/ EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKLEWVGRIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGDEYVSWFAYWGQGTLLTVSS/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 599)

Chain 3 - PSMA-H_L1.78 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYYASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQHNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
600)

>XENP34241 PSMA-H H1 L1.81 Fab-PSMA-H H1 L1.81 Fab (G4S) [CD3] L1.47 H1.89 scFv(GKPGS)4-IgG1 pl(-)
_Isosteric A /PVA /S267K/L368D/K370S-Fc(216) IgG1 C220S/PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 601)

Figure 30H

Chain 2 - PSMA-**H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/
QAVVTQEPSTLVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKGLEWVGRIIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGDEYVSWFAYWGQGTLLTVSS/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 602)

Chain 3 - PSMA-H_L1.81 Light Chain

DIIVMTQSPDLSAVSLGERATLSRASQDVGTAVDWYQQKPDQSPKLLIYHASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQHNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
603)

**>XENP34242 PSMA-H_H1_L1.84 Fab-PSMA-H_H1_L1.84 Fab (G4S) [CD3]_L1.47_H1.89_scFv(GKPGS)4-IgG1_pl(-)
) Isosteric A/PVA_/S267K/L368D/K370S-Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q**

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEYNYSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 604)

Chain 2 - PSMA-**H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q**

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/
QAVVTQEPSTLVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKGLEWVGRIIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGDEYVSWFAYWGQGTLLTVSS/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 605)

Chain 3 - PSMA-H_L1.84 Light Chain

DIIVMTQSPDLSAVSLGERATLSRASQDVGTAVDWYQQKPDQSPKLLIYASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQHNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
606)

Figure 30I

>XENP34629 PSMA-H H1L1 Fab-PSMA-H H1L1 Fab (G4S) [CD3] L1.47 H1.89 scFv(GKPGS)4-IgG1 pl(-)
Isosteric A /PVA /S267K/L368D/K370S/M428L/N434S-
Fc(216) IgG1 C220S/PVA /S267K/S364K/E357Q/M428L/N434S

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S/M428L/N434S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSGSSFFLYSKLTVDKSRWEQGDVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 607)

Chain 2 - PSMA-

H_H1L1_Fab_(G4S)_[CD3]_L1.47_H1.89_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q/M428L/N434S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/SGGGSGGGGS/
QAVVTQEPISLTVPGGTITLTCGSSSTGAVTTSNYANWVQKPKGKSPRGLIGGTNKRAPGVPARFSGSLGKKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/ EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKLEWVGRIIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGDEYVSWFAYWGQGTLLTVSS/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREMQMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSGSSFFLYSKLTVDKSRWQGN
VFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 608)

Chain 3 - PSMA-H_L1 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQKPKDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
609)

>XENP34630 PSMA-H H1 L1.58 Fab-PSMA-H H1 L1.58 Fab (G4S) [CD3] L1.47 H1.89 scFv(GKPGS)4-IgG1 pl(-)
Isosteric A /PVA /S267K/L368D/K370S/M428L/N434S-
Fc(216) IgG1 C220S/PVA /S267K/S364K/E357Q/M428L/N434S

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S/M428L/N434S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSGSSFFLYSKLTVDKSRWEQGDVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 610)

Figure 30J

Chain 2 - PSMA-

H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q/M428L/N434S
EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/SGGGSGGGGS/
QAVVTQEPSTLVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGDEYVSWFAYWGQGLTVTVSS/EPKSSDKHTCPGPCAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVCLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 611)

Chain 3 - PSMA-H_L1.58 Light Chain

DIVMTQSPDLSAVSLGERATLSRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYQSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYERHKVYACEVTHQGLSPVTKSFNRGEC (SEQ ID NO:
612)

**>XENP34631 PSMA-H_H1_L1.24 Fab-PSMA-H_H1_L1.24 Fab (G4S) [CD3]_L1.47_H1.89_scFv(GKPGS)4-IgG1 pl(-)
Isosteric A_/PVA_/S267K/L368D/K370S/M428L/N434S-
Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q/M428L/N434S**

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric A_/PVA_/S267K/L368D/K370S/M428L/N434S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPGPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQP
ENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 613)

Chain 2 - PSMA-

H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q/M428L/N434S
EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGLTVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/SGGGSGGGGS/
QAVVTQEPSTLVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGDEYVSWFAYWGQGLTVTVSS/EPKSSDKHTCPGPCAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVCLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 614)

Chain 3 - PSMA-H_L1.24 Light Chain

DIVMTQSPDLSAVSLGERATLSRASQDVGTAVDWYQQKPDQSPKLLIYASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYERHKVYACEVTHQGLSPVTKSFNRGEC (SEQ ID NO:
615)

Figure 30K

>XENP34632 PSMA-H H1 L1.29 Fab-PSMA-H H1 L1.29 Fab (G4S) [CD3] L1.47 H1.89 scFv(GKPGS)4-IgG1 pl(-)
) Isosteric A /PVA /S267K/L368D/K370S/M428L/N434S-
Fc(216) IgG1 C220S/PVA /S267K/S364K/E357Q/M428L/N434S

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S/M428L/N434S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGGQP
ENNYKTPPVLDSDGSFFLYSKLTVDKSRWEEQGVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 616)

Chain 2 - PSMA-

H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q/M428L/N434S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSSLRSED TAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/SGGGSGGGGS/
QAVVTQEPSTLVSPGGTVTLTSGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGQTKLTVL/GKPGSGKPGSGKPGSGKPGS/ EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKGLEWVGRIIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGDEYVSWFAYWGQGTLLTVSS/EPKSSDKTHTCPPCPAPFVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVKLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 617)

Chain 3 - PSMA-H_L1.29 Light Chain

DIVMTQSPDSLAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYTASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
618)

Figure 31

>XENP31854 PSMA-H H1L1 Fab-PSMA-H H1L1 CH1 (G4S)2 [CD3] H1.33 L1.47 scFv(GKPGS)4 (G4S)2-IgG1_pl(-) Isosteric A /PVA /S267K/L368D/K370S-Fc(222) IgG1 PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTPPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 619)

Chain 2 - PSMA-H_H1_CH1_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/
EVQLVESGGGLVQPGGSLRLSCAASGFTFTSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRAEDTAVYYCVRHGNFGDSYVSWFDYWGQGTLLTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQE
PSLTVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGKAALTISGAQPED
EADYYCALWYSNHWVFGGGTKLTVL/GGGGSGGGGS/KTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSR
WQQGNVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 620)

Chain 3 - PSMA-H_L1 Light Chain

DIVMTQSPDLSAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
621)

Figure 32

>XENP31857 PSMA-H H1L1 Fab-PSMA-H H1L1 CH1 (G4S)2 [CD3] L1.47 H1.33 scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A /PVA /S267K/L368D/K370S-Fc(222) IgG1 PVA /S267K/S364K/E357Q

Chain 1 - PSMA-H_H1_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAP
PVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLH
QDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGP
ENNYKTPPVLDSDGSFFLYSKLTVDKSRWEEQGDVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 622)

Chain 2 - PSMA-H_H1_CH1_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q

EVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTIHWVRQAPGQSLEWMGNINPNNGGTTYNQKFQGRVTITVDKSTS
TAYMELSSLRSEDTAVYYCAAGWNFDYWGQGTLLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGSGGGGS/
QAVVTQEPSTLTVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNGFDSYVSWFDYWGQGTLLVTVSS/GGGGSGGGGS/KTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPSREQMTKNQVKLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSR
WQQGNVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 623)

Chain 3 - PSMA-H_L1 Light Chain

DIVMTQSPDLSAVSLGERATLSCRASQDVGTAVDWYQQKPDQSPKLLIYWASTRHTGVPDRFTGSGSGTDFTLTIS
LQAEDVAVYFCQQYNSYPLTFGAGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKKHVKYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
624)

Figure 33

XENP34282

EVQLVESGGGLVQPGGSLTSLCAASRFMISEYHMHVVRQAPGKGLEWVSTINPAGTTDYAESVKGRFTISRDNAKNT
LYLQMNSLKPEDTAVYYCDSYGYRGQGTQVTVSSGGGGSGGGSEVQLVESGGGLVQPGNSLRSLCAASGFTFSKFGM
SWVRQAPGKGLEWVSSISGSGRDTLYADSVKGRFTISRDNAKTTLYLQMNSLRPEDTAVYYCTIGGSLSVSSQGTLV
TVSSGGGGSGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLEWVARIRSKYNNYATYYAD
QVKDRFTISRDDSKNTAYLQMNNLKTEDTAVYYCVRHANFGNSYISYWAYWGQGTTLTVSS/GGGGSGGGGS/GGGG
SQTVVVTQEPSLTVSPGGTVTLTCASSTGAVTSGNYPNWVQQKPGQAPRGLIGGTFKFLVPGTPARFSGSLLGGKAALT
LSGVQPEDEAEYYCTLWYSNRWVFGGGTKLTVLHHHHHH (SEQ ID NO: 625)

XENP34283

QVQLVESGGGLVKPGESLRSLCAASGFTFSDDYYMYWVRQAPGKGLEWVAIISDGGYYTYYSDIKGRFTISRDNAKN
SLYLQMNSLKAEDTAVYYCARGFPLLRHGAMDYWGQGTTLTVSS/GGGGSGGGGS/GGGGSDIQMTQSPSSLSASVG
DRVITITCKASQNVDTNVAWYQQKPGQAPKSLIYSASYRYSADVPSRFSGSASGTDFTLTISVQSEDFATYYCQQYDS
YPYTFGGGKLEIKSGGGGSEVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAMNWVRQAPGKGLEWVARIRSKYNN
YATYYADSVKDRFTISRDDSKNTAYLQMNNLKTEDTAVYYCVRHGNFGNSYISYWAYWGQGTTLTVSS/GGGGSGGG
GS/GGGGSQTVVTQEPSLTVSPGGTVTLTCGSSTGAVTSGNYPNWVQQKPGQAPRGLIGGTFKFLAPGTPARFSGSLL
GGKAALTLSGVQPEDEAEYYCVLWYSNRWVFGGGTKLTVLHHHHHH (SEQ ID NO: 626)

Figure 34

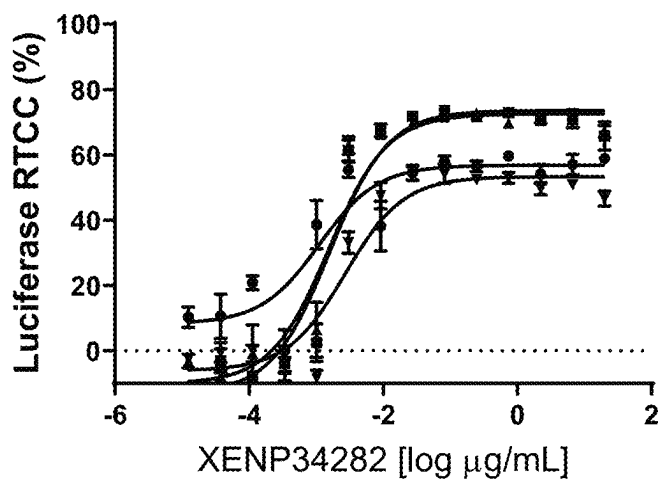
	<u>Alias</u>	<u>VL Variant</u>	<u>Response</u>	<u>K_D (M)</u>	<u>k_a (1/Ms)</u>	<u>k_e (1/s)</u>
XENP14484	PSMA-H_H1L1	WT				
XENP33755	PSMA-H_H1_L1.58	N92Q	1.8682	6.55E-09	1.22E+05	7.98E-04
XENP33756	PSMA-H_H1_L1.11	A32S	1.7929	1.82E-08	8.22E+04	1.50E-03
XENP33757	PSMA-H_H1_L1.24	W50Y	1.4435	3.83E-08	8.95E+04	3.43E-03
XENP33758	PSMA-H_H1_L1.26	W50H	1.4485	3.35E-08	9.78E+04	3.27E-03
XENP33759	PSMA-H_H1_L1.75	L96Q	1.2069	4.05E-08	8.46E+04	3.43E-03
XENP33760	PSMA-H_H1_L1.68	Y94K	1.2353	6.74E-08	1.41E+05	9.50E-03
XENP33761	PSMA-H_H1_L1.29	W50T	1.0526	8.38E-08	8.01E+04	6.71E-03
XENP33762	PSMA-H_H1_L1.52	Y91H	1.0485	9.37E-08	6.21E+04	5.82E-03
XENP34234	PSMA-H_H1_L1.78	W50Y Y91H	0.8422	1.16E-07	7.45E+04	8.63E-03
XENP34235	PSMA-H_H1_L1.81	W50H Y91H	1.2466	4.65E-08	7.30E+04	3.40E-03
XENP34236	PSMA-H_H1_L1.84	W50T Y91H	1.0943	5.58E-08	6.17E+04	3.44E-03

Figure 35

	<u>Alias</u>	<u>VL Variant</u>	<u>Response</u>	<u>K_D (M)</u>	<u>k_a (1/Ms)</u>	<u>k_e (1/s)</u>
XENP14484	PSMA-H_H1L1	WT	1.8513	4.75E-09	9.24E+04	4.39E-04
XENP33755	PSMA-H_H1_L1.58	N92Q	1.7798	7.41E-09	1.20E+05	8.92E-04
XENP33756	PSMA-H_H1_L1.11	A32S	1.6794	8.80E-08	6.05E+04	5.32E-03
XENP33757	PSMA-H_H1_L1.24	W50Y	1.4148	4.37E-08	8.51E+04	3.72E-03
XENP33758	PSMA-H_H1_L1.26	W50H	1.3752	5.05E-08	7.99E+04	4.04E-03
XENP33759	PSMA-H_H1_L1.75	L96Q	1.4129	NA	NA	NA
XENP33760	PSMA-H_H1_L1.68	Y94K	1.5594	1.48E-08	1.17E+05	1.73E-03
XENP33761	PSMA-H_H1_L1.29	W50T	1.0724	1.52E-07	6.56E+04	9.96E-03
XENP33762	PSMA-H_H1_L1.52	Y91H	0.9257	NA	NA	NA
XENP34234	PSMA-H_H1_L1.78	W50Y Y91H	0.76	1.26E-07	8.60E+04	1.08E-02
XENP34235	PSMA-H_H1_L1.81	W50H Y91H	1.2655	5.75E-08	7.94E+04	4.56E-03
XENP34236	PSMA-H_H1_L1.84	W50T Y91H	1.0696	7.56E-08	6.41E+04	4.85E-03

Figure 36

A)

IHC Score 2

● PC3 (~100k)

IHC Score 1

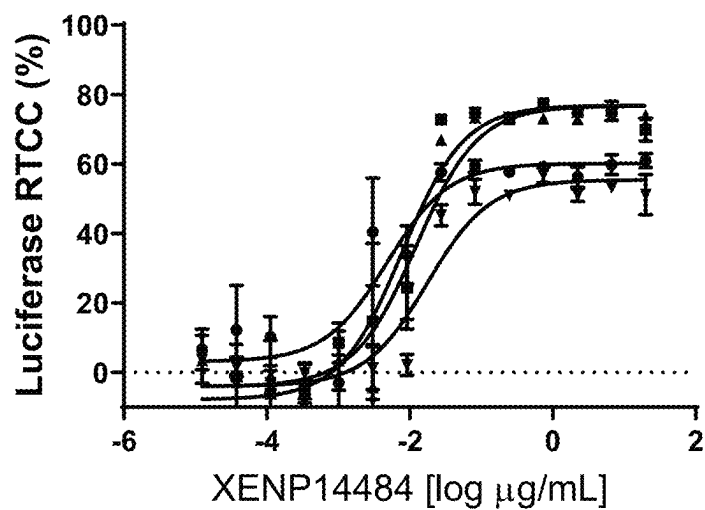
■ PC3 (~50k)

▲ PC3 (~32k)

IHC Score 0

▼ PC3 (~3k)

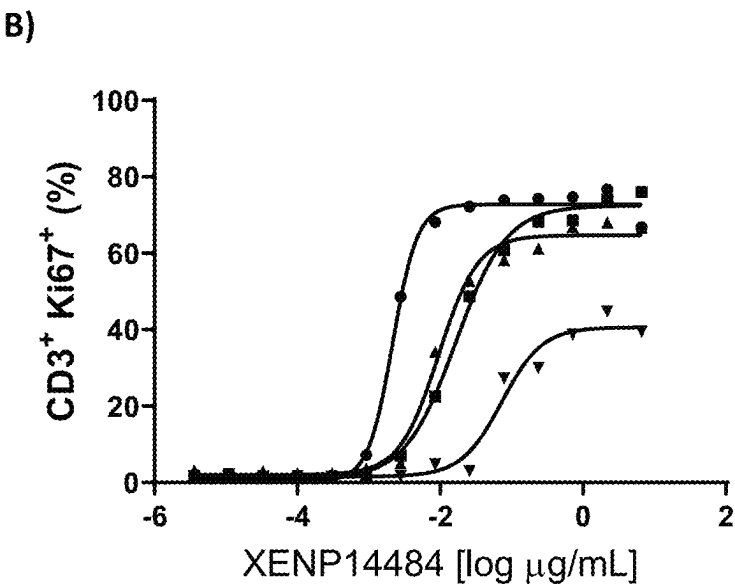
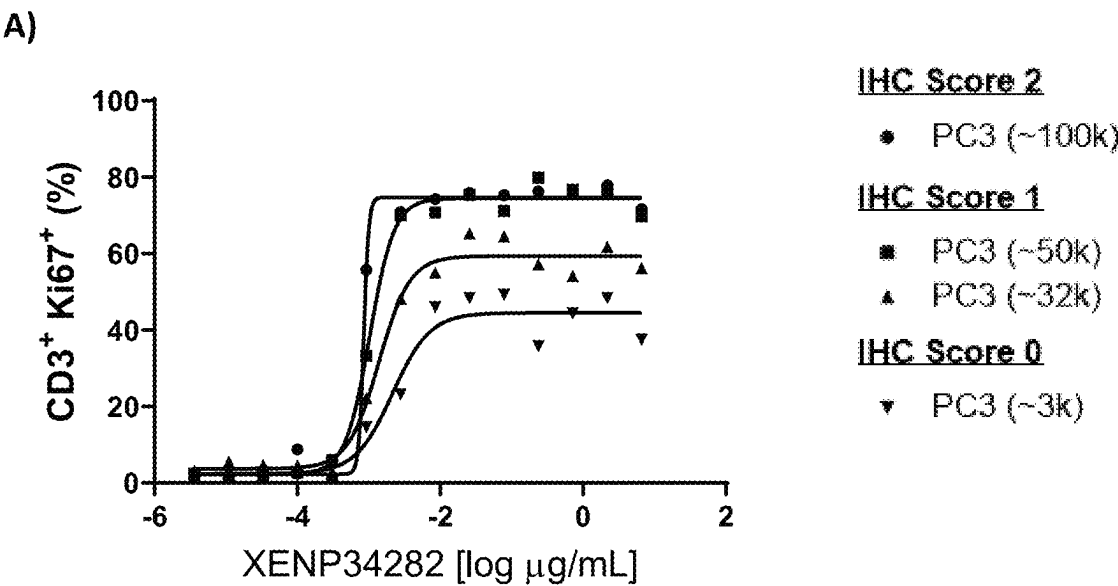
B)



Luciferase RTCC
E:T; 1:1
48 hr. mAb

	PC3 (~100K) EC ₅₀ (ng/mL)	PC3 (~50K) EC ₅₀ (ng/mL)	PC3 (~32K) EC ₅₀ (ng/mL)	PC3 (~3K) EC ₅₀ (ng/mL)
XENP34282	1	2	2	3
XENP14484	5	8	12	18

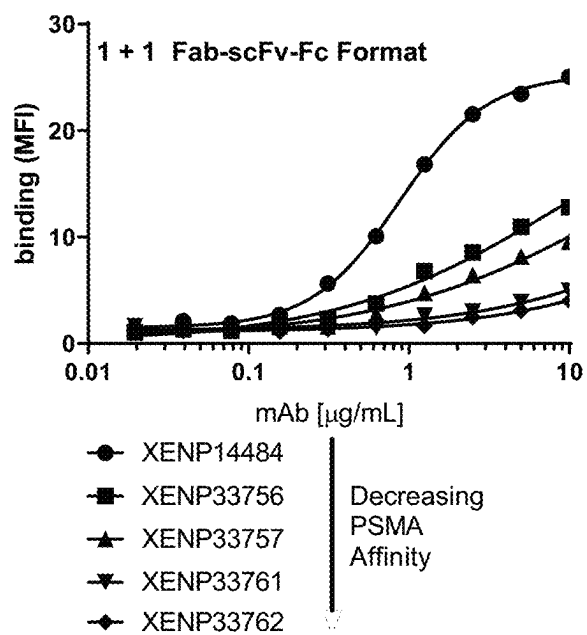
Figure 37



CD3 ⁺ Ki67 ⁺ E:T; 1:1 72 hr. mAb	PC3 (~100K)	PC3 (~50K)	PC3 (~32K)	PC3 (~3K)
	EC ₅₀ (ng/mL)	EC ₅₀ (ng/mL)	EC ₅₀ (ng/mL)	EC ₅₀ (ng/mL)
XENP34282	1	1	1	2
XENP14484B	2	17	10	73

Figure 38

A



B

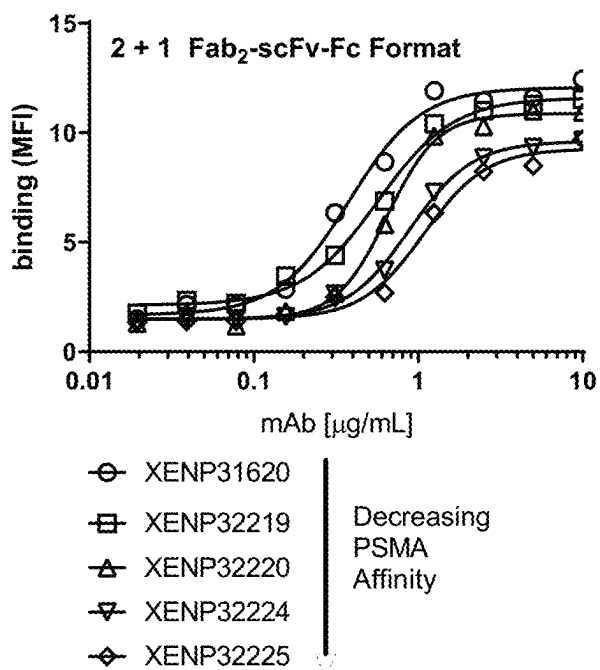


Figure 39

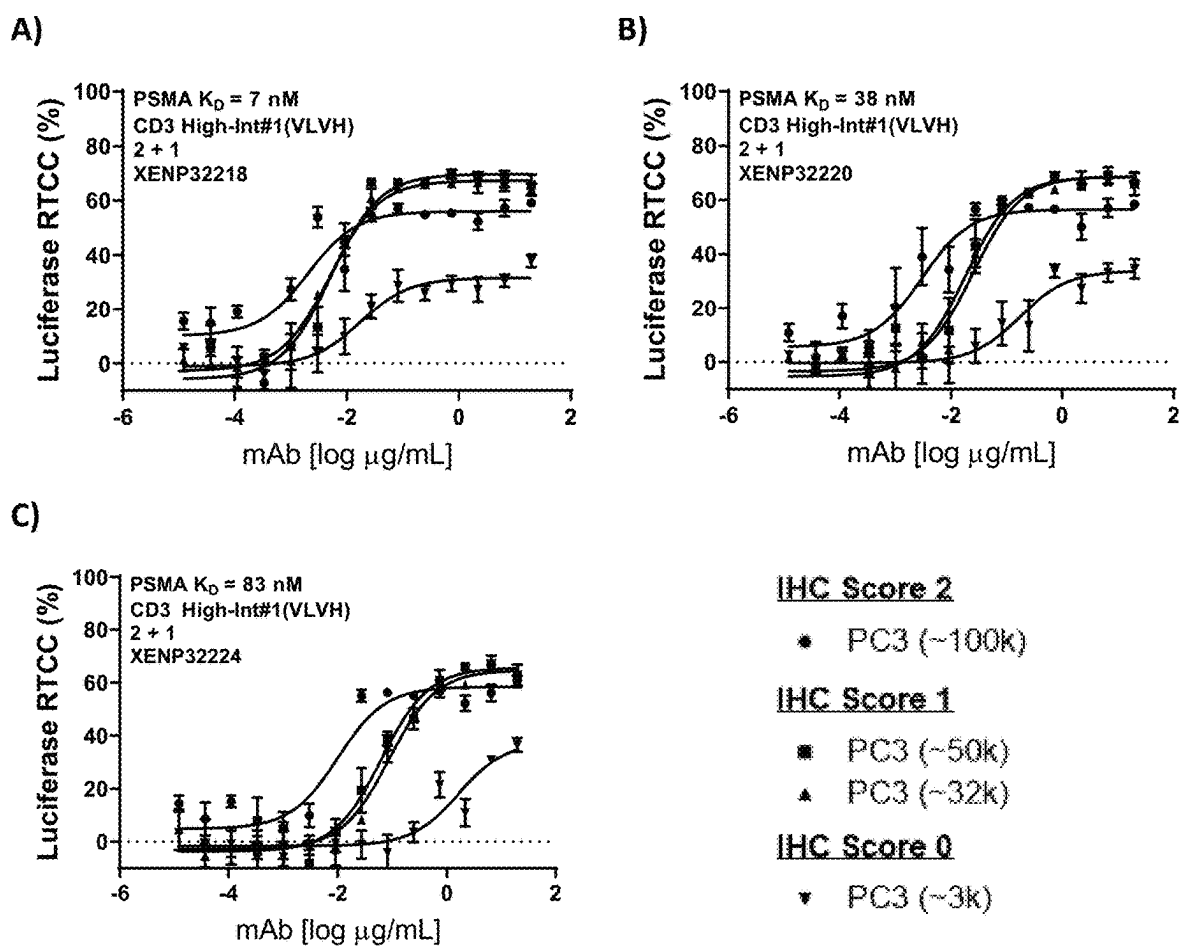


Figure 40

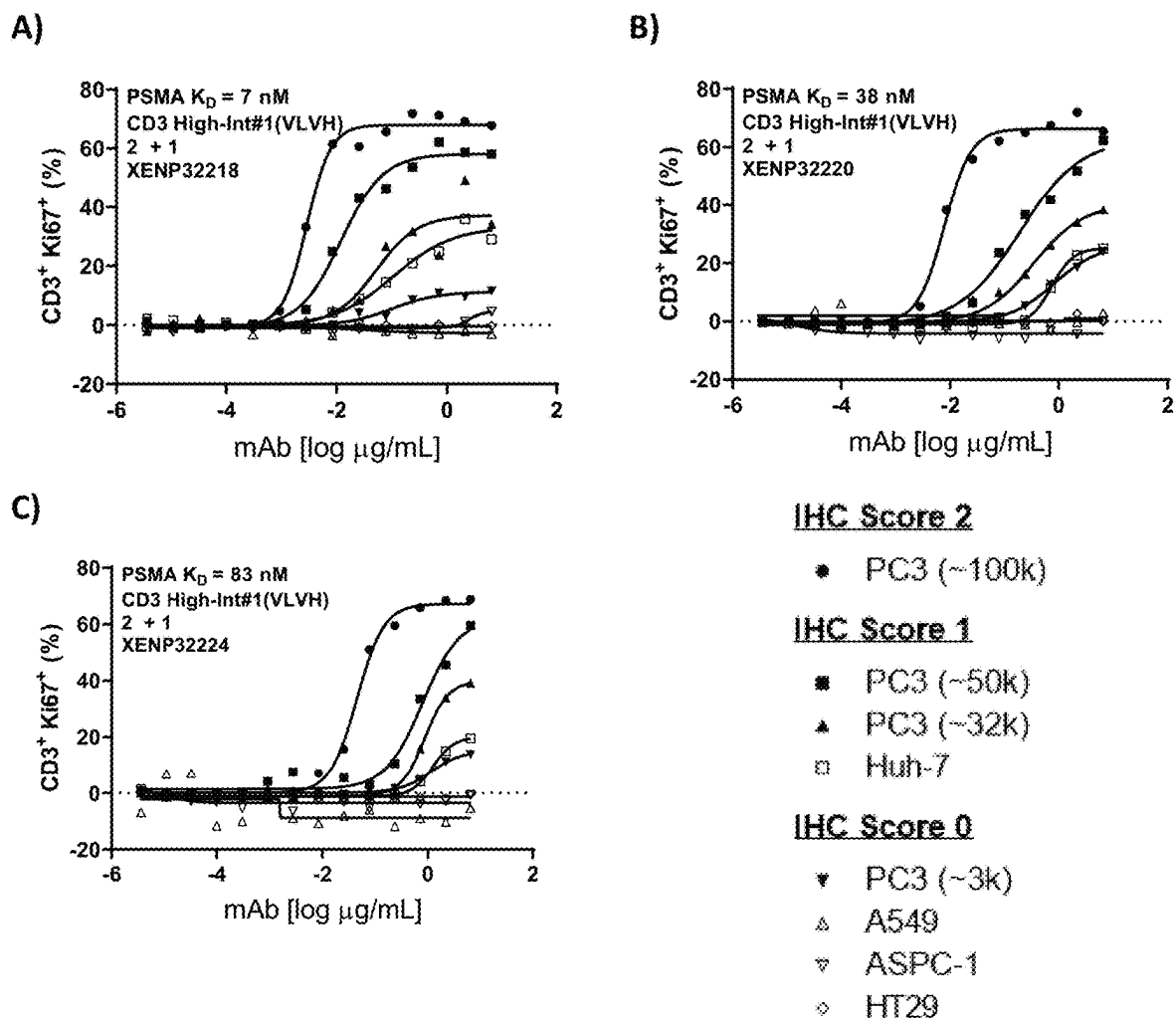


Figure 41

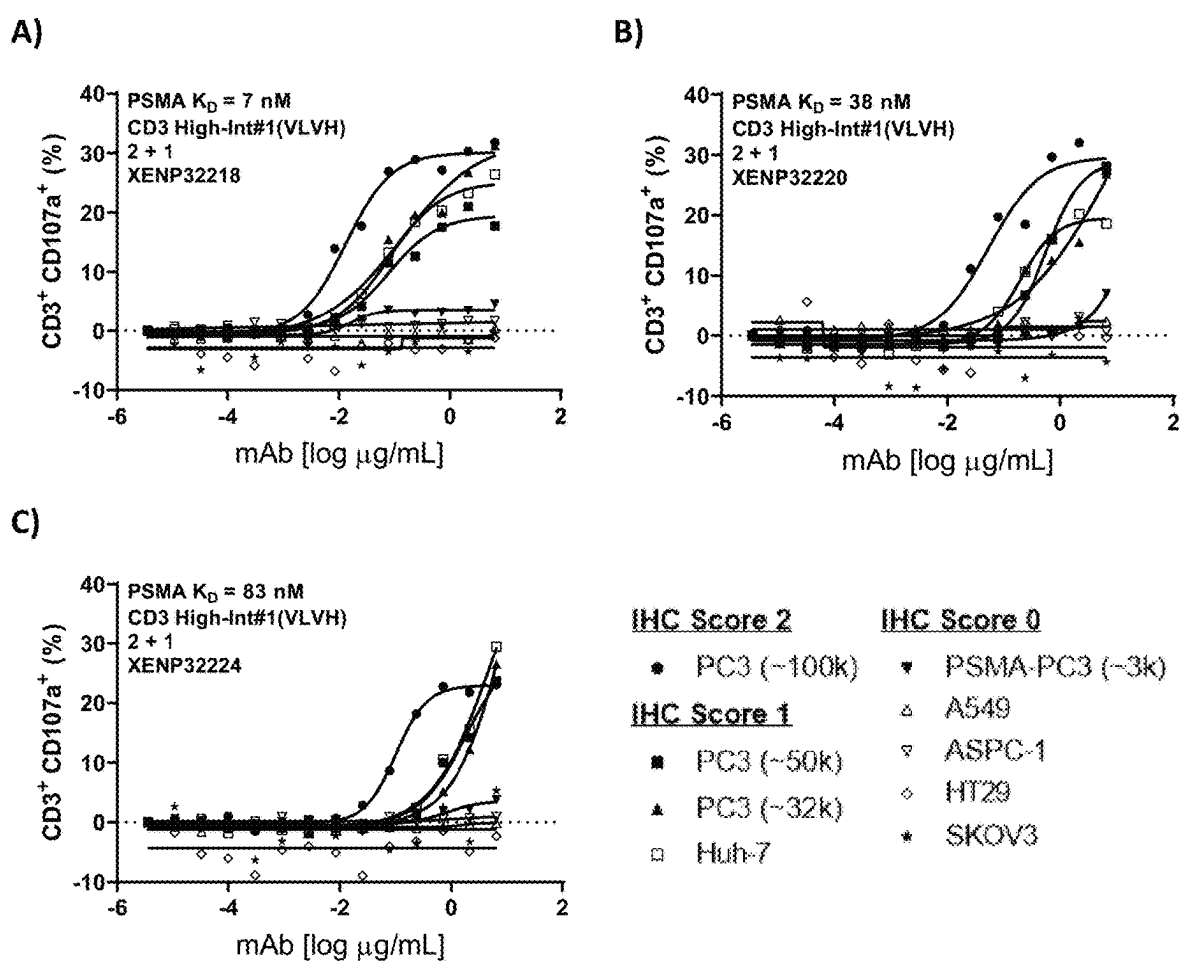


Figure 42

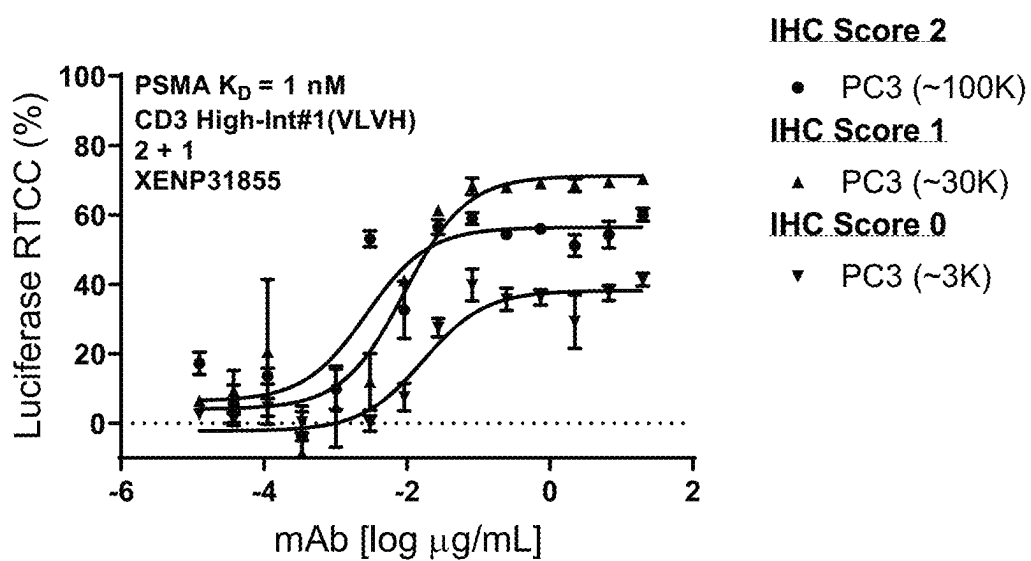


Figure 43

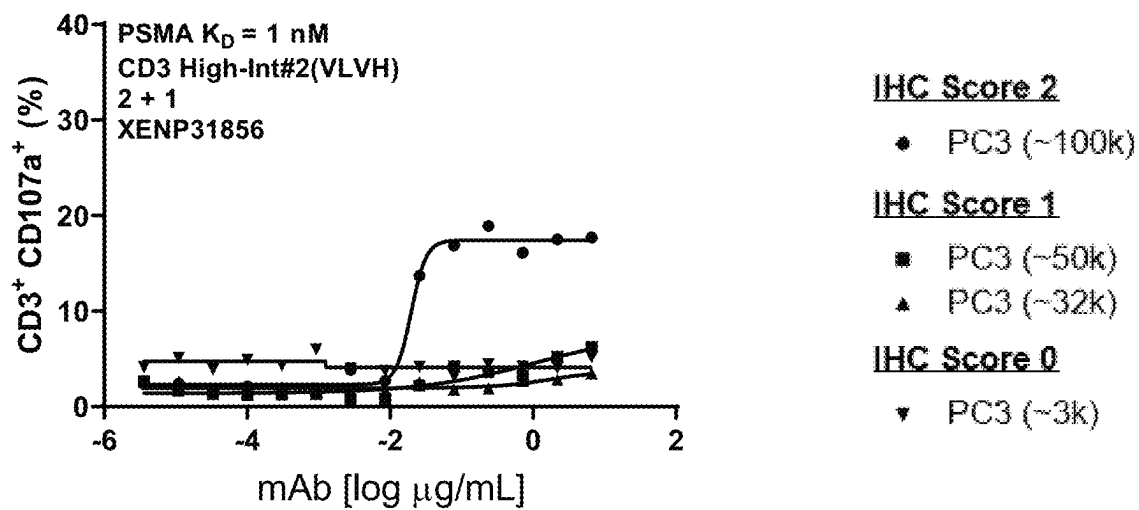
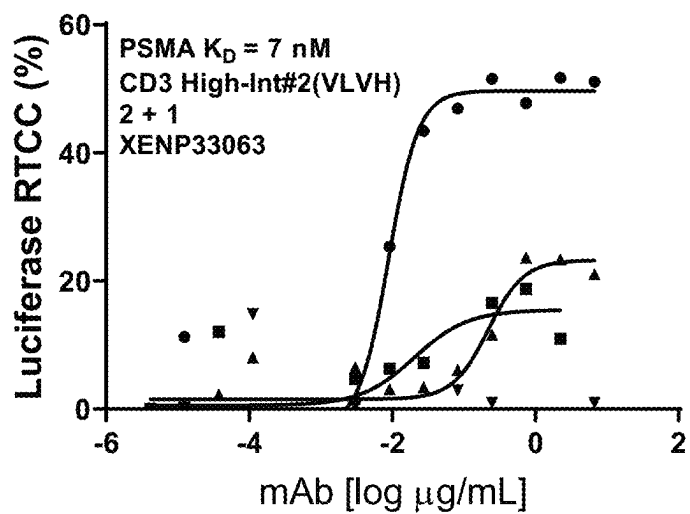


Figure 44

IHC Score 2

- PC3 (~100k)

IHC Score 1

- PC3 (~50k)
- ▲ PC3 (~32k)

IHC Score 0

- ▼ PC3 (~3k)

Figure 45

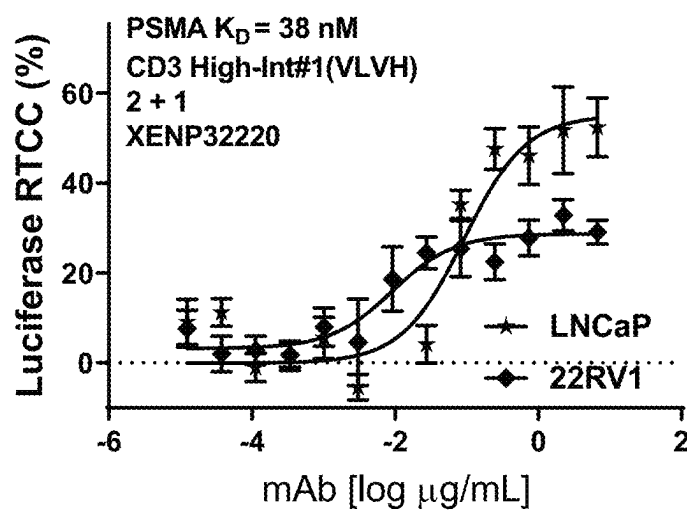


Figure 46

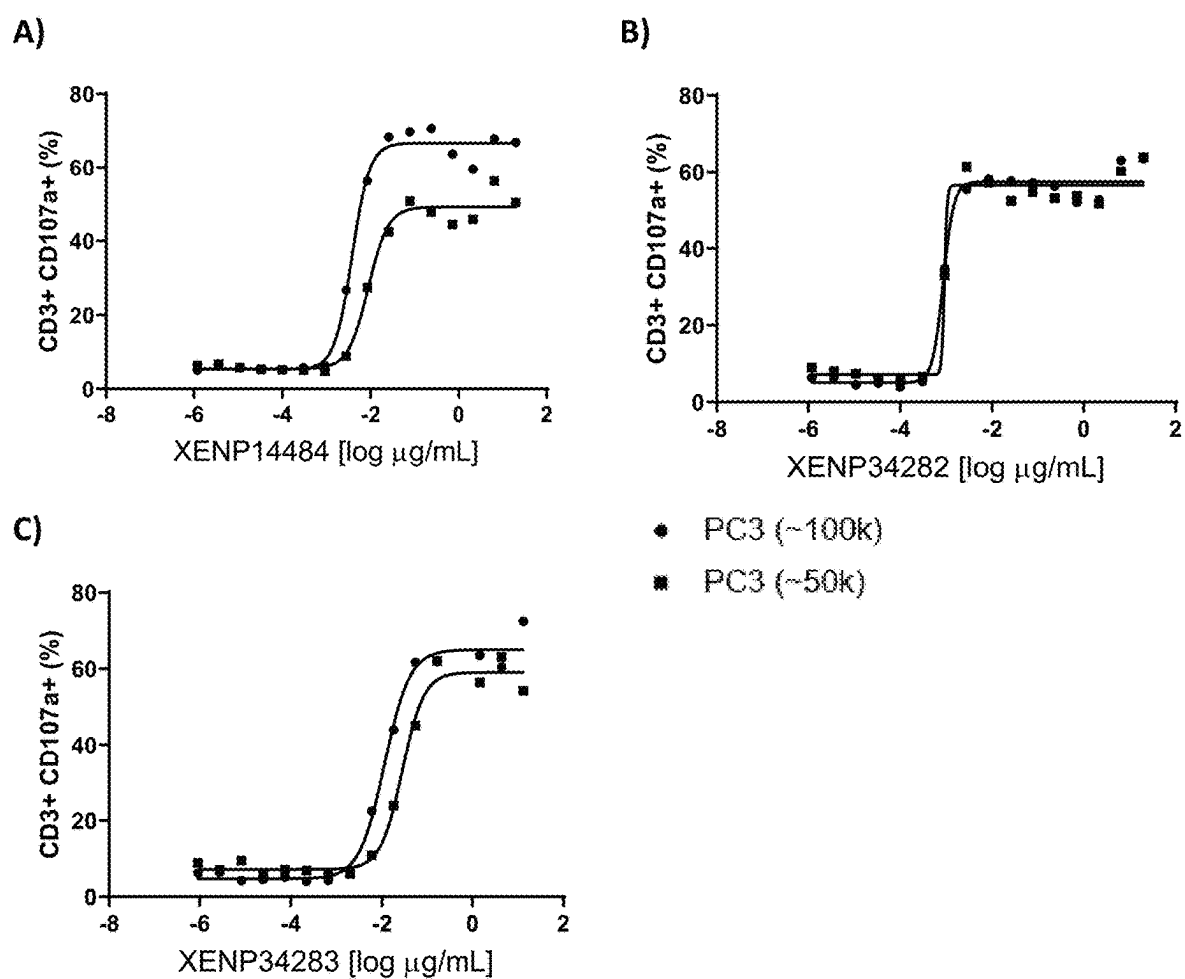


Figure 47

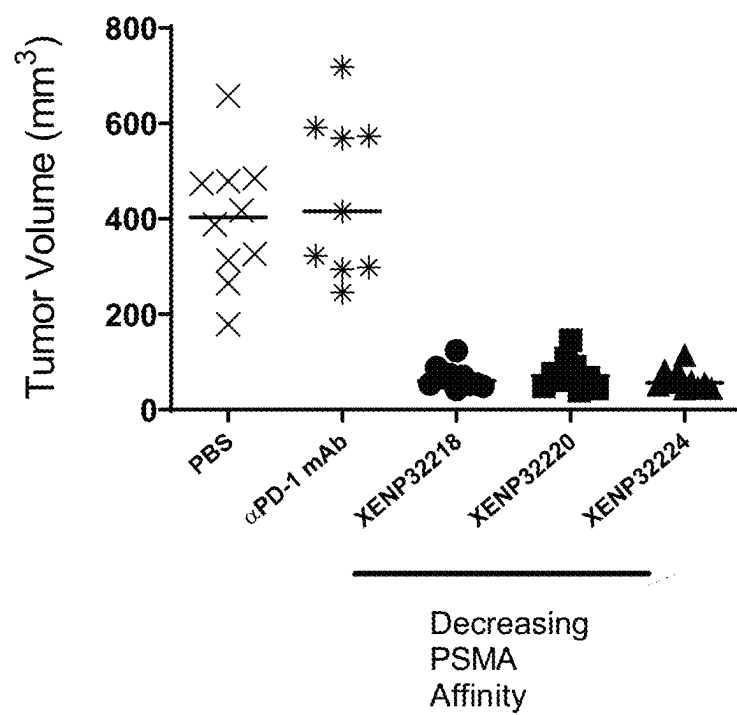


Figure 48

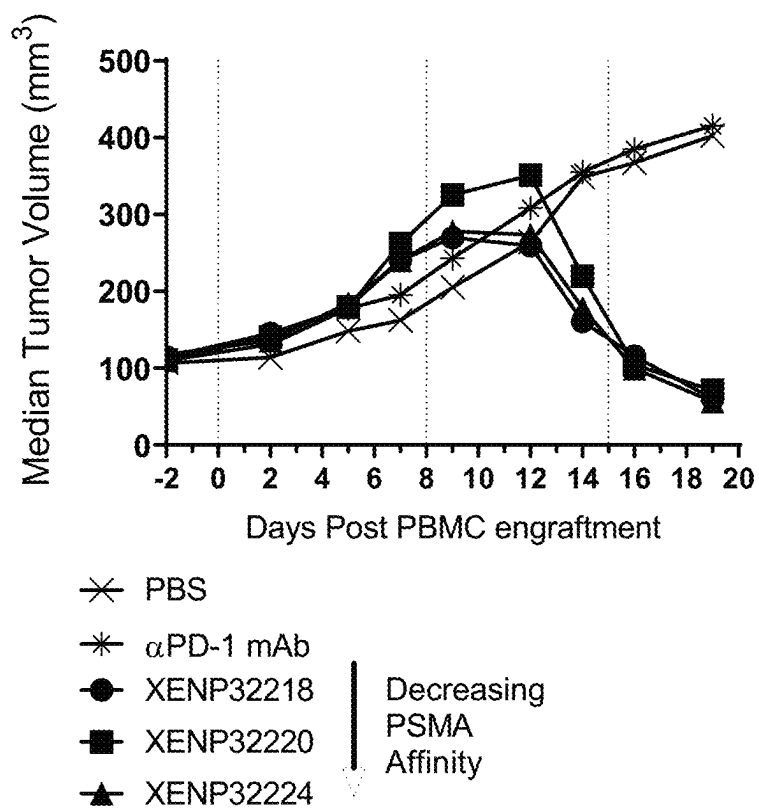


Figure 49A

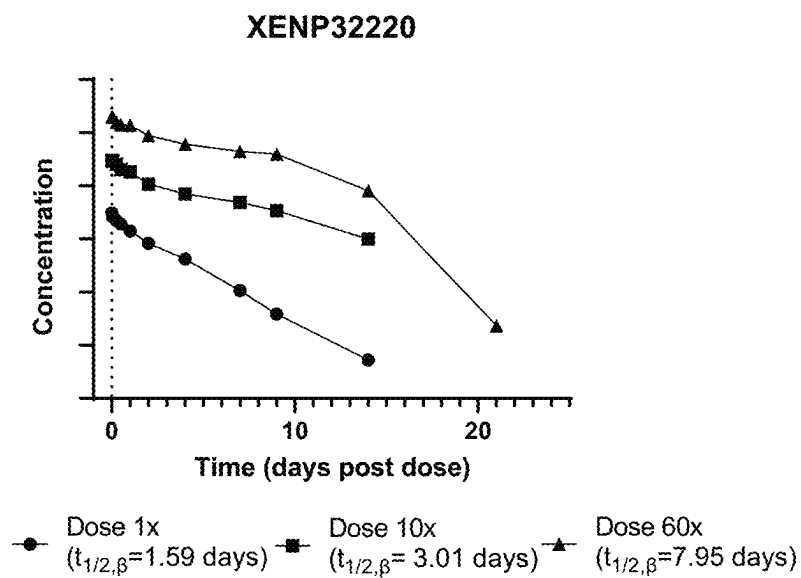
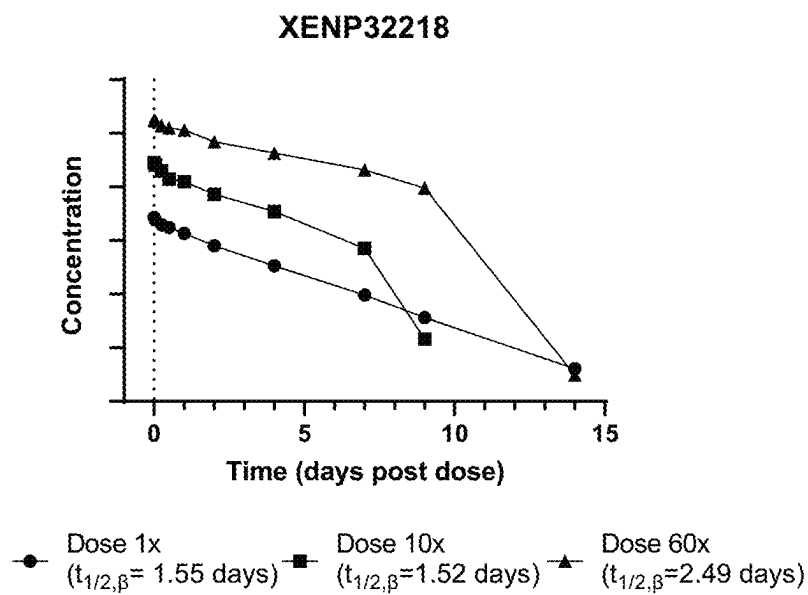


Figure 49B

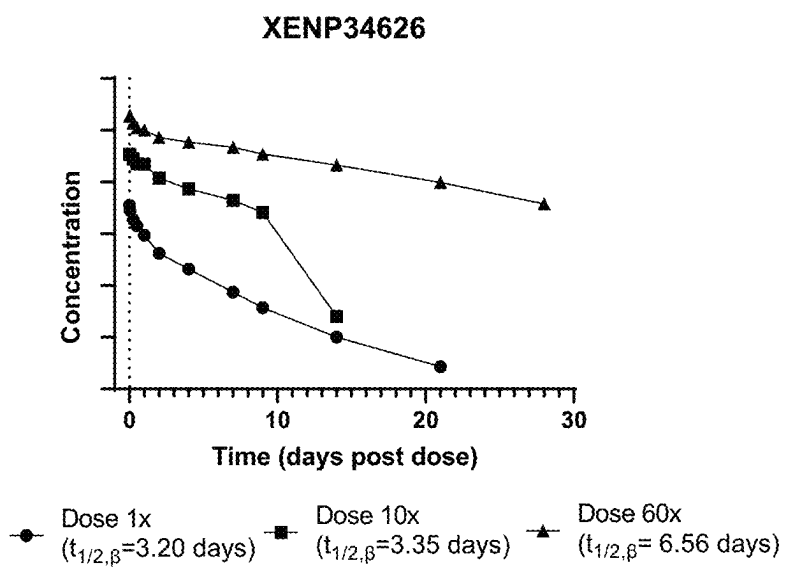
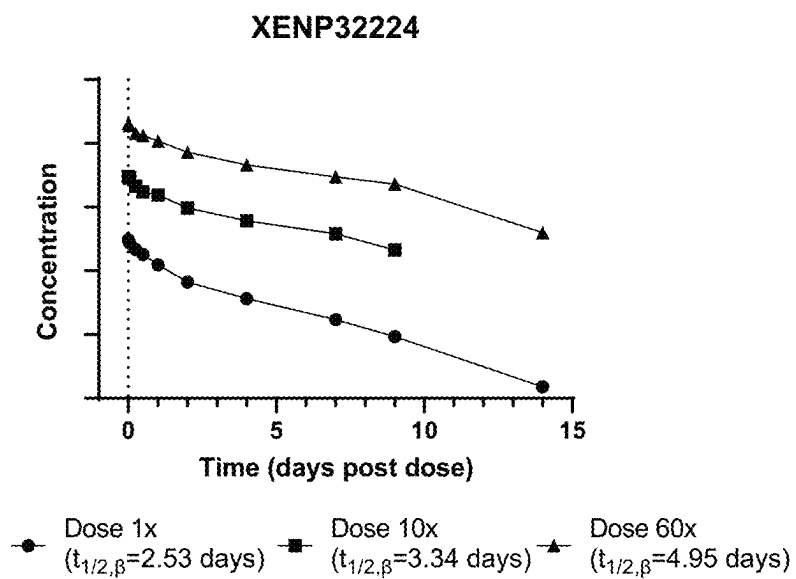


Figure 49C

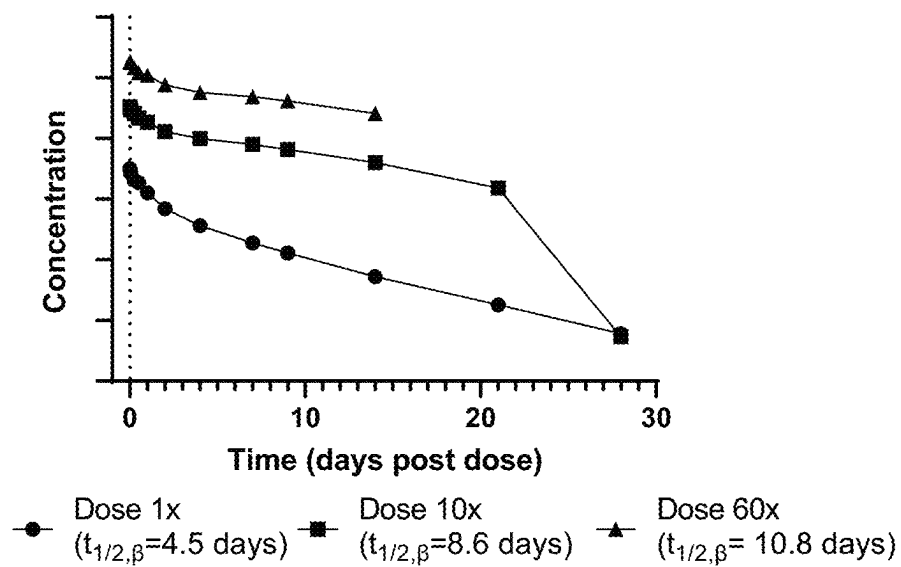
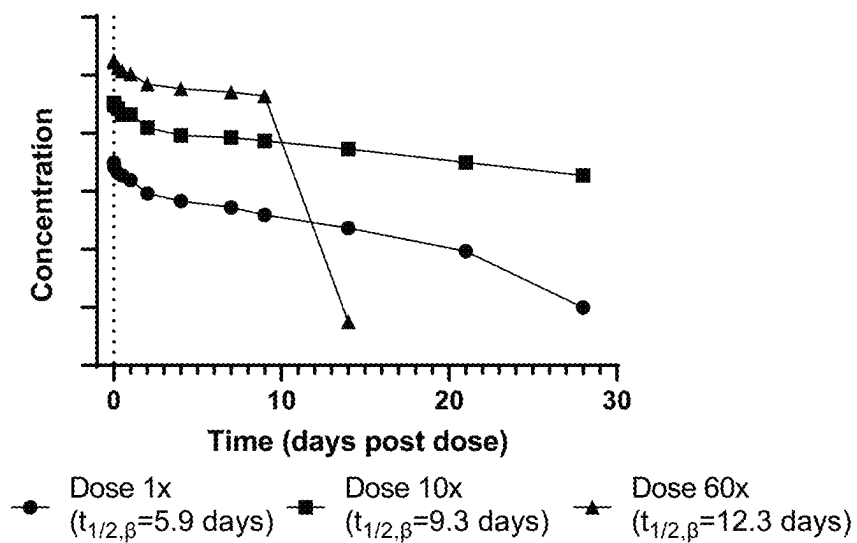
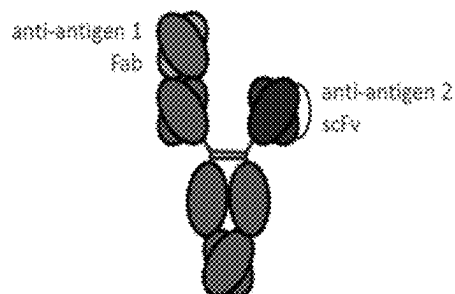
XENP34627**XENP34628**

Figure 50A-50F

Figure 50A



Bottle Opener or
1+1 Fab-scFv-Fc

Figure 50B

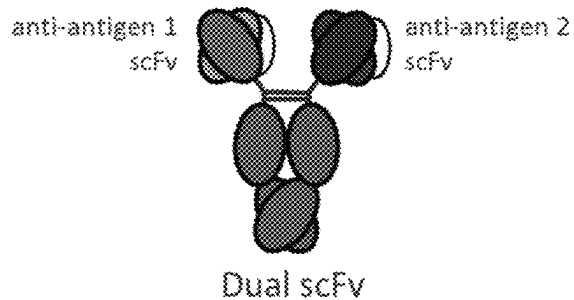
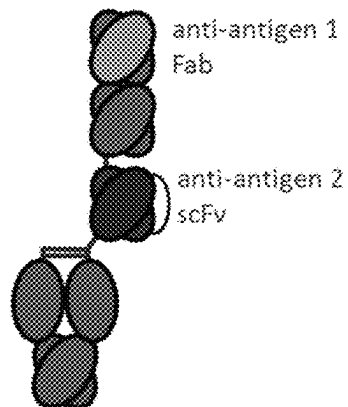
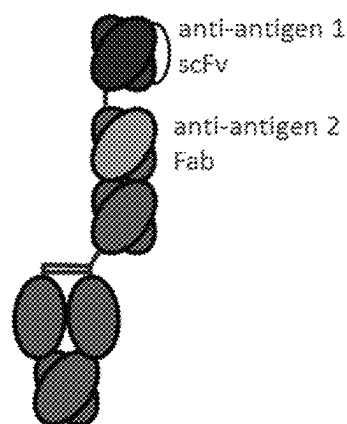


Figure 50C



One-arm central-scFv

Figure 50D



One-arm scFv-mAb

Figure 50E

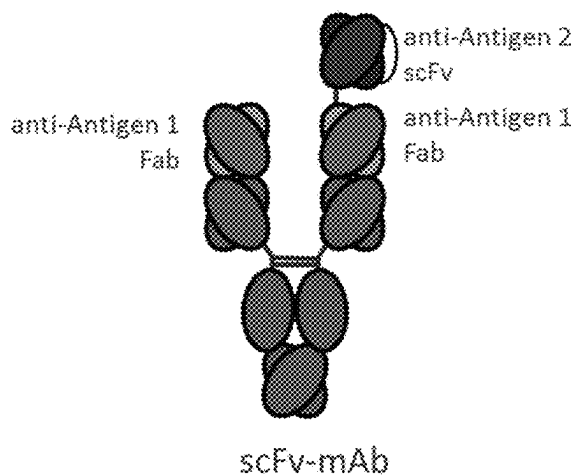


Figure 50F

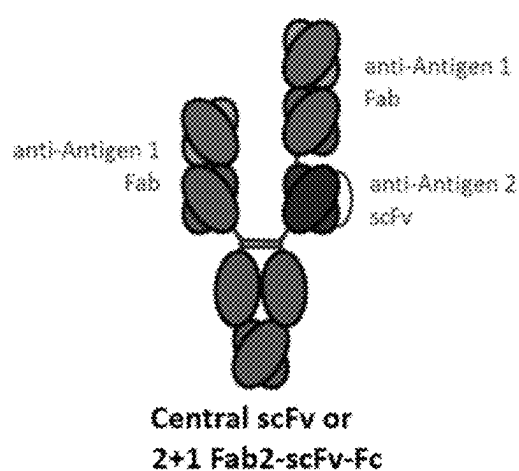
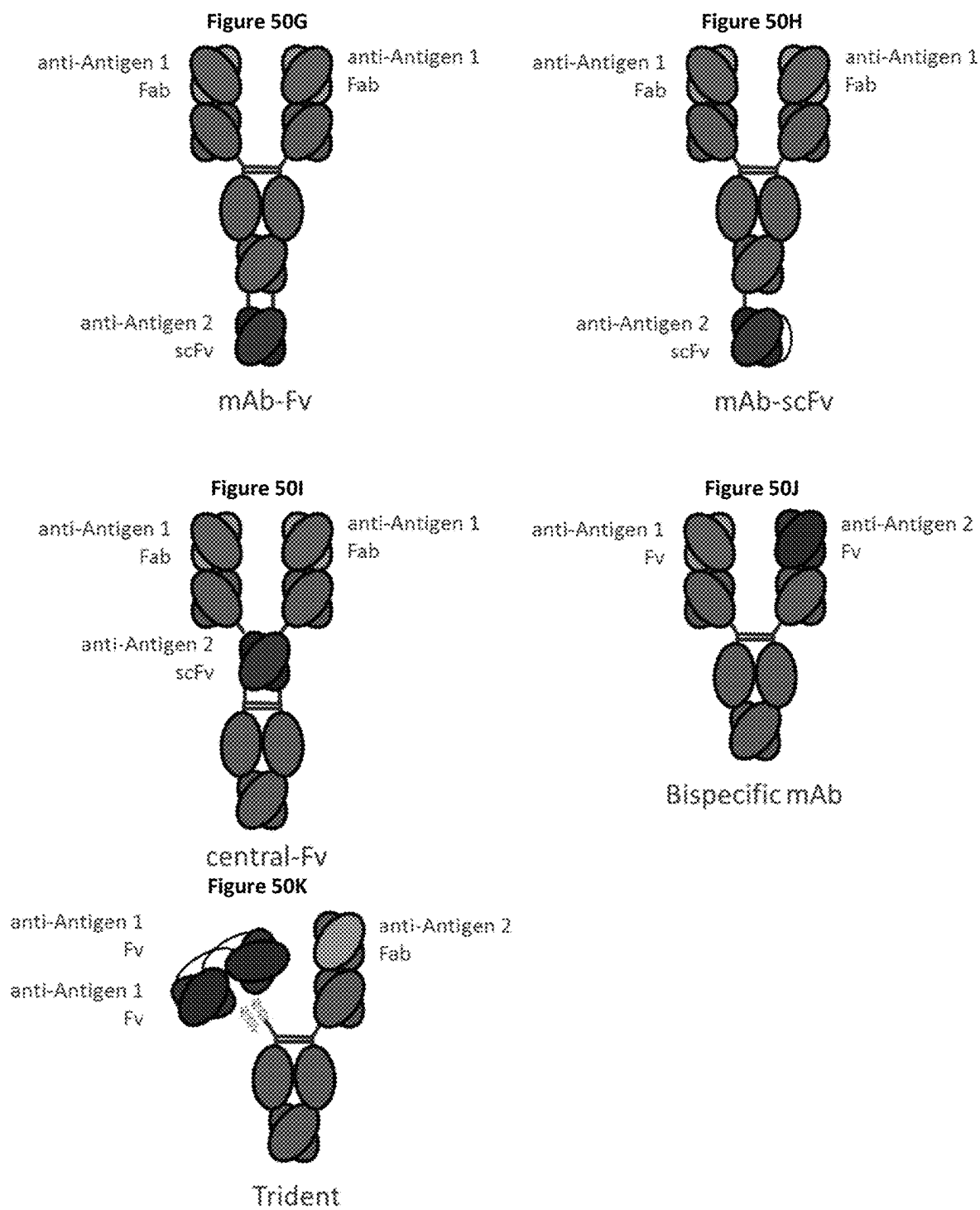


Figure 50G-50K



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HETERODIMERIC ANTIBODIES THAT BIND PROSTATE SPECIFIC MEMBRANE ANTIGEN (PSMA) AND CD3

PRIORITY

This application is a continuation of U.S. patent application Ser. No. 17/321,325, filed May 14, 2021, which claims the benefit of U.S. Provisional Patent Application Nos. 63/025,082, filed May 14, 2020, and 63/042,315, filed Jun. 22, 2020, which are hereby incorporated by reference in their entireties.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. Said XML copy, created on Jan. 17, 2024, is named 067461-5269-US01 SL.xml and is 1,199,689 bytes in size.

BACKGROUND

Antibody-based therapeutics have been used successfully to treat a variety of diseases, including cancer. An increasingly prevalent avenue being explored is the engineering of single immunoglobulin molecules that co-engage two different antigens. Such alternate antibody formats that engage two different antigens are often referred to as bispecific antibodies. Because the considerable diversity of the antibody variable region (Fv) makes it possible to produce an Fv that recognizes virtually any molecule, the typical approach to bispecific antibody generation is the introduction of new variable regions into the antibody.

A particularly useful approach for bispecific antibodies is to engineer a first binding domain which engages CD3 and a second binding domain which engages an antigen associated with or upregulated on cancer cells so that the bispecific antibody redirects CD3⁺ T cells to destroy the cancer cells. Prostate cancer (PC) is one of the most prevalent cancers in men, and end stage (castration-resistant prostate cancer) has no curative treatment option. Prostate Specific Membrane Antigen (PSMA), a type II transmembrane protein with a large extracellular domain, has long generated interest as a therapeutic target. It is highly overexpressed in PC compared to normal tissue, and its expression has been shown to correlate with malignancy. Previous attempts to target PSMA include antibody-based radiotherapy and antibody drug conjugates, which have shown some success but can be hampered by the inherent toxicity of the modality. Thus, there is a need for additional anti-PSMA antibodies for the treatment of PSMA-related cancers include, for example, prostate cancer.

BRIEF SUMMARY

Provided herein are novel bispecific antibodies to CD3 and PSMA that are capable of localizing CD3⁺ effector T cells to PSMA expressing tumors such as in prostate cancer. The anti-PSMA antibodies provided herein include PSMA binding domain with binding affinities and valencies that allow for the advantageous selectivity for cells expressing high levels of PSMA while minimizing reactivity on low PSMA expressing cells. In some embodiments, such anti-PSMA antibodies include CD3 binding domains with binding affinity that further contribute to selective targeting of high-PSMA expressing cells lines. Such PSMA antibodies

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are useful, for example, for cancers that express high levels of PSMA including, for example, prostate cancer.

In one aspect, provided herein is a composition that includes a Prostate Specific Membrane Antigen (PSMA) binding domain. The PSMA binding domain includes: a) a variable heavy domain that includes the variable heavy complementary determining regions 1-3 (vhCDR1-3) of PSMA-H variable heavy domain H1 (FIG. 17); and b) a variable light domain that includes the variable light complementary determining regions (vlCDR1-3) of a PSMA-H variable light domain selected from PSMA-H variable light domains L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In some embodiments, the vhCDR1-3 and vlCDR1-3 are selected from the vhCDR1-3 and vlCDR1-3 sequences of any one of the anti-PSMA binding domains provided in FIGS. 19A-19X.

Also provided herein are nucleic acid compositions that include polynucleotide(s) encoding the subject PSMA binding domains, expression vectors that include such polynucleotides and host cells that include such expression vectors. Also provided herein are methods of making such PSMA binding domains.

In another aspect, provided herein is composition that includes a Prostate Specific Membrane Antigen (PSMA) binding domain. The PSMA binding domain includes: a) a variable heavy domain, wherein the variable heavy domain is the PSMA-H variable heavy domain H1 (FIG. 17); and b) a variable light domain selected from PSMA-H variable light domains L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E).

In another aspect, provided herein is an anti-PSMA antibody that includes a Prostate Specific Membrane Antigen (PSMA) binding domain. The PSMA binding domain includes: a) a variable heavy domain that includes the variable heavy complementary determining regions 1-3 (vhCDR1-3) of PSMA-H variable heavy domain H1 (FIG. 17); and b) a variable light domain that includes the variable light complementary determining regions (vlCDR1-3) of a PSMA-H variable light domain selected from PSMA-H variable light domains L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In some embodiments, the vhCDR1-3 and vlCDR1-3 are selected from the vhCDR1-3 and vlCDR1-3 of selected from any one of the anti-PSMA binding domains provided in FIGS. 19A-19X.

In one aspect, provided herein is a Prostate Specific Membrane Antigen (PSMA) binding domain. The PSMA binding domain includes: a) a variable heavy domain, wherein the variable heavy domain is the PSMA-H variable heavy domain H1 (FIG. 17); and b) a variable light domain selected from PSMA-H variable light domains L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In some embodiments, the antibody includes: a) a first monomer that includes a first antigen binding domain and a first constant domain; and b) a second monomer that includes a second antigen binding domain and a second constant domain, wherein either of the first antigen binding domain or second antigen binding domain is the PSMA binding domain. In certain embodiments, the first antigen binding domain and the second antigen binding domain bind different antigens.

In some embodiments, the first antigen binding domain is the PSMA binding domain and the second antigen binding domain is a CD3 binding domain. In particular embodiments, the CD3 binding domain includes the vhCDR1-3, and vlCDR1-3 of any of the following CD3 binding domains: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F). In cer-

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tain embodiments, the v hCDR1-3 and v LCDR1-3 of the CD3 binding domain are selected from the v hCDR1-3 and v LCDR1-3 in FIGS. 10A-10F. In some embodiments, the CD3 binding domain includes the variable heavy domain and variable light domain of any of the following CD3 binding domains: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F). In certain embodiments, the CD3 binding domain is an anti-CD3 scFv. In some embodiments, the scFv includes a charged scFv linker. In some embodiments, the first and second constant domains each include CH2-CH3. In certain embodiments, the first and second constant domains each include CH1-hinge-CH2-CH3.

In particular embodiments, the first and second constant domains each are a variant constant domain. In some embodiments, the first and second monomers include a set of heterodimerization variants selected from the group consisting of those depicted in FIGS. 1A-1E. In exemplary embodiments, the set of heterodimerization variants selected is from the group consisting of S364K/E357Q:L368D/K370S; S364K:L368D/K370S; S364K:L368E/K370S; D401K: T411E/K360E/Q362E; and T366W:T366S/L368A/Y407V. In some embodiments, the first and second monomers each further include an ablation variant. In exemplary embodiments, the ablation variant is E233P/L234V/L235A/G236del/S267K.

In certain embodiments, at least one of the first or second monomer further includes one or more pI variants. In exemplary embodiments, the one or more pI variants is N208D/Q295E/N384D/Q418E/N421D.

In another aspect, provided is a heterodimeric antibody that includes: a) a first monomer, b) a second monomer; and c) a light chain. The first monomer includes: i) an anti-CD3 scFv that includes a first variable heavy domain, an scFv linker and a first variable light domain; and ii) a first Fc domain, wherein the scFv is covalently attached to the N-terminus of the first Fc domain using a domain linker. The second monomer includes a VH2-CH1-hinge-CH2-CH3 monomer, wherein VH is a second variable heavy domain and CH2-CH3 is a second Fc domain; and the light chain includes a second variable light domain. The second variable heavy domain and the second variable light domain form an PSMA binding domain.

In some embodiments of the heterodimeric antibody, the second variable heavy domain includes the variable heavy complementary determining regions 1-3 (v hCDR1-3) of PSMA-H variable heavy domain H1 (FIG. 17); and the second variable light domain includes the variable light complementary determining regions (v LCDR1-3) of a PSMA-H variable light domain selected from PSMA-H variable light domains L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In exemplary embodiments, the v hCDR1-3 of the second variable heavy domain and the v LCDR1-3 of the second variable light domain are selected from any one of the anti-PSMA binding domains provided in FIGS. 19A-19X. In some embodiments, the second heavy variable domain is PSMA-H variable heavy domain H1 (FIG. 17); and the second variable light domain is selected from PSMA-H variable light domains L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E).

In certain embodiments, the anti-CD3 scFv includes the v hCDR1-3 and the v LCDR1-3 of any of the following CD3 binding domains: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89,

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L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F). In some embodiments, the v hCDR1-3 and v LCDR1-3 of the anti-CD3 scFv are selected from the v hCDR1-3 and v LCDR1-3 in FIGS. 10A-10F. In exemplary embodiments, the first variable heavy domain and the first variable light domain are the variable heavy domain and variable light domain, respectively, of any of the following CD3 binding domains: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F).

In one embodiment, the first variable light domain is covalently attached to the N-terminus of the first Fc domain using a domain linker. In some embodiments, the first variable heavy domain is covalently attached to the N-terminus of the first Fc domain using a domain linker.

In some embodiments of the heterodimeric antibody, the scFv linker is a charged scFv linker. In exemplary embodiments, the scFv linker is a charged scFv linker having the amino acid sequence (GKPGS)₄ (SEQ ID NO: 1).

In certain embodiments, the first and second Fc domains are variant Fc domains. In some embodiments, the first and second monomers include a set of heterodimerization variants selected from the group consisting of those depicted in FIGS. 1A-1E. In exemplary embodiments, the set of heterodimerization variants selected is from the group consisting of S364K/E357Q:L368D/K370S; S364K:L368D/K370S; S364K:L368E/K370S; D401K: T411E/K360E/Q362E; and T366W:T366S/L368A/Y407V, wherein numbering is according to EU numbering. In some embodiments, the first and second monomers further includes an ablation variant. In exemplary embodiments, the ablation variant is E233P/L234V/L235A/G236del/S267K, wherein numbering is according to EU numbering.

In some embodiments, one of the first or second monomer includes one or more pI variants. In particular embodiments, the one or more pI variants are N208D/Q295E/N384D/Q418E/N421D, wherein numbering is according to EU numbering.

In exemplary embodiments of the heterodimeric antibody, the first monomer includes amino acid variants S364K/E357Q/E233P/L234V/L235A/G236del/S267K, the second monomer includes amino acid variants L368D/K370S/N208D/Q295E/N384D/Q418E/N421D/E233P/L234V/L235A/G236del/S267K, and the numbering is according to EU numbering.

In certain embodiments of the heterodimeric antibody, the first and second monomers each further include amino acid variants 428/434S.

In exemplary embodiments, the heterodimeric antibody is one of the following heterodimeric antibodies: XENP14484, XENP33755, XENP33756, XENP33757, XENP33758, XENP33759, XENP33760, XENP33761, XENP33762, XENP34234, XENP34235, XENP34236, XENP16873, XENP16874, and XENP19722.

In another aspect, provided herein is a heterodimeric antibody that includes: a) a first monomer; b) a second monomer; and c) a light chain. The first monomer includes, from N-terminus to C-terminus, a scFv-linker-CH2-CH3, wherein scFv is an anti-CD3 scFv and CH2-CH3 is a first Fc domain. The second monomer includes, from N-terminus to C-terminus, a VH-CH1-hinge-CH2-CH3, wherein CH2-CH3 is a second Fc domain. The light chain includes a VL-CL. The first variant Fc domain includes amino acid variants S364K/E357Q, the second variant Fc domain includes amino acid variants L368D/K370S, the first and second variant Fc domains each include amino acid variants

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E233P/L234V/L235A/G236del/S267K, and the hinge-CH2-CH3 of the second monomer includes amino acid variants N208D/Q295E/N384D/Q418E/N421D. The VH and VL form an PSMA binding domain that includes the variable heavy domain and the variable light domain, respectively, of an PSMA binding domain selected from PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; and PSMA-H H1_L1.13. Further, the anti-CD3 scFv includes the variable heavy domain and the variable light domain of a CD3 binding domain selected from H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31. In such heterodimeric antibodies, the numbering of the amino acid variants is according to EU numbering.

In some embodiments of this heterodimeric antibody, the scFv includes a charged scFv linker having the amino acid sequence (GKPGS)₄ (SEQ ID NO: 1). In certain embodiments, the first and second variant Fc domains each further include amino acid variants 428/434S, wherein numbering is according to EU numbering.

In another aspect, provided herein is a heterodimeric antibody that includes: a) a first monomer; b) a second monomer; and c) a common light chain. The first monomer includes, from N-terminus to C-terminus, a VH1-CH1-linker 1-scFv-linker 2-CH2-CH3, wherein VH1 is a first variable heavy domain, scFv is an anti-CD3 scFv, linker 1 and linker 2 are a first domain linker and second domain linker, respectively, and CH2-CH3 is a first Fc domain. The second monomer includes, from N-terminus to C-terminus, a VH2-CH1-hinge-CH2-CH3, wherein VH2 is a second variable heavy domain and CH2-CH3 is a second Fc domain. The common light chain includes a variable light domain. The first variable heavy domain and the variable light domain form a first PSMA binding domain, and the second variable heavy domain and the variable light domain form a second PSMA binding domain. In some embodiments, the first and second PSMA binding domains each include the variable heavy complementary determining regions 1-3 (vhCDR1-3) of PSMA-H variable heavy domain H1 (FIG. 17); and the variable light complementary determining regions (vLCDR1-3) of a PSMA-H variable light domain selected from PSMA-H variable light domains L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In exemplary embodiments, the vhCDR1-3 and vLCDR1-3 of the first and second PSMA binding domains are selected from the vhCDR1-3 and vLCDR1-3 provided in FIGS. 17 and 18A-18E. In some embodiments, the first and second variable heavy domain each is a PSMA-H variable heavy domain H1 (FIG. 17), and the variable light domain of the common light chain is selected from PSMA-H variable light domains L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E).

In certain embodiments, the scFv includes the vhCDR1-3 and the vLCDR1-3 of any of the following CD3 binding domains: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F). In exemplary embodiments, the vhCDR1-3 and vLCDR1-3 of the scFv are selected from the vhCDR1-3 and vLCDR1-3 in FIGS. 10A-10F. In some embodiments, the scFv includes the variable heavy domain and variable light domain of any of the following CD3 binding domains: H1.30_L1.47,

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H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F).

In certain embodiments, the scFv includes an scFv variable heavy domain, an scFv variable light domain and an scFv linker that connects the scFv variable heavy domain and the scFv variable light domain. In some embodiments, the scFv variable heavy domain is attached to the C-terminus of the CH1 of the first monomer using the first domain linker and the scFv variable light domain is covalently attached to the N-terminus of the first Fc domain using the second domain linker. In other embodiments, the scFv variable light domain is attached to the C-terminus of the CH1 of the first monomer using the first domain linker and the scFv variable heavy domain is covalently attached to the N-terminus of the first Fc domain using the second domain linker. In certain embodiments of this heterodimeric antibody, the scFv linker is a charged scFv linker. In exemplary embodiments, the scFv linker is a charged scFv linker having the amino acid sequence (GKPGS)₄ (SEQ ID NO: 1).

In certain embodiments of this heterodimeric antibody, the first and second Fc domains are variant Fc domains. In some embodiments, the first and second monomers include a set of heterodimerization variants selected from the group consisting of those depicted in FIGS. 1A-1E. In exemplary embodiments, the set of heterodimerization variants selected is from the group consisting of S364K/E357Q/L368D/K370S; S364K:L368D/K370S; S364K:L368E/K370S; D401K: T411E/K360E/Q362E; and T366W:T366S/L368A/Y407V, wherein numbering is according to EU numbering. In some embodiments, the first and second monomers further include an ablation variant. In exemplary embodiments, the ablation variant is E233P/L234V/L235A/G236del/S267K, wherein numbering is according to EU numbering.

In some embodiments, one of the first or second monomer includes one or more pI variants. In particular embodiments, the one or more pI variants are N208D/Q295E/N384D/Q418E/N421D, wherein numbering is according to EU numbering.

In exemplary embodiments of the heterodimeric antibody, the first monomer includes amino acid variants S364K/E357Q/E233P/L234V/L235A/G236del/S267K, the second monomer includes amino acid variants L368D/K370S/N208D/Q295E/N384D/Q418E/N421D/E233P/L234V/L235A/G236del/S267K, and the numbering is according to EU numbering.

In certain embodiments of the heterodimeric antibody, the first and second monomers each further include amino acid variants 428/434S.

In exemplary embodiments, this heterodimeric antibody is one of the following: XENP31602, XENP31603, XENP31855, XENP32218, XENP32219, XENP32220, XENP32221, XENP32222, XENP32223, XENP32224, XENP32225, XENP32226, XENP34237, XENP34238, XENP34239, XENP34625, XENP34626, XENP34627, XENP34628, XENP31853, XENP31856, XENP33063, XENP33064, XENP33065, XENP33066, XENP33067, XENP33068, XENP33069, XENP33070, XENP33071, XENP34240, XENP34241, XENP34242, XENP34629, XENP34630, XENP34631, XENP34632, XENP31854, and XENP31857.

In yet another aspect, provided herein is a heterodimeric antibody that includes: a) a first monomer; b) a second monomer; and c) a common light chain. The first monomer includes, from N-terminus to C-terminus, a VH-CH1-linker 1-scFv-linker 2-CH2-CH3, wherein scFv is an anti-CD3

scFv and CH2-CH3 is a first Fc domain. The b) a second monomer includes, from N-terminus to C-terminus a VH-CH1-hinge-CH2-CH3, wherein CH2-CH3 is a second Fc domain. The common light chain includes VL-CL. The first variant Fc domain includes amino acid variants S364K/E357Q, the second variant Fc domain includes amino acid variants L368D/K370S. The first and second variant Fc domains each include amino acid variants E233P/L234V/L235A/G236del/S267K, the hinge-CH2-CH3 of the second monomer includes amino acid variants N208D/Q295E/N384D/Q418E/N421D. The VH of this heterodimeric antibody is PSMA-H variable heavy domain H1 (FIG. 17), and the VL is a variable light domain selected from PSMA-H variable light domains L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). Further, the anti-CD3 scFv includes the variable heavy domain and the variable light domain of a CD3 binding domain selected from H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F). In such heterodimeric antibodies, the numbering of the amino acid variants is according to EU numbering.

In some embodiments, the scFv includes a charged scFv linker having the amino acid sequence (GKPGS)₄ (SEQ ID NO: 1). In certain embodiments, the first and second variant Fc domains each further include amino acid variants 428/434S.

In another aspect, provided herein are heterodimeric anti-PSMA×anti-CD3 antibodies XENP14484, XENP33755, XENP33756, XENP33757, XENP33758, XENP33759, XENP33760, XENP33761, XENP33762, XENP34234, XENP34235, XENP34236, XENP16873, XENP16874, and XENP19722.

In yet another aspect, provided herein are heterodimeric anti-PSMA×anti-CD3 antibodies XENP31602, XENP31603, XENP31855, XENP32218, XENP32219, XENP32220, XENP32221, XENP32222, XENP32223, XENP32224, XENP32225, XENP32226, XENP34237, XENP34238, XENP34239, XENP34625, XENP34626, XENP34627, XENP34628, XENP31853, XENP31856, XENP33063, XENP33064, XENP33065, XENP33066, XENP33067, XENP33068, XENP33069, XENP33070, XENP33071, XENP34240, XENP34241, XENP34242, XENP34629, XENP34630, XENP34631, XENP34632, XENP31854, and XENP31857.

Also provided herein are nucleic acid compositions that include polynucleotide(s) encoding the subject anti-PSMA antibodies, expression vectors that include such polynucleotides and host cells that include such expression vectors. Further provided herein are methods of making such anti-PSMA antibodies, wherein a subject host cell is cultured under conditions wherein the anti-PSMA antibody is expressed, and recovering the anti-PSMA antibody.

In another aspect, provided herein is a method of treating a cancer that includes administering to a patient in need thereof any one of the anti-PSMA antibody described herein.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A-1E depict useful pairs of Fc heterodimerization variant sets (including skew and pI variants). There are variants for which there are no corresponding “monomer 2” variants; these are pI variants which can be used alone on either monomer.

FIG. 2 depicts a list of isosteric variant antibody constant regions and their respective substitutions. pI₍₋₎ indicates lower pI variants, while pI₍₊₎ indicates higher pI variants.

These can be optionally and independently combined with other heterodimerization variants of the antibodies described herein (and other variant types as well, as outlined herein).

FIG. 3 depicts useful ablation variants that ablate FcγR binding (sometimes referred to as “knock outs” or “KO” variants). Generally, ablation variants are found on both monomers, although in some cases they may be on only one monomer.

FIG. 4 depicts particularly useful embodiments of “non-Fv” components of the antibodies described herein.

FIG. 5 depicts a number of charged scFv linkers that find use in increasing or decreasing the pI of the subject heterodimeric bsAbs that utilize one or more scFv as a component, as described herein. The (+H) positive linker finds particular use herein, particularly with anti-CD3 V_L and V_H sequences shown herein. A single prior art scFv linker with a single charge is referenced as “Whitlow”, from Whitlow et al., Protein Engineering 6(8):989-995 (1993). It should be noted that this linker was used for reducing aggregation and enhancing proteolytic stability in scFvs. Such charged scFv linkers can be used in any of the subject antibody formats disclosed herein that include scFvs (e.g., 1+1 Fab-scFv-Fc and 2+1 Fab₂-scFv-Fc formats).

FIG. 6 depicts a number of exemplary domain linkers. In some embodiments, these linkers find use linking a single-chain Fv to an Fc chain. In some embodiments, these linkers may be combined. For example, a GGGGS linker (SEQ ID NO: 2) may be combined with a “half hinge” linker.

FIGS. 7A-7D depict the sequences of several useful 1+1 Fab-scFv-Fc bispecific antibody format heavy chain backbones based on human IgG1, without the Fv sequences (e.g. the scFv and the VH for the Fab side). Backbone 1 is based on human IgG1 (356E/358M allotype), and includes the S364K/E357Q:L368D/K370S skew variants, C220S on the chain with the S364K/E357Q skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 2 is based on human IgG1 (356E/358M allotype), and includes S364K:L368D/K370S skew variants, C220S on the chain with the S364K skew variant, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants, and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 3 is based on human IgG1 (356E/358M allotype), and includes S364K:L368E/K370S skew variants, C220S on the chain with the S364K skew variant, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368E/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 4 is based on human IgG1 (356E/358M allotype), and includes D401K:K360E/Q362E/T411E skew variants, C220S on the chain with the D401K skew variant, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with K360E/Q362E/T411E skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 5 is based on human IgG1 (356D/358L allotype), and includes S364K/E357Q:L368D/K370S skew variants, C220S on the chain with the S364K/E357Q skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 6 is based on human IgG1 (356E/358M allotype), and includes S364K/E357Q:L368D/K370S skew variants, C220S on the chain with the S364K/E357Q skew variants, N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew

variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains, as well as an N297A variant on both chains. Backbone 7 is identical to 6 except the mutation is N297S. Backbone 8 is based on human IgG4, and includes the S364K/E357Q:L368D/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants, as well as a S228P (EU numbering, this is S241P in Kabat) variant on both chains that ablates Fab arm exchange as is known in the art. Backbone 9 is based on human IgG2, and includes the S364K/E357Q:L368D/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants. Backbone 10 is based on human IgG2, and includes the S364K/E357Q:L368D/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants as well as a S267K variant on both chains. Backbone 11 is identical to backbone 1, except it includes M428L/N434S Xtend mutations. Backbone 12 is based on human IgG1 (356E/358M allotype), and includes S364K/E357Q:L368D/K370S skew variants, C220S and the P217R/P229R/N276K pI variants on the chain with S364K/E357Q skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Included within each of these backbones are sequences that are 90, 95, 98 and 99% identical (as defined herein) to the recited sequences, and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 additional amino acid substitutions (as compared to the “parent” of the Figure, which, as will be appreciated by those in the art, already contain a number of amino acid modifications as compared to the parental human IgG1 (or IgG2 or IgG4, depending on the backbone). That is, the recited backbones may contain additional amino acid modifications (generally amino acid substitutions) in addition to the skew, pI and ablation variants contained within the backbones of this figure.

FIGS. 8A-8C depict the sequences of several useful 2+1 Fab₂-scFv-Fc bispecific antibody format heavy chain backbones based on human IgG1, without the Fv sequences (e.g. the scFv and the VH for the Fab side). Backbone 1 is based on human IgG1 (356E/358M allotype), and includes the S364K/E357Q:L368D/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 2 is based on human IgG1 (356E/358M allotype), and includes S364K:L368D/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants, and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 3 is based on human IgG1 (356E/358M allotype), and includes S364K:L368E/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368E/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 4 is based on human IgG1 (356E/358M allotype), and includes D401K:K360E/Q362E/T411E skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with K360E/Q362E/T411E skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 5 is based on human IgG1 (356D/358L allotype), and includes S364K/E357Q:L368D/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 6 is based on human IgG1 (356E/358M allotype), and includes S364K/

E357Q:L368D/K370S skew variants, N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains, as well as an N297A variant on both chains. Backbone 7 is identical to 6 except the mutation is N297S. Backbone 8 is identical to backbone 1, except it includes M428L/N434S Xtend mutations. Backbone 9 is based on human IgG1 (356E/358M allotype), and includes S364K/E357Q:L368D/K370S skew variants, the P217R/P229R/N276K pI variants on the chain with S364K/E357Q skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Included within each of these backbones are sequences that are 90, 95, 98 and 99% identical (as defined herein) to the recited sequences, and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 additional amino acid substitutions (as compared to the “parent” of the Figure, which, as will be appreciated by those in the art, already contain a number of amino acid modifications as compared to the parental human IgG1 (or IgG2 or IgG4, depending on the backbone). That is, the recited backbones may contain additional amino acid modifications (generally amino acid substitutions) in addition to the skew, pI and ablation variants contained within the backbones of this figure.

FIG. 9 depicts the sequences of several useful constant light domain backbones based on human IgG1, without the Fv sequences (e.g. the scFv or the Fab). Included herein are constant light backbone sequences that are 90, 95, 98 and 99% identical (as defined herein) to the recited sequences, and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 additional amino acid modifications.

FIGS. 10A-10F depict sequences for exemplary anti-CD3 scFvs suitable for use in the bispecific antibodies described herein. The CDRs are underlined, the scFv linker is double underlined (in the sequences, the scFv linker is a positively charged scFv (GKPGS)₄ linker (SEQ ID NO: 1), although as will be appreciated by those in the art, this linker can be replaced by other linkers, including uncharged or negatively charged linkers, some of which are depicted in FIG. 5), and the slashes indicate the border(s) of the variable domains. In addition, the naming convention illustrates the orientation of the scFv from N- to C-terminus. As noted herein and is true for every sequence herein containing CDRs, the exact identification of the CDR locations may be slightly different depending on the numbering used as is shown in Table 2, and thus included herein are not only the CDRs that are underlined but also CDRs included within the VH and V_L domains using other numbering systems. Furthermore, as for all the sequences in the Figures, these VH and V_L sequences can be used either in a scFv format or in a Fab format.

FIGS. 11A and 11B depict the antigen sequences for PSMA, including human, mouse and cyno, to facilitate the development of antigen binding domains that bind to both for ease of clinical development.

FIG. 12 depicts illustrative IHC of biopsy cores of prostate cancer and adjacent normal tissue showing PSMA expression.

FIG. 13 depicts breakdown of IHC scores of 192 biopsy cores showing PSMA expression.

FIG. 14 depicts antigen density (determined using Quick-Cal protocol) on cancer cell lines LnCAP, 22Rv1, Huh-7, A549, ASPC-1, HT29, and SKOV3 as well as PSMA-transfected PC3 cells.

FIG. 15 depicts illustrative IHC of cancer cell lines and PSMA-transfected PC3 cells showing PSMA expression.

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FIG. 16 depicts PSMA expression on cancer cell lines and PSMA-transfected PC3 cell lines as determined by IHC and flow cytometry.

FIG. 17 depicts the variable heavy and variable light chain sequences for an exemplary humanized PSMA binding domain referred to herein as PSMA-H, as well as the sequences for XENP31858 and XENP31604, anti-PSMA mAbs based on PSMA-H and IgG1 backbone. CDRs are underlined and slashes indicate the border(s) between the variable regions and constant domain. As noted herein and is true for every sequence herein containing CDRs, the exact identification of the CDR locations may be slightly different depending on the numbering used as is shown in Table 2, and thus included herein are not only the CDRs that are underlined but also CDRs included within the V_H and V_L domains using other numbering systems. Furthermore, as for all the sequences in the Figures, these V_H and V_L sequences can be used either in a scFv format or in a Fab format.

FIGS. 18A-18E depict the variable light chain sequences for PSMA-H variants engineered with the aim to tune binding affinity for human PSMA. CDRs are underlined and slashes indicate the border(s) between the variable regions and constant domain. As noted herein and is true for every sequence herein containing CDRs, the exact identification of the CDR locations may be slightly different depending on the numbering used as is shown in Table 2, and thus included herein are not only the CDRs that are underlined but also CDRs included within V_L domains using other numbering systems. Further, as for all the sequences in the Figures, these V_L sequences can be used either in a scFv format or in a Fab format. Each of the variable light domains depicted herein can be paired with any other α PSMA variable heavy domain.

FIGS. 19A-19Y depict the amino acid sequences for PSMA-H variants engineered with the aim to tune binding affinity for human PSMA formatted as bivalent anti-PSMA mAbs. CDRs are underlined and slashes indicate the border(s) between the variable regions and constant domain. As noted herein and is true for every sequence herein containing CDRs, the exact identification of the CDR locations may be slightly different depending on the numbering used as is shown in Table 2, and thus included herein are not only the CDRs that are underlined but also CDRs included within the V_H and V_L domains using other numbering systems.

FIG. 20A-20B depicts BLI-response, apparent dissociation constant (K_{Dapp}), association rate (k_a), and dissociation rate (k_d) of affinity-engineered PSMA-H variants (in bivalent IgG1 format) as determined by Octet. Substitutions in variable light regions are based on Kabat numbering.

FIG. 21A-21B depicts a couple of formats of the present invention. FIG. 21A depicts the "1+1 Fab-scFv-Fc" format, with a first Fab arm binding PSMA and a second scFv arm binding CD3. FIG. 21B depicts the "2+1 Fab₂-scFv-Fc" format, with a first Fab arm binding PSMA and a second Fab-scFv arm, wherein the Fab binds PSMA and the scFv binds CD3.

FIGS. 22A-22F depict the sequences for illustrative α PSMA $\times\alpha$ CD3 bsAbs in the 1+1 Fab-scFv-Fc format and comprising a H1.30_L1.47 anti-CD3 scFv (a.k.a. CD3 High [VHVL]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times\alpha$ CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can

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include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

FIG. 23 depicts the sequences for illustrative α PSMA $\times\alpha$ CD3 bsAbs in the 1+1 Fab-scFv-Fc format and comprising a H1.33_L1.47 anti-CD3 scFv (a.k.a. CD3 Intermediate [VHVL]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times\alpha$ CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

FIG. 24 depicts the sequences for illustrative α PSMA $\times\alpha$ CD3 bsAbs in the 1+1 Fab-scFv-Fc format and comprising a H1.31_L1.47 anti-CD3 scFv (a.k.a. CD3 High-Int [VHVL]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times\alpha$ CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

FIG. 25 depicts the sequences for illustrative α PSMA $\times\alpha$ CD3 bsAbs in the 1+1 Fab-scFv-Fc format and comprising a H1.32_L1.47 anti-CD3 scFv (a.k.a. CD3 High-Int #1[VHVL]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times\alpha$ CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

FIG. 26 depicts the sequences for illustrative α PSMA $\times\alpha$ CD3 bsAbs in the 2+1 Fab₂-scFv-Fc format and comprising a H1.30_L1.47 anti-CD3 scFv (a.k.a. CD3 High [VHVL]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times\alpha$ CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

FIG. 27 depicts the sequences for illustrative α PSMA $\times\alpha$ CD3 bsAbs in the 2+1 Fab₂-scFv-Fc format and comprising a H1.32_L1.47 anti-CD3 scFv (a.k.a. CD3 High-Int #1[VHVL]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times\alpha$ CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences

that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

FIGS. 28A-28K depict the sequences for illustrative α PSMA $\times$ α CD3 bsAbs in the 2+1 Fab₂-scFv-Fc format and comprising a L1.47_H1.32 anti-CD3 scFv (a.k.a. CD3 High-Int #1[VLVH]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times$ α CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

FIG. 29 depicts the sequences for illustrative α PSMA $\times$ α CD3 bsAbs in the 2+1 Fab₂-scFv-Fc format and comprising a H1.89_L1.47 anti-CD3 scFv (a.k.a. CD3 High-Int #2[VHVL]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times$ α CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

FIGS. 30A-30K depicts the sequences for illustrative α PSMA $\times$ α CD3 bsAbs in the 2+1 Fab₂-scFv-Fc format and comprising a L1.47_H1.89 anti-CD3 scFv (a.k.a. CD3 High-Int #2[VLVH]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times$ α CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

FIG. 31 depicts the sequences for illustrative α PSMA $\times$ α CD3 bsAbs in the 2+1 Fab-scFv-Fc format and comprising a H1.33_L1.47 anti-CD3 scFv (a.k.a. CD3 Intermediate [VHVL]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times$ α CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

FIG. 32 depicts the sequences for illustrative α PSMA $\times$ α CD3 bsAbs in the 2+1 Fab-scFv-Fc format and comprising a L1.47_H1.33 anti-CD3 scFv (a.k.a. CD3 Intermediate [VLVH]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain

components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times$ α CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

FIG. 33 depicts the sequences for illustrative prototypic α PSMA $\times$ α CD3 bsAbs in alternative formats.

FIG. 34 depicts BLI-response, dissociation constant (K_D), association rate (k_a), and dissociation rate (k_d) of affinity-engineered PSMA-H variants (in the context of PSMA $\times$ CD3 bispecifics in the 1+1 Fab-scFv-Fc format with a H1.30_L1.47 anti-CD3 scFv) for human PSMA as determined by Octet. Substitutions in variable light regions are based on Kabat numbering.

FIG. 35 depicts BLI-response, dissociation constant (K_D), association rate (k_a), and dissociation rate (k_d) of affinity-engineered PSMA-H variants (in the context of PSMA $\times$ CD3 bispecifics in the 1+1 Fab-scFv-Fc format with a H1.30_L1.47 anti-CD3 scFv) for cynomolgus PSMA as determined by Octet. Substitutions in variable light regions are based on Kabat numbering. A couple of the bsAbs are listed with NA due to odd sensorgrams.

FIG. 36 depicts induction of RTCC on luciferase-transduced PC3 cells with varying surface PSMA densities by A) XENP34282 and B) XENP14484. The data show that the two prototypic 1+1 anti-PSMA $\times$ anti-CD3 induced RTCC of cell lines expressing high and low PSMA levels, including PC3 (~3K) which represent normal tissues, with similar potency.

FIG. 37 depicts induction of T cell proliferation (as indicated by percentage T cells expressing Ki67) by A) XENP34282 and B) XENP14484 in the presence of PC3 cells with varying surface PSMA densities. The data show that the two prototypic 1+1 anti-PSMA $\times$ anti-CD3 induced T cell proliferation in the presence of cell lines expressing high and low PSMA levels, including PC3 (~3K) which represent normal tissues, with similar potency.

FIG. 38 depicts the binding to PSMA-transfected PC3 (~32K) cells (representative of low PSMA-expressing on-target cells) by A) 1+1 Fab-scFv-Fc bispecific antibodies XENP14484, XENP33756, XENP33757, XENP33761, and XENP337652 and by B) 2+1 Fab₂-scFv-Fc bispecific antibodies XENP31620, XENP32218, XENP32220, XENP32224, and XENP32225. The data show that as monovalent PSMA binding affinity is decreased in the 1+1 Fab-scFv-Fc bispecific antibodies, their binding to PC3 (~32K) cells is drastically reduced. Notably, as monovalent PSMA binding affinity is decreased in the 2+1 Fab₂-scFv-Fc bispecific antibodies, their binding to PC3 (~32K) cells is retained.

FIG. 39 depicts induction of RTCC on transduced PC3 luciferase-transfected cancer cells with varying surface PSMA densities by A) XENP32218, B) XENP32220, and C) XENP32224.

FIG. 40 depicts induction of T cell proliferation (as indicated by percentage T cells expressing Ki67) by A) XENP32218, B) XENP32220, and C) XENP32224 in the presence of cancer cells with varying surface PSMA densities.

FIG. 41 depicts induction of T cell degranulation (as indicated by percentage T cells expressing CD107a) by A)

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XENP32218, B) XENP32220, and C) XENP32224 in the presence of cancer cells with varying surface PSMA densities.

FIG. 42 depicts induction of RTCC on luciferase-transduced PC3 cancer cells with varying surface PSMA densities by XENP31855 (1 nM K_D PSMA+CD3 High-Int #1(VLVH) in 2+1 Fab-scFv-Fc format).

FIG. 43 depicts induction of T cell degranulation (as indicated by percentage T cells expressing CD107a) by XENP31856 (1 nM K_D PSMA+CD3 High-Int #2(VLVH)) in the presence of PC3 cancer cells transfected with varying surface PSMA densities. The data show that XENP31856 was highly selective for high PSMA expressing PC3 (~100K) cell line; however, XENP31856 induced little to no degranulation in the presence of all the low PSMA expressing PC3 cell lines.

FIG. 44 depicts induction of RTCC on luciferase-transduced PC3 cancer cells with varying surface PSMA densities by XENP33063 (7 nM K_D PSMA+CD3 High-Int #2(VLVH)). The data show that XENP33063 was highly selective for high PSMA expressing PC3 (~100K) cell line; however, XENP31856 induced little to no killing on all the low PSMA expressing PC3 cell lines.

FIG. 45 depicts induction of RTCC on luciferase-transduced LNCaP cancer cells and 22Rv1 cancer cells by XENP32220 (38 nM K_D PSMA+CD3 High-Int #1(VLVH)). The data show that XENP32220 was able to induce cell kill on both LNCaP and 22Rv1 cancer cells.

FIG. 46 depicts induction of T cell degranulation (as indicated by percentage T cells expressing CD107a) by A) XENP14484, B) XENP34282, and C) XENP34283 in the presence of cancer cells with varying surface PSMA densities. The data show that the three prototypic 1+1 anti-PSMA×anti-CD3 induced T cell degranulation in the presence of cell lines expressing higher and lower PSMA levels with similar potency.

FIG. 47 depicts tumor volume on Day 19 in PSMA-transfected PC3 (~100K) and huPBMc-engrafted mice following first dose with PBS, bivalent anti-PD1 mAb, XENP32218, XENP32220, or XENP32224. Each of the tuned PSMA×CD3 bispecific antibodies significantly enhanced ($p < 0.05$ vs. PBS or α PD-1 mAb) anti-tumor activity (as indicated by tumor volume; statistics performed on baseline corrected data using unpaired t-test).

FIG. 48 depicts the change in tumor volume (as determined by caliper measurements) over time in PSMA-transfected PC3 (~100K) and huPBMc-engrafted mice dosed with PBS, bivalent anti-PD1 mAb, XENP32218, XENP32220, or XENP32224.

FIGS. 49A-49C depict the pharmacokinetic data from a study in which each healthy male cynomolgus was administered by IV either a 1× dose, 10× dose, or 60× dose of the indicated test article. Xtend variants XENP34262, XENP34267, and XENP34628 showed improved pharmacokinetics over non-Xtend variants XENP32218, XENP32220, and XENP32224. All test articles were tolerated at each dose level.

FIGS. 50A-50K depict several formats for use in the anti-PSMA×anti-CD3 bispecific antibodies disclosed herein. The first is the “1+1 Fab-scFv-Fc” format (also referred to as the “bottle opener” or “Triple F” format), with a first antigen binding domain that is a Fab domain and a second anti-antigen binding domain that is an scFv domain (FIG. 1A). Additionally, “mAb-Fv,” “mAb-scFv,” “2+1 Fab2-scFv-Fc” (also referred to as the “central scFv” or “central-scFv” format), “central-Fv,” “one armed central-scFv,” “one scFv-mAb,” “scFv-mAb,” “dual scFv,” “tri-

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dent,” and non-heterodimeric bispecific formats are all shown. The scFv domains depicted in FIGS. 10A-10F can be either, from N- to C-terminus orientation: variable heavy-(optional linker)-variable light, or variable light-(optional linker)-variable heavy. In addition, for the one armed scFv-mAb, the scFv can be attached either to the N-terminus of a heavy chain monomer or to the N-terminus of the light chain. In certain embodiments, “Anti-antigen 1” in FIG. 50 refers to a PSMA binding domain. In certain embodiments, “Anti-antigen 1” in FIG. 50 refers to a CD3 binding domain. In certain embodiments, “Anti-antigen 2” in FIG. 50 refers to a PSMA binding domain. In certain embodiments “Anti-antigen 2” in FIG. 50 refers to a CD3 binding domain. In some embodiments, “Anti-antigen 1” in FIG. 50 refers to a PSMA binding domain and “Anti-antigen 2” in FIG. 50 refers to a CD3 binding domain. In some embodiments, “Anti-antigen 1” in FIG. 50 refers to a CD3 binding domain and “Anti-antigen 2” in FIG. 50 refers to a PSMA binding domain. Any of the PSMA binding domains and CD3 binding domains disclosed can be included in the bispecific formats of FIG. 50.

DETAILED DESCRIPTION

I. Overview

Provided herein are novel anti-CD3× anti-PSMA (also referred to as anti-PSMA×anti-CD3, α CD3× α PSMA, or α PSMA× α CD3) heterodimeric bispecific antibodies and methods of using such antibodies for the treatment of cancers. In particular, provided herein are anti-CD3, anti-PSMA bispecific antibodies in a variety of formats such as those depicted in FIGS. 21A and 21B. These bispecific antibodies are useful for the treatment of cancers, particularly those with increased PSMA expression such as prostate cancers. Such antibodies are used to direct CD3+ effector T cells to PSMA+ tumors, thereby allowing the CD3+ effector T cells to attack and lyse the PSMA+ tumors.

The anti-PSMA antibodies provided herein include PSMA binding domain with binding affinities and valencies that allow for the advantageous selectivity for cells expressing high levels of PSMA while minimizing reactivity on low PSMA expressing cells. Such PSMA antibodies are useful, for example, for cancers that express high levels of PSMA including, for example, prostate cancer.

In some embodiments, such anti-PSMA antibodies include CD3 binding domains with binding affinity that further contribute to selective targeting of high-PSMA expressing cells lines. Such bispecific antibodies that have different binding affinities to human CD3 that can alter or reduce the potential side effects of anti-CD3 therapy. That is, in some embodiments the antibodies described herein provide antibody constructs comprising anti-CD3 antigen binding domains that are “strong” or “high affinity” binders to CD3 (e.g. one example are heavy and light variable domains depicted as H1.30_L1.47 (optionally including a charged linker as appropriate)) and also bind to PSMA. In other embodiments, the antibodies described herein provide antibody constructs comprising anti-CD3 antigen binding domains that are “lite” or “lower affinity” binders to CD3. Additional embodiments provides antibody constructs comprising anti-CD3 antigen binding domains that have intermediate or “medium” affinity to CD3 that also bind to PSMA. While a very large number of anti-CD3 antigen binding domains (ABDs) can be used, particularly useful embodiments use 6 different anti-CD3 ABDs, although they

can be used in two scFv orientations as discussed herein. Affinity is generally measured using a Biacore assay.

It should be appreciated that the “high, medium, low” anti-CD3 sequences provided herein can be used in a variety of heterodimerization formats as depicted in FIGS. 21A, 21B. In general, due to the potential side effects of T cell recruitment, exemplary embodiments utilize formats that only bind CD3 monovalently, such as depicted in FIGS. 21A and 21B, and in the formats depicted herein, it is the CD3 ABD that is a scFv as more fully described herein. In contrast, the subject bispecific antibodies can bind PSMA either monovalently (e.g. FIG. 21A) or bivalently (e.g. FIG. 21B).

Accordingly, in one aspect, provided herein are heterodimeric antibodies that bind to two different antigens, e.g. the antibodies are “bispecific”, in that they bind two different target antigens, generally PSMA and CD3 as described herein. These heterodimeric antibodies can bind these target antigens either monovalently (e.g. there is a single antigen binding domain such as a variable heavy and variable light domain pair) or bivalently (there are two antigen binding domains that each independently bind the antigen). In some embodiments, the heterodimeric antibody provided herein includes one CD3 binding domain and one PSMA binding domain (e.g., heterodimeric antibodies in the “1+1 Fab-scFv-Fc” format described herein). In other embodiments, the heterodimeric antibody provided herein includes one CD3 binding domain and two PSMA binding domains (e.g., heterodimeric antibodies in the “2+1 Fab₂-scFv-Fc” formats described herein). The heterodimeric antibodies provided herein are based on the use different monomers which contain amino acid substitutions that “skew” formation of heterodimers over homodimers, as is more fully outlined below, coupled with “pI variants” that allow simple purification of the heterodimers away from the homodimers, as is similarly outlined below. The heterodimeric bispecific antibodies provided generally rely on the use of engineered or variant Fc domains that can self-assemble in production cells to produce heterodimeric proteins, and methods to generate and purify such heterodimeric proteins.

II. Nomenclature

The antibodies provided herein are listed in several different formats. In some instances, each monomer of a particular antibody is given a unique “XENP” number, although as will be appreciated in the art, a longer sequence might contain a shorter one. For example, a “scFv-Fc” monomer of a 1+1 Fab-scFv-Fc format antibody may have a first XENP number, while the scFv domain itself will have a different XENP number. Some molecules have three polypeptides, so the XENP number, with the components, is used as a name. Thus, the molecule XENP31602, which is in 2+1 Fab₂-scFv-Fc format, comprises three sequences (see FIG. 26): 1) a “Fab-Fc Heavy Chain” monomer; 2) a “Fab-scFv-Fc Heavy Chain” monomer; and 3) a “Light Chain” monomer or equivalents, although one of skill in the art would be able to identify these easily through sequence alignment. These XENP numbers are in the sequence listing as well as identifiers, and used in the Figures. In addition, one molecule, comprising the three components, gives rise to multiple sequence identifiers. For example, the listing of the Fab includes, the full heavy chain sequence, the variable heavy domain sequence and the three CDRs of the variable heavy domain sequence, the full light chain sequence, a variable light domain sequence and the three CDRs of the variable light domain sequence. A Fab-scFv-Fc monomer

includes a full length sequence, a variable heavy domain sequence, 3 heavy CDR sequences, and an scFv sequence (include scFv variable heavy domain sequence, scFv variable light domain sequence and scFv linker). Note that some molecules herein with a scFv domain use a single charged scFv linker (+H), although others can be used. In addition, the naming nomenclature of particular antigen binding domains (e.g., PSMA and CD3 binding domains) use a “Hx.xx_Ly.yy” type of format, with the numbers being unique identifiers to particular variable chain sequences. Thus, the PSMA binding domain PSMA-H H1_L1 (e.g., FIG. 17A) is “H1_L1”, which indicates that the variable heavy domain, H1, was combined with the light domain L1. In the case that these sequences are used as scFvs, the designation “H1_L1”, indicates that the variable heavy domain, H1 is combined with the light domain, L1, and is in VH-linker-VL orientation, from N- to C-terminus. This molecule with the identical sequences of the heavy and light variable domains but in the reverse order (VL-linker-VH orientation, from N- to C-terminus) would be designated “L1_H1.1”. Similarly, different constructs may “mix and match” the heavy and light chains as will be evident from the sequence listing and the figures.

III. Definitions

In order that the application may be more completely understood, several definitions are set forth below. Such definitions are meant to encompass grammatical equivalents.

By “PSMA” or “Prostate Specific Membrane Antigen” (e.g., Genbank Accession Number NP 005012.2) herein is meant a type II transmembrane protein that is expressed in all prostatic tissues, including primary prostate adenocarcinomas, metastatic prostate cancer, and in the tumor neovasculature of many solid tumors. In prostate cancer (PCa), PSMA is highly expressed in poorly differentiated, highly metastatic prostatic cells and in castrate-resistant models. PSMA sequences are disclosed in FIGS. 11A and 11B.

By “ablation” herein is meant a decrease or removal of activity. Thus for example, “ablating FcγR binding” means the Fc region amino acid variant has less than 50% starting binding as compared to an Fc region not containing the specific variant, with more than 70-80-90-95-98% loss of activity being preferred, and in general, with the activity being below the level of detectable binding in a Biacore, SPR or BLI assay. Of particular use in the ablation of FcγR binding are those shown in FIG. 5, which generally are added to both monomers.

By “ADCC” or “antibody dependent cell-mediated cytotoxicity” as used herein is meant the cell-mediated reaction wherein nonspecific cytotoxic cells that express FcγRs recognize bound antibody on a target cell and subsequently cause lysis of the target cell. ADCC is correlated with binding to FcγRIIIa; increased binding to FcγRIIIa leads to an increase in ADCC activity.

By “ADCP” or antibody dependent cell-mediated phagocytosis as used herein is meant the cell-mediated reaction wherein nonspecific phagocytic cells that express FcγRs recognize bound antibody on a target cell and subsequently cause phagocytosis of the target cell.

As used herein, term “antibody” is used generally. Antibodies described herein can take on a number of formats as described herein, including traditional antibodies as well as antibody derivatives, fragments and mimetics, described herein.

Traditional immunoglobulin (Ig) antibodies are “Y” shaped tetramers. Each tetramer is typically composed of two identical pairs of polypeptide chains, each pair having one “light chain” monomer (typically having a molecular weight of about 25 kDa) and one “heavy chain” monomer (typically having a molecular weight of about 50-70 kDa).

Other useful antibody formats include, but are not limited to, the 1+1 Fab-scFv-Fc format and 2+1 Fab₂-scFv-Fc antibody formats described herein, as well as “mAb-Fv,” “mAb-scFv,” “central-Fv,” “one armed scFv-mAb,” “scFv-mAb,” “dual scFv,” and “trident” format antibodies, as shown in FIG. 50.

Antibody heavy chains typically include a variable heavy (VH) domain, which includes vhCDR1-3, and an Fc domain, which includes a CH2-CH3 monomer. In some embodiments, antibody heavy chains include a hinge and CH1 domain. Traditional antibody heavy chains are monomers that are organized, from N- to C-terminus: VH-CH1-hinge-CH2-CH3. The CH1-hinge-CH2-CH3 is collectively referred to as the heavy chain “constant domain” or “constant region” of the antibody, of which there are five different categories or “isotypes”: IgA, IgD, IgG, IgE and IgM. Thus, “isotype” as used herein is meant any of the subclasses of immunoglobulins defined by the chemical and antigenic characteristics of their constant regions. It should be understood that therapeutic antibodies can also comprise hybrids of isotypes and/or subclasses. For example, as shown in US Publication 2009/0163699, incorporated by reference, the antibodies described herein include the use of human IgG1/G2 hybrids.

In some embodiments, the antibodies provided herein include IgG isotype constant domains, which has several subclasses, including, but not limited to IgG1, IgG2, IgG3, and IgG4. In the IgG subclass of immunoglobulins, there are several immunoglobulin domains in the heavy chain. By “immunoglobulin (Ig) domain” herein is meant a region of an immunoglobulin having a distinct tertiary structure. Of interest in the antibodies described herein are the heavy chain domains, including, the constant heavy (CH) domains and the hinge domains. In the context of IgG antibodies, the IgG isotypes each have three CH regions. Accordingly, “CH” domains in the context of IgG are as follows: “CH1” refers to positions 118-220 according to the EU index as in Kabat. “CH2” refers to positions 237-340 according to the EU index as in Kabat, and “CH3” refers to positions 341-447 according to the EU index as in Kabat. As shown herein and described below, the pI variants can be in one or more of the CH regions, as well as the hinge region, discussed below.

It should be noted that IgG1 has different allotypes with polymorphisms at 356 (D or E) and 358 (L or M). The sequences depicted herein use the 356D/358M allotype, however the other allotype is included herein. That is, any sequence inclusive of an IgG1 Fc domain included herein can have 356E/358L replacing the 356D/358M allotype. It should be understood that therapeutic antibodies can also comprise hybrids of isotypes and/or subclasses. For example, as shown in US Publication 2009/0163699, incorporated by reference, the present antibodies, in some embodiments, include IgG1/IgG2 hybrids.

By “Fc” or “Fc region” or “Fc domain” as used herein is meant the polypeptide comprising the constant region of an antibody, in some instances, excluding all of the first constant region immunoglobulin domain (e.g., CH1) or a portion thereof, and in some cases, optionally including all or part of the hinge. For IgG, the Fc domain comprises immunoglobulin domains CH2 and CH3 (Cγ2 and Cγ3), and

optionally all or a portion of the hinge region between CH1 (Cγ1) and CH2 (Cγ2). Thus, in some cases, the Fc domain includes, from N- to C-terminal, CH2-CH3 and hinge-CH2-CH3. In some embodiments, the Fc domain is that from IgG1, IgG2, IgG3 or IgG4, with IgG1 hinge-CH2-CH3 and IgG4 hinge-CH2-CH3 finding particular use in many embodiments. Additionally, in the case of human IgG1 Fc domains, frequently the hinge includes a C220S amino acid substitution. Furthermore, in the case of human IgG4 Fc domains, frequently the hinge includes a S228P amino acid substitution. Although the boundaries of the Fc region may vary, the human IgG heavy chain Fc region is usually defined to include residues E216, C226, or A231 to its carboxyl-terminal, wherein the numbering is according to the EU index as in Kabat. In some embodiments, as is more fully described below, amino acid modifications are made to the Fc region, for example to alter binding to one or more FcγR or to the FcRn.

By “heavy chain constant region” herein is meant the CH1-hinge-CH2-CH3 portion of an antibody (or fragments thereof), excluding the variable heavy domain; in EU numbering of human IgG1 this is amino acids 118-447. By “heavy chain constant region fragment” herein is meant a heavy chain constant region that contains fewer amino acids from either or both of the N- and C-termini but still retains the ability to form a dimer with another heavy chain constant region.

Another type of Ig domain of the heavy chain is the hinge region. By “hinge” or “hinge region” or “antibody hinge region” or “hinge domain” herein is meant the flexible polypeptide comprising the amino acids between the first and second constant domains of an antibody. Structurally, the IgG CH1 domain ends at EU position 215, and the IgG CH2 domain begins at residue EU position 231. Thus for IgG the antibody hinge is herein defined to include positions 216 (E216 in IgG1) to 230 (p230 in IgG1), wherein the numbering is according to the EU index as in Kabat. In some cases, a “hinge fragment” is used, which contains fewer amino acids at either or both of the N- and C-termini of the hinge domain. As noted herein, pI variants can be made in the hinge region as well. Many of the antibodies herein have at least one the cysteines at position 220 according to EU numbering (hinge region) replaced by a serine. Generally, this modification is on the “scFv monomer” side for most of the sequences depicted herein, although it can also be on the “Fab monomer” side, or both, to reduce disulfide formation. Specifically included within the sequences herein are one or both of these cysteines replaced (C220S).

As will be appreciated by those in the art, the exact numbering and placement of the heavy constant region domains can be different among different numbering systems. A useful comparison of heavy constant region numbering according to EU and Kabat is as below, see Edelman et al., 1969, Proc Natl Acad Sci USA 63:78-85 and Kabat et al., 1991, Sequences of Proteins of Immunological Interest, 5th Ed., United States Public Health Service, National Institutes of Health, Bethesda, entirely incorporated by reference.

TABLE 1

	EU Numbering	Kabat Numbering
CH1	118-215	114-223
Hinge	216-230	226-243
CH2	231-340	244-360
CH3	341-447	361-478

The antibody light chain generally comprises two domains: the variable light domain (VL), which includes light chain CDRs vLCDR1-3, and a constant light chain region (often referred to as CL or C κ). The antibody light chain is typically organized from N- to C-terminus: VL-CL.

By "antigen binding domain" or "ABD" herein is meant a set of six Complementary Determining Regions (CDRs) that, when present as part of a polypeptide sequence, specifically binds a target antigen (e.g., PSMA or CD3) as discussed herein. As is known in the art, these CDRs are generally present as a first set of variable heavy CDRs (vHCDRs or VHCDRs) and a second set of variable light CDRs (vLCDRs or VLCDRs), each comprising three CDRs: vHCDR1, vHCDR2, vHCDR3 variable heavy CDRs and vLCDR1, vLCDR2 and vLCDR3 variable light CDRs. The CDRs are present in the variable heavy domain (vHCDR1-3) and variable light domain (vLCDR1-3). The variable heavy domain and variable light domain from an Fv region.

The antibodies described herein provide a large number of different CDR sets. In this case, a "full CDR set" comprises the three variable light and three variable heavy CDRs, e.g., a vLCDR1, vLCDR2, vLCDR3, vHCDR1, vHCDR2 and vHCDR3. These can be part of a larger variable light or variable heavy domain, respectfully. In addition, as more fully outlined herein, the variable heavy and variable light domains can be on separate polypeptide chains, when a heavy and light chain is used (for example when Fabs are used), or on a single polypeptide chain in the case of scFv sequences.

As will be appreciated by those in the art, the exact numbering and placement of the CDRs can be different among different numbering systems. However, it should be understood that the disclosure of a variable heavy and/or variable light sequence includes the disclosure of the associated (inherent) CDRs. Accordingly, the disclosure of each variable heavy region is a disclosure of the vHCDRs (e.g., vHCDR1, vHCDR2 and vHCDR3) and the disclosure of each variable light region is a disclosure of the vLCDRs (e.g., vLCDR1, vLCDR2 and vLCDR3). A useful comparison of CDR numbering is as below, see Lafranc et al., *Dev. Comp. Immunol.* 27(1):55-77 (2003):

TABLE 2

	Kabat + Chothia	IMGT	Kabat	AbM	Chothia	Contact	Xencor
vHCDR1	26-35	27-38	31-35	26-35	26-32	30-35	27-35
vHCDR2	50-65	56-65	50-65	50-58	52-56	47-58	54-61
vHCDR3	95-102	105-117	95-102	95-102	95-102	93-101	103-116
vLCDR1	24-34	27-38	24-34	24-34	24-34	30-36	27-38
vLCDR2	50-56	56-65	50-56	50-56	50-56	46-55	56-62
vLCDR3	89-97	105-117	89-97	89-97	89-97	89-96	97-105

Throughout the present specification, the Kabat numbering system is generally used when referring to a residue in the variable domain (approximately, residues 1-107 of the light chain variable region and residues 1-113 of the heavy chain variable region) and the EU numbering system for Fc regions (e.g., Kabat et al., supra (1991)).

The CDRs contribute to the formation of the antigen-binding, or more specifically, epitope binding site of the antigen binding domains and antibodies. "Epitope" refers to a determinant that interacts with a specific antigen binding site in the variable region of an antibody molecule known as a paratope. Epitopes are groupings of molecules such as amino acids or sugar side chains and usually have specific

structural characteristics, as well as specific charge characteristics. A single antigen may have more than one epitope.

The epitope may comprise amino acid residues directly involved in the binding (also called immunodominant component of the epitope) and other amino acid residues, which are not directly involved in the binding, such as amino acid residues which are effectively blocked by the specifically antigen binding peptide; in other words, the amino acid residue is within the footprint of the specifically antigen binding peptide.

Epitopes may be either conformational or linear. A conformational epitope is produced by spatially juxtaposed amino acids from different segments of the linear polypeptide chain. A linear epitope is one produced by adjacent amino acid residues in a polypeptide chain. Conformational and nonconformational epitopes may be distinguished in that the binding to the former but not the latter is lost in the presence of denaturing solvents.

An epitope typically includes at least 3, and more usually, at least 5 or 8-10 amino acids in a unique spatial conformation. Antibodies that recognize the same epitope can be verified in a simple immunoassay showing the ability of one antibody to block the binding of another antibody to a target antigen, for example "binning." As outlined below, the disclosure not only includes the enumerated antigen binding domains and antibodies herein, but those that compete for binding with the epitopes bound by the enumerated antigen binding domains.

In some embodiments, the six CDRs of the antigen binding domain are contributed by a variable heavy and a variable light domain. In a "Fab" format, the set of 6 CDRs are contributed by two different polypeptide sequences, the variable heavy domain (vh or VH; containing the vHCDR1, vHCDR2 and vHCDR3) and the variable light domain (vl or VL; containing the vLCDR1, vLCDR2 and vLCDR3), with the C-terminus of the vh domain being attached to the N-terminus of the CH1 domain of the heavy chain and the C-terminus of the vl domain being attached to the N-terminus of the constant light domain (and thus forming the light chain). In a scFv format, the vh and vl domains are covalently attached, generally through the use of a linker (a "scFv linker") as outlined herein, into a single polypeptide

sequence, which can be either (starting from the N-terminus) vh-linker-vl or vl-linker-vh, with the former being generally preferred (including optional domain linkers on each side, depending on the format used (e.g., from FIG. 1). In general, the C-terminus of the scFv domain is attached to the N-terminus of the hinge in the second monomer.

By "variable region" or "variable domain" as used herein is meant the region of an immunoglobulin that comprises one or more Ig domains substantially encoded by any of the V κ , V δ , and/or VH genes that make up the kappa, lambda, and heavy chain immunoglobulin genetic loci respectively, and contains the CDRs that confer antigen specificity. Thus, a "variable heavy domain" pairs with a "variable light domain" to form an antigen binding domain ("ABD"). In

addition, each variable domain comprises three hypervariable regions ("complementary determining regions," "CDRs") (VHCDR1, VHCDR2 and VHCDR3 for the variable heavy domain and VLCDR1, VLCDR2 and VLCDR3 for the variable light domain) and four framework (FR) regions, arranged from amino-terminus to carboxy-terminus in the following order: FR1-CDR1-FR2-CDR2-FR3-CDR3-FR4. The hypervariable region generally encompasses amino acid residues from about amino acid residues 24-34 (LCDR1; "L" denotes light chain), 50-56 (LCDR2) and 89-97 (LCDR3) in the light chain variable region and around about 31-35B (HCDR1; "H" denotes heavy chain), 50-65 (HCDR2), and 95-102 (HCDR3) in the heavy chain variable region; Kabat et al., SEQUENCES OF PROTEINS OF IMMUNOLOGICAL INTEREST, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, Md. (1991) and/or those residues forming a hypervariable loop (e.g. residues 26-32 (LCDR1), 50-52 (LCDR2) and 91-96 (LCDR3) in the light chain variable region and 26-32 (HCDR1), 53-55 (HCDR2) and 96-101 (HCDR3) in the heavy chain variable region; Chothia and Lesk (1987) J. Mol. Biol. 196:901-917. Specific CDRs of the invention are described in Table 2.

By "Fab" or "Fab region" as used herein is meant the polypeptide that comprises the VH, CH1, VL, and CL immunoglobulin domains, generally on two different polypeptide chains (e.g. VH-CH1 on one chain and VL-CL on the other). Fab may refer to this region in isolation, or this region in the context of a bispecific antibody described herein. In the context of a Fab, the Fab comprises an Fv region in addition to the CH1 and CL domains.

By "Fv" or "Fv fragment" or "Fv region" as used herein is meant a polypeptide that comprises the VL and VH domains of an ABD. Fv regions can be formatted as both Fabs (as discussed above, generally two different polypeptides that also include the constant regions as outlined above) and scFvs, where the VL and VH domains are combined (generally with a linker as discussed herein) to form an scFv.

By "single chain Fv" or "scFv" herein is meant a variable heavy domain covalently attached to a variable light domain, generally using a scFv linker as discussed herein, to form a scFv or scFv domain. A scFv domain can be in either orientation from N- to C-terminus (VH-linker-VL or VL-linker-VH). In the sequences depicted in the sequence listing and in the figures, the order of the VH and VL domain is indicated in the name, e.g. H.X_L.Y means N- to C-terminal is VH-linker-VL, and L.Y_H.X is VL-linker-VH.

Some embodiments of the subject antibodies provided herein comprise at least one scFv domain, which, while not naturally occurring, generally includes a variable heavy domain and a variable light domain, linked together by a scFv linker. As outlined herein, while the scFv domain is generally from N- to C-terminus oriented as VH-scFv linker-VL, this can be reversed for any of the scFv domains (or those constructed using vh and vl sequences from Fabs), to VL-scFv linker-VH, with optional linkers at one or both ends depending on the format.

By "modification" herein is meant an amino acid substitution, insertion, and/or deletion in a polypeptide sequence or an alteration to a moiety chemically linked to a protein. For example, a modification may be an altered carbohydrate or PEG structure attached to a protein. By "amino acid modification" herein is meant an amino acid substitution, insertion, and/or deletion in a polypeptide sequence. For clarity, unless otherwise noted, the amino acid modification

is always to an amino acid coded for by DNA, e.g. the 20 amino acids that have codons in DNA and RNA.

By "amino acid substitution" or "substitution" herein is meant the replacement of an amino acid at a particular position in a parent polypeptide sequence with a different amino acid. In particular, in some embodiments, the substitution is to an amino acid that is not naturally occurring at the particular position, either not naturally occurring within the organism or in any organism. For example, the substitution E272Y refers to a variant polypeptide, in this case an Fc variant, in which the glutamic acid at position 272 is replaced with tyrosine. For clarity, a protein which has been engineered to change the nucleic acid coding sequence but not change the starting amino acid (for example exchanging CGG (encoding arginine) to CGA (still encoding arginine) to increase host organism expression levels) is not an "amino acid substitution"; that is, despite the creation of a new gene encoding the same protein, if the protein has the same amino acid at the particular position that it started with, it is not an amino acid substitution.

By "amino acid insertion" or "insertion" as used herein is meant the addition of an amino acid sequence at a particular position in a parent polypeptide sequence. For example, -233E or 233E designates an insertion of glutamic acid after position 233 and before position 234. Additionally, -233ADE or A233ADE designates an insertion of AlaAsp-Glu after position 233 and before position 234.

By "amino acid deletion" or "deletion" as used herein is meant the removal of an amino acid sequence at a particular position in a parent polypeptide sequence. For example, E233- or E233 #, E233() or E233del designates a deletion of glutamic acid at position 233. Additionally, EDA233- or EDA233 #designates a deletion of the sequence GluAspAla that begins at position 233.

By "variant protein" or "protein variant", or "variant" as used herein is meant a protein that differs from that of a parent protein by virtue of at least one amino acid modification. The protein variant has at least one amino acid modification compared to the parent protein, yet not so many that the variant protein will not align with the parental protein using an alignment program such as that described below. In general, variant proteins (such as variant Fc domains, etc., outlined herein, are generally at least 75, 80, 85, 90, 91, 92, 93, 94, 95, 96, 97, 98 or 99% identical to the parent protein, using the alignment programs described below, such as BLAST. "Variant" as used herein also refers to particular amino acid modifications that confer particular function (e.g., a "heterodimerization variant," "pI variant," "ablation variant," etc.).

As described below, in some embodiments the parent polypeptide, for example an Fc parent polypeptide, is a human wild type sequence, such as the heavy constant domain or Fc region from IgG1, IgG2, IgG3 or IgG4, although human sequences with variants can also serve as "parent polypeptides", for example the IgG1/2 hybrid of US Publication 2006/0134105 can be included. The protein variant sequence herein will preferably possess at least about 80% identity with a parent protein sequence, and most preferably at least about 90% identity, more preferably at least about 95-98-99% identity. Accordingly, by "antibody variant" or "variant antibody" as used herein is meant an antibody that differs from a parent antibody by virtue of at least one amino acid modification, "IgG variant" or "variant IgG" as used herein is meant an antibody that differs from a parent IgG (again, in many cases, from a human IgG sequence) by virtue of at least one amino acid modification, and "immunoglobulin variant" or "variant immunoglobulin"

as used herein is meant an immunoglobulin sequence that differs from that of a parent immunoglobulin sequence by virtue of at least one amino acid modification. "Fc variant" or "variant Fc" as used herein is meant a protein comprising an amino acid modification in an Fc domain as compared to an Fc domain of human IgG1, IgG2 or IgG4.

"Fc variant" or "variant Fc" as used herein is meant a protein comprising an amino acid modification in an Fc domain. The modification can be an addition, deletion, or substitution. The Fc variants are defined according to the amino acid modifications that compose them. Thus, for example, N434S or 434S is an Fc variant with the substitution for serine at position 434 relative to the parent Fc polypeptide, wherein the numbering is according to the EU index. Likewise, M428L/N434S defines an Fc variant with the substitutions M428L and N434S relative to the parent Fc polypeptide. The identity of the WT amino acid may be unspecified, in which case the aforementioned variant is referred to as 428L/434S. It is noted that the order in which substitutions are provided is arbitrary, that is to say that, for example, 428L/434S is the same Fc variant as 434S/428L, and so on. For all positions discussed herein that relate to antibodies or derivatives and fragments thereof (e.g., Fc domains), unless otherwise noted, amino acid position numbering is according to the EU index. The "EU index" or "EU index as in Kabat" or "EU numbering" scheme refers to the numbering of the EU antibody (Edelman et al., 1969, *Proc Natl Acad Sci USA* 63:78-85, hereby entirely incorporated by reference).

In general, variant Fc domains have at least about 80, 85, 90, 95, 97, 98 or 99 percent identity to the corresponding parental human IgG Fc domain (using the identity algorithms discussed below, with one embodiment utilizing the BLAST algorithm as is known in the art, using default parameters). Alternatively, the variant Fc domains can have from 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 amino acid modifications as compared to the parental Fc domain. Alternatively, the variant Fc domains can have up to 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 amino acid modifications as compared to the parental Fc domain. Additionally, as discussed herein, the variant Fc domains described herein still retain the ability to form a dimer with another Fc domain as measured using known techniques as described herein, such as non-denaturing gel electrophoresis.

By "protein" herein is meant at least two covalently attached amino acids, which includes proteins, polypeptides, oligopeptides and peptides. In addition, polypeptides that make up the antibodies described herein may include synthetic derivatization of one or more side chains or termini, glycosylation, PEGylation, circular permutation, cyclization, linkers to other molecules, fusion to proteins or protein domains, and addition of peptide tags or labels.

By "residue" as used herein is meant a position in a protein and its associated amino acid identity. For example, Asparagine 297 (also referred to as Asn297 or N297) is a residue at position 297 in the human antibody IgG1.

By "IgG subclass modification" or "isotype modification" as used herein is meant an amino acid modification that converts one amino acid of one IgG isotype to the corresponding amino acid in a different, aligned IgG isotype. For example, because IgG1 comprises a tyrosine and IgG2 a phenylalanine at EU position 296, a F296Y substitution in IgG2 is considered an IgG subclass modification.

By "non-naturally occurring modification" as used herein is meant an amino acid modification that is not isotypic. For

example, because none of the human IgGs comprise a serine at position 434, the substitution 434S in IgG1, IgG2, IgG3, or IgG4 (or hybrids thereof) is considered a non-naturally occurring modification.

By "amino acid" and "amino acid identity" as used herein is meant one of the 20 naturally occurring amino acids that are coded for by DNA and RNA.

By "effector function" as used herein is meant a biochemical event that results from the interaction of an antibody Fc region with an Fc receptor or ligand. Effector functions include but are not limited to ADCC, ADCP, and CDC.

By "IgG Fc ligand" as used herein is meant a molecule, preferably a polypeptide, from any organism that binds to the Fc region of an IgG antibody to form an Fc/Fc ligand complex. Fc ligands include but are not limited to FcγRI, FcγRII, FcγRIII, FcRn, C1q, C3, mannan binding lectin, mannose receptor, staphylococcal protein A, streptococcal protein G, and viral FcγR. Fc ligands also include Fc receptor homologs (FcRH), which are a family of Fc receptors that are homologous to the FcγRs (Davis et al., 2002, *Immunological Reviews* 190:123-136, entirely incorporated by reference). Fc ligands may include undiscovered molecules that bind Fc. Particular IgG Fc ligands are FcRn and Fc gamma receptors. By "Fc ligand" as used herein is meant a molecule, preferably a polypeptide, from any organism that binds to the Fc region of an antibody to form an Fc/Fc ligand complex.

By "Fc gamma receptor", "FcγR" or "FcγgammaR" as used herein is meant any member of the family of proteins that bind the IgG antibody Fc region and is encoded by an FcγR gene. In humans this family includes but is not limited to FcγRI (CD64), including isoforms FcγRIa, FcγRIb, and FcγRIc; FcγRII (CD32), including isoforms FcγRIIa (including allotypes H131 and R131), FcγRIIb (including FcγRIIb-1 and FcγRIIb-2), and FcγRIIc; and FcγRIII (CD16), including isoforms FcγRIIIa (including allotypes V158 and F158) and FcγRIIIb (including allotypes FcγRIIb-NA1 and FcγRIIb-NA2) (Jefferis et al., 2002, *Immunol Lett* 82:57-65, entirely incorporated by reference), as well as any undiscovered human FcγRs or FcγR isoforms or allotypes. An FcγR may be from any organism, including but not limited to humans, mice, rats, rabbits, and monkeys. Mouse FcγRs include but are not limited to FcγRI (CD64), FcγRII (CD32), FcγRIII (CD16), and FcγRIII-2 (CD16-2), as well as any undiscovered mouse FcγRs or FcγR isoforms or allotypes.

By "FcRn" or "neonatal Fe Receptor" as used herein is meant a protein that binds the IgG antibody Fc region and is encoded at least in part by an FcRn gene. The FcRn may be from any organism, including but not limited to humans, mice, rats, rabbits, and monkeys. As is known in the art, the functional FcRn protein comprises two polypeptides, often referred to as the heavy chain and light chain. The light chain is beta-2-microglobulin and the heavy chain is encoded by the FcRn gene. Unless otherwise noted herein, FcRn or an FcRn protein refers to the complex of FcRn heavy chain with beta-2-microglobulin. A variety of FcRn variants used to increase binding to the FcRn receptor, and in some cases, to increase serum half-life. An "FcRn variant" is one that increases binding to the FcRn receptor, and suitable FcRn variants are shown below.

By "parent polypeptide" as used herein is meant a starting polypeptide that is subsequently modified to generate a variant. The parent polypeptide may be a naturally occurring polypeptide, or a variant or engineered version of a naturally occurring polypeptide. Accordingly, by "parent immunoglobulin" as used herein is meant an unmodified immuno-

globulin polypeptide that is modified to generate a variant, and by “parent antibody” as used herein is meant an unmodified antibody that is modified to generate a variant antibody. It should be noted that “parent antibody” includes known commercial, recombinantly produced antibodies as outlined below. In this context, a “parent Fc domain” will be relative to the recited variant; thus, a “variant human IgG1 Fc domain” is compared to the parent Fc domain of human IgG1, a “variant human IgG4 Fc domain” is compared to the parent Fc domain human IgG4, etc.

By “position” as used herein is meant a location in the sequence of a protein. Positions may be numbered sequentially, or according to an established format, for example the EU index for antibody numbering.

By “target antigen” as used herein is meant the molecule that is bound specifically by the antigen binding domain comprising the variable regions of a given antibody.

By “strandedness” in the context of the monomers of the heterodimeric antibodies described herein is meant that, similar to the two strands of DNA that “match”, heterodimerization variants are incorporated into each monomer so as to preserve the ability to “match” to form heterodimers. For example, if some pI variants are engineered into monomer A (e.g. making the pI higher) then steric variants that are “charge pairs” that can be utilized as well do not interfere with the pI variants, e.g. the charge variants that make a pI higher are put on the same “strand” or “monomer” to preserve both functionalities. Similarly, for “skew” variants that come in pairs of a set as more fully outlined below, the skilled artisan will consider pI in deciding into which strand or monomer one set of the pair will go, such that pI separation is maximized using the pI of the skews as well.

By “target cell” as used herein is meant a cell that expresses a target antigen.

By “host cell” in the context of producing a bispecific antibody according to the antibodies described herein is meant a cell that contains the exogenous nucleic acids encoding the components of the bispecific antibody and is capable of expressing the bispecific antibody under suitable conditions. Suitable host cells are discussed below.

By “wild type or WT” herein is meant an amino acid sequence or a nucleotide sequence that is found in nature, including allelic variations. A WT protein has an amino acid sequence or a nucleotide sequence that has not been intentionally modified.

Provided herein are a number of antibody domains that have sequence identity to human antibody domains. Sequence identity between two similar sequences (e.g., antibody variable domains) can be measured by algorithms such as that of Smith, T. F. & Waterman, M. S. (1981) “Comparison Of Biosequences,” *Adv. Appl. Math.* 2:482 [local homology algorithm]; Needleman, S. B. & Wunsch, CD. (1970) “A General Method Applicable To The Search For Similarities In The Amino Acid Sequence Of Two Proteins,” *J. Mol. Biol.* 48:443 [homology alignment algorithm]; Pearson, W. R. & Lipman, D. J. (1988) “Improved Tools For Biological Sequence Comparison,” *Proc. Natl. Acad. Sci. (U.S.A.)* 85:2444 [search for similarity method]; or Altschul, S. F. et al, (1990) “Basic Local Alignment Search Tool,” *J. Mol. Biol.* 215:403-10, the “BLAST” algorithm, see <https://blast.ncbi.nlm.nih.gov/Blast.cgi>. When using any of the aforementioned algorithms, the default parameters (for Window length, gap penalty, etc) are used. In one embodiment, sequence identity is done using the BLAST algorithm, using default parameters

The antibodies described herein are generally isolated or recombinant. “Isolated,” when used to describe the various

polypeptides disclosed herein, means a polypeptide that has been identified and separated and/or recovered from a cell or cell culture from which it was expressed. Ordinarily, an isolated polypeptide will be prepared by at least one purification step. An “isolated antibody,” refers to an antibody that is substantially free of other antibodies having different antigenic specificities. “Recombinant” means the antibodies are generated using recombinant nucleic acid techniques in exogenous host cells, and they can be isolated as well.

“Specific binding” or “specifically binds to” or is “specific for” a particular antigen or an epitope means binding that is measurably different from a non-specific interaction. Specific binding can be measured, for example, by determining binding of a molecule compared to binding of a control molecule, which generally is a molecule of similar structure that does not have binding activity. For example, specific binding can be determined by competition with a control molecule that is similar to the target.

Specific binding for a particular antigen or an epitope can be exhibited, for example, by an antibody having a KD for an antigen or epitope of at least about 10^{-4} M, at least about 10^{-5} M, at least about 10^{-6} M, at least about 10^{-7} M, at least about 10^{-8} M, at least about 10^{-9} M, alternatively at least about 10^{-10} M, at least about 10^{-11} M, at least about 10^{-12} M, or greater, where KD refers to a dissociation rate of a particular antibody-antigen interaction. Typically, an antibody that specifically binds an antigen will have a KD that is 20-, 50-, 100-, 500-, 1000-, 5,000-, 10,000- or more times greater for a control molecule relative to the antigen or epitope.

Also, specific binding for a particular antigen or an epitope can be exhibited, for example, by an antibody having a KA or Ka for an antigen or epitope of at least 20-, 50-, 100-, 500-, 1000-, 5,000-, 10,000- or more times greater for the epitope relative to a control, where KA or Ka refers to an association rate of a particular antibody-antigen interaction. Binding affinity is generally measured using a Biacore, SPR or BLI assay.

IV. PSMA Binding Domains

In one aspect, provided herein are PSMA antigen binding domains (ABDs) and compositions that include such PSMA antigen binding domains (ABDs), including anti-PSMA antibodies. Subject antibodies that include such PSMA antigen binding domains (e.g., anti-PSMA×anti-CD3 bispecific antibodies) advantageously target cells that express high levels of PSMA over those that express levels of PSMA (e.g., normal cells). Such PSMA binding domains and related antibodies find use, for example, in the treatment of PSMA associated cancers, such as prostate cancer.

As will be appreciated by those in the art, suitable PSMA binding domains can comprise a set of 6 CDRs as depicted in the sequence listing and FIGS. 17-19, either as the CDRs are underlined or, in the case where a different numbering scheme is used as described herein and as shown in Table 2, as the CDRs that are identified using other alignments within the variable heavy (VH) domain and variable light domain (VL) sequences of those depicted in FIGS. 17-19 and the Sequence Listing (see Table 2). Suitable PSMA ABDs can also include the entire VH and VL sequences as depicted in these sequences and figures, used as scFvs or as Fab domains.

In one embodiment, the PSMA antigen binding domain includes the 6 CDRs (i.e., vHCDR1-3 and vLCDR1-3) of a PSMA ABD described herein, including the Figures and sequence listing. In some embodiments, the PSMA ABD

includes the vhCDR1-3 of PSMA-H H1 (FIG. 17) and the vlCDR1-3 of a PSMA variable light domain selected from PSMA-H L1 (FIG. 17) and L1.1-L1.84 (FIG. 18A-E). In exemplary embodiments, the PSMA ABD is one of the following PSMA ABDs: PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; and PSMA-H H1_L1.13.

In addition to the parental CDR sets disclosed in the figures and sequence listing that form an ABD to PSMA, provided herein are variant PSMA ABDs having CDRs that include at least one modification of the PSMA ABD CDRs disclosed herein. In one embodiment, the PSMA ABD includes a set of 6 CDRs with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 amino acid modifications as compared to the 6 CDRs of a PSMA ABD described herein, including the figures and sequence listing. In exemplary embodiments, the PSMA ABD includes a set of 6 CDRs with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 amino acid modifications as compared to the 6 CDRs of one of the following PSMA ABDs: PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; and PSMA-H H1_L1.13. In certain embodiments, the variant PSMA ABD is capable of binding PSMA antigen, as measured by at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g., Octet assay) assay, with the latter finding particular use in many embodiments. In particular embodiments, the PSMA ABD is capable of binding human PSMA antigen (see Example 5).

In one embodiment, the PSMA ABD includes 6 CDRs that are at least 90, 95, 97, 98 or 99% identical to the 6 CDRs of a PSMA ABD as described herein, including the figures and sequence listing. In exemplary embodiments, the PSMA ABD includes 6 CDRs that are at least 90, 95, 97, 98 or 99% identical to the 6 CDRs of one of the following PSMA ABDs: PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; and PSMA-H H1_L1.13. In certain embodiments, the PSMA ABD is capable of binding to PSMA antigen, as measured by at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g., Octet assay) assay, with the latter finding particular use in many embodiments. In particular embodiments, the PSMA ABD is capable of binding human PSMA antigen (see FIG. 2).

In another exemplary embodiment, the PSMA ABD include the variable heavy (VH) domain and variable light (VL) domain of any one of the PSMA ABDs described herein, including the figures and sequence listing. In some embodiments, the PSMA ABD includes the PSMA-H H1 variable heavy domain (FIG. 17) and a variable light domain selected from PSMA-H L1 (FIG. 17) and L1.1-L1.84 (FIG. 18A-E). In exemplary embodiments, the PSMA ABD is one of the following PSMA ABDs: PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84 (FIGS. 17-19). In exemplary embodiments, the PSMA ABD is PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H

H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; or PSMA-H H1_L1.13 (FIGS. 17-19).

In addition to the parental PSMA variable heavy and variable light domains disclosed herein, provided herein are PSMA ABDs that include a variable heavy domain and/or a variable light domain that are variants of a PSMA ABD VH and VL domain disclosed herein. In one embodiment, the variant VH domain and/or VL domain has from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid changes from a VH and/or VL domain of a PSMA ABD described herein, including the figures and sequence listing. In exemplary embodiments, the variant VH domain and/or VL domain has from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid changes from a VH and/or VL domain of one of the following PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84 (FIGS. 17-19). In exemplary embodiments, the PSMA ABD is PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; or PSMA-H H1_L1.13 (FIGS. 17-19). In certain embodiments, the PSMA ABD is capable of binding to PSMA, as measured at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g., Octet assay) assay, with the latter finding particular use in many embodiments. In particular embodiments, the PSMA ABD is capable of binding human PSMA antigen (see Example 5).

In one embodiment, the variant VH and/or VL domain is at least 90, 95, 97, 98 or 99% identical to the VH and/or VL of a PSMA ABD as described herein, including the figures and sequence listing. In exemplary embodiments, the variant VH and/or VL domain is at least 90, 95, 97, 98 or 99% identical to the VH and/or VL of one of the following PSMA ABDs: PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84 (FIGS. 17-19). In exemplary embodiments, the PSMA ABD is PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; or PSMA-H H1_L1.13 (FIGS. 17-19). In certain embodiments, the PSMA ABD is capable of binding to the PSMA, as measured by at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g., Octet assay) assay, with the latter finding particular use in many embodiments. In particular embodiments, the PSMA ABD is capable of binding human PSMA antigen (see Example 5).

V. Antibodies

In one aspect, provided herein are antibodies that bind to PSMA (e.g., anti-PSMA antibodies). In certain embodiments, the antibody binds to human PSMA (FIG. 11A). Subject anti-PSMA antibodies include monospecific PSMA antibodies, as well as multi-specific (e.g., bispecific) anti-PSMA antibodies. In certain embodiments, the anti-PSMA antibody has a format according to any one of the antibody formats depicted in FIGS. 21A and 21B.

In some embodiments, the subject compositions include a PSMA binding domain. In some embodiments, the composition includes an antibody having a PSMA binding domain. Antibodies provided herein include one, two, three, four, and five or more PSMA binding domains. In certain embodiments, the PSMA binding domain includes any one of the vhCDR1, vhCDR2, vhCDR3, vlCDR1, vlCDR2 and vlCDR3 sequences of an PSMA binding domain selected

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from those depicted in FIGS. 17-19. In some embodiments, the PSMA binding domain includes the underlined vhCDR1, vhCDR2, vhCDR3, vlCDR1, vlCDR2 and vlCDR3 sequences of a PSMA binding domain selected from those depicted in FIGS. 17-19. In some embodiments, the PSMA binding domain includes the variable heavy domain and variable light domain of a PSMA binding domain selected from those depicted in FIGS. 17-19. PSMA binding domains depicted in FIGS. 12, 13A-13B, and 14A-14I include: PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84.

In one aspect, provided herein are bispecific antibodies that bind to PSMA and CD3, in various formats as outlined below, and generally depicted in FIGS. 21A and 21B. These bispecific, heterodimeric antibodies include a PSMA binding domain. In certain embodiments, the PSMA binding domain includes the VHCDR1, VHCDR2, VHCDR3, VLCDR1, VLCDR2 and VLCDR3 sequences of an PSMA binding domain selected from the group consisting of those depicted in FIGS. 17-19. In some embodiments, the PSMA binding domain includes the underlined VHCDR1, VHCDR2, VHCDR3, VLCDR1, VLCDR2 and VLCDR3 sequences of an PSMA binding domain selected from those depicted in FIGS. 17-19.

These bispecific heterodimeric antibodies bind PSMA and CD3. Such antibodies include a CD3 binding domain and at least one PSMA binding domain. Any suitable PSMA binding domain can be included in the anti-PSMA×anti-CD3 bispecific antibody. In some embodiments, the anti-PSMA×anti-CD3 bispecific antibody includes one, two, three, four or more PSMA binding domains, including but not limited to those depicted in FIGS. 17-19. In certain embodiments, the anti-PSMA×anti-CD3 antibody includes an PSMA binding domain that includes the VHCDR1, VHCDR2, VHCDR3, VLCDR1, VLCDR2 and VLCDR3 sequences of an PSMA binding domain selected from the group consisting of PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84 (FIGS. 17-19). In some embodiments, the anti-PSMA×anti-CD3 antibody includes a PSMA binding domain that includes the underlined VHCDR1, VHCDR2, VHCDR3, VLCDR1, VLCDR2 and VLCDR3 sequences of an PSMA binding domain selected from the group consisting of PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84 (FIGS. 17-19). In some embodiments, the anti-PSMA×anti-CD3 antibody includes a PSMA binding domain that includes the variable heavy domain and variable light domain of an PSMA binding domain selected from the group consisting of PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84 (FIGS. 17-19).

The anti-PSMA×anti-CD3 antibody provided herein can include any suitable CD3 binding domain. In certain embodiments, the anti-PSMA×anti-CD3 antibody includes a CD3 binding domain that includes the VHCDR1, VHCDR2, VHCDR3, VLCDR1, VLCDR2 and VLCDR3 sequences of a CD3 binding domain selected from the group consisting of those depicted in FIG. 10A-F. In some embodiments, the anti-PSMA×anti-CD3 antibody includes a CD3 binding domain that includes the underlined VHCDR1, VHCDR2, VHCDR3, VLCDR1, VLCDR2 and VLCDR3 sequences of a CD3 binding domain selected from the group consisting of those depicted in FIG. 10A-10F. In some embodiments, the anti-PSMA×anti-CD3 antibody includes a CD3 binding domain that includes the variable heavy domain and variable light domain of a CD3 binding domain selected from the group consisting of those depicted in FIG. 10A-10F. In some embodiments, the CD3 binding domain is selected from H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30,

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L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F). As outlined herein, these anti-CD3 antigen binding domains (CD3-ABDs) can be used in scFv formats in either orientation (e.g. from N- to C-terminal, VH-scFv linker-VL or VL-scFv linker-VH).

The antibodies provided herein include different antibody domains. As described herein and known in the art, the antibodies described herein include different domains within the heavy and light chains, which can be overlapping as well. These domains include, but are not limited to, the Fc domain, the CH1 domain, the CH2 domain, the CH3 domain, the hinge domain, the heavy constant domain (CH1-hinge-Fc domain or CH1-hinge-CH2-CH3), the variable heavy domain, the variable light domain, the light constant domain, Fab domains and scFv domains.

As shown herein, there are a number of suitable linkers (for use as either domain linkers or scFv linkers) that can be used to covalently attach the recited domains (e.g., scFvs, Fabs, Fc domains, etc.), including traditional peptide bonds, generated by recombinant techniques. Exemplary linkers to attach domains of the subject antibody to each other are depicted in FIG. 6. In some embodiments, the linker peptide may predominantly include the following amino acid residues: Gly, Ser, Ala, or Thr. The linker peptide should have a length that is adequate to link two molecules in such a way that they assume the correct conformation relative to one another so that they retain the desired activity. In one embodiment, the linker is from about 1 to 50 amino acids in length, preferably about 1 to 30 amino acids in length. In one embodiment, linkers of 1 to 20 amino acids in length may be used, with from about 5 to about 10 amino acids finding use in some embodiments. Useful linkers include glycine-serine polymers, including for example (GS)_n, (GSGGS)_n (SEQ ID NO: 3), (GGGGGS)_n (SEQ ID NO: 627), and (GGGS)_n (SEQ ID NO: 4), where n is an integer of at least one (and generally from 3 to 4), glycine-alanine polymers, alanine-serine polymers, and other flexible linkers, some of which are shown in FIG. 5 and FIG. 6. Alternatively, a variety of nonproteinaceous polymers, including but not limited to polyethylene glycol (PEG), polypropylene glycol, polyoxyalkylenes, or copolymers of polyethylene glycol and polypropylene glycol, may find use as linkers.

Other linker sequences may include any sequence of any length of CL/CH1 domain but not all residues of CL/CH1 domain; for example the first 5-12 amino acid residues of the CL/CH1 domains. Linkers can be derived from immunoglobulin light chain, for example C_κ or C_λ. Linkers can be derived from immunoglobulin heavy chains of any isotype, including for example C_γ1, C_γ2, C_γ3, C_γ4, C_α1, C_α2, C_δ, C_ε, and C_μ. Linker sequences may also be derived from other proteins such as Ig-like proteins (e.g. TCR, FcR, KIR), hinge region-derived sequences, and other natural sequences from other proteins.

In some embodiments, the linker is a "domain linker", used to link any two domains as outlined herein together. For example, in FIG. 21B, there may be a domain linker that attaches the C-terminus of the CH1 domain of the Fab to the N-terminus of the scFv, with another optional domain linker attaching the C-terminus of the scFv to the CH2 domain (although in many embodiments the hinge is used as this domain linker). While any suitable linker can be used, many embodiments utilize a glycine-serine polymer as the domain linker, including for example (GS)_n, (GSGGS)_n (SEQ ID NO: 3), (GGGGGS)_n (SEQ ID NO: 627), and (GGGS)_n (SEQ ID NO: 4), where n is an integer of at least one (and generally from 3 to 4 to 5) as well as any peptide sequence that allows for recombinant attachment of the two domains

with sufficient length and flexibility to allow each domain to retain its biological function. In some cases, and with attention being paid to “strandedness”, as outlined below, charged domain linkers, as used in some embodiments of scFv linkers can be used. Exemplary useful domain linkers are depicted in FIG. 6.

With particular reference to the domain linker used to attach the scFv domain to the Fc domain in the “2+1” format, there are several domain linkers that find particular use, including “full hinge C220S variant,” “flex half hinge,” “charged half hinge 1,” and “charged half hinge 2” as shown in FIG. 6.

In some embodiments, the linker is a “scFv linker”, used to covalently attach the VH and VL domains as discussed herein. In many cases, the scFv linker is a charged scFv linker, a number of which are shown in FIG. 5. Accordingly, in some embodiments, the antibodies described herein further provide charged scFv linkers, to facilitate the separation in pI between a first and a second monomer. That is, by incorporating a charged scFv linker, either positive or negative (or both, in the case of scaffolds that use scFvs on different monomers), this allows the monomer comprising the charged linker to alter the pI without making further changes in the Fc domains. These charged linkers can be substituted into any scFv containing standard linkers. Again, as will be appreciated by those in the art, charged scFv linkers are used on the correct “strand” or monomer, according to the desired changes in pI. For example, as discussed herein, to make 1+1 Fab-scFv-Fc format heterodimeric antibody, the original pI of the Fv region for each of the desired antigen binding domains are calculated, and one is chosen to make an scFv, and depending on the pI, either positive or negative linkers are chosen.

Charged domain linkers can also be used to increase the pI separation of the monomers of the antibodies described herein as well, and thus those included in FIG. 5 can be used in any embodiment herein where a linker is utilized.

In particular, the formats depicted in FIGS. 21A and 21B are antibodies, usually referred to as “heterodimeric antibodies”, meaning that the protein has at least two associated Fc sequences self-assembled into a heterodimeric Fc domain and at least two Fv regions, whether as Fabs or as scFvs.

The PSMA binding domains provided can be included in any useful antibody format including, for example, canonical immunoglobulin, as well as the 1+1 Fab-scFv-Fc and 2+1 Fab2-scFv-Fv formats provided herein. Other useful antibody formats include, but are not limited to, “mAb-Fv,” “mAb-scFv,” “central-Fv,” “one armed scFv-mAb,” “scFv-mAb,” “dual scFv,” and “trident” format antibodies, as disclosed in FIGS. 50A-50K.

In some embodiments, the subject antibody includes one or more of the PSMA ABDs provided herein. In some embodiments, the antibody includes one PSMA ABD. In other embodiments, the antibody includes two PSMA ABDs. In exemplary embodiments, the PSMA ABD includes the variable heavy domain and variable light domain of one of the following PSMA ABDs: PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84 (FIGS. 17-19). In some embodiments, the PSMA ABD is one of the following PSMA ABDs: PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84 (FIGS. 17-19).

In an exemplary embodiment, the antibody is a bispecific antibody that includes one or two PSMA ABDs, including any of the PSMA ABDs provided herein. Bispecific antibody that include such PSMA ABDs include, for example, 1+1 Fab-scFv-Fc and 2+1 Fab₂-scFv-Fc bispecifics format antibodies. In exemplary embodiments, the PSMA ABD is

one of the following PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84 (FIGS. 17-19). In exemplary embodiments the PSMA binding domains is a Fab. In some embodiments, such bispecific antibodies are heterodimeric bispecific antibodies that include any of the heterodimerization skew variants, pI variants and/or ablation variants described herein.

A. Chimeric and Humanized Antibodies

In certain embodiments, the antibodies described herein comprise a heavy chain variable region from a particular germline heavy chain immunoglobulin gene and/or a light chain variable region from a particular germline light chain immunoglobulin gene. For example, such antibodies may comprise or consist of a human antibody comprising heavy or light chain variable regions that are “the product of” or “derived from” a particular germline sequence. A human antibody that is “the product of” or “derived from” a human germline immunoglobulin sequence can be identified as such by comparing the amino acid sequence of the human antibody to the amino acid sequences of human germline immunoglobulins and selecting the human germline immunoglobulin sequence that is closest in sequence (i.e., greatest % identity) to the sequence of the human antibody (using the methods outlined herein). A human antibody that is “the product of” or “derived from” a particular human germline immunoglobulin sequence may contain amino acid differences as compared to the germline sequence, due to, for example, naturally-occurring somatic mutations or intentional introduction of site-directed mutation. However, a humanized antibody typically is at least 90% identical in amino acids sequence to an amino acid sequence encoded by a human germline immunoglobulin gene and contains amino acid residues that identify the antibody as being derived from human sequences when compared to the germline immunoglobulin amino acid sequences of other species (e.g., murine germline sequences). In certain cases, a humanized antibody may be at least 95, 96, 97, 98 or 99%, or even at least 96%, 97%, 98%, or 99% identical in amino acid sequence to the amino acid sequence encoded by the germline immunoglobulin gene. Typically, a humanized antibody derived from a particular human germline sequence will display no more than 10-20 amino acid differences from the amino acid sequence encoded by the human germline immunoglobulin gene (prior to the introduction of any skew, pI and ablation variants herein; that is, the number of variants is generally low, prior to the introduction of the variants described herein). In certain cases, the humanized antibody may display no more than 5, or even no more than 4, 3, 2, or 1 amino acid difference from the amino acid sequence encoded by the germline immunoglobulin gene (again, prior to the introduction of any skew, pI and ablation variants herein; that is, the number of variants is generally low, prior to the introduction of the variants described herein).

In one embodiment, the parent antibody has been affinity matured, as is known in the art. Structure-based methods may be employed for humanization and affinity maturation, for example as described in U.S. Ser. No. 11/004,590. Selection based methods may be employed to humanize and/or affinity mature antibody variable regions, including but not limited to methods described in Wu et al., 1999, J. Mol. Biol. 294:151-162; Baca et al., 1997, J. Biol. Chem. 272(16):10678-10684; Rosok et al., 1996, J. Biol. Chem. 271(37): 22611-22618; Rader et al., 1998, Proc. Natl. Acad. Sci. USA 95: 8910-8915; Krauss et al., 2003, Protein Engineering 16(10):753-759, all entirely incorporated by reference. Other humanization methods may involve the

grafting of only parts of the CDRs, including but not limited to methods described in U.S. Ser. No. 09/810,510; Tan et al., 2002, J. Immunol. 169:1119-1125; De Pascalis et al., 2002, J. Immunol. 169:3076-3084, all entirely incorporated by reference.

B. Heterodimeric Antibodies

In exemplary embodiments, the bispecific antibodies provided herein are heterodimeric bispecific antibodies that include two variant Fc domain sequences. Such variant Fc domains include amino acid modifications to facilitate the self-assembly and/or purification of the heterodimeric antibodies.

An ongoing problem in antibody technologies is the desire for “bispecific” antibodies that bind to two different antigens simultaneously, in general thus allowing the different antigens to be brought into proximity and resulting in new functionalities and new therapies. In general, these antibodies are made by including genes for each heavy and light chain into the host cells. This generally results in the formation of the desired heterodimer (A-B), as well as the two homodimers (A-A and B-B (not including the light chain heterodimeric issues)). However, a major obstacle in the formation of bispecific antibodies is the difficulty in biasing the formation of the desired heterodimeric antibody over the formation of the homodimers and/or purifying the heterodimeric antibody away from the homodimers.

There are a number of mechanisms that can be used to generate the subject heterodimeric antibodies. In addition, as will be appreciated by those in the art, these different mechanisms can be combined to ensure high heterodimerization. Amino acid modifications that facilitate the production and purification of heterodimers are collectively referred to generally as “heterodimerization variants.” As discussed below, heterodimerization variants include “skew” variants (e.g., the “knobs and holes” and the “charge pairs” variants described below) as well as “pI variants,” which allow purification of heterodimers from homodimers. As is generally described in U.S. Pat. No. 9,605,084, hereby incorporated by reference in its entirety and specifically as below for the discussion of heterodimerization variants, useful mechanisms for heterodimerization include “knobs and holes” (“KIH”) as described in U.S. Pat. No. 9,605,084, “electrostatic steering” or “charge pairs” as described in U.S. Pat. No. 9,605,084, pI variants as described in U.S. Pat. No. 9,605,084, and general additional Fc variants as outlined in U.S. Pat. No. 9,605,084 and below.

Heterodimerization variants that are useful for the formation and purification of the subject heterodimeric antibody (e.g., bispecific antibodies) are further discussed in detailed below.

1. Skew Variants

In some embodiments, the heterodimeric antibody includes skew variants which are one or more amino acid modifications in a first Fc domain (A) and/or a second Fc domain (B) that favor the formation of Fc heterodimers (Fc dimers that include the first and the second Fc domain; (A-B) over Fc homodimers (Fc dimers that include two of the first Fc domain or two of the second Fc domain; A-A or B-B). Suitable skew variants are included in the FIG. 29 of US Publ. App. No. 2016/0355608, hereby incorporated by reference in its entirety and specifically for its disclosure of skew variants, as well as in FIGS. 1A-1E and FIG. 4.

One mechanism is generally referred to in the art as “knobs and holes”, referring to amino acid engineering that creates steric influences to favor heterodimeric formation and disfavor homodimeric formation can also optionally be used; this is sometimes referred to as “knobs and holes”, as

described in U.S. Ser. No. 61/596,846, Ridgway et al., Protein Engineering 9(7):617 (1996); Atwell et al., J. Mol. Biol. 1997 270:26; U.S. Pat. No. 8,216,805, all of which are hereby incorporated by reference in their entirety. The Figures identify a number of “monomer A-monomer B” pairs that rely on “knobs and holes”. In addition, as described in Merchant et al., Nature Biotech. 16:677 (1998), these “knobs and hole” mutations can be combined with disulfide bonds to skew formation to heterodimerization.

An additional mechanism that finds use in the generation of heterodimers is sometimes referred to as “electrostatic steering” as described in Gunasekaran et al., J. Biol. Chem. 285(25):19637 (2010), hereby incorporated by reference in its entirety. This is sometimes referred to herein as “charge pairs”. In this embodiment, electrostatics are used to skew the formation towards heterodimerization. As those in the art will appreciate, these may also have an effect on pI, and thus on purification, and thus could in some cases also be considered pI variants. However, as these were generated to force heterodimerization and were not used as purification tools, they are classified as “steric variants”. These include, but are not limited to, D221E/P228E/L368E paired with D221R/P228R/K409R (e.g. these are “monomer corresponding sets) and C220E/P228E/368E paired with C220R/E224R/P228R/K409R.

In some embodiments, the skew variants advantageously and simultaneously favor heterodimerization based on both the “knobs and holes” mechanism as well as the “electrostatic steering” mechanism. In some embodiments, the heterodimeric antibody includes one or more sets of such heterodimerization skew variants. These variants come in “pairs” of “sets”. That is, one set of the pair is incorporated into the first monomer and the other set of the pair is incorporated into the second monomer. It should be noted that these sets do not necessarily behave as “knobs in holes” variants, with a one-to-one correspondence between a residue on one monomer and a residue on the other. That is, these pairs of sets may instead form an interface between the two monomers that encourages heterodimer formation and discourages homodimer formation, allowing the percentage of heterodimers that spontaneously form under biological conditions to be over 90%, rather than the expected 50% (25% homodimer A/A:50% heterodimer A/B:25% homodimer B/B). Exemplary heterodimerization “skew” variants are depicted in FIG. 4. In exemplary embodiments, the heterodimeric antibody includes a S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L; K370S:S364K/E357Q; or a T366S/L368A/Y407V: T366W (optionally including a bridging disulfide, T366S/L368A/Y407V/Y349C:T366W/S354C) “skew” variant amino acid substitution set. In an exemplary embodiment, the heterodimeric antibody includes a “S364K/E357Q: L368D/K370S” amino acid substitution set. In terms of nomenclature, the pair “S364K/E357Q:L368D/K370S” means that one of the monomers includes an Fc domain that includes the amino acid substitutions S364K and E357Q and the other monomer includes an Fc domain that includes the amino acid substitutions L368D and K370S; as above, the “strandedness” of these pairs depends on the starting pI.

In some embodiments, the skew variants provided herein can be optionally and independently incorporated with any other modifications, including, but not limited to, other skew variants (see, e.g., in FIG. 37 of US Publ. App. No. 2012/0149876, hereby incorporated by reference, particularly for its disclosure of skew variants), pI variants, isotypic variants, FcRn variants, ablation variants, etc. into one or both

of the first and second Fc domains of the heterodimeric antibody. Further, individual modifications can also independently and optionally be included or excluded from the subject the heterodimeric antibody.

Additional monomer A and monomer B variants that can be combined with other variants, optionally and independently in any amount, such as pI variants outlined herein or other steric variants that are shown in FIG. 37 of US 2012/0149876, the figure and legend and SEQ ID NOs of which are incorporated expressly by reference herein.

In some embodiments, the steric variants outlined herein can be optionally and independently incorporated with any pI variant (or other variants such as Fc variants, FcRn variants, etc.) into one or both monomers, and can be independently and optionally included or excluded from the proteins of the antibodies described herein.

A list of suitable skew variants is found in FIGS. 1A-IE, with FIG. 4 showing some pairs of particular utility in many embodiments. Of particular use in many embodiments are the pairs of sets including, but not limited to, S364K/E357Q; L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L and K370S:S364K/E357Q. In terms of nomenclature, the pair "S364K/E357Q:L368D/K370S" means that one of the monomers has the double variant set S364K/E357Q and the other has the double variant set L368D/K370S.

2. pI (Isoelectric Point) Variants for Heterodimers

In some embodiments, the heterodimeric antibody includes purification variants that advantageously allow for the separation of heterodimeric antibody (e.g., anti-PSMA $\times$ anti-CD3 bispecific antibody) from homodimeric proteins.

There are several basic mechanisms that can lead to ease of purifying heterodimeric antibodies. For example, modifications to one or both of the antibody heavy chain monomers A and B such that each monomer has a different pI allows for the isoelectric purification of heterodimeric A-B antibody from monomeric A-A and B-B proteins. Alternatively, some scaffold formats, such as the "1+1 Fab-scFv-Fc" format and the "2+1 Fab₂-scFv-Fc" format, also allows separation on the basis of size. As described above, it is also possible to "skew" the formation of heterodimers over homodimers using skew variants. Thus, a combination of heterodimerization skew variants and pI variants find particular use in the heterodimeric antibodies provided herein.

Additionally, as more fully outlined below, depending on the format of the heterodimeric antibody, pI variants either contained within the constant region and/or Fc domains of a monomer, and/or domain linkers can be used. In some embodiments, the heterodimeric antibody includes additional modifications for alternative functionalities that can also create pI changes, such as Fc, FcRn and KO variants.

In some embodiments, the subject heterodimeric antibodies provided herein include at least one monomer with one or more modifications that alter the pI of the monomer (i.e., a "pI variant"). In general, as will be appreciated by those in the art, there are two general categories of pI variants: those that increase the pI of the protein (basic changes) and those that decrease the pI of the protein (acidic changes). As described herein, all combinations of these variants can be done: one monomer may be wild type, or a variant that does not display a significantly different pI from wild-type, and the other can be either more basic or more acidic. Alternatively, each monomer is changed, one to more basic and one to more acidic.

Depending on the format of the heterodimer antibody, pI variants can be either contained within the constant and/or

Fc domains of a monomer, or charged linkers, either domain linkers or scFv linkers, can be used. That is, antibody formats that utilize scFv(s) such as "1+1 Fab-scFv-Fc", format can include charged scFv linkers (either positive or negative), that give a further pI boost for purification purposes. As will be appreciated by those in the art, some 1+1 Fab-scFv-Fc formats are useful with just charged scFv linkers and no additional pI adjustments, although the antibodies described herein do provide pI variants that are on one or both of the monomers, and/or charged domain linkers as well. In addition, additional amino acid engineering for alternative functionalities may also confer pI changes, such as Fc, FcRn and KO variants.

In subject heterodimeric antibodies that utilizes pI as a separation mechanism to allow the purification of heterodimeric proteins, amino acid variants are introduced into one or both of the monomer polypeptides. That is, the pI of one of the monomers (referred to herein for simplicity as "monomer A") can be engineered away from monomer B, or both monomer A and B change be changed, with the pI of monomer A increasing and the pI of monomer B decreasing. As is outlined more fully below, the pI changes of either or both monomers can be done by removing or adding a charged residue (e.g., a neutral amino acid is replaced by a positively or negatively charged amino acid residue, e.g., glycine to glutamic acid), changing a charged residue from positive or negative to the opposite charge (aspartic acid to lysine) or changing a charged residue to a neutral residue (e.g., loss of a charge; lysine to serine). A number of these variants are shown in the FIGS. 3 and 4.

Thus, in some embodiments, the subject heterodimeric antibody includes amino acid modifications in the constant regions that alter the isoelectric point (pI) of at least one, if not both, of the monomers of a dimeric protein to form "pI antibodies" by incorporating amino acid substitutions ("pI variants" or "pI substitutions") into one or both of the monomers. As shown herein, the separation of the heterodimers from the two homodimers can be accomplished if the pIs of the two monomers differ by as little as 0.1 pH unit, with 0.2, 0.3, 0.4 and 0.5 or greater all finding use in the antibodies described herein.

As will be appreciated by those in the art, the number of pI variants to be included on each or both monomer(s) to get good separation will depend in part on the starting pI of the components, for example in the 1+1 Fab-scFv-Fc and 2+1 Fab₂-scFv-Fc formats, the starting pI of the scFv and Fab(s) of interest. That is, to determine which monomer to engineer or in which "direction" (e.g., more positive or more negative), the Fv sequences of the two target antigens are calculated and a decision is made from there. As is known in the art, different Fvs will have different starting pIs which are exploited in the antibodies described herein. In general, as outlined herein, the pIs are engineered to result in a total pI difference of each monomer of at least about 0.1 logs, with 0.2 to 0.5 being preferred as outlined herein.

In the case where pI variants are used to achieve heterodimerization, by using the constant region(s) of the heavy chain(s), a more modular approach to designing and purifying bispecific proteins, including antibodies, is provided. Thus, in some embodiments, heterodimerization variants (including skew and pI heterodimerization variants) are not included in the variable regions, such that each individual antibody must be engineered. In addition, in some embodiments, the possibility of immunogenicity resulting from the pI variants is significantly reduced by importing pI variants from different IgG isotypes such that pI is changed without introducing significant immunogenicity. Thus, an additional

problem to be solved is the elucidation of low pI constant domains with high human sequence content, e.g., the minimization or avoidance of non-human residues at any particular position. Alternatively or in addition to isotopic substitutions, the possibility of immunogenicity resulting from the pI variants is significantly reduced by utilizing isosteric substitutions (e.g. Asn to Asp; and Gln to Glu).

As discussed below, a side benefit that can occur with this pI engineering is also the extension of serum half-life and increased FcRn binding. That is, as described in US Publ. App. No. US 2012/0028304 (incorporated by reference in its entirety), lowering the pI of antibody constant domains (including those found in antibodies and Fc fusions) can lead to longer serum retention in vivo. These pI variants for increased serum half-life also facilitate pI changes for purification.

In addition, it should be noted that the pI variants give an additional benefit for the analytics and quality control process of bispecific antibodies, as the ability to either eliminate, minimize and distinguish when homodimers are present is significant. Similarly, the ability to reliably test the reproducibility of the heterodimeric antibody production is important.

In general, embodiments of particular use rely on sets of variants that include skew variants, which encourage heterodimerization formation over homodimerization formation, coupled with pI variants, which increase the pI difference between the two monomers to facilitate purification of heterodimers away from homodimers.

Exemplary combinations of pI variants are shown in FIGS. 4 and 5, and FIG. 30 of US Publ. App. No. 2016/0355608, all of which are herein incorporated by reference in its entirety and specifically for the disclosure of pI variants. Preferred combinations of pI variants are shown in FIGS. 1 and 2. As outlined herein and shown in the figures, these changes are shown relative to IgG1, but all isotypes can be altered this way, as well as isotype hybrids. In the case where the heavy chain constant domain is from IgG2-4, R133E and R133Q can also be used.

In one embodiment, a preferred combination of pI variants has one monomer (the negative Fab side) comprising 208D/295E/384D/418E/421D variants (N208D/Q295E/N384D/Q418E/N421D when relative to human IgG1) and a second monomer (the positive scFv side) comprising a positively charged scFv linker, including (GKPGS)₄ (SEQ ID NO: 1). However, as will be appreciated by those in the art, the first monomer includes a CH1 domain, including position 208. Accordingly, in constructs that do not include a CH1 domain (for example for antibodies that do not utilize a CH1 domain on one of the domains), a preferred negative pI variant Fc set includes 295E/384D/418E/421D variants (Q295E/N384D/Q418E/N421D when relative to human IgG1).

Accordingly, in some embodiments, one monomer has a set of substitutions from FIG. 2 and the other monomer has a charged linker (either in the form of a charged scFv linker because that monomer comprises an scFv or a charged domain linker, as the format dictates, which can be selected from those depicted in FIG. 5).

In some embodiments, modifications are made in the hinge of the Fc domain, including positions 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, and 230 based on EU numbering. Thus, pI mutations and particularly substitutions can be made in one or more of positions 216-230, with 1, 2, 3, 4 or 5 mutations finding use. Again, all possible combinations are contemplated, alone or with other pI variants in other domains.

Specific substitutions that find use in lowering the pI of hinge domains include, but are not limited to, a deletion at position 221, a non-native valine or threonine at position 222, a deletion at position 223, a non-native glutamic acid at position 224, a deletion at position 225, a deletion at position 235 and a deletion or a non-native alanine at position 236. In some cases, only pI substitutions are done in the hinge domain, and in others, these substitution(s) are added to other pI variants in other domains in any combination.

In some embodiments, mutations can be made in the CH2 region, including positions 233, 234, 235, 236, 274, 296, 300, 309, 320, 322, 326, 327, 334 and 339, based on EU numbering. It should be noted that changes in 233-236 can be made to increase effector function (along with 327A) in the IgG2 backbone. Again, all possible combinations of these 14 positions can be made; e.g., =may include a variant Fc domain with 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 CH2 pI substitutions.

Specific substitutions that find use in lowering the pI of CH2 domains include, but are not limited to, a non-native glutamine or glutamic acid at position 274, a non-native phenylalanine at position 296, a non-native phenylalanine at position 300, a non-native valine at position 309, a non-native glutamic acid at position 320, a non-native glutamic acid at position 322, a non-native glutamic acid at position 326, a non-native glycine at position 327, a non-native glutamic acid at position 334, a non-native threonine at position 339, and all possible combinations within CH2 and with other domains.

In this embodiment, the modifications can be independently and optionally selected from position 355, 359, 362, 384, 389, 392, 397, 418, 419, 444 and 447 (EU numbering) of the CH3 region. Specific substitutions that find use in lowering the pI of CH3 domains include, but are not limited to, a non-native glutamine or glutamic acid at position 355, a non-native serine at position 384, a non-native asparagine or glutamic acid at position 392, a non-native methionine at position 397, a non-native glutamic acid at position 419, a non-native glutamic acid at position 359, a non-native glutamic acid at position 362, a non-native glutamic acid at position 389, a non-native glutamic acid at position 418, a non-native glutamic acid at position 444, and a deletion or non-native aspartic acid at position 447.

In general, as will be appreciated by those in the art, there are two general categories of pI variants: those that increase the pI of the protein (basic changes) and those that decrease the pI of the protein (acidic changes). As described herein, all combinations of these variants can be done: one monomer may be wild type, or a variant that does not display a significantly different pI from wild-type, and the other can be either more basic or more acidic. Alternatively, each monomer is changed, one to more basic and one to more acidic.

Preferred combinations of pI variants are shown in FIG. 4. As outlined herein and shown in the figures, these changes are shown relative to IgG1, but all isotypes can be altered this way, as well as isotype hybrids. In the case where the heavy chain constant domain is from IgG2-4, R133E and R133Q can also be used.

In one embodiment, for example in the FIGS. 21A and 21B formats, a preferred combination of pI variants has one monomer (the negative Fab side) comprising 208D/295E/384D/418E/421D variants (N208D/Q295E/N384D/Q418E/N421D when relative to human IgG1) and a second monomer (the positive scFv side) comprising a positively charged scFv linker, including (GKPGS)₄ (SEQ ID NO: 1). However, as will be appreciated by those in the art, the first

monomer includes a CH1 domain, including position 208. Accordingly, in constructs that do not include a CH1 domain (for example for antibodies that do not utilize a CH1 domain on one of the domains, for example in a dual scFv format or a “one armed” format such as those depicted in FIG. 42B, C or D), a preferred negative pI variant Fc set includes 295E/384D/418E/421D variants (Q295E/N384D/Q418E/N421D when relative to human IgG1).

Accordingly, in some embodiments, one monomer has a set of substitutions from FIG. 4 and the other monomer has a charged linker (either in the form of a charged scFv linker because that monomer comprises an scFv or a charged domain linker, as the format dictates, which can be selected from those depicted in FIG. 5).

3. Isotypic Variants

In addition, many embodiments of the antibodies described herein rely on the “importation” of pI amino acids at particular positions from one IgG isotype into another, thus reducing or eliminating the possibility of unwanted immunogenicity being introduced into the variants. A number of these are shown in FIG. 21 of US Publ. 2014/0370013, hereby incorporated by reference. That is, IgG1 is a common isotype for therapeutic antibodies for a variety of reasons, including high effector function. However, the heavy constant region of IgG1 has a higher pI than that of IgG2 (8.10 versus 7.31). By introducing IgG2 residues at particular positions into the IgG1 backbone, the pI of the resulting monomer is lowered (or increased) and additionally exhibits longer serum half-life. For example, IgG1 has a glycine (pI 5.97) at position 137, and IgG2 has a glutamic acid (pI 3.22); importing the glutamic acid will affect the pI of the resulting protein. As is described below, a number of amino acid substitutions are generally required to significant affect the pI of the variant antibody. However, it should be noted as discussed below that even changes in IgG2 molecules allow for increased serum half-life.

In other embodiments, non-isotypic amino acid changes are made, either to reduce the overall charge state of the resulting protein (e.g. by changing a higher pI amino acid to a lower pI amino acid), or to allow accommodations in structure for stability, etc. as is more further described below.

In addition, by pI engineering both the heavy and light constant domains, significant changes in each monomer of the heterodimer can be seen. As discussed herein, having the pIs of the two monomers differ by at least 0.5 can allow separation by ion exchange chromatography or isoelectric focusing, or other methods sensitive to isoelectric point.

4. Calculating pI

The pI of each monomer can depend on the pI of the variant heavy chain constant domain and the pI of the total monomer, including the variant heavy chain constant domain and the fusion partner. Thus, in some embodiments, the change in pI is calculated on the basis of the variant heavy chain constant domain, using the chart in the FIG. 19 of US Pub. 2014/0370013. As discussed herein, which monomer to engineer is generally decided by the inherent pI of the Fv and scaffold regions. Alternatively, the pI of each monomer can be compared.

5. pI Variants that Also Confer Better FcRn In Vivo Binding

In the case where the pI variant decreases the pI of the monomer, they can have the added benefit of improving serum retention in vivo.

Although still under examination, Fc regions are believed to have longer half-lives in vivo, because binding to FcRn at pH 6 in an endosome sequesters the Fc (Ghetie and Ward, 1997 Immunol Today. 18(12): 592-598, entirely incorpo-

rated by reference). The endosomal compartment then recycles the Fc to the cell surface. Once the compartment opens to the extracellular space, the higher pH, ~7.4, induces the release of Fc back into the blood. In mice, Dall'Acqua et al. showed that Fc mutants with increased FcRn binding at pH 6 and pH 7.4 actually had reduced serum concentrations and the same half-life as wild-type Fc (Dall'Acqua et al. 2002, J. Immunol. 169:5171-5180, entirely incorporated by reference). The increased affinity of Fc for FcRn at pH 7.4 is thought to forbid the release of the Fc back into the blood. Therefore, the Fc mutations that will increase Fc's half-life in vivo will ideally increase FcRn binding at the lower pH while still allowing release of Fc at higher pH. The amino acid histidine changes its charge state in the pH range of 6.0 to 7.4. Therefore, it is not surprising to find His residues at important positions in the Fc/FcRn complex.

Recently it has been suggested that antibodies with variable regions that have lower isoelectric points may also have longer serum half-lives (Igawa et al., 2010 PEDS. 23(5): 385-392, entirely incorporated by reference). However, the mechanism of this is still poorly understood. Moreover, variable regions differ from antibody to antibody. Constant region variants with reduced pI and extended half-life would provide a more modular approach to improving the pharmacokinetic properties of antibodies, as described herein.

C. Additional Fc Variants for Additional Functionality

In addition to the heterodimerization variants discussed above, there are a number of useful Fc amino acid modification that can be made for a variety of reasons, including, but not limited to, altering binding to one or more FcγR receptors, altered binding to FcRn receptors, etc. as discussed below.

Accordingly, the antibodies provided herein (heterodimeric, as well as homodimeric) can include such amino acid modifications with or without the heterodimerization variants outlined herein (e.g., the pI variants and steric variants). Each set of variants can be independently and optionally included or excluded from any particular heterodimeric protein.

1. FcγR Variants

Accordingly, there are a number of useful Fc substitutions that can be made to alter binding to one or more of the FcγR receptors. In certain embodiments, the subject antibody includes modifications that alter the binding to one or more FcγR receptors (i.e., “FcγR variants”). Substitutions that result in increased binding as well as decreased binding can be useful. For example, it is known that increased binding to FcγRIIIa generally results in increased ADCC (antibody dependent cell-mediated cytotoxicity; the cell-mediated reaction wherein nonspecific cytotoxic cells that express FcγRs recognize bound antibody on a target cell and subsequently cause lysis of the target cell). Similarly, decreased binding to FcγRIIb (an inhibitory receptor) can be beneficial as well in some circumstances. Amino acid substitutions that find use in the antibodies described herein include those listed in U.S. Pat. No. 8,188,321 (particularly FIG. 41) and U.S. Pat. No. 8,084,582, and US Publ. App. Nos. 20060235208 and 20070148170, all of which are expressly incorporated herein by reference in their entirety and specifically for the variants disclosed therein. Particular variants that find use include, but are not limited to, 236A, 239D, 239E, 332E, 332D, 239D/332E, 267D, 267E, 328F, 267E/328F, 236A/332E, 239D/332E/330Y, 239D/332E/330L, 243A, 243L, 264A, 264V and 299T.

In addition, there are additional Fc substitutions that find use in increased binding to the FcRn receptor and increased serum half-life, as specifically disclosed in U.S. Ser. No.

12/341,769, hereby incorporated by reference in its entirety, including, but not limited to, 434S, 434A, 428L, 308F, 259I, 428L/434S, 259I/308F, 436I/428L, 436I or V/434S, 436V/428L and 259I/308F/428L. Such modification may be included in one or both Fc domains of the subject antibody.

2. Ablation Variants

Similarly, another category of functional variants are “FcγR ablation variants” or “Fc knock out (FcKO or KO)” variants. In these embodiments, for some therapeutic applications, it is desirable to reduce or remove the normal binding of the Fc domain to one or more or all of the Fcγ receptors (e.g. FcγR1, FcγRIIa, FcγRIIb, FcγRIIIa, etc.) to avoid additional mechanisms of action. That is, for example, in many embodiments, particularly in the use of bispecific antibodies that bind CD3 monovalently it is generally desirable to ablate FcγRIIIa binding to eliminate or significantly reduce ADCC activity, wherein one of the Fc domains comprises one or more Fcγ receptor ablation variants. These ablation variants are depicted in FIG. 14, and each can be independently and optionally included or excluded, with preferred aspects utilizing ablation variants selected from the group consisting of G236R/L328R, E233P/L234V/L235A/G236del/S239K, E233P/L234V/L235A/G236del/S267K, E233P/L234V/L235A/G236del/S239K/A327G, E233P/L234V/L235A/G236del/S267K/A327G and E233P/L234V/L235A/G236del. It should be noted that the ablation variants referenced herein ablate FcγR binding but generally not FcRn binding.

As is known in the art, the Fc domain of human IgG1 has the highest binding to the Fcγ receptors, and thus ablation variants can be used when the constant domain (or Fc domain) in the backbone of the heterodimeric antibody is IgG1. Alternatively, or in addition to ablation variants in an IgG1 background, mutations at the glycosylation position 297 (generally to A or S) can significantly ablate binding to FcγRIIIa, for example. Human IgG2 and IgG4 have naturally reduced binding to the Fcγ receptors, and thus those backbones can be used with or without the ablation variants.

D. Combination of Heterodimeric and Fc Variants

As will be appreciated by those in the art, all of the recited heterodimerization variants (including skew and/or pI variants) can be optionally and independently combined in any way, as long as they retain their “strandedness” or “monomer partition”. In some embodiments, the heterodimeric antibodies provided herein include the combination of heterodimerization skew variants, isosteric pI substitutions and FcKO variants as depicted in FIG. 4. In addition, all of these variants can be combined into any of the heterodimerization formats.

In the case of pI variants, while embodiments finding particular use are shown in the Figures, other combinations can be generated, following the basic rule of altering the pI difference between two monomers to facilitate purification.

In addition, any of the heterodimerization variants, skew and pI, are also independently and optionally combined with Fc ablation variants, Fc variants, FcRn variants, as generally outlined herein.

Exemplary combination of variants that are included in some embodiments of the heterodimeric 1+1 Fab-scFv-Fc and 2+1 Fab₂-scFv-Fc format antibodies are included in FIG. 4. In certain embodiments, the antibody is a heterodimeric 1+1 Fab-scFv-Fc or 2+1 Fab₂-scFv-Fc format antibody as shown in FIGS. 21A and 21B.

E. Anti-PSMA×Anti-CD3 Bispecific Antibodies

In another aspect, provided herein are anti-PSMA×anti-CD3 (also referred to herein as “αPSMA×αCD3”) bispecific antibodies. Such antibodies include at least one PSMA

binding domain and at least one CD3 binding domain. In some embodiments, bispecific αPSMA×αCD3 provided herein immune responses selectively in tumor sites that express PSMA.

Note that unless specified herein, the order of the antigen list in the name does not confer structure; that is a PSMA×CD3 1+1 Fab-scFv-Fc antibody can have the scFv bind to PSMA or CD3, although in some cases, the order specifies structure as indicated.

As is more fully outlined herein, these combinations of ABDs can be in a variety of formats, as outlined below, generally in combinations where one ABD is in a Fab format and the other is in an scFv format. Exemplary formats that are used in the bispecific antibodies provided herein include the 1+1 Fab-scFv-Fc and 2+1 Fab₂-scFv-Fv formats (see, e.g., FIGS. 15A and 15B). Other useful antibody formats include, but are not limited to, “mAb-Fv,” “mAb-scFv,” “central-Fv,” “one armed scFv-mAb,” “scFv-mAb,” “dual scFv,” and “trident” format antibodies, as disclosed in FIG. 50A-50K.

In addition, in general, one of the ABDs comprises a scFv as outlined herein, in an orientation from N- to C-terminus of VH-scFv linker-VL or VL-scFv linker-VH. One or both of the other ABDs, according to the format, generally is a Fab, comprising a VH domain on one protein chain (generally as a component of a heavy chain) and a VL on another protein chain (generally as a component of a light chain).

As will be appreciated by those in the art, any set of 6 CDRs or VH and VL domains can be in the scFv format or in the Fab format, which is then added to the heavy and light constant domains, where the heavy constant domains comprise variants (including within the CH1 domain as well as the Fc domain). The scFv sequences contained in the sequence listing utilize a particular charged linker, but as outlined herein, uncharged or other charged linkers can be used, including those depicted in FIG. 5 and FIG. 6.

In addition, as discussed above, the numbering used in the Sequence Listing for the identification of the CDRs is Kabat, however, different numbering can be used, which will change the amino acid sequences of the CDRs as shown in Table 2.

For all of the variable heavy and light domains listed herein, further variants can be made. As outlined herein, in some embodiments the set of 6 CDRs can have from 0, 1, 2, 3, 4 or 5 amino acid modifications (with amino acid substitutions finding particular use), as well as changes in the framework regions of the variable heavy and light domains, as long as the frameworks (excluding the CDRs) retain at least about 80, 85 or 90% identity to a human germline sequence selected from those listed in FIG. 1 of U.S. Pat. No. 7,657,380, which Figure and Legend is incorporated by reference in its entirety herein. Thus, for example, the identical CDRs as described herein can be combined with different framework sequences from human germline sequences, as long as the framework regions retain at least 80, 85 or 90% identity to a human germline sequence selected from those listed in FIG. 1 of U.S. Pat. No. 7,657,380. Alternatively, the CDRs can have amino acid modifications (e.g., from 1, 2, 3, 4 or 5 amino acid modifications in the set of CDRs (that is, the CDRs can be modified as long as the total number of changes in the set of 6 CDRs is less than 6 amino acid modifications, with any combination of CDRs being changed; e.g., there may be one change in vLCDR1, two in vHCDR2, none in vHCDR3, etc.)), as well as having framework region changes, as long as the framework regions retain at least 80, 85 or 90% identity to

a human germline sequence selected from those listed in FIG. 1 of U.S. Pat. No. 7,657,380.

The anti-PSMA×anti-CD3 bispecific antibody can include any suitable CD3 ABD, including those described herein (see, e.g., FIGS. 10A-10F). In some embodiments, the CD3 ABD of the anti-PSMA×anti-CD3 bispecific antibody includes the variable heavy domain and variable light domain of a CD3 ABD provided herein, including those described in FIGS. 10A-10F and the sequence listing. In some embodiments, the CD3 ABD includes the variable heavy domain and variable light domain of one of the following CD3 ABDs: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F). In exemplary embodiments, the CD3 ABD is one of the following CD3 ABDs: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F) or a variant thereof. The anti-PSMA×anti-CD3 bispecific antibody can include any suitable PSMA ABD, including those described herein (see, e.g., FIGS. 12, 13A-13B, and 14A-14I). In some embodiments, the PSMA ABD of the anti-PSMA×anti-CD3 bispecific antibody includes the variable heavy domain and variable light domain of a PSMA ABD provided herein, including those described in FIGS. 12, 13A-13B, and 14A-14I and the sequence listing. In some embodiments, the PSMA ABD includes the variable heavy domain and variable light domain of one of the following PSMA ABDs: PSMA-H L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In exemplary embodiments, the PSMA ABD is one of the following PSMA ABDs: PSMA-H L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E) or variants thereof.

F. Useful Formats

As will be appreciated by those in the art and discussed more fully below, the heterodimeric bispecific antibodies provided herein can take on a wide variety of configurations, as are generally depicted in FIG. 1. Some figures depict “single ended” configurations, where there is one type of specificity on one “arm” of the molecule and a different specificity on the other “arm”. Other figures depict “dual ended” configurations, where there is at least one type of specificity at the “top” of the molecule and one or more different specificities at the “bottom” of the molecule. Thus, in some embodiments, the antibodies described herein are directed to novel immunoglobulin compositions that co-engage a different first and a second antigen.

As will be appreciated by those in the art, the heterodimeric formats of the antibodies described herein can have different valencies as well as be bispecific. That is, heterodimeric antibodies of the antibodies described herein can be bivalent and bispecific, wherein one target tumor antigen (e.g. CD3) is bound by one binding domain and the other target tumor antigen (e.g. PSMA) is bound by a second binding domain. The heterodimeric antibodies can also be trivalent and bispecific, wherein the first antigen is bound by two binding domains and the second antigen by a second binding domain. As is outlined herein, when CD3 is one of the target antigens, it is preferable that the CD3 is bound only monovalently, to reduce potential side effects.

The antibodies described herein utilize anti-CD3 antigen binding domains in combination with anti-PSMA binding domains. As will be appreciated by those in the art, any collection of anti-CD3 CDRs, anti-CD3 variable light and variable heavy domains, Fabs and scFvs as depicted in any

of the Figures can be used. Similarly, any of the anti-PSMA antigen binding domains can be used, whether CDRs, variable light and variable heavy domains, Fabs and scFvs as depicted in any of the Figures (e.g., FIGS. 17, 18A-E and 19A-X) can be used, optionally and independently combined in any combination.

1. 1+1 Fab-scFv-Fc Format

One heterodimeric scaffold that finds particular use in the antibodies described herein is the “1+1 Fab-scFv-Fc” or “bottle-opener” format as shown in FIG. 21A with an exemplary combination of a CD3 binding domain and a tumor target antigen (PSMA) binding domain. In this embodiment, one heavy chain monomer of the antibody contains a single chain Fv (“scFv”, as defined below) and an Fc domain. The scFv includes a variable heavy domain (VH1) and a variable light domain (VL1), wherein the VH1 is attached to the VL1 using an scFv linker that can be charged (see, e.g., FIG. 5). The scFv is attached to the heavy chain using a domain linker (see, e.g., FIG. 6). The other heavy chain monomer is a “regular” heavy chain (VH-CH1-hinge-CH2-CH3). The 1+1 Fab-scFv-Fc also includes a light chain that interacts with the VH-CH1 to form a Fab. This structure is sometimes referred to herein as the “bottle-opener” format, due to a rough visual similarity to a bottle-opener. The two heavy chain monomers are brought together by the use of amino acid variants (e.g., heterodimerization variants, discussed above) in the constant regions (e.g., the Fc domain, the CH1 domain and/or the hinge region) that promote the formation of heterodimeric antibodies as is described more fully below.

There are several distinct advantages to the present “1+1 Fab-scFv-Fc” format. As is known in the art, antibody analogs relying on two scFv constructs often have stability and aggregation problems, which can be alleviated in the antibodies described herein by the addition of a “regular” heavy and light chain pairing. In addition, as opposed to formats that rely on two heavy chains and two light chains, there is no issue with the incorrect pairing of heavy and light chains (e.g. heavy 1 pairing with light 2, etc.).

Many of the embodiments outlined herein rely in general on the 1+1 Fab-scFv-Fc or “bottle opener” format antibody that comprises a first monomer comprising an scFv, comprising a variable heavy and a variable light domain, covalently attached using an scFv linker (charged, in many but not all instances), where the scFv is covalently attached to the N-terminus of a first Fc domain usually through a domain linker (i.e., from N- to C-terminus scFv-linker-CH2-CH3). In some embodiments, the variable light domain of the scFv is attached to the first Fc domain. In other embodiments, the variable heavy domain of the scFv is attached to the first Fc domain. The domain linker can be either charged or uncharged and exogenous or endogenous (e.g., all or part of the native hinge domain). Any suitable linker can be used to attach the scFv to the N-terminus of the first Fc domain. In some embodiments, the domain linker is chosen from the domain linkers in FIG. 6. The second monomer of the 1+1 Fab-scFv-Fc format or “bottle opener” format is a heavy chain (i.e., from N- to C-terminus VH-CH1-hinge-CH2-CH3), and the composition further comprises a light chain.

In addition, the Fc domains of the antibodies described herein generally include skew variants (e.g. a set of amino acid substitutions as shown in FIGS. 1 and 4, with particularly useful skew variants being selected from the group consisting of S364K/E357Q:L368D/K370S; L368D/K370S: S364K; L368E/K370S:S364K; T411T/E360E/Q362E: D401K; L368D/K370S:S364K/E357L; K370S:S364K/E357Q; T366S/L368A/Y407V:T366W and T366S/L368A/

Y407V/Y349C:T366W/S354C), optionally ablation variants (including those shown in FIG. 3), optionally charged scFv linkers (including those shown in FIG. 5) and the heavy chain comprises pl variants (including those shown in FIG. 2).

In general, in many preferred embodiments, the scFv is the domain that binds to the CD3, and the Fab forms a PSMA binding domain. An exemplary anti-PSMA×anti-CD3 bispecific antibody in the 1+1 Fab-scFv-Fc format is depicted in FIG. 21A. Exemplary anti-PSMA×anti-CD3 bispecific antibody in the 1+1 Fab-scFv-Fc format is depicted in FIGS. 22-25. Exemplary variable heavy and light domains of the scFv that binds to CD3 are included in FIG. 10A-10F. Exemplary variable heavy and light domains of the Fv that binds to PSMA are included in FIGS. 17 and 18. In an exemplary embodiment, the PSMA binding domain of the 1+1 Fab-scFv-Fc PSMA×CD3 bispecific antibody includes the VH of PSMA-H H1 (FIG. 17) and VL of one of the following PSMA binding domains: PSMA-H L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In one embodiment, the CD3 binding domain of the 1+1 Fab-scFv-Fc PSMA×CD3 bispecific antibody includes the VH and VL of one of the following CD3 binding domains: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F). Particularly useful PSMA and CD3 combinations for use in the 1+1 Fab-scFv-Fc PSMA×CD3 bispecific antibody format are disclosed in FIGS. 22-25 and include: a) CD3H1.30_L1.47×PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; or PSMA-H H1_L1.13; b) CD3H1.31_L1.47, CD3H1.32_L1.47, or CD3H1.33_L1.47×PSMA-H H1L1.

In certain embodiments, the 1+1 Fab-scFv-Fc scaffold format includes a first monomer that includes a scFv-domain linker-CH2-CH3 monomer, a second monomer that includes a first variable heavy domain-CH1-hinge-CH2-CH3 monomer and a third monomer that includes a first variable light domain. In some embodiments, the CH2-CH3 of the first monomer is a first variant Fc domain and the CH2-CH3 of the second monomer is a second variant Fc domain. In some embodiments, the scFv includes a scFv variable heavy domain and a scFv variable light domain that form a CD3 binding moiety. In certain embodiments, the scFv variable heavy domain and scFv variable light domain are covalently attached using an scFv linker (charged, in many but not all instances. See, e.g., FIG. 5). In some embodiments, the first variable heavy domain and first variable light domain form a PSMA binding domain. CD3 binding domain sequences finding particular use in these embodiments include, but are not limited to, H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (see FIGS. 10A-10F). PSMA binding domain sequences that are of particular use in these embodiments include, but are not limited to, PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; and PSMA-H H1_L1.13. Particularly useful PSMA and CD3 combinations for use in the 1+1 Fab-scFv-Fc PSMA×CD3 bispecific

antibody format are disclosed in FIGS. 22-25 and include: a) CD3H1.30_L1.47×PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; or PSMA-H H1_L1.13; b) CD3H1.31_L1.47, CD3H1.32_L1.47, or CD3H1.33_L1.47×PSMA-H H1L1.

In some embodiments, the 1+1 Fab-scFv-Fc format includes skew variants, pl variants, and ablation variants. Accordingly, some embodiments include 1+1 Fab-scFv-Fc formats that comprise: a) a first monomer (the “scFv monomer”) that comprises a charged scFv linker (with the +H sequence of FIG. 5 being preferred in some embodiments), the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and an scFv that binds to CD3 as outlined herein; b) a second monomer (the “Fab monomer”) that comprises the skew variants L368D/K370S, the pl variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain; and c) a light chain that includes a variable light domain light domain (VL) and a constant light domain (CL), wherein numbering is according to EU numbering. The variable heavy domain and variable light domain make up a PSMA binding moiety. CD3 binding domain sequences finding particular use in these embodiments include, but are not limited to, H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (see FIGS. 10A-10F). PSMA binding domain sequences that are of particular use in these embodiments include, but are not limited to, PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; and PSMA-H H1_L1.13. Particularly useful PSMA and CD3 combinations for use in the 1+1 Fab-scFv-Fc PSMA×CD3 bispecific antibody format are disclosed in FIGS. 22-25 and include: a) CD3H1.30_L1.47×PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; or PSMA-H H1_L1.13; b) CD3H1.31_L1.47, CD3H1.32_L1.47, or CD3H1.33_L1.47×PSMA-H H1L1.

In some embodiments, the 1+1 Fab-scFv-Fc format includes skew variants, pl variants, ablation variants and FcRn variants. Accordingly, some embodiments include 1+1 Fab-scFv-Fc formats that comprise: a) a first monomer (the “scFv monomer”) that comprises a charged scFv linker (with the +H sequence of FIG. 6 being preferred in some embodiments), the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and an scFv that binds to CD3 as outlined herein; b) a second monomer (the “Fab monomer”) that comprises the skew variants L368D/K370S, the pl variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S, and a variable heavy domain; and c) a light chain that includes a variable light domain (VL) and a constant light domain (CL), wherein numbering is according to EU numbering. The variable heavy domain and variable light domain make up a PSMA binding domain. CD3 binding domain sequences finding particular use in

these embodiments include, but are not limited to, H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (see FIGS. 10A-10F). PSMA binding domain sequences that are of particular use in these embodiments include, but are not limited to, PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; and PSMA-H H1_L1.13. Particularly useful PSMA and CD3 combinations for use in the 1+1 Fab-scFv-Fc PSMA×CD3 bispecific antibody format are disclosed in FIGS. 22-25 and include: a) CD3H1.30_L1.47× PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; or PSMA-H H1_L1.13; b) CD3H1.31_L1.47, CD3H1.32_L1.47, or CD3H1.33_L1.47×PSMA-H H1L1.

FIGS. 7A-7D show some exemplary Fc domain sequences that are useful in the 1+1 Fab-scFv-Fc format antibodies. The “monomer 1” sequences depicted in FIGS. 7A-7D typically refer to the Fc domain of the “Fab-Fc heavy chain” and the “monomer 2” sequences refer to the Fc domain of the “scFv-Fc heavy chain.” Further, FIG. 9 provides useful CL sequences that can be used with this format. In some embodiments, any of the VH and VL sequences depicted herein (including all VH and VL sequences depicted in the Figures and Sequence Listings, including those directed to PSMA) can be added to the bottle opener backbone formats of FIG. 7A-7D as the “Fab side”, using any of the anti-CD3 scFv sequences shown in the Figures and Sequence Listings. For bottle opener backbone 1 from FIG. 7A, (optionally including the 428L/434S variants), CD binding domain sequences finding particular use in these embodiments include, but are not limited to, CD3 binding domain anti-CD3H1.30_L1.47, anti-CD3H1.32_L1.47, anti-CD3H1.89_L1.47, anti-CD3H1.90_L1.47, anti-CD3H1.33_L1.47 and anti-CD3H1.31_L1.47, as well as those depicted in FIG. 10A-10F, attached as the scFv side of the backbones shown in FIGS. 7A-7D. Particularly useful PSMA and CD3 sequence combinations (optionally including the 428L/434S variants) and exemplary anti-CD3× anti-PSMA antibodies in the 1+1 Fab-scFv-Fc format are depicted in FIGS. 22-25.

2. mAb-Fv

One heterodimeric scaffold that finds particular use in the antibodies described herein is the mAb-Fv format (FIG. 50G). In this embodiment, the format relies on the use of a C-terminal attachment of an “extra” variable heavy domain to one monomer and the C-terminal attachment of an “extra” variable light domain to the other monomer, thus forming a third antigen binding domain, wherein the Fab portions of the two monomers bind a PSMA and the “extra” scFv domain binds CD3.

In this embodiment, the first monomer comprises a first heavy chain, comprising a first variable heavy domain and a first constant heavy domain comprising a first Fc domain, with a first variable light domain covalently attached to the C-terminus of the first Fc domain using a domain linker (VH1-CH1-hinge-CH2-CH3-[optional linker]-VL2). The second monomer comprises a second variable heavy domain of the second constant heavy domain comprising a second Fc domain, and a third variable heavy domain covalently

attached to the C-terminus of the second Fc domain using a domain linker (vh1-CH1-hinge-CH2-CH3-[optional linker]-VH2). The two C-terminally attached variable domains make up a Fv that binds CD3 (as it is less preferred to have bivalent CD3 binding). This embodiment further utilizes a common light chain comprising a variable light domain and a constant light domain that associates with the heavy chains to form two identical Fabs that bind a PSMA. As for many of the embodiments herein, these constructs include skew variants, pI variants, ablation variants, additional Fc variants, etc. as desired and described herein.

The antibodies described herein provide mAb-Fv formats where the CD3 binding domain sequences are as shown in FIG. 10A-10F. The antibodies described herein provide mAb-Fv formats wherein the PSMA binding domain sequences are as shown in FIGS. 17 and 18.

In addition, the Fc domains of the mAb-Fv format comprise skew variants (e.g. a set of amino acid substitutions as shown in FIGS. 1 and 4, with particularly useful skew variants being selected from the group consisting of S364K/E357Q/L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L, K370S:S364K/E357Q, T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C: T366W/S354C), optionally ablation variants (including those shown in FIG. 3), optionally charged scFv linkers (including those shown in FIG. 5) and the heavy chain comprises pI variants (including those shown in FIG. 2).

In some embodiments, the mAb-Fv format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments include mAb-Fv formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a first variable heavy domain that, with the first variable light domain of the light chain, makes up an Fv that binds to PSMA, and a second variable heavy domain; b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and a first variable heavy domain that, with the first variable light domain, makes up the Fv that binds to PSMA as outlined herein, and a second variable light chain, that together with the second variable heavy domain forms an Fv (ABD) that binds to CD3; and c) a light chain comprising a first variable light domain and a constant light domain.

In some embodiments, the mAb-Fv format includes skew variants, pI variants, ablation variants and FcRn variants. Accordingly, some embodiments include mAb-Fv formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a first variable heavy domain that, with the first variable light domain of the light chain, makes up an Fv that binds to PSMA, and a second variable heavy domain; b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a first variable heavy domain that, with the first variable light domain, makes up the Fv that binds to PSMA as outlined herein, and a second variable light chain, that together with the second variable heavy domain of the first monomer forms an Fv (ABD) that binds CD3; and c) a light chain comprising a first variable light domain and a constant light domain.

3. mAb-scFv

One heterodimeric scaffold that finds particular use in the antibodies described herein is the mAb-scFv format (FIG. 50H). In this embodiment, the format relies on the use of a C-terminal attachment of a scFv to one of the monomers, thus forming a third antigen binding domain, wherein the Fab portions of the two monomers bind PSMA and the “extra” scFv domain binds CD3. Thus, the first monomer comprises a first heavy chain (comprising a variable heavy domain and a constant domain), with a C-terminally covalently attached scFv comprising a scFv variable light domain, an scFv linker and a scFv variable heavy domain in either orientation (VH1-CH1-hinge-CH2-CH3-[optional linker]-VH2-scFv linker-VL2 or VH1-CH1-hinge-CH2-CH3-[optional linker]-VL2-scFv linker-VH2). This embodiment further utilizes a common light chain comprising a variable light domain and a constant light domain, that associates with the heavy chains to form two identical Fabs that bind PSMA. As for many of the embodiments herein, these constructs include skew variants, pl variants, ablation variants, additional Fc variants, etc. as desired and described herein.

The antibodies described herein provide mAb-scFv formats where the CD binding domain sequences are as shown in FIGS. 10A-10F and the PSMA binding domain sequences are as shown in FIGS. 17 and 18A-18E.

In addition, the Fc domains of the mAb-scFv format comprise skew variants (e.g. a set of amino acid substitutions as shown in FIG. 1, with particularly useful skew variants being selected from the group consisting of S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L, K370S:S364K/E357Q, T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C: T366W/S354C), optionally ablation variants (including those shown in FIG. 3), optionally charged scFv linkers (including those shown in FIG. 5) and the heavy chain comprises pl variants (including those shown in FIG. 2).

In some embodiments, the mAb-scFv format includes skew variants, pl variants, and ablation variants. Accordingly, some embodiments include mAb-scFv formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that comprises the skew variants L368D/K370S, the pl variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein; and c) a common light chain comprising a variable light domain and a constant light domain.

In some embodiments, the mAb-scFv format includes skew variants, pl variants, ablation variants and FcRn variants. Accordingly, some embodiments include mAb-scFv formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that comprises the skew variants L368D/K370S, the pl variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light

domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein; and c) a common light chain comprising a variable light domain and a constant light domain.

4. 2+1 Fab₂-scFv-Fc Format

One heterodimeric scaffold that finds particular use in the antibodies described herein is the “2+1 Fab₂-scFv-Fc” format (also referred to in previous related filings as “central-scFv format”) shown in FIG. 21B with an exemplary combination of a CD3 binding domain and two tumor target antigen (PSMA) binding domains. In this embodiment, the format relies on the use of an inserted scFv domain thus forming a third antigen binding domain, wherein the Fab portions of the two monomers bind PSMA and the “extra” scFv domain binds CD3. The scFv domain is inserted between the Fc domain and the CH1-Fv region of one of the monomers, thus providing a third antigen binding domain. As described, PSMA×CD3 bispecific antibodies having the 2+1 Fab₂-scFv-Fc format are potent in inducing redirected T cell cytotoxicity in cellular environments that express low levels of PSMA. Moreover, as shown in the examples, PSMA×CD3 bispecific antibodies having the 2+1 Fab₂-scFv-Fc format allow for the “fine tuning” of immune responses as such antibodies exhibit a wide variety of different properties, depending on the PSMA and/or CD3 binding domains used. For example, such antibodies exhibit differences in selectivity for cells with different PSMA expression, potencies for PSMA expressing cells, ability to elicit cytokine release, and sensitivity to soluble PSMA. These PSMA antibodies find use, for example, in the treatment of PSMA associated cancers.

In this embodiment, one monomer comprises a first heavy chain comprising a first variable heavy domain, a CH1 domain (and optional hinge) and Fc domain, with a scFv comprising a scFv variable light domain, an scFv linker and a scFv variable heavy domain. The scFv is covalently attached between the C-terminus of the CH1 domain of the heavy constant domain and the N-terminus of the first Fc domain using optional domain linkers (N- to C-terminus: VH1-CH1-[optional linker]-VH2-scFv linker-VL2-[optional linker including the hinge]-CH2-CH3, or the opposite orientation for the scFv, N- to C-terminus: VH1-CH1-[optional linker]-VL2-scFv linker-VH2-[optional linker including the hinge]-CH2-CH3). The optional linkers can be any suitable peptide linkers, including, for example, the domain linkers included in FIG. 6. In some embodiments, the optional linker is a hinge or a fragment thereof. The other monomer is a standard Fab side (i.e., VH1-CH1-hinge-CH2-CH3). This embodiment further utilizes a common light chain comprising a variable light domain and a constant light domain that associates with the heavy chains to form two identical Fabs that bind PSMA. As for many of the embodiments herein, these constructs include skew variants, pl variants, ablation variants, additional Fc variants, etc. as desired and described herein.

In one embodiment, the 2+1 Fab₂-scFv-Fc format antibody includes an scFv with the VH and VL of a CD3 binding domain sequence depicted in FIG. 10A-10F or the Sequence Listing. In one embodiment, the 2+1 Fab₂-scFv-Fc format antibody includes two Fabs having the VH and VL of a PSMA binding domain as shown in FIGS. 17 and 18A-18E and the Sequence Listing. In an exemplary embodiment, the PSMA binding domain of the 2+1 Fab₂-scFv-Fc PSMA×CD3 bispecific antibody includes the VH of PSMA-H H1 (FIG. 17) and VL of one of the following PSMA binding domains: PSMA-H L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In one embodiment, the CD3 binding domain of the

2+1 Fab₂-scFv-Fc format antibody includes the VH and VL of one of the following CD3 binding domains: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F). Particularly useful PSMA and CD3 combinations for use in the 2+1 Fab₂-scFv-Fc format antibody format are disclosed in FIGS. 26-32 and include: a) CD3H1.30_L1.47, CD3H1.32_L1.47, CD3 L1.47_H1.32, CD3H1.89_L1.47, CD3 L1.47_H1.89, CD3H1.33_L1.47, CD3 L1.47_H1.32xb) PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84

In addition, the Fc domains of the 2+1 Fab₂-scFv-Fc format comprise skew variants (e.g. a set of amino acid substitutions as shown in FIG. 1, with particularly useful skew variants being selected from the group consisting of S364K/E357Q/L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L, K370S:S364K/E357Q, T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C:T366W/S354C), optionally ablation variants (including those shown in FIG. 3), optionally charged scFv linkers (including those shown in FIG. 5) and the heavy chain comprises pl variants (including those shown in FIG. 2).

In some embodiments, the 2+1 Fab₂-scFv-Fc format antibody includes skew variants, pl variants, and ablation variants. Accordingly, some embodiments include 2+1 Fab₂-scFv-Fc formats that comprise: a) a first monomer (the Fab-scFv-Fc side) that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein, and an scFv domain that binds to CD3; b) a second monomer (the Fab-Fc side) that comprises the skew variants L368D/K370S, the pl variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein; and c) a common light chain comprising the variable light domain and a constant light domain, where numbering is according to EU numbering. In some embodiments, the common light chain and variable heavy domains on each monomer for PSMA binding domains. CD3 binding domain sequences finding particular use in these embodiments include, but are not limited to, H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 as well as those depicted in FIGS. 10A-10F. PSMA binding domain sequences that are of particular use in these embodiments include, but are not limited to, PSMA-H H1_L1, and PSMA-H H1_L1.1-L1.84 as depicted in FIGS. 17-19.

In some embodiments, the 2+1 Fab₂-scFv-Fc format antibody includes skew variants, pl variants, ablation variants and FcRn variants. Accordingly, some embodiments include 2+1 Fab₂-scFv-Fc formats that comprise: a) a first monomer (the Fab-scFv-Fc side) that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light

domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein, and an scFv domain that binds to CD3; b) a second monomer (the Fab-Fc side) that comprises the skew variants L368D/K370S, the pl variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein; and c) a common light chain comprising a variable light domain and a constant light domain, where numbering is according to EU numbering.

In some embodiments, the common light chain and variable heavy domains on each monomer for PSMA binding domains. CD3 binding domain sequences finding particular use in these embodiments include, but are not limited to, H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 as well as those depicted in FIGS. 10A-10F. PSMA binding domain sequences that are of particular use in these embodiments include, but are not limited to, PSMA-H H1_L1, and PSMA-H H1_L1.1-L1.84 as depicted in FIGS. 17-19.

FIGS. 8A-8C shows some exemplary Fc domain sequences that are useful with the 2+1 Fab₂-scFv-Fc format. The "monomer 1" sequences depicted in FIGS. 8A-8C typically refer to the Fc domain of the "Fab-Fc heavy chain" and the "monomer 2" sequences refer to the Fc domain of the "Fab-scFv-Fc heavy chain." Further, FIG. 9 provides useful CL sequences that can be used with this format.

Exemplary anti-CD3× anti-PSMA antibodies in the 2+1 Fab₂-scFv-Fc format are depicted in FIGS. 26-32.

5. Central-Fv

One heterodimeric scaffold that finds particular use in the antibodies described herein is the central-Fv format (FIG. 50I). In this embodiment, the format relies on the use of an inserted Fv domain (i.e., the central Fv domain) thus forming an "extra" third antigen binding domain, wherein the Fab portions of the two monomers bind a PSMA and the "extra" central Fv domain binds CD3. The "extra" central Fv domain is inserted between the Fc domain and the CH1-Fv region of the monomers, thus providing a third antigen binding domain (i.e., the "extra" central Fv domain), wherein each monomer contains a component of the "extra" central Fv domain (i.e., one monomer comprises the variable heavy domain and the other a variable light domain of the "extra" central Fv domain).

In this embodiment, one monomer comprises a first heavy chain comprising a first variable heavy domain, a CH1 domain, and Fc domain and an additional variable light domain. The light domain is covalently attached between the C-terminus of the CH1 domain of the heavy constant domain and the N-terminus of the first Fc domain using domain linkers (VH1-CH1-[optional linker]-VL2-hinge-CH2-CH3). The other monomer comprises a first heavy chain comprising a first variable heavy domain, a CH1 domain and Fc domain and an additional variable heavy domain (VH1-CH1-[optional linker]-VH2-hinge-CH2-CH3). The light domain is covalently attached between the C-terminus of the CH1 domain of the heavy constant domain and the N-terminus of the first Fc domain using domain linkers.

This embodiment further utilizes a common light chain comprising a variable light domain and a constant light domain that associates with the heavy chains to form two identical Fabs that each bind a PSMA. As for many of the

embodiments herein, these constructs include skew variants, pI variants, ablation variants, additional Fc variants, etc. as desired and described herein.

The antibodies described herein provide central-Fv formats where the CD3 binding domain sequences are as shown in 10A-10F and the PSMA binding domain sequences (VH, VL and CDRs) are as shown in FIGS. 17 and 18.

6. One Armed Central-scFv

One heterodimeric scaffold that finds particular use in the antibodies described herein is the one armed central-scFv format (FIG. 50C). In this embodiment, one monomer comprises just an Fc domain, while the other monomer includes a Fab domain (a first antigen binding domain), a scFv domain (a second antigen binding domain) and an Fc domain, where the scFv domain is inserted between the Fc domain and the Fc domain. In this format, the Fab portion binds one receptor target and the scFv binds another. In this format, either the Fab portion binds a PSMA and the scFv binds CD3 or vice versa.

In this embodiment, one monomer comprises a first heavy chain comprising a first variable heavy domain, a CH1 domain and Fc domain, with a scFv comprising a scFv variable light domain, an scFv linker and a scFv variable heavy domain. The scFv is covalently attached between the C-terminus of the CH1 domain of the heavy constant domain and the N-terminus of the first Fc domain using domain linkers, in either orientation, VH1-CH1-[optional domain linker]-VH2-scFv linker-VL2-[optional domain linker]-CH2-CH3 or VH1-CH1-[optional domain linker]-VL2-scFv linker-VH2-[optional domain linker]-CH2-CH3. The second monomer comprises an Fc domain (CH2-CH3). This embodiment further utilizes a light chain comprising a variable light domain and a constant light domain that associates with the heavy chain to form a Fab.

As for many of the embodiments herein, these constructs include skew variants, pI variants, ablation variants, additional Fc variants, etc. as desired and described herein.

The antibodies described herein provide central-Fv formats where the CD3 binding domain sequences are as shown in FIGS. 10A-10F and the PSMA binding domain sequences (VH, VL and CDRs) are as shown in FIGS. 17 and 18.

In addition, the Fc domains of the one armed central-scFv format generally include skew variants (e.g. a set of amino acid substitutions as shown in FIG. 1, with particularly useful skew variants being selected from the group consisting of S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L, K370S:S364K/E357Q, T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C:T366W/S354C), optionally ablation variants (including those shown in FIG. 3), optionally charged scFv linkers (including those shown in FIG. 5) and the heavy chain comprises pI variants (including those shown in FIG. 2).

In some embodiments, the one armed central-scFv format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments of the one armed central-scFv formats comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the light chain, makes up an Fv that binds to PSMA as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that includes an Fc domain having the skew variants L368D/K370S, the pI variants N208D/Q295E/

N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K; and c) a light chain comprising a variable light domain and a constant light domain.

In some embodiments, the one armed central-scFv format includes skew variants, pI variants, ablation variants and FcRn variants. Accordingly, some embodiments of the one armed central-scFv formats comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the light chain, makes up an Fv that binds to PSMA as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that includes an Fc domain having the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and the FcRn variants M428L/N434S; and c) a light chain comprising a variable light domain and a constant light domain.

7. One Armed scFv-mAb

One heterodimeric scaffold that finds particular use in the antibodies described herein is the one armed scFv-mAb format (FIG. 50D). In this embodiment, one monomer comprises just an Fc domain, while the other monomer uses a scFv domain attached at the N-terminus of the heavy chain, generally through the use of a linker: VH-scFv linker-VL-[optional domain linker]-CH1-hinge-CH2-CH3 or (in the opposite orientation) VL-scFv linker-VH-[optional domain linker]-CH1-hinge-CH2-CH3. In this format, the Fab portions each bind PSMA and the scFv binds CD3. This embodiment further utilizes a light chain comprising a variable light domain and a constant light domain that associates with the heavy chain to form a Fab. As for many of the embodiments herein, these constructs include skew variants, pI variants, ablation variants, additional Fc variants, etc. as desired and described herein.

The antibodies described herein provide one armed scFv-mAb formats where the CD3 binding domain sequences are as shown in 10A-10F and wherein the PSMA binding domain sequences (VH, VL and CDRs) are as shown in FIGS. 17 and 18.

In addition, the Fc domains of the one armed scFv-mAb format generally include skew variants (e.g. a set of amino acid substitutions as shown in FIG. 1, with particularly useful skew variants being selected from the group consisting of S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L, K370S:S364K/E357Q, T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C:T366W/S354C), optionally ablation variants (including those shown in FIG. 3), optionally charged scFv linkers (including those shown in FIG. 5) and the heavy chain comprises pI variants (including those shown in FIG. 2).

In some embodiments, the one armed scFv-mAb format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments of the one armed scFv-mAb formats comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the light chain, makes up an Fv that binds to PSMA as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that includes an Fc domain having the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/

L235A/G236del/S267K; and c) a light chain comprising a variable light domain and a constant light domain.

In some embodiments, the one armed scFv-mAb format includes skew variants, pI variants, ablation variants and FcRn variants. Accordingly, some embodiments one armed scFv-mAb formats comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the light chain, makes up an Fv that binds to PSMA as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that includes an Fc domain having the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and the FcRn variants M428L/N434S; and c) a light chain comprising a variable light domain and a constant light domain.

8. scFv-mAb

One heterodimeric scaffold that finds particular use in the antibodies described herein is the mAb-scFv format (FIG. 50E). In this embodiment, the format relies on the use of a N-terminal attachment of a scFv to one of the monomers, thus forming a third antigen binding domain, wherein the Fab portions of the two monomers bind PSMA and the “extra” scFv domain binds CD3.

In this embodiment, the first monomer comprises a first heavy chain (comprising a variable heavy domain and a constant domain), with a N-terminally covalently attached scFv comprising a scFv variable light domain, an scFv linker and a scFv variable heavy domain in either orientation ((VH1-scFv linker-VL1-[optional domain linker]-VH2-CH1-hinge-CH2-CH3) or (with the scFv in the opposite orientation) ((VL1-scFv linker-VH1-[optional domain linker]-VH2-CH1-hinge-CH2-CH3)). This embodiment further utilizes a common light chain comprising a variable light domain and a constant light domain that associates with the heavy chains to form two identical Fabs that bind PSMA. As for many of the embodiments herein, these constructs include skew variants, pI variants, ablation variants, additional Fc variants, etc. as desired and described herein.

The antibodies described herein provide scFv-mAb formats where the CD3 binding domain sequences are as shown in 10A-10F and wherein the PSMA binding domain sequences (VH, VL and CDRs) are as shown in FIGS. 17 and 18.

In addition, the Fc domains of the scFv-mAb format generally include skew variants (e.g. a set of amino acid substitutions as shown in FIG. 1, with particularly useful skew variants being selected from the group consisting of S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L, K370S:S364K/E357Q, T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C:T366W/S354C), optionally ablation variants (including those shown in FIG. 3), optionally charged scFv linkers (including those shown in FIG. 5) and the heavy chain comprises pI variants (including those shown in FIG. 2).

In some embodiments, the scFv-mAb format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments include scFv-mAb formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein, and

a scFv domain that binds to CD3; b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein; and c) a common light chain comprising a variable light domain and a constant light domain.

In some embodiments, the scFv-mAb format includes skew variants, pI variants, ablation variants and FcRn variants. Accordingly, some embodiments include scFv-mAb formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein; and c) a common light chain comprising a variable light domain and a constant light domain.

9. Dual scFv Formats

The antibodies described herein also provide dual scFv formats (FIG. 50B) as are known in the art. In this embodiment, the PSMA×CD3 heterodimeric bispecific antibody is made up of two scFv-Fc monomers (both in either (VH-scFv linker-VL-[optional domain linker]-CH2-CH3) format or (VL-scFv linker-VH-[optional domain linker]-CH2-CH3) format, or with one monomer in one orientation and the other in the other orientation.

The antibodies described herein provide dual scFv formats where the CD3 binding domain sequences are as shown in FIGS. 10A-10F and wherein the PSMA binding domain sequences (VH, VL and CDRs) are as shown in FIGS. 17 and 18. In some embodiments, the dual scFv format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments include dual scFv formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a first scFv that binds either CD3 or PSMA; and b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and a second scFv that binds either CD3 or PSMA. In some embodiments, the dual scFv format includes skew variants, pI variants, ablation variants and FcRn variants. In some embodiments, the dual scFv format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments include dual scFv formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a first scFv that binds either CD3 or PSMA; and b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a second scFv that binds either CD3 or PSMA.

10. Non-Heterodimeric Bispecific Antibodies

As will be appreciated by those in the art, the PSMA and CD3 Fv sequences outlined herein can also be used in both

monospecific antibodies (e.g., “traditional monoclonal antibodies”) or non-heterodimeric bispecific formats.

CD3 binding domain sequences finding particular use include, but are not limited to H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F).

PSMA binding domain sequences that are of particular use include, but are not limited to: PSMA-H L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E).

Suitable non-heterodimeric bispecific formats are known in the art, and include a number of different formats as generally depicted in Spiess et al., *Molecular Immunology* (67):95-106 (2015) and Kontermann, *mAbs* 4:2, 182-197 (2012), both of which are expressly incorporated by reference and in particular for the figures, legends and citations to the formats therein.

11. Trident Format

In some embodiments, the bispecific antibodies described herein are in the “Trident” format (FIG. 50K) as generally described in WO2015/184203, hereby expressly incorporated by reference in its entirety and in particular for the Figures, Legends, definitions and sequences of “Heterodimer-Promoting Domains” or “HPDs”, including “K-coil” and “E-coil” sequences. Tridents rely on using two different HPDs that associate to form a heterodimeric structure as a component of the structure, see FIG. 1K. In this embodiment, the Trident format include a “traditional” heavy and light chain (e.g., VH1-CH1-hinge-CH2-CH3 and VL1-CL), a third chain comprising a first “diabody-type binding domain” or “DART®”, VH2-(linker)-VL3-HPD1 and a fourth chain comprising a second DART®, VH3-(linker)-(linker)-VL2-HPD2. The VH1 and VL1 form a first ABD, the VH2 and VL2 form a second ABD, and the VH3 and VL3 form a third ABD. In some cases, as is shown in FIG. 1K, the second and third ABDs bind the same antigen, in this instance generally PSMA, e.g., bivalently, with the first ABD binding a CD3 monovalently.

12. Monospecific, Monoclonal Antibodies

As will be appreciated by those in the art, the novel Fv sequences outlined herein can also be used in both monospecific antibodies (e.g., “traditional monoclonal antibodies”) or non-heterodimeric bispecific formats. Accordingly, in some embodiments, the antibodies described herein provide monoclonal (monospecific) antibodies comprising the 6 CDRs and/or the vh and vl sequences from the figures, generally with IgG1, IgG2, IgG3 or IgG4 constant regions, with IgG1, IgG2 and IgG4 (including IgG4 constant regions comprising a S228P amino acid substitution) finding particular use in some embodiments. That is, any sequence herein with a “H_L” designation can be linked to the constant region of a human IgG1 antibody.

In some embodiments, the monospecific antibody is a PSMA monospecific antibody. In certain embodiments, the monospecific anti-PSMA antibody includes the 6 CDRs of any of the anti-PSMA binding domains selected from: PSMA-H L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In certain embodiments, the monospecific anti-PSMA antibody includes the variable heavy domain (VH) and variable light domain (VL) of any of the anti-PSMA binding domains selected from: PSMA-H L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E).

G. Antigen Binding Domains

As discussed herein, the subject heterodimeric antibodies include two antigen binding domains (ABDs), each of which bind to PSMA or CD3. As outlined herein, these heterodi-

meric antibodies can be bispecific and bivalent (each antigen is bound by a single ABD, for example, in the format depicted in FIG. 21A), or bispecific and trivalent (one antigen is bound by a single ABD and the other is bound by two ABDs, for example as depicted in FIG. 21B).

In addition, in general, one of the ABDs comprises a scFv as outlined herein, in an orientation from N- to C-terminus of VH-scFv linker-VL or VL-scFv linker-VH. One or both of the other ABDs, according to the format, generally is a Fab, comprising a VH domain on one protein chain (generally as a component of a heavy chain) and a VL on another protein chain (generally as a component of a light chain).

The disclosure provides a number of ABDs that bind to a number of different checkpoint proteins, as outlined below. As will be appreciated by those in the art, any set of 6 CDRs or VH and VL domains can be in the scFv format or in the Fab format, which is then added to the heavy and light constant domains, where the heavy constant domains comprise variants (including within the CH1 domain as well as the Fc domain). The scFv sequences contained in the sequence listing utilize a particular charged linker, but as outlined herein, uncharged or other charged linkers can be used, including those depicted in FIG. 6.

In addition, as discussed above, the numbering used in the Sequence Listing for the identification of the CDRs is Kabat, however, different numbering can be used, which will change the amino acid sequences of the CDRs as shown in Table 2.

For all of the variable heavy and light domains listed herein, further variants can be made. As outlined herein, in some embodiments the set of 6 CDRs can have from 0, 1, 2, 3, 4 or 5 amino acid modifications (with amino acid substitutions finding particular use), as well as changes in the framework regions of the variable heavy and light domains, as long as the frameworks (excluding the CDRs) retain at least about 80, 85 or 90% identity to a human germline sequence selected from those listed in FIG. 1 of U.S. Pat. No. 7,657,380, which Figure and Legend is incorporated by reference in its entirety herein. Thus, for example, the identical CDRs as described herein can be combined with different framework sequences from human germline sequences, as long as the framework regions retain at least 80, 85 or 90% identity to a human germline sequence selected from those listed in FIG. 1 of U.S. Pat. No. 7,657,380. Alternatively, the CDRs can have amino acid modifications (e.g. from 1, 2, 3, 4 or 5 amino acid modifications in the set of CDRs (that is, the CDRs can be modified as long as the total number of changes in the set of 6 CDRs is less than 6 amino acid modifications, with any combination of CDRs being changed; e.g. there may be one change in VLCDR1, two in VHCDR2, none in VHCDR3, etc.)), as well as having framework region changes, as long as the framework regions retain at least 80, 85 or 90% identity to a human germline sequence selected from those listed in FIG. 1 of U.S. Pat. No. 7,657,380.

1. PSMA Antigen Binding Domains

Provided herein are PSMA antigen binding domain and antibodies that include such binding domains. Suitable sets of 6 CDRs and/or VH and VL domains included in PSMA binding domains are depicted in FIG. 17 (anti-PSMA vhCDRs and VH) and FIGS. 17 and 18A-E (anti-PSMA vlCDRs and VL). In some embodiments, the heterodimeric antibody is a 1+1 Fab-scFv-Fc or 2+1 Fab₂-scFv-Fv format antibody (see, e.g., FIGS. 21A and 21B).

In one embodiment, the PSMA antigen binding domain includes the 6 CDRs (i.e., vhCDR1-3 and vlCDR1-3) of a PSMA ABD described herein, including the figures and

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sequence listing. In certain embodiments the PSMA antigen binding domain includes a variable heavy domain that includes v hCDR1-3 of PSMA-H H1 (FIG. 17) and a variable light domain that includes the v LCDR1-3 of any of PSMA-H L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In exemplary embodiments, the PSMA antigen binding domain includes the 6 CDRs of one of the following PSMA antigen binding domains: PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; and PSMA-H H1_L1.13.

In one embodiment, the PSMA antigen binding domain includes the variable heavy and variable light domain of a PSMA ABD described herein, including the figures and sequence listing. In certain embodiments the PSMA antigen binding domain includes a variable heavy domain that is PSMA-H H1 (FIG. 17) and a variable light domain selected from PSMA-H L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In exemplary embodiments, the PSMA antigen binding domain includes the variable heavy domain and variable light domain of one of the following PSMA antigen binding domains: PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; and PSMA-H H1_L1.13.

As will be appreciated by those in the art, suitable PSMA binding domains can comprise a set of 6 CDRs as depicted in the Figures, either as they are underlined or, in the case where a different numbering scheme is used as described herein and as shown in Table 2, as the CDRs that are identified using other alignments within the VH and VL sequences of those depicted in FIGS. 17 and 18A-E. Suitable ABDs can also include the entire VH and VL sequences as depicted in these sequences and Figures, used as scFvs or as Fabs. In many of the embodiments herein that contain an Fv to PSMA, it is the Fab monomer that binds PSMA.

In addition to the parental CDR sets disclosed in the figures and sequence listing that form an ABD to PSMA, the disclosure provides variant CDR sets. In one embodiment, a set of 6 CDRs can have 1, 2, 3, 4 or 5 amino acid changes from the parental CDRs, as long as the PSMA ABD is still able to bind to the target antigen, as measured by at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g., Octet assay) assay, with the latter finding particular use in many embodiments.

In addition to the parental variable heavy and variable light domains disclosed herein that form an ABD to PSMA, the disclosure provides variant VH and VL domains. In one embodiment, the variant VH and VL domains each can have from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid changes from the parental VH and VL domain, as long as the ABD is still able to bind to the target antigen, as measured at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g. Octet assay) assay, with the latter finding particular use in many embodiments. In another embodiment, the variant VH and VL are at least 90, 95, 97, 98 or 99% identical to the respective parental VH or VL, as long as the ABD is still able to bind to the target antigen, as measured by at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g. Octet assay) assay, with the latter finding particular use in many embodiments.

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2. CD3 Antigen Binding Domains

In some embodiments, one of the ABDs binds CD3. Suitable sets of 6 CDRs and/or VH and VL domains, as well as scFv sequences, are depicted in FIGS. 10A-10F and the Sequence Listing. CD3 binding domain sequences that are of particular use include, but are not limited to, anti-CD3H1.30_L1.47, anti-CD3H1.32, anti-CD3 L1.47, anti-CD3H1.89_L1.47, anti-CD3H1.90_L1.47, anti-CD3H1.33_L1.47, anti-CD3H1.31_L1.47, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.32, anti-CD3 L1.47_H1.89, anti-CD3 L1.47_H1.90, anti-CD3 L1.47_H1.33, and anti-CD3 L1.47_H1.31 as depicted in FIGS. 10A-10F.

As will be appreciated by those in the art, suitable CD3 binding domains can comprise a set of 6 CDRs as depicted in FIGS. 10A-10F, either as they are underlined or, in the case where a different numbering scheme is used as described herein and as shown in Table 2, as the CDRs that are identified using other alignments within the VH and VL sequences of those depicted in FIGS. 10A-10F. Suitable ABDs can also include the entire VH and VL sequences as depicted in these sequences and Figures, used as scFvs or as Fabs. In many of the embodiments herein that contain an Fv to CD3, it is the scFv monomer that binds CD3.

In addition to the parental CDR sets disclosed in the figures and sequence listing that form an ABD to CD3, the disclosure provides variant CDR sets. In one embodiment, a set of 6 CDRs can have 1, 2, 3, 4 or 5 amino acid changes from the parental CDRs, as long as the CD3 ABD is still able to bind to the target antigen, as measured by at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g. Octet assay) assay, with the latter finding particular use in many embodiments.

In addition to the parental variable heavy and variable light domains disclosed herein that form an ABD to CD3, the disclosure provides variant VH and VL domains. In one embodiment, the variant VH and VL domains each can have from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid changes from the parental VH and VL domain, as long as the ABD is still able to bind to the target antigen, as measured by at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g. Octet assay) assay, with the latter finding particular use in many embodiments. In another embodiment, the variant VH and VL are at least 90, 95, 97, 98 or 99% identical to the respective parental VH or VL, as long as the ABD is still able to bind to the target antigen, as measured by at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g. Octet assay) assay, with the latter finding particular use in many embodiments.

VI. Nucleic Acids

The disclosure further provides nucleic acid compositions encoding the anti-PSMA antibodies provided herein, including, but not limited to, anti-PSMA $\times$ anti-CD3 bispecific antibodies and anti-PSMA monospecific antibodies.

As will be appreciated by those in the art, the nucleic acid compositions will depend on the format and scaffold of the heterodimeric protein. Thus, for example, when the format requires three amino acid sequences, such as for the 1+1 Fab-scFv-Fc format (e.g. a first amino acid monomer comprising an Fc domain and a scFv, a second amino acid monomer comprising a heavy chain and a light chain), three nucleic acid sequences can be incorporated into one or more expression vectors for expression. Similarly, some formats

(e.g. dual scFv formats such as disclosed in FIG. 1) only two nucleic acids are needed; again, they can be put into one or two expression vectors.

As is known in the art, the nucleic acids encoding the components of the antibodies described herein can be incorporated into expression vectors as is known in the art, and depending on the host cells used to produce the heterodimeric antibodies described herein. Generally, the nucleic acids are operably linked to any number of regulatory elements (promoters, origin of replication, selectable markers, ribosomal binding sites, inducers, etc.). The expression vectors can be extra-chromosomal or integrating vectors.

The nucleic acids and/or expression vectors of the antibodies described herein are then transformed into any number of different types of host cells as is well known in the art, including mammalian, bacterial, yeast, insect and/or fungal cells, with mammalian cells (e.g. CHO cells), finding use in many embodiments.

In some embodiments, nucleic acids encoding each monomer and the optional nucleic acid encoding a light chain, as applicable depending on the format, are each contained within a single expression vector, generally under different or the same promoter controls. In embodiments of particular use in the antibodies described herein, each of these two or three nucleic acids are contained on a different expression vector. As shown herein and in 62/025,931, hereby incorporated by reference, different vector ratios can be used to drive heterodimer formation. That is, surprisingly, while the proteins comprise first monomer:second monomer:light chains (in the case of many of the embodiments herein that have three polypeptides comprising the heterodimeric antibody) in a 1:1:2 ratio, these are not the ratios that give the best results.

The heterodimeric antibodies described herein are made by culturing host cells comprising the expression vector(s) as is well known in the art. Once produced, traditional antibody purification steps are done, including an ion exchange chromatography step. As discussed herein, having the pIs of the two monomers differ by at least 0.5 can allow separation by ion exchange chromatography or isoelectric focusing, or other methods sensitive to isoelectric point. That is, the inclusion of pI substitutions that alter the isoelectric point (pI) of each monomer so that each monomer has a different pI and the heterodimer also has a distinct pI, thus facilitating isoelectric purification of the "1+1 Fab-scFv-Fc" and "2+1" heterodimers (e.g., anionic exchange columns, cationic exchange columns). These substitutions also aid in the determination and monitoring of any contaminating dual scFv-Fc and mAb homodimers post-purification (e.g., IEF gels, cIEF, and analytical IEX columns).

VII. Biological and Biochemical Functionality of the Heterodimeric Bispecific Antibodies

Generally the bispecific PSMA×CD3 antibodies described herein are administered to patients with cancer, and efficacy is assessed, in a number of ways as described herein. Thus, while standard assays of efficacy can be run, such as cancer load, size of tumor, evaluation of presence or extent of metastasis, etc., immuno-oncology treatments can be assessed on the basis of immune status evaluations as well. This can be done in a number of ways, including both in vitro and in vivo assays.

VIII. Treatments

Once made, the compositions of the antibodies described herein find use in a number of applications. PSMA is highly

expressed in prostate cancer. Accordingly, the heterodimeric compositions of the antibodies described herein find use in the treatment of such PSMA positive cancers.

IX. Antibody Compositions for In Vivo Administration

Formulations of the antibodies used in accordance with the antibodies described herein are prepared for storage by mixing an antibody having the desired degree of purity with optional pharmaceutically acceptable carriers, excipients or stabilizers (Remington's Pharmaceutical Sciences 16th edition, Osol, A. Ed. [1980]), in the form of lyophilized formulations or aqueous solutions. Acceptable carriers, excipients, or stabilizers are nontoxic to recipients at the dosages and concentrations employed, and include buffers such as phosphate, citrate, and other organic acids; antioxidants including ascorbic acid and methionine; preservatives (such as octadecyldimethylbenzyl ammonium chloride; hexamethonium chloride; benzalkonium chloride, benzethonium chloride; phenol, butyl or benzyl alcohol; alkyl parabens such as methyl or propyl paraben; catechol; resorcinol; cyclohexanol; 3-pentanol; and m-cresol); low molecular weight (less than about 10 residues) polypeptides; proteins, such as serum albumin, gelatin, or immunoglobulins; hydrophilic polymers such as polyvinylpyrrolidone; amino acids such as glycine, glutamine, asparagine, histidine, arginine, or lysine; monosaccharides, disaccharides, and other carbohydrates including glucose, mannose, or dextrans; chelating agents such as EDTA; sugars such as sucrose, mannitol, trehalose or sorbitol; salt-forming counter-ions such as sodium; metal complexes (e.g. Zn-protein complexes); and/or non-ionic surfactants such as TWEEN™ PLURONICS™ or polyethylene glycol (PEG).

X. Administrative Modalities

The antibodies and chemotherapeutic agents described herein are administered to a subject, in accord with known methods, such as intravenous administration as a bolus or by continuous infusion over a period of time.

XI. Treatment Modalities

In the methods described herein, therapy is used to provide a positive therapeutic response with respect to a disease or condition. By "positive therapeutic response" is intended an improvement in the disease or condition, and/or an improvement in the symptoms associated with the disease or condition. For example, a positive therapeutic response would refer to one or more of the following improvements in the disease: (1) a reduction in the number of neoplastic cells; (2) an increase in neoplastic cell death; (3) inhibition of neoplastic cell survival; (5) inhibition (i.e., slowing to some extent, preferably halting) of tumor growth; (6) an increased patient survival rate; and (7) some relief from one or more symptoms associated with the disease or condition.

Positive therapeutic responses in any given disease or condition can be determined by standardized response criteria specific to that disease or condition. Tumor response can be assessed for changes in tumor morphology (i.e., overall tumor burden, tumor size, and the like) using screening techniques such as magnetic resonance imaging (MRI) scan, x-radiographic imaging, computed tomographic (CT) scan, bone scan imaging, endoscopy, and tumor biopsy sampling including bone marrow aspiration (BMA) and counting of tumor cells in the circulation.

In addition to these positive therapeutic responses, the subject undergoing therapy may experience the beneficial effect of an improvement in the symptoms associated with the disease.

Treatment according to the disclosure includes a “therapeutically effective amount” of the medicaments used. A “therapeutically effective amount” refers to an amount effective, at dosages and for periods of time necessary, to achieve a desired therapeutic result.

A therapeutically effective amount may vary according to factors such as the disease state, age, sex, and weight of the individual, and the ability of the medicaments to elicit a desired response in the individual. A therapeutically effective amount is also one in which any toxic or detrimental effects of the antibody or antibody portion are outweighed by the therapeutically beneficial effects.

A “therapeutically effective amount” for tumor therapy may also be measured by its ability to stabilize the progression of disease. The ability of a compound to inhibit cancer may be evaluated in an animal model system predictive of efficacy in human tumors.

Alternatively, this property of a composition may be evaluated by examining the ability of the compound to inhibit cell growth or to induce apoptosis by in vitro assays known to the skilled practitioner. A therapeutically effective amount of a therapeutic compound may decrease tumor size, or otherwise ameliorate symptoms in a subject. One of ordinary skill in the art would be able to determine such amounts based on such factors as the subject’s size, the severity of the subject’s symptoms, and the particular composition or route of administration selected.

Dosage regimens are adjusted to provide the optimum desired response (e.g., a therapeutic response). For example, a single bolus may be administered, several divided doses may be administered over time or the dose may be proportionally reduced or increased as indicated by the exigencies of the therapeutic situation. Parenteral compositions may be formulated in dosage unit form for ease of administration and uniformity of dosage. Dosage unit form as used herein refers to physically discrete units suited as unitary dosages for the subjects to be treated; each unit contains a predetermined quantity of active compound calculated to produce the desired therapeutic effect in association with the required pharmaceutical carrier.

The specification for the dosage unit forms of the disclosure are dictated by and directly dependent on (a) the unique characteristics of the active compound and the particular therapeutic effect to be achieved, and (b) the limitations inherent in the art of compounding such an active compound for the treatment of sensitivity in individuals.

The efficient dosages and the dosage regimens for the bispecific antibodies described herein depend on the disease or condition to be treated and may be determined by the persons skilled in the art.

An exemplary, non-limiting range for a therapeutically effective amount of an bispecific antibody used in the antibodies described herein is about 0.1-100 mg/kg.

All cited references are herein expressly incorporated by reference in their entirety.

Whereas particular embodiments of the disclosure have been described above for purposes of illustration, it will be appreciated by those skilled in the art that numerous variations of the details may be made without departing from the invention as described in the appended claims.

EXAMPLES

Examples are provided below to illustrate the antibodies described herein. These examples are not meant to constrain

the antibodies described herein to any particular application or theory of operation. For all constant region positions discussed in the antibodies described herein, numbering is according to the EU index as in Kabat (Kabat et al., 1991, Sequences of Proteins of Immunological Interest, 5th Ed., United States Public Health Service, National Institutes of Health, Bethesda, entirely incorporated by reference). Those skilled in the art of antibodies will appreciate that this convention consists of nonsequential numbering in specific regions of an immunoglobulin sequence, enabling a normalized reference to conserved positions in immunoglobulin families. Accordingly, the positions of any given immunoglobulin as defined by the EU index will not necessarily correspond to its sequential sequence.

General and specific scientific techniques are outlined in US Publications 2015/0307629, 2014/0288275 and WO2014/145806, all of which are expressly incorporated by reference in their entirety and particularly for the techniques outlined therein.

Example 1: Generating Biologically Relevant Surrogates for PSMA-Expressing Tumor Cells

To ensure cell lines with biologically valid PSMA antigen densities were used to inform the development of the novel anti-PSMA $\times$ anti-CD3 bispecific antibodies of the invention, IHC was conducted on paraffin embedded arrays of 160 biopsy cores of prostate cancer, 16 tumor adjacent cores, 16 normal prostate cores, and several cancer cell lines. Illustrative IHC of biopsy cores are depicted in FIG. 12, and samples were qualitatively scored in-house on a scale of 0-3 with 0 representing little to no PSMA expression and 3 representing high PSMA expression (herein referred to as IHC score; breakdown of score for each core is depicted in FIG. 13). Based on the results, it was determined that the novel bispecific antibodies of the invention should target cell lines having IHC scores of 3, 2, and 1 as they represent 97% of the patient segment and should not target score 0 cell lines as they resemble normal tissue. Upon matching the staining intensity between the sample types, cell lines were identified that could serve as surrogates of tumor and normal tissues. LNCaP cancer cells were found to stain as intensely as cancer tumors expressing high amounts of PSMA; 22Rv1 cancer cells were found to stain similar to some tumors expressing moderate amounts of PSMA as well as some normal prostate cores; Huh-7 cancer cells were found to express low levels of PSMA; and A549, ASPC-1, HT29, SKOV3, and PC3 cancer cells were found to express little to no PSMA.

PC3 cell lines expressing varying PSMA antigen densities were also generated in order to expand the selection of surrogate cell lines having varying antigen density levels to tune the antibodies of the invention. Cell-surface PSMA antigen density levels on the above cancer cell lines and the PSMA-transfected PC3 cell lines were estimated by FACS using fluorescently-labeled beads as advised by the Quick-Cal protocol (Bangs Laboratories, Inc., Fishers, IN). 50,000 cells per well and MESF beads were combined with A647-conjugated anti-PSMA mAb (J591) for 30 minutes at 4° C. Cells were then washed and fixed in 1% PFA. Flow cytometry was performed to determine antibody binding, and antigen density calculations were carried out using Quick-Cal® V.2.3. software (Bangs Laboratories, Inc., Fishers, IN). Data depicting PSMA density on the various cell lines are depicted in FIGS. 14-16. PSMA-transfected PC3 cell lines

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are hereon referred to by their PSMA density (i.e. PC3 (~100 k) has a MESF score of 100,000).

Example 2: Antigen Binding Domains

2A: CD3 Binding Domains

Sequences for CD3 binding domains having different CD3 binding affinities are depicted in FIG. 10.

2B: PSMA Binding Domains

The variable regions of a mouse anti-human PSMA binding domain were humanized using string content optimization (see, e.g., U.S. Pat. No. 7,657,380, issued Feb. 2, 2010). Sequences for the humanized PSMA binding domain, hereon referred to as PSMA-H, are depicted in FIG. 17.

2B(a): Tuning PSMA-H Binding Affinity for PSMA

Variants of PSMA-H were engineered by introducing point substitutions into the variable light region (VL). In a first round, 75 variant VLs were engineered designated as L1.1-L1.75, sequences for which are depicted in FIG. 18. The variant VLs were paired with the wild-type humanized variable heavy region (VH) of PSMA-H and produced as bivalent mAbs (sequences for which are depicted in FIG. 19). Binding affinity of the variants for human PSMA was screened using Octet, a BioLayer Interferometry (BLI)-based method. Experimental steps for Octet generally include the following: Immobilization (capture of ligand to a biosensor); Association (dipping of ligand-coated biosensors into wells containing the analyte); and Dissociation (returning of biosensors to well containing buffer). In particular, anti-human Fc sensors were used to capture the bivalent mAbs and dipped into human PSMA antigen. The resulting apparent dissociation constant (K_{Dapp} due to avidity concerns), association rate (k_a), dissociation rate (k_d), as well as sensorgram response are depicted in FIG. 20.

Based on the above, useful variant VLs were identified on the basis of change in binding affinity for PSMA, reversion of CDRs to human germline sequence (to reduce immunogenic potential), and introduction of negative charge (to aid in longer serum half-life by reducing non-specific clearance). In some instances, substitutions from suitable variant VLs were combined to generate additional VL variants (sequences for which are depicted in FIG. 18 as L1.76-L1.84). The variant VLs were paired with the wild-type humanized VH and produced as anti-PSMA $\times$ anti-CD3 bsAbs (sequences for which are depicted in FIG. 22) and screened for binding affinity for human and cynomolgus PSMA using Octet as generally described above. The resulting dissociation constant (K_D), association rate (k_a), and dissociation rate (k_d) are depicted in FIGS. 34-35. A range of affinities from 6.55 nM to 116 nM were obtained. Notably, affinity for human and cynomolgus PSMA did not track across all the variants. For example, XENP33756 bound human PSMA with K_D of 18 nM while binding cynomolgus PSMA with K_D of 88 nM. For ease of clinical development, it is advantageous for the PSMA binding domain to bind human and cynomolgus PSMA with similar affinity.

2C: Engineering Anti-PSMA $\times$ Anti-CD3 Bispecific Antibodies (bsAbs)

A number of formats for α PSMA $\times$ α CD3 bispecific antibodies (bsAbs) were conceived, illustrative formats for which are outlined below and in FIG. 21.

One such format is the 1+1 Fab-scFv-Fc format which comprises a single-chain Fv ("scFv") covalently attached to a first heterodimeric Fc domain, a heavy chain variable region (VH) covalently attached to a complementary second

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heterodimeric Fc domain, and a light chain (LC) transfected separately so that a Fab domain is formed with the variable heavy domain.

Another format is the 2+1 Fab2-scFv-Fc format which comprises a VH domain covalently attached to a CH1 domain covalently attached to an scFv covalently attached to a first heterodimeric Fc domain (VH-CH1-scFv-Fc), a VH domain covalently attached to a complementary second heterodimeric Fc domain, and a LC transfected separately so that Fab domains are formed with the VH domains.

DNA encoding chains of the α PSMA $\times$ α CD3 bsAbs were generated by standard gene synthesis followed by isothermal cloning (Gibson assembly) or subcloning into a pTT5 expression vector containing fusion partners (e.g. domain linkers as depicted in FIG. 6 and/or backbones as depicted in FIGS. 7-9). DNA was transfected into HEK293E cells for expression. Sequences for illustrative α PSMA $\times$ α CD3 bsAbs (based on CD3 binding domains as described in Example 2A and PSMA binding domains as described in Example 2B) in the 1+1 Fab-scFv-Fc format and in the 2+1 Fab2-scFv-Fc format are depicted respectively in FIGS. 22-32.

Example 3: Prototypic 1+1 Anti-PSMA $\times$ Anti-CD3 bsAbs Indiscriminately Kill Cell Lines Expressing Low Levels of PSMA

To benchmark the activity of the novel anti-PSMA $\times$ anti-CD3 bsAbs of the invention, the activity prototypic 1+1 anti-PSMA $\times$ anti-CD3 bsAbs XENP14484, XENP34282, and XENP34283 (sequences for which are respectively depicted in FIGS. 22 and 33) were first investigated in redirected T-cell cytotoxicity (RTCC) assays.

PC3 cell lines expressing various PSMA densities (as described in Example 1) were transduced to constitutively express luciferase. Luciferase released from dead cancer cells rapidly degrade in assay media, so live target cells can be quantified based on luminescence readout. The cancer cells were incubated with freshly enriched CD3⁺ T cells at an effector to target ratio of 1:1 for 24 hours. Next, the bispecific antibodies were added to the cells at the indicated concentrations. In a first experiment, 48 hours after addition of the bsAbs, Bio-Glo Luciferase reagent (Promega, Madison, WI.) was added and plates were read with the Envision Reader on luminescence setting. % RTCC was calculated by $1 - (\text{Raw Value}/\text{PBS AVG}) \times 100$. In a second experiment, 72 hours after addition of the bsAbs, cells were assayed via flow cytometry for Ki67 expression on T cells (as an indicator of proliferation).

Data showing cell kill are depicted in FIG. 36, and data showing T cell proliferation are depicted in FIG. 37. The data show that the two prototypic 1+1 anti-PSMA $\times$ anti-CD3 induced RTCC on cell lines expressing high and low PSMA levels, including PC3 (~3K) which represent normal tissues. In fact, in an in vivo study (data not shown), cynomolgus monkeys treated with XENP14484 experienced dose-limiting toxicity (DLT; as indicated by high IL-6 release). Accordingly, activity on the PC3 (~3K) cell line may be a surrogate for DLT, and novel bispecific antibodies of the invention should be designed to avoid activity on PC3 (~3K) cells.

In another experiment, CD107a T cell degranulation was investigated as an indicator of activity by the prototypic bispecific antibodies. PC3 cell lines expressing various PSMA densities (~100K vs ~50K) were incubated with freshly enriched CD3⁺ T cells at an effector to target ratio of 10:1 for 24 hours. Next, the bispecific antibodies were added

to the cells at the indicated concentrations. 18 hours after addition of the bsAbs, cells were assayed via flow cytometry for CD107a degranulation on T cells, data for which are shown in FIG. 46. The data show that each of the prototypic bispecific antibodies induced T cell degranulation with very similar potency in the presence of both cells having higher and lower PSMA densities.

Example 4: Tuning Anti-PSMA×Anti-CD3 bsAbs to Enhance Selectivity and Therapeutic Index

4A: Tuning PSMA Binding Valency and Binding Affinity

In order to encourage avid binding and strong activity on high PSMA expressing cells (e.g. tumors) while minimizing reactivity on low expressing cells (e.g. normal tissues), the anti-PSMA×anti-CD3 bsAbs were tuned for PSMA binding valency and affinity as well as CD3 binding affinity. Towards this, anti-PSMA×anti-CD3 bsAbs were engineered in the 2+1 Fab₂-scFv-Fc format with a range of PSMA binding affinities and reduced CD3 binding affinities and the following illustrative bsAbs were investigated in cell binding and RTCC assays: XENP31855 (having 1 nM K_D PSMA-H₁L1 and CD3 High-Int #1[VLVH]), XENP32218 (having 7 nM PSMA-H₁L1.58 and CD3 High-Int #1[VLVH]), XENP32220 (having 38 nM PSMA-H₁L1.24 and CD3 High-Int #1[VLVH]), and XENP32224 (having 83 nM PSMA-H₁L1.29 and CD3 High-Int #1[VLVH]).

First, the effect of reduced monovalent PSMA binding affinity on cell binding was investigated. PSMA-transfected PC3 (~32K) cancer cells were treated with the indicated concentrations of the indicated test articles. Binding was detected using anti-human Fc mAb, data for which are shown in FIG. 38. The data show that in comparison the monovalent 1+1 Fab-scFv-Fc format, the bivalent 2+1 Fab₂-scFv-Fc format retains cell binding despite reduced monovalent PSMA binding affinity due to avidity.

Next, cell lines expressing various PSMA densities were transduced to constitutively express luciferase. The cells were incubated with freshly enriched CD3⁺ T cells at an effector to target ratio of 1:1 for 24 hours. Next, the bispecific antibodies were added to the cells at the indicated concentrations. 18 hours after addition of the bsAbs, cells were assayed via flow cytometry for CD107a degranulation on T cells (as an indicator of T cell activity). 48 hours after addition of the bsAbs, Bio-Glo Luciferase reagent (Promega, Madison, WI.) was added and plates were read with the Envision Reader on luminescence setting. % RTCC was calculated by 1-(Raw Value/PBS AVG))*100. 72 hours after addition of the bsAbs, cells were assayed via flow cytometry for Ki67 expression on T cells (as an indicator of proliferation). Data showing cell kill are depicted in FIG. 39, data showing T cell proliferation are depicted in FIG. 40, and data showing CD107a degranulation are depicted in FIG. 41. Collectively, the data show that as PSMA binding affinity is reduced, selectivity for higher antigen density cell lines over lower antigen density cell lines is improved. Notably as shown in FIGS. 42 and 39, with low PSMA binding affinity (e.g. 38 nM and 83 nM) it is possible to achieve efficacious killing of cell lines exhibiting IHC Score 2 and Score 1 while inducing little to no killing of cell line exhibiting IHC Score 0.

4B: Further Tuning CD3 Binding Affinity

Another approach explored for optimizing the anti-PSMA×anti-CD3 bispecific antibodies was tuning CD3 binding affinity. According, anti-PSMA×anti-CD3 bsAbs in the 2+1 Fab₂-scFv-Fc format having CD3 High-Int #2[VLVH] binding domain and either 1 nM KD or 7 nM KD PSMA binding domains, respectively XENP31856 and XENP33063, were investigated in assays as generally

described above. Data as depicted in FIGS. 43 and 44 show that reducing the CD3 binding affinity may also convey selective targeting to high PSMA expressing cell lines (even with higher affinity PSMA binding).

5 4C: Tuned Anti-PSMA×Anti-CD3 Bispecific Antibodies are Able to Kill Clinically Relevant PSMA+ Prostate Cancer Cells

To confirm that the tuned anti-PSMA×anti-CD3 bispecific antibodies of the invention are able to kill clinically relevant PSMA⁺ prostate cancer cells, their ability to induce RTCC on LNCaP (human prostate adenocarcinoma cells having IHC score 3 and ~140K PSMA density) and 22Rv1 (human prostate carcinoma cells having IHC score 2 and ~115K PSMA density).

Cancer cells were incubated with freshly enriched CD3⁺ T cells at an effector to target ratio of 10:1 for 24 hours. Next, the bispecific antibodies were added to the cells at the indicated concentrations. 72 hours after addition of the bsAbs, Bio-Glo Luciferase reagent (Promega, Madison, WI.) was added and plates were read with the Envision Reader on luminescence setting. % RTCC was calculated by 1-(Raw Value/PBS AVG))*100. Data depicted in FIG. 45 show that tuned anti-PSMA×anti-CD3 bispecific mAb XENP32220 was able to induce cell kill on both LNCaP and 22Rv1 cancer cells.

Example 5: Tuned PSMA×CD3 Bispecific Antibodies are Active In Vivo

Next, the in vivo anti-tumor effect of the tune anti-PSMA×anti-CD3 bispecific antibodies of the invention was investigated. NOD-SCID gamma (NSG) mice were engrafted intradermally with 1×10⁶ PC3 (~100K) cells in the right flank on Day -16. On Day -1, mice were engrafted intraperitoneally with 5×10⁶ human PBMCs. Mice (n=10) were then treated on Days 0, 8, 15, and 22 with 3 mg/kg XENP32218, XENP32220, or XENP32224. Controls (N=10) used were PBS and 3 mg/kg anti-PD-1 mAb (a checkpoint inhibitor which enhances anti-tumor activity by de-repressing the engrafted human T cells). Tumor volumes were monitored by caliper measurements, data for which are shown in FIG. 47 for Day 19 and change over time in FIG. 48. The data show that each of the tuned PSMA×CD3 bispecific antibodies significantly enhanced (p<0.05 vs. PBS or αPD-1 mAb) anti-tumor activity (as indicated by tumor volume; statistics performed on baseline corrected data using unpaired t-test) despite their reduced PSMA binding affinity.

Example 6: Tuned PSMA×CD3 bsAbs Demonstrate Favorable Tolerability and Pharmacokinetics in Cynomolgus

In a cynomolgus study, each healthy male cynomolgus (n=1) was administered by IV either a 1× dose, 10× dose, or 60× dose of tuned PSMA×CD3 bsAbs XENP32218, XENP32220, or XENP32224, or the analogs of these three molecules that additionally contain the Xtend mutation (M428L/N434S for enhanced serum half-life) in the Fc domain-XENP34626, XENP34627, or XENP34628, respectively. All XENPs were generally well tolerated (i.e., no dose limiting side effects) up to the highest 60× dose (data not shown). As seen in FIG. 49, the variants with the Xtend mutations resulted in improved pharmacokinetics, particularly at the 1× and 10× doses. For example, the terminal serum half-lives of XENP32220 were 1.59, 3.01, or 7.95 days at each relative dose level, while its Xtend analog XENP34627 had serum half-lives of 4.5, 8.6, and 10.8 days across each dose level. Additionally, the lower PSMA affinity XENP32224 had half-lives of 2.53 or 3.34 days at the 1× and 10× doses, while its Xtend analog XENP34628 had half-lives of 5.9 or 9.3 days at the 1× and 10× dose levels.

binding affinity, demonstrated the fastest clearance, and XENP32224, having the lowest PSMA binding affinity, demonstrated the slowest clearance. Notably, the 60× dose was high enough to clear the sink effect so that half lives at that dose were no longer affinity-dependent.

At the lowest 1× dose level, serum clearance was PSMA affinity-dependent. XENP32218, having the highest PSMA

SEQUENCE LISTING

Sequence total quantity: 627		
SEQ ID NO: 1	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
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SEQUENCE: 1		
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SEQ ID NO: 2	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
source	note = Description of Artificial Sequence: Synthetic peptide 1..5 mol_type = protein organism = synthetic construct	
SEQUENCE: 2		
GGGS		5
SEQ ID NO: 3	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
REGION	note = Description of Artificial Sequence: Synthetic peptide 1..5 note = See specification as filed for detailed description of substitutions and preferred embodiments	
source	1..5 mol_type = protein organism = synthetic construct	
SEQUENCE: 3		
GSGGS		5
SEQ ID NO: 4	moltype = AA length = 4	
FEATURE	Location/Qualifiers	
REGION	1..4	
REGION	note = Description of Artificial Sequence: Synthetic peptide 1..4 note = See specification as filed for detailed description of substitutions and preferred embodiments	
source	1..4 mol_type = protein organism = synthetic construct	
SEQUENCE: 4		
GGGS		4
SEQ ID NO: 5	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
source	note = Description of Artificial Sequence: Synthetic peptide 1..15 mol_type = protein organism = synthetic construct	
SEQUENCE: 5		
GGGSGGGGS GGGGS		15
SEQ ID NO: 6	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
REGION	1..18	
source	note = Description of Artificial Sequence: Synthetic peptide 1..18 mol_type = protein organism = synthetic construct	
SEQUENCE: 6		
GSTSGSGKPG SGEGSTKG		18
SEQ ID NO: 7	moltype = AA length = 14	

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FEATURE	Location/Qualifiers	
REGION	1..14	
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source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 7		
IRPRAIGGSK PRVA		14
SEQ ID NO: 8	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 8		
GKGGSGKGGG GKGGG		15
SEQ ID NO: 9	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 9		
GKGGSGGKGGG GKGGG		15
SEQ ID NO: 10	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 10		
GGGKSGGGKS GGGKS		15
SEQ ID NO: 11	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 11		
GKGGSGGKGGG GKGGG		15
SEQ ID NO: 12	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 12		
GGGKSGGKGGG GKGGG		15
SEQ ID NO: 13	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 13		
GKPGSGKPGS GKPGS		15
SEQ ID NO: 14	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 14		
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SEQ ID NO: 15	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
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	note = Description of Artificial Sequence: Synthetic peptide	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 15		
GGGSGGGGS GGGSGGGGS		20
SEQ ID NO: 16	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 16		
STAGDTHLGG EDFD		14
SEQ ID NO: 17	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 17		
GEGSGEGGS GEGGS		15
SEQ ID NO: 18	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 18		
GGEGSGEGGS GEGGS		15
SEQ ID NO: 19	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 19		
GGGESGGGES GGGES		15
SEQ ID NO: 20	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 20		
GEGESGEGES GEGES		15
SEQ ID NO: 21	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 21		
GGGESGGEGGS GEGGS		15
SEQ ID NO: 22	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 22		
GEGESGEGES GEGESGEGES		20

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SEQ ID NO: 23	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 23		
PRGASKSGSA SQTGSAPGS		19
SEQ ID NO: 24	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 24		
GTAAAGAGAA GGAAAGAAG		19
SEQ ID NO: 25	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 25		
GTSGSSGSGS GGSGSGGGG		19
SEQ ID NO: 26	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
REGION	1..10	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 26		
GGGSGGGGS		10
SEQ ID NO: 27	moltype = AA length = 25	
FEATURE	Location/Qualifiers	
REGION	1..25	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..25	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 27		
GGGSGGGGS GGGSGGGGS GGGGS		25
SEQ ID NO: 28	moltype = AA length = 30	
FEATURE	Location/Qualifiers	
REGION	1..30	
	note = Description of Artificial Sequence: Synthetic polypeptide	
source	1..30	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 28		
GGGSGGGGS GGGSGGGGS GGGSGGGGS		30
SEQ ID NO: 29	moltype = AA length = 35	
FEATURE	Location/Qualifiers	
REGION	1..35	
	note = Description of Artificial Sequence: Synthetic polypeptide	
source	1..35	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 29		
GGGSGGGGS GGGSGGGGS GGGSGGGGS GGGGS		35
SEQ ID NO: 30	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..5	
	mol_type = protein	

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SEQUENCE: 30	organism = synthetic construct	
GGGGA		5
SEQ ID NO: 31	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
REGION	1..10	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 31		
GGGGAGGGGA		10
SEQ ID NO: 32	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 32		
GGGGAGGGGA GGGGA		15
SEQ ID NO: 33	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 33		
GGGGAGGGGA GGGGAGGGGA		20
SEQ ID NO: 34	moltype = AA length = 25	
FEATURE	Location/Qualifiers	
REGION	1..25	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..25	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 34		
GGGGAGGGGA GGGGAGGGGA GGGGA		25
SEQ ID NO: 35	moltype = AA length = 30	
FEATURE	Location/Qualifiers	
REGION	1..30	
	note = Description of Artificial Sequence: Synthetic polypeptide	
source	1..30	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 35		
GGGGAGGGGA GGGGAGGGGA GGGGAGGGGA		30
SEQ ID NO: 36	moltype = AA length = 35	
FEATURE	Location/Qualifiers	
REGION	1..35	
	note = Description of Artificial Sequence: Synthetic polypeptide	
source	1..35	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 36		
GGGGAGGGGA GGGGAGGGGA GGGGAGGGGA GGGGA		35
SEQ ID NO: 37	moltype = AA length = 30	
FEATURE	Location/Qualifiers	
REGION	1..30	
	note = Description of Artificial Sequence: Synthetic polypeptide	
source	1..30	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 37		
DPALVHQRPA PPGGGGSGGG GSGGGGSGGG		30
SEQ ID NO: 38	moltype = AA length = 5	
FEATURE	Location/Qualifiers	

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REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 38		
GKPGS		5
SEQ ID NO: 39	moltype = AA length = 25	
FEATURE	Location/Qualifiers	
REGION	1..25	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..25	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 39		
GKPGSGKPGS GKPGSGKPGS GKPGS		25
SEQ ID NO: 40	moltype = AA length = 30	
FEATURE	Location/Qualifiers	
REGION	1..30	
	note = Description of Artificial Sequence: Synthetic polypeptide	
source	1..30	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 40		
GKPGSGKPGS GKPGSGKPGS GKPGSGKPGS		30
SEQ ID NO: 41	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 41		
GGGES		5
SEQ ID NO: 42	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 42		
KTHTCPPCP		9
SEQ ID NO: 43	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 43		
EPKSSDKTHT CPPCP		15
SEQ ID NO: 44	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 44		
GGGSGGGGS KTHTCPPCP		19
SEQ ID NO: 45	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 45		
GKPGSGKPGS KTHTCPPCP		19

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FEATURE Location/Qualifiers
 REGION 1..329
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..329
 note = 1 + 1 Fab-scFv-Fc Backbone 3 Fab-Fc Side

source 1..329
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 51

ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	60
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	DTKVDKKVEP	KSCDKTHTCP	PCPAPPVAGP	120
SVFLFPPKPK	DTLMISRTPE	VTCTVVVDVKH	EDPEVKFNWY	VDGVEVHNAK	TKPREEEYNS	180
TYRVVSVLTV	LHQDWLNGKE	YKCKVSNKAL	PAPIEKTISK	AKGQPREPQV	YTLPPSREEM	240
TKNQVSLTCE	VSGFYPSTIA	VEWESDGQPE	NNYKTTPPVL	DSGGSFFLYS	KLTVDKSRWE	300
QGDVFSCSV	HEALHNHYTQ	KSLSLSPGK				329

SEQ ID NO: 52 moltype = AA length = 231
 FEATURE Location/Qualifiers
 REGION 1..231
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..231
 note = 1 + 1 Fab-scFv-Fc Backbone 3 scFv-Fc Side

source 1..231
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 52

EPKSSDKTHT	CPPCPAPPVA	GPSVFLFPPK	PKDTLMISRT	PEVTCVVVDV	KHEDPEVKFN	60
WYVDGVEVHN	AKTKPREEQY	NSTYRVVSVL	TVLHQDWLNG	KEYKCKVSNK	ALPAPIEKTI	120
SKAKGQPREP	QVYTLPPSRE	EMTKNQVSLT	CLVKGFPYPS	IAVEWESNGQ	PENNYKTTPP	180
VLDSDGSFFL	YSKLTVDKSR	WQQGNVFSCS	VMHEALHNHY	TQKSLSLSPG	K	231

SEQ ID NO: 53 moltype = AA length = 329
 FEATURE Location/Qualifiers
 REGION 1..329
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..329
 note = 1 + 1 Fab-scFv-Fc Backbone 4 Fab-Fc Side

source 1..329
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 53

ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	60
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	DTKVDKKVEP	KSCDKTHTCP	PCPAPPVAGP	120
SVFLFPPKPK	DTLMISRTPE	VTCTVVVDVKH	EDPEVKFNWY	VDGVEVHNAK	TKPREEEYNS	180
TYRVVSVLTV	LHQDWLNGKE	YKCKVSNKAL	PAPIEKTISK	AKGQPREPQV	YTLPPSREEM	240
TENEVSLTCL	VKGFYPSTIA	VEWESDGQPE	NNYKTTPPVL	DSGGSFFLYS	KLEVDKSRWE	300
QGDVFSCSV	HEALHNHYTQ	KSLSLSPGK				329

SEQ ID NO: 54 moltype = AA length = 231
 FEATURE Location/Qualifiers
 REGION 1..231
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..231
 note = 1 + 1 Fab-scFv-Fc Backbone 4 scFv-Fc Side

source 1..231
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 54

EPKSSDKTHT	CPPCPAPPVA	GPSVFLFPPK	PKDTLMISRT	PEVTCVVVDV	KHEDPEVKFN	60
WYVDGVEVHN	AKTKPREEQY	NSTYRVVSVL	TVLHQDWLNG	KEYKCKVSNK	ALPAPIEKTI	120
SKAKGQPREP	QVYTLPPSRE	EMTKNQVSLT	CLVKGFPYPS	IAVEWESNGQ	PENNYKTTPP	180
VLDSDGSFFL	YSKLTVDKSR	WQQGNVFSCS	VMHEALHNHY	TQKSLSLSPG	K	231

SEQ ID NO: 55 moltype = AA length = 329
 FEATURE Location/Qualifiers
 REGION 1..329
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..329
 note = 1 + 1 Fab-scFv-Fc Backbone 5 Fab-Fc Side

source 1..329
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 55

ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	60
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GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 180
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSRDEL 240
 TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE 300
 QGDVFSCSVM HEALHNHYTQ KSLSLSPGK 329

SEQ ID NO: 56 moltype = AA length = 231
 FEATURE Location/Qualifiers
 REGION 1..231
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..231
 note = 1 + 1 Fab-scFv-Fc Backbone 5 scFv-Fc Side
 source 1..231
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 56
 EPKSSDKTHT CPPCPAPPVA GPSVFLFPPK PKDTLMISRT PEVTCVVVDV KHEDPEVKFN 60
 WYVDGVEVHN AKTKPREEQY NSTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI 120
 SKAKGQPREP QVYTLPPSRD QLTKNQVKLT CLVKGIFYPSD IAVEWESNGQ PENNYKTTPP 180
 VLDSGDSFFL YSKLTVDKSR WQQGNVFSCS VMHEALHNHY TQKSLSLSPG K 231

SEQ ID NO: 57 moltype = AA length = 329
 FEATURE Location/Qualifiers
 REGION 1..329
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..329
 note = 1 + 1 Fab-scFv-Fc Backbone 6 Fab-Fc Side
 source 1..329
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 57
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYAS 180
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
 TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE 300
 QGDVFSCSVM HEALHNHYTQ KSLSLSPGK 329

SEQ ID NO: 58 moltype = AA length = 231
 FEATURE Location/Qualifiers
 REGION 1..231
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..231
 note = 1 + 1 Fab-scFv-Fc Backbone 6 scFv-Fc Side
 source 1..231
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 58
 EPKSSDKTHT CPPCPAPPVA GPSVFLFPPK PKDTLMISRT PEVTCVVVDV KHEDPEVKFN 60
 WYVDGVEVHN AKTKPREEQY ASTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI 120
 SKAKGQPREP QVYTLPPSRE QMTKNQVKLT CLVKGIFYPSD IAVEWESNGQ PENNYKTTPP 180
 VLDSGDSFFL YSKLTVDKSR WQQGNVFSCS VMHEALHNHY TQKSLSLSPG K 231

SEQ ID NO: 59 moltype = AA length = 329
 FEATURE Location/Qualifiers
 REGION 1..329
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..329
 note = 1 + 1 Fab-scFv-Fc Backbone 7 Fab-Fc Side
 source 1..329
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 59
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYSS 180
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
 TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE 300
 QGDVFSCSVM HEALHNHYTQ KSLSLSPGK 329

SEQ ID NO: 60 moltype = AA length = 231
 FEATURE Location/Qualifiers
 REGION 1..231
 note = Description of Artificial Sequence: Synthetic
 polypeptide

-continued

REGION 1..231
 note = 1 + 1 Fab-scFv-Fc Backbone 7 scFv-Fc Side
 source 1..231
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 60
 EPKSSDKTHT CPPCPAPPVA GPSVFLFPPK PKDTLMISRT PEVTCVVVDV KHEDPEVKFN 60
 WYVDGVEVHN AKTKPREEQY SSTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI 120
 SKAKGQPREP QVYTLPPSRE QMTKNQVCLT CLVKGFPYSD IAVEWESNGQ PENNYKTTTP 180
 VLDSGGSFPL YSKLTVDKSR WQQGNVFSKS VMHEALHNHY TQKSLSLSPG K 231

SEQ ID NO: 61 moltype = AA length = 327
 FEATURE Location/Qualifiers
 REGION 1..327
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..327
 note = 1 + 1 Fab-scFv-Fc Backbone 8 Fab-Fc Side
 source 1..327
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 61
 ASTKGPSVFP LAPCSRSTSE STAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 60
 GLYSLSVVTV VPSSSLGTKT YTCNVDPKPS DTKVDKRVES KYGPPCPPCP APEFLGGPSV 120
 FLFPKPKKDT LMISRTPEVT CVVVDVSQED PEVQFNWYVD GVEVHNATK PREEFNSTY 180
 RVVSVLTVLH QDWLNGKEYK CKVSNKGLPS SIEKTISKAK GQPREPQVYT LPPSQEEMTK 240
 NQVSLTCDVS GFYPSDIAVE WESDGGQFENN YKTPPVLDSD DGSFFLYSRL TVDKSRWEEG 300
 DVFSCSVMEH ALHNHYTQKS LSLSLGK 327

SEQ ID NO: 62 moltype = AA length = 229
 FEATURE Location/Qualifiers
 REGION 1..229
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..229
 note = 1 + 1 Fab-scFv-Fc Backbone 8 scFv-Fc Side
 source 1..229
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 62
 ESKYGPCCPP CPAPEFLGGP SVFLFPPKPK DTLMISRTPE VTCVVVDVSQ EDPEVQFNWY 60
 VDGVEVHNATK TKPREEQFNS TYRVVSVLTV LHQDWLNGKE YKCKVSNKGL PSSIEKTISK 120
 AKGQPREPQV YTLPPSQEQM TKNQVKLTCL VKGFPYSDIA VEWESNGQPE NNYKTTTPVL 180
 DSDGSFFLYS RLTVDKSRWQ EGNVFSKSVH HEALHNHYTQ KSLSLSLGK 229

SEQ ID NO: 63 moltype = AA length = 326
 FEATURE Location/Qualifiers
 REGION 1..326
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..326
 note = 1 + 1 Fab-scFv-Fc Backbone 9 Fab-Fc Side
 source 1..326
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 63
 ASTKGPSVFP LAPCSRSTSE STAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 60
 GLYSLSVVTV VPSSNFGTQT YTCNVDPKPS DTKVDKTVR KCCVECPPCP APPVAGPSVF 120
 LFPKPKKDTL MISRTPEVTC VVVDVSHEDP EVQFNWYVDG VEVHNATKP REEFNSTFR 180
 VVSVLTVVHQ DWLNGKEYK KVSNGKLPAP IEKTISKTKG QPREPQVYTL PPSREEMTKN 240
 QVSLTCDVSG FYPDSIAVEW ESDGQFENNY KTHPPMLDSD GSFFLYSKLT VDKSRWEQGD 300
 VFSCSVMEHA LHNHYTQKSL SLSPGK 326

SEQ ID NO: 64 moltype = AA length = 228
 FEATURE Location/Qualifiers
 REGION 1..228
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..228
 note = 1 + 1 Fab-scFv-Fc Backbone 9 scFv-Fc Side
 source 1..228
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 64
 ERKCCVECP CPAPPVAGPS VFLFPPKPKD TLMISRTPEV TCVVVDVSHE DPEVQFNWYV 60
 DGVEVHNATK KPREEQFNST FRVSVLTVV HQDWLNGKEY KCKVSNKGLP APIEKTISK 120
 KGQPREPQVY TLPPSREQMT KNQVKLTCLV KGFYPSDIAV EWESNGQFEN NYKTHPPMLD 180
 SDGSFFLYSK LTVDKSRWQ GNVFSCSVMH EALHNHYTQK SLSLSPGK 228

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SEQ ID NO: 65      moltype = AA length = 326
FEATURE           Location/Qualifiers
REGION            1..326
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..326
                  note = 1 + 1 Fab-scFv-Fc Backbone 10 Fab-Fc Side
source            1..326
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 65
ASTKGPSVFP LAPCSRSTSE STAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 60
GLYSLSSVVT VPSSNFGTQT YTCNVDHKPS DTKVDKTVR KCCVECPPCP APPVAGPSVF 120
LFPKPCKDTL MISRTPEVTC VVVDVKHEDP EVQFNWYVDG VEVHNAKTKP REEEFNSTFR 180
VVSVLTVVHQ DWLNGKEYKC KVSNGKGLPAP IEKTISKTKG QPREPQVYTL PPSREEMTKN 240
QVSLTCDVSG FYPSDIAVEW ESDGQPENNY KTTTPMLDSD GSFFLYSKLT VDKSRWEQGD 300
VFSCSVMHEA LHNHYTQKSL SLSPGK 326

SEQ ID NO: 66      moltype = AA length = 228
FEATURE           Location/Qualifiers
REGION            1..228
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..228
                  note = 1 + 1 Fab-scFv-Fc Backbone 10 scFv-Fc Side
source            1..228
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 66
ERKCCVECP CPAPPVAGPS VFLEPPKPKD TLMISRTPEV TCVVVDVKHE DPEVQFNWYV 60
DGVEVHNAKT KPREEQFNST FRVVSVLTVV HQDWLNGKEY KCKVSNKGLP APIEKTISK 120
KGQPREPQVY TLPPSREQMT KNQVKLTCLV KGFYPSDIAV EWESNGQPEN NYKTTTPMLD 180
SDGSFFLYSK LTVDKSRWQQ GNVFSCSVMH EALHNHYTQK SLSPGK 228

SEQ ID NO: 67      moltype = AA length = 329
FEATURE           Location/Qualifiers
REGION            1..329
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..329
                  note = 1 + 1 Fab-scFv-Fc Backbone 11 Fab-Fc Side
source            1..329
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 67
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
SVFLPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 180
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWE 300
QGDVFSCSVL HEALHSHYTQ KSLSLSPGK 329

SEQ ID NO: 68      moltype = AA length = 231
FEATURE           Location/Qualifiers
REGION            1..231
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..231
                  note = 1 + 1 Fab-scFv-Fc Backbone 11 scFv-Fc Side
source            1..231
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 68
EPKSSDKTHT CPPCPAPPVA GPSVFLPPPK PKDTLMISRT PEVTCVVVDV KHEDPEVKFN 60
WYVDGVEVHN AKTKPREEQY NSTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI 120
SKAKGQPREP QVYTLPPSRE QMTKNQVCLT CLVKGFPYPS IAVEWESNGQ PENNYKTTTP 180
VLDSGGSFFL YSKLTVDKSR WQQGNVFSCS VLHEALHSHY TQKSLSLSPG K 231

SEQ ID NO: 69      moltype = AA length = 329
FEATURE           Location/Qualifiers
REGION            1..329
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..329
                  note = 1 + 1 Fab-scFv-Fc Backbone 12 Fab-Fc Side
source            1..329
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 69

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ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 180
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
TKNQVSLTCD VAGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ 300
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 329

SEQ ID NO: 70      moltype = AA length = 231
FEATURE           Location/Qualifiers
REGION            1..231
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..231
                  note = 1 + 1 Fab-scFv-Fc Backbone 12 scFv-Fc Side
source            1..231
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 70
ERKSSDKTHT CPPRPAPPVA GPSVFLFPPK PKDTLMISRT PEVTCVVVDV KHEDPEVKFK 60
WYVDGVEVHN AKTKPREEQY NSTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI 120
SKAKGQPREP QVYTLPPSRE QMTKNQVQLT CLVKGFPYPS IAVEWESNGQ PENNYKTTPP 180
VLDSGGSFFL YSKLTVDKSR WQQGNVFSCS VMHEALHNHY TQKSLSLSPG K 231

SEQ ID NO: 71      moltype = AA length = 329
FEATURE           Location/Qualifiers
REGION            1..329
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..329
                  note = 2 + 1 Fab2-scFv-Fc Backbone 1 Fab-Fc Side
source            1..329
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 71
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 180
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE 300
QGDVFSCSVM HEALHNHYTQ KSLSLSPGK 329

SEQ ID NO: 72      moltype = AA length = 216
FEATURE           Location/Qualifiers
REGION            1..216
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..216
                  note = 2 + 1 Fab2-scFv-Fc Backbone 1 Fab-scFv-Fc Side
source            1..216
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 72
APPVAGPSVF LFPPKPKDTL MISRTPEVTC VVVDVKHEDP EVKFNWYVDG VEVHNAKTKP 60
REEQYNSTYR VVSVLTVLHQ DWLNGKEYKC KVSNKALPAP IEKTISKAKG QPREPQVYTL 120
PPSRREQMTKN QVKLTCLVKG FYPSDIAVEW ESNQOPENNY KTTTPVLDSG GSFFLYSKLT 180
VDKSRWQQGN VFSCSVMHEA LHNHYTQKSL SLSPGK 216

SEQ ID NO: 73      moltype = AA length = 329
FEATURE           Location/Qualifiers
REGION            1..329
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..329
                  note = 2 + 1 Fab2-scFv-Fc Backbone 2 Fab-Fc Side
source            1..329
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 73
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 180
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE 300
QGDVFSCSVM HEALHNHYTQ KSLSLSPGK 329

SEQ ID NO: 74      moltype = AA length = 216
FEATURE           Location/Qualifiers
REGION            1..216
                  note = Description of Artificial Sequence: Synthetic

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REGION          polypeptide
                  1..216
note = 2 + 1 Fab2-scFv-Fc Backbone 2 Fab-scFv-Fc Side
source          1..216
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 74
APPVAGPSVF LFPPKPKDTL MISRTPEVTC VVVDVKHEDP EVKFNWYVDG VEVHNAKTKP 60
REEQYNSTYR VVSVLTVLHQ DWLNGKEYKC KVSNKALPAP IEKTISKAKG QPREPQVYTL 120
PPSREEMTKN QVKLTCLVKG FYPSDIAVEW ESNQOPENNY KTTTPVLDSG GSFFLYSKLT 180
VDKSRWQQGN VFSCSVMEHA LHNHYTQKSL SLSPGK 216

SEQ ID NO: 75    moltype = AA length = 329
FEATURE          Location/Qualifiers
REGION          1..329
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION          1..329
                  note = 2 + 1 Fab2-scFv-Fc Backbone 3 Fab-Fc Side
source          1..329
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 75
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 180
TYRVSVSLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
TKNQVSLTCE VSGFYPSDIA VEWESDGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWE 300
QGDVFSQSVH HEALHNHYTQ KSLSLSPGK 329

SEQ ID NO: 76    moltype = AA length = 216
FEATURE          Location/Qualifiers
REGION          1..216
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION          1..216
                  note = 2 + 1 Fab2-scFv-Fc Backbone 3 Fab-scFv-Fc Side
source          1..216
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 76
APPVAGPSVF LFPPKPKDTL MISRTPEVTC VVVDVKHEDP EVKFNWYVDG VEVHNAKTKP 60
REEQYNSTYR VVSVLTVLHQ DWLNGKEYKC KVSNKALPAP IEKTISKAKG QPREPQVYTL 120
PPSREEMTKN QVKLTCLVKG FYPSDIAVEW ESNQOPENNY KTTTPVLDSG GSFFLYSKLT 180
VDKSRWQQGN VFSCSVMEHA LHNHYTQKSL SLSPGK 216

SEQ ID NO: 77    moltype = AA length = 329
FEATURE          Location/Qualifiers
REGION          1..329
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION          1..329
                  note = 2 + 1 Fab2-scFv-Fc Backbone 4 Fab-Fc Side
source          1..329
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 77
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 180
TYRVSVSLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
TENEVSLTCL VKGFYPSDIA VEWESDGQPE NNYKTTTPVL DSDGSFFLYS KLEVDSRWE 300
QGDVFSQSVH HEALHNHYTQ KSLSLSPGK 329

SEQ ID NO: 78    moltype = AA length = 216
FEATURE          Location/Qualifiers
REGION          1..216
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION          1..216
                  note = 2 + 1 Fab2-scFv-Fc Backbone 4 Fab-scFv-Fc Side
source          1..216
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 78
APPVAGPSVF LFPPKPKDTL MISRTPEVTC VVVDVKHEDP EVKFNWYVDG VEVHNAKTKP 60
REEQYNSTYR VVSVLTVLHQ DWLNGKEYKC KVSNKALPAP IEKTISKAKG QPREPQVYTL 120
PPSREEMTKN QVSLTCLVKG FYPSDIAVEW ESNQOPENNY KTTTPVLDSK GSFFLYSKLT 180
VDKSRWQQGN VFSCSVMEHA LHNHYTQKSL SLSPGK 216

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SEQ ID NO: 79	moltype = AA length = 329	
FEATURE	Location/Qualifiers	
REGION	1..329	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..329	
	note = 2 + 1 Fab2-scFv-Fc Backbone 5 Fab-Fc Side	
source	1..329	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 79		
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK DYPFPEPTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
SVFLFPPKPK	DTLMISRTP	VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 180
TYRVVSVLTV	LHQDWLNGKE	YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSRDEL 240
TKNQVSLTCD	VSGFYPSDIA	VEWESDQQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWE 300
QGDVFSCSV	HEALHNHYTQ	KSLSLSPGK 329
SEQ ID NO: 80	moltype = AA length = 216	
FEATURE	Location/Qualifiers	
REGION	1..216	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..216	
	note = 2 + 1 Fab2-scFv-Fc Backbone 5 Fab-scFv-Fc Side	
source	1..216	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 80		
APPVAGPSVF	LFPPPKPDKTL	MISRTPEVTC VVVDVKHEDP EVKFNWYVDG VEVHNAKTKP 60
REEQYNSTYR	VVSVLTVLHQ	DWLNGKEYKC KVSNKALPAP IEKTISKAKG QPREPQVYTL 120
PPSRDQLTKN	QVKLTCLVKG	FYPSDIAVEW ESNQGPENNY KTTTPVLDSG GSFFLYSKLT 180
VDKSRWQQGN	VFSCSVMHEA	LHNHYTQKSL SLSPGK 216
SEQ ID NO: 81	moltype = AA length = 329	
FEATURE	Location/Qualifiers	
REGION	1..329	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..329	
	note = 2 + 1 Fab2-scFv-Fc Backbone 6 Fab-Fc Side	
source	1..329	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 81		
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK DYPFPEPTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
SVFLFPPKPK	DTLMISRTP	VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYAS 180
TYRVVSVLTV	LHQDWLNGKE	YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
TKNQVSLTCD	VSGFYPSDIA	VEWESDQQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWE 300
QGDVFSCSV	HEALHNHYTQ	KSLSLSPGK 329
SEQ ID NO: 82	moltype = AA length = 216	
FEATURE	Location/Qualifiers	
REGION	1..216	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..216	
	note = 2 + 1 Fab2-scFv-Fc Backbone 6 Fab-scFv-Fc Side	
source	1..216	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 82		
APPVAGPSVF	LFPPPKPDKTL	MISRTPEVTC VVVDVKHEDP EVKFNWYVDG VEVHNAKTKP 60
REEQYASTYR	VVSVLTVLHQ	DWLNGKEYKC KVSNKALPAP IEKTISKAKG QPREPQVYTL 120
PPSRQMTKN	QVKLTCLVKG	FYPSDIAVEW ESNQGPENNY KTTTPVLDSG GSFFLYSKLT 180
VDKSRWQQGN	VFSCSVMHEA	LHNHYTQKSL SLSPGK 216
SEQ ID NO: 83	moltype = AA length = 329	
FEATURE	Location/Qualifiers	
REGION	1..329	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..329	
	note = 2 + 1 Fab2-scFv-Fc Backbone 7 Fab-Fc Side	
source	1..329	
	mol_type = protein	
	organism = synthetic construct	

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SEQUENCE: 83

ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	60
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	DTKVDKKVEP	KSCDKTHTCP	PCPAPPVAGP	120
SVFLFPPKPK	DTLMISRTPE	VTCVVVDVKH	EDPEVKFNWY	VDGVEVHNAK	TKPREEEYSS	180
TYRVVSVLTV	LHQDWLNGKE	YKCKVSNKAL	PAPIEKTISK	AKGQPREPQV	YTLPPSREEM	240
TKNQVSLTCD	VSGFYPSDIA	VEWESDGQPE	NNYKTTPPVL	DSGGSFFLYS	KLTVDKSRWE	300
QGDVFSCSV	HEALHNYHTQ	KSLSLSPGK				329

SEQ ID NO: 84 moltype = AA length = 216

FEATURE Location/Qualifiers

REGION 1..216

 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..216

 note = 2 + 1 Fab2-scFv-Fc Backbone 7 Fab-scFv-Fc Side

source 1..216

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 84

APPVAGPSVF	LFPPKPKDTL	MISRTPEVTC	VVVDVKHEDP	EVKFNWYVDG	VEVHNAKTKP	60
REEQYSSTYR	VVSVLTVLHQ	DWLNGKEYKC	KVSNKALPAP	IEKTISKAKG	QPREPQVYTL	120
PPSRQMTKN	QVKLTCLVKG	FYPSDIAVEW	ESNGQPENNY	KTPPVLDSD	GSFFLYSKLT	180
VDKSRWQQGN	VFSCSVMHEA	LHNHYTQKSL	SLSPGK			216

SEQ ID NO: 85 moltype = AA length = 329

FEATURE Location/Qualifiers

REGION 1..329

 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..329

 note = 2 + 1 Fab2-scFv-Fc Backbone 8 Fab-Fc Side

source 1..329

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 85

ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	60
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	DTKVDKKVEP	KSCDKTHTCP	PCPAPPVAGP	120
SVFLFPPKPK	DTLMISRTPE	VTCVVVDVKH	EDPEVKFNWY	VDGVEVHNAK	TKPREEEYNS	180
TYRVVSVLTV	LHQDWLNGKE	YKCKVSNKAL	PAPIEKTISK	AKGQPREPQV	YTLPPSREEM	240
TKNQVSLTCD	VSGFYPSDIA	VEWESDGQPE	NNYKTTPPVL	DSGGSFFLYS	KLTVDKSRWE	300
QGDVFSCSVL	HEALHSHYTQ	KSLSLSPGK				329

SEQ ID NO: 86 moltype = AA length = 216

FEATURE Location/Qualifiers

REGION 1..216

 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..216

 note = 2 + 1 Fab2-scFv-Fc Backbone 8 Fab-scFv-Fc Side

source 1..216

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 86

APPVAGPSVF	LFPPKPKDTL	MISRTPEVTC	VVVDVKHEDP	EVKFNWYVDG	VEVHNAKTKP	60
REEQYNSTYR	VVSVLTVLHQ	DWLNGKEYKC	KVSNKALPAP	IEKTISKAKG	QPREPQVYTL	120
PPSRQMTKN	QVKLTCLVKG	FYPSDIAVEW	ESNGQPENNY	KTPPVLDSD	GSFFLYSKLT	180
VDKSRWQQGN	VFSCSVLHEA	LHSHYTQKSL	SLSPGK			216

SEQ ID NO: 87 moltype = AA length = 329

FEATURE Location/Qualifiers

REGION 1..329

 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..329

 note = 2 + 1 Fab2-scFv-Fc Backbone 9 Fab-Fc Side

source 1..329

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 87

ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	60
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	NTKVDKKVEP	KSCDKTHTCP	PCPAPPVAGP	120
SVFLFPPKPK	DTLMISRTPE	VTCVVVDVKH	EDPEVKFNWY	VDGVEVHNAK	TKPREEQYNS	180
TYRVVSVLTV	LHQDWLNGKE	YKCKVSNKAL	PAPIEKTISK	AKGQPREPQV	YTLPPSREEM	240
TKNQVSLTCD	VAGFYPSDIA	VEWESNGQPE	NNYKTTPPVL	DSGGSFFLYS	KLTVDKSRWQ	300
QGNVFSCSV	HEALHNYHTQ	KSLSLSPGK				329

SEQ ID NO: 88 moltype = AA length = 216

FEATURE Location/Qualifiers

REGION 1..216

-continued

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note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION      1..216
note = 2 + 1 Fab2-scFv-Fc Backbone 9 Fab-scFv-Fc Side
source      1..216
mol_type = protein
organism = synthetic construct

SEQUENCE: 88
APPVAGPSVF LFPPKPKDTL MISRTPEVTC VVVDVKHEDP EVKFKWYVDG VEVHNAKTKP   60
REEQYNSTYR VVSVLTVLHQ DWLNGKEYKC KVSNKALPAP IEKTISKAKG QPREPQVYTL   120
PPSREQMTKN QVKLTCLVKG FYPSDIAVEW ESNGQPENNY KTTTPVLDSG GSFFLYSKLT   180
VDKSRWQQGN VFSCSVMHEA LHNHYTQKSL SLSPGK                               216

SEQ ID NO: 89      moltype = AA length = 107
FEATURE           Location/Qualifiers
REGION           1..107
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION           1..107
note = Constant Light Domain - Kappa
source           1..107
mol_type = protein
organism = synthetic construct

SEQUENCE: 89
RTVAAPSVFI FPPSDEQLKS GTASVVCLLN NFYPREAKVQ WKVDNALQSG NSQESVTEQD   60
SKDSTYSLSS TLTLSKADYE KHKVYACEVT HQGLSSPVTK SFNRGEC                     107

SEQ ID NO: 90      moltype = AA length = 106
FEATURE           Location/Qualifiers
REGION           1..106
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION           1..106
note = Constant Light Domain - Lambda
source           1..106
mol_type = protein
organism = synthetic construct

SEQUENCE: 90
GQPKAAPSVT LFPPSSEELQ ANKATLVCLI SDFYPGAVTV AWKADSSPVK AGVETTPPSK   60
QSNKYAASS YLSLTPEQWK SHRSYSCQVT HEGSTVEKTV APTECS                     106

SEQ ID NO: 91      moltype = AA length = 254
FEATURE           Location/Qualifiers
REGION           1..254
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION           1..254
note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv scFv
source           1..254
mol_type = protein
organism = synthetic construct

SEQUENCE: 91
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT   60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGTL   120
VTYSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSLTVSPG GTVTLTCGSS TGAVTTSNYA   180
NHWVQKPKGS PRGLIGGTNK RAPGVPARFS GSSLGGKAAL TISGAQPEDE ADYYCALWYS   240
NHWVFGGGTK LTVL                               254

SEQ ID NO: 92      moltype = AA length = 125
FEATURE           Location/Qualifiers
REGION           1..125
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION           1..125
note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv Variable
      Heavy (vh) Domain
source           1..125
mol_type = protein
organism = synthetic construct

SEQUENCE: 92
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT   60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGTL   120
VTYSS                               125

SEQ ID NO: 93      moltype = AA length = 5
FEATURE           Location/Qualifiers
REGION           1..5
note = Description of Artificial Sequence: Synthetic peptide
REGION           1..5

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source	note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv vHCDR1 1..5 mol_type = protein organism = synthetic construct	
SEQUENCE: 93 TYAMN		5
SEQ ID NO: 94 FEATURE REGION	moltype = AA length = 19 Location/Qualifiers 1..19 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..19 note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv vHCDR2	
source	1..19 mol_type = protein organism = synthetic construct	
SEQUENCE: 94 RIRSKYNNYA TYYADSVKG		19
SEQ ID NO: 95 FEATURE REGION	moltype = AA length = 14 Location/Qualifiers 1..14 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..14 note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv vHCDR3	
source	1..14 mol_type = protein organism = synthetic construct	
SEQUENCE: 95 HGNFGDSYVS WFAY		14
SEQ ID NO: 96 FEATURE REGION	moltype = AA length = 109 Location/Qualifiers 1..109 note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..109 note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv Variable Light (v1) Domain	
source	1..109 mol_type = protein organism = synthetic construct	
SEQUENCE: 96 QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVL		60 109
SEQ ID NO: 97 FEATURE REGION	moltype = AA length = 14 Location/Qualifiers 1..14 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..14 note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv vLCDR1	
source	1..14 mol_type = protein organism = synthetic construct	
SEQUENCE: 97 GSSTGAVTTS NYAN		14
SEQ ID NO: 98 FEATURE REGION	moltype = AA length = 7 Location/Qualifiers 1..7 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7 note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv vLCDR2	
source	1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 98 GTNKRAP		7
SEQ ID NO: 99 FEATURE REGION	moltype = AA length = 9 Location/Qualifiers 1..9 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9 note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv vLCDR3	
source	1..9 mol_type = protein organism = synthetic construct	

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SEQUENCE: 99
ALWYSNHWV                                     9

SEQ ID NO: 100      moltype = AA  length = 20
FEATURE             Location/Qualifiers
REGION              1..20
                    note = Description of Artificial Sequence: Synthetic peptide
REGION              1..20
                    note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv Linker
source              1..20
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 100
GKPGSGKPGS GKPGSGKPGS                         20

SEQ ID NO: 101      moltype = AA  length = 254
FEATURE             Location/Qualifiers
REGION              1..254
                    note = Description of Artificial Sequence: Synthetic
                        polypeptide
REGION              1..254
                    note = CD3 High-Int #1 - [anti-CD3]_H1.32_L1.47_scFv scFv
source              1..254
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 101
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKANNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGTL 120
VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSLTVSPG GTVTLTCGSS TGAVTTSNYA 180
NHWQQKPGKS PRGLIGGTNK RAPGVPARFS GSSLGGKAAL TISGAQPEDE ADYYCALWYS 240
NHWVFGGGTK LTVL                                     254

SEQ ID NO: 102      moltype = AA  length = 125
FEATURE             Location/Qualifiers
REGION              1..125
                    note = Description of Artificial Sequence: Synthetic
                        polypeptide
REGION              1..125
                    note = CD3 High-Int #1 - [anti-CD3]_H1.32_L1.47_scFv
                        Variable Heavy (vh) Domain
source              1..125
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 102
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKANNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGTL 120
VTVSS                                     125

SEQ ID NO: 103      moltype = AA  length = 5
FEATURE             Location/Qualifiers
REGION              1..5
                    note = Description of Artificial Sequence: Synthetic peptide
REGION              1..5
                    note = CD3 High-Int #1 - [anti-CD3]_H1.32_L1.47_scFv vhCDR1
source              1..5
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 103
TYAMN                                     5

SEQ ID NO: 104      moltype = AA  length = 19
FEATURE             Location/Qualifiers
REGION              1..19
                    note = Description of Artificial Sequence: Synthetic peptide
REGION              1..19
                    note = CD3 High-Int #1 - [anti-CD3]_H1.32_L1.47_scFv vhCDR2
source              1..19
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 104
RIRSKANNYA TYYADSVKG                         19

SEQ ID NO: 105      moltype = AA  length = 14
FEATURE             Location/Qualifiers
REGION              1..14
                    note = Description of Artificial Sequence: Synthetic peptide
REGION              1..14
                    note = CD3 High-Int #1 - [anti-CD3]_H1.32_L1.47_scFv vhCDR3
source              1..14

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mol_type = protein
organism = synthetic construct

SEQUENCE: 105
HGNFGDSYVS WPAY 14

SEQ ID NO: 106
FEATURE
REGION
REGION
source
SEQUENCE: 106
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV 60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVL 109

SEQ ID NO: 107
FEATURE
REGION
REGION
source
SEQUENCE: 107
GSSTGAVTTS NYAN 14

SEQ ID NO: 108
FEATURE
REGION
REGION
source
SEQUENCE: 108
GTNKRAP 7

SEQ ID NO: 109
FEATURE
REGION
REGION
source
SEQUENCE: 109
ALWYSNHWV 9

SEQ ID NO: 110
FEATURE
REGION
REGION
source
SEQUENCE: 110
GKPGSGKPGS GKPGSGKPGS 20

SEQ ID NO: 111
FEATURE
REGION
REGION
source
SEQUENCE: 111

```

mol_type = protein
organism = synthetic construct

moltype = AA length = 109
Location/Qualifiers
1..109
note = Description of Artificial Sequence: Synthetic polypeptide
1..109
note = CD3 High-Int #1 - [anti-CD3]_H1.32_L1.47_scFv Variable Light (v1) Domain
1..109
mol_type = protein
organism = synthetic construct

moltype = AA length = 14
Location/Qualifiers
1..14
note = Description of Artificial Sequence: Synthetic peptide
1..14
note = CD3 High-Int #1 - [anti-CD3]_H1.32_L1.47_scFv v1CDR1
1..14
mol_type = protein
organism = synthetic construct

moltype = AA length = 7
Location/Qualifiers
1..7
note = Description of Artificial Sequence: Synthetic peptide
1..7
note = CD3 High-Int #1 - [anti-CD3]_H1.32_L1.47_scFv v1CDR2
1..7
mol_type = protein
organism = synthetic construct

moltype = AA length = 9
Location/Qualifiers
1..9
note = Description of Artificial Sequence: Synthetic peptide
1..9
note = CD3 High-Int #1 - [anti-CD3]_H1.32_L1.47_scFv v1CDR3
1..9
mol_type = protein
organism = synthetic construct

moltype = AA length = 20
Location/Qualifiers
1..20
note = Description of Artificial Sequence: Synthetic peptide
1..20
note = CD3 High-Int #1 - [anti-CD3]_H1.32_L1.47_scFv Linker
1..20
mol_type = protein
organism = synthetic construct

moltype = AA length = 254
Location/Qualifiers
1..254
note = Description of Artificial Sequence: Synthetic polypeptide
1..254
note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv scFv
1..254
mol_type = protein
organism = synthetic construct

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EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWVRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDEYVS WFAYWGQGTL 120
VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSLTVSPG GTVTLTCGSS TGAVTTSNYA 180
NWVQQKPGKS PRGLIGGTNK RAPGVPARFS GSSLGGAAL TISGAQPEDE ADYYCALWYS 240
NHWVFGGGTK LTVL 254

SEQ ID NO: 112      moltype = AA  length = 125
FEATURE            Location/Qualifiers
REGION            1..125
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..125
                  note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv
                  Variable Heavy (vh) Domain
source            1..125
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 112
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWVRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDEYVS WFAYWGQGTL 120
VTVSS 125

SEQ ID NO: 113      moltype = AA  length = 5
FEATURE            Location/Qualifiers
REGION            1..5
                  note = Description of Artificial Sequence: Synthetic peptide
REGION            1..5
                  note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv vhCDR1
source            1..5
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 113
TYAMN 5

SEQ ID NO: 114      moltype = AA  length = 19
FEATURE            Location/Qualifiers
REGION            1..19
                  note = Description of Artificial Sequence: Synthetic peptide
REGION            1..19
                  note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv vhCDR2
source            1..19
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 114
RIRSKYNNYA TTYADSVKG 19

SEQ ID NO: 115      moltype = AA  length = 14
FEATURE            Location/Qualifiers
REGION            1..14
                  note = Description of Artificial Sequence: Synthetic peptide
REGION            1..14
                  note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv vhCDR3
source            1..14
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 115
HGNFGDEYVS WFAY 14

SEQ ID NO: 116      moltype = AA  length = 109
FEATURE            Location/Qualifiers
REGION            1..109
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..109
                  note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv
                  Variable Light (vl) Domain
source            1..109
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 116
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV 60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVL 109

SEQ ID NO: 117      moltype = AA  length = 14
FEATURE            Location/Qualifiers
REGION            1..14
                  note = Description of Artificial Sequence: Synthetic peptide
REGION            1..14
                  note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv v1CDR1

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source                1..14
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 117
GSSTGAVTTS NYAN                                14

SEQ ID NO: 118      moltype = AA length = 7
FEATURE             Location/Qualifiers
REGION              1..7
                      note = Description of Artificial Sequence: Synthetic peptide
REGION              1..7
                      note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv v1CDR2
source              1..7
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 118
GTNKRAP                                              7

SEQ ID NO: 119      moltype = AA length = 9
FEATURE             Location/Qualifiers
REGION              1..9
                      note = Description of Artificial Sequence: Synthetic peptide
REGION              1..9
                      note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv v1CDR3
source              1..9
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 119
ALWYSNHWV                                           9

SEQ ID NO: 120      moltype = AA length = 20
FEATURE             Location/Qualifiers
REGION              1..20
                      note = Description of Artificial Sequence: Synthetic peptide
REGION              1..20
                      note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv Linker
source              1..20
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 120
GKPGSGKPGS GKPGSGKPGS                             20

SEQ ID NO: 121      moltype = AA length = 254
FEATURE             Location/Qualifiers
REGION              1..254
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..254
                      note = CD3 High-Int - [anti-CD3]_H1.90_L1.47_scFv scFv
source              1..254
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 121
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDPYVS WFAYWGQGTL 120
VTSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSLTVSPG GTVTLTCGSS TGAVTTSNYA 180
NWVQKPGKS PRGLIGGTNK RAPGVPARFS GSSLGGKAAL TISGAQPEDE ADYICALWYS 240
NHWVFGGGTK LTVL                                     254

SEQ ID NO: 122      moltype = AA length = 125
FEATURE             Location/Qualifiers
REGION              1..125
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..125
                      note = CD3 High-Int - [anti-CD3]_H1.90_L1.47_scFv Variable
                      Heavy (vh) Domain
source              1..125
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 122
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDPYVS WFAYWGQGTL 120
VTSS                                             125

SEQ ID NO: 123      moltype = AA length = 5
FEATURE             Location/Qualifiers
REGION              1..5
                      note = Description of Artificial Sequence: Synthetic peptide

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REGION	1..5	
	note = CD3 High-Int - [anti-CD3]_H1.90_L1.47_scFv vhCDR1	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 123		
TYAMN		5
SEQ ID NO: 124	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..19	
	note = CD3 High-Int - [anti-CD3]_H1.90_L1.47_scFv vhCDR2	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 124		
RIRSKYNNYA TYYADSVKG		19
SEQ ID NO: 125	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..14	
	note = CD3 High-Int - [anti-CD3]_H1.90_L1.47_scFv vhCDR3	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 125		
HGNFGDPYVS WPAY		14
SEQ ID NO: 126	moltype = AA length = 109	
FEATURE	Location/Qualifiers	
REGION	1..109	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..109	
	note = CD3 High-Int - [anti-CD3]_H1.90_L1.47_scFv Variable Light (v1) Domain	
source	1..109	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 126		
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTKRAPGV		60
PARFSGSLLG GKAAALTISGA QPEDEADYYC ALWYSNHWFV GGGTKLTVL		109
SEQ ID NO: 127	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..14	
	note = CD3 High-Int - [anti-CD3]_H1.90_L1.47_scFv v1CDR1	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 127		
GSSTGAVTTS NYAN		14
SEQ ID NO: 128	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7	
	note = CD3 High-Int - [anti-CD3]_H1.90_L1.47_scFv v1CDR2	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 128		
GTNKRAP		7
SEQ ID NO: 129	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9	
	note = CD3 High-Int - [anti-CD3]_H1.90_L1.47_scFv v1CDR3	
source	1..9	
	mol_type = protein	

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organism = synthetic construct

SEQUENCE: 129
ALWYSNHWV                                     9

SEQ ID NO: 130      moltype = AA  length = 20
FEATURE            Location/Qualifiers
REGION            1..20
                  note = Description of Artificial Sequence: Synthetic peptide
REGION            1..20
                  note = CD3 High-Int - [anti-CD3]_H1.90_L1.47_scFv Linker
source            1..20
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 130
GKPGSGKPGS GKPGSGKPGS                       20

SEQ ID NO: 131      moltype = AA  length = 254
FEATURE            Location/Qualifiers
REGION            1..254
                  note = Description of Artificial Sequence: Synthetic
                        polypeptide
REGION            1..254
                  note = Anti-CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv
                        scFv
source            1..254
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 131
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT   60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFDYWGQGTL  120
VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSLTVSPG GTVTLTCGSS TGAVTTSNYA  180
NWVQQKPGKS PRGLIGGTNK RAPGVPARFS GSLGGKAAL TISGAQPEDE ADYYCALWYS   240
NHWVFGGGTK LTVL                                     254

SEQ ID NO: 132      moltype = AA  length = 125
FEATURE            Location/Qualifiers
REGION            1..125
                  note = Description of Artificial Sequence: Synthetic
                        polypeptide
REGION            1..125
                  note = Anti-CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv
                        Variable Heavy (vh) Domain
source            1..125
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 132
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT   60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFDYWGQGTL  120
VTVSS                                     125

SEQ ID NO: 133      moltype = AA  length = 5
FEATURE            Location/Qualifiers
REGION            1..5
                  note = Description of Artificial Sequence: Synthetic peptide
REGION            1..5
                  note = Anti-CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv
                        vHCDR1
source            1..5
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 133
TYAMN                                     5

SEQ ID NO: 134      moltype = AA  length = 19
FEATURE            Location/Qualifiers
REGION            1..19
                  note = Description of Artificial Sequence: Synthetic peptide
REGION            1..19
                  note = Anti-CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv
                        vHCDR2
source            1..19
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 134
RIRSKYNNYA TYYADSVKG                       19

SEQ ID NO: 135      moltype = AA  length = 14
FEATURE            Location/Qualifiers
REGION            1..14

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REGION	note = Description of Artificial Sequence: Synthetic peptide 1..14	
	note = Anti-CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv vhCDR3	
source	1..14 mol_type = protein organism = synthetic construct	
SEQUENCE: 135		
HGNFGDSYVS WFDY		14
SEQ ID NO: 136	moltype = AA length = 109	
FEATURE	Location/Qualifiers	
REGION	1..109	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..109	
	note = Anti-CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv Variable Light (vl) Domain	
source	1..109 mol_type = protein organism = synthetic construct	
SEQUENCE: 136		
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV		60
PARFSGSLLG GKAALTISGA QPDEADYYC ALWYSNHWVF GGGTKLTVL		109
SEQ ID NO: 137	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..14	
	note = Anti-CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv vLCDR1	
source	1..14 mol_type = protein organism = synthetic construct	
SEQUENCE: 137		
GSSTGAVTTS NYAN		14
SEQ ID NO: 138	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7	
	note = Anti-CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv vLCDR2	
source	1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 138		
GTNKRAP		7
SEQ ID NO: 139	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9	
	note = Anti-CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv vLCDR3	
source	1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 139		
ALWYSNHWV		9
SEQ ID NO: 140	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..20	
	note = Anti-CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv Linker	
source	1..20 mol_type = protein organism = synthetic construct	
SEQUENCE: 140		
GKPGSGKPGS GKPGSGKPGS		20
SEQ ID NO: 141	moltype = AA length = 254	
FEATURE	Location/Qualifiers	

-continued

REGION 1..254
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..254
 note = CD3 High-Int - [anti-CD3]_H1.31_L1.47_scFv scFv

source 1..254
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 141
 EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMSWVRQA PGKGLEWVGR IRSKYNNYAT 60
 YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAIWGQGTL 120
 VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSLTVSPG GTVTLTCGSS TGAVTTSNYA 180
 NWWQQKPGKS PRGLIGGTNK RAPGVPARFS GSLLGKKAAL TISGAQPEDE ADYYCALWYS 240
 NHHVFGGGTK LTVL 254

SEQ ID NO: 142 moltype = AA length = 125
 FEATURE Location/Qualifiers
 REGION 1..125
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..125
 note = CD3 High-Int - [anti-CD3]_H1.31_L1.47_scFv Variable Heavy (vh) Domain

source 1..125
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 142
 EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMSWVRQA PGKGLEWVGR IRSKYNNYAT 60
 YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAIWGQGTL 120
 VTVSS 125

SEQ ID NO: 143 moltype = AA length = 5
 FEATURE Location/Qualifiers
 REGION 1..5
 note = Description of Artificial Sequence: Synthetic peptide

REGION 1..5
 note = CD3 High-Int - [anti-CD3]_H1.31_L1.47_scFv vHCDR1

source 1..5
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 143
 TYAMS 5

SEQ ID NO: 144 moltype = AA length = 19
 FEATURE Location/Qualifiers
 REGION 1..19
 note = Description of Artificial Sequence: Synthetic peptide

REGION 1..19
 note = CD3 High-Int - [anti-CD3]_H1.31_L1.47_scFv vHCDR2

source 1..19
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 144
 RIRSKYNNYA TYYADSVKG 19

SEQ ID NO: 145 moltype = AA length = 14
 FEATURE Location/Qualifiers
 REGION 1..14
 note = Description of Artificial Sequence: Synthetic peptide

REGION 1..14
 note = CD3 High-Int - [anti-CD3]_H1.31_L1.47_scFv vHCDR3

source 1..14
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 145
 HGNFGDSYVS WFAI 14

SEQ ID NO: 146 moltype = AA length = 109
 FEATURE Location/Qualifiers
 REGION 1..109
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..109
 note = CD3 High-Int - [anti-CD3]_H1.31_L1.47_scFv Variable Light (vl) Domain

source 1..109
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 146

QAVVTQEPSTL TVSPGGTCTL	TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV	60
PARFSGSLLG GKAALTISGA	QPEDEADYYC ALWYSNHWVF GGGTKLTVL	109
SEQ ID NO: 147	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..14	
	note = CD3 High-Int - [anti-CD3]_H1.31_L1.47_scFv v1CDR1	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 147		
GSSTGAVTTS NYAN		14
SEQ ID NO: 148	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7	
	note = CD3 High-Int - [anti-CD3]_H1.31_L1.47_scFv v1CDR2	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 148		
GTNKRAP		7
SEQ ID NO: 149	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9	
	note = CD3 High-Int - [anti-CD3]_H1.31_L1.47_scFv v1CDR3	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 149		
ALWYSNHWV		9
SEQ ID NO: 150	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..20	
	note = CD3 High-Int - [anti-CD3]_H1.31_L1.47_scFv Linker	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 150		
GKPGSGKPGS GKPGSGKPGS		20
SEQ ID NO: 151	moltype = AA length = 254	
FEATURE	Location/Qualifiers	
REGION	1..254	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..254	
	note = CD3 High[VL-VH] - [anti-CD3]_L1.47_H1.30_scFv scFv	
source	1..254	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 151		
QAVVTQEPSTL TVSPGGTCTL	TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV	60
PARFSGSLLG GKAALTISGA	QPEDEADYYC ALWYSNHWVF GGGTKLTVLG KPGSGKPGSG	120
KPGSGKPGSE VQLVESGGGL	VQPGGSLRLS CAASGFTFST YAMNWRQAP GKGLEWVGRI	180
RSKYNNYATY YADSVKGRFT	ISRDDSKNTL YLQMNSLRAE DTAVYYCVRH GNFGDSYVSW	240
FAYWGQGLTV TVSS		254
SEQ ID NO: 152	moltype = AA length = 109	
FEATURE	Location/Qualifiers	
REGION	1..109	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..109	
	note = CD3 High[VL-VH] - [anti-CD3]_L1.47_H1.30_scFv Variable Light (v1) Domain	
source	1..109	
	mol_type = protein	
	organism = synthetic construct	

SEQ ID NO: 159	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..14	
	note = CD3 High[VL-VH] - [anti-CD3]_L1.47_ H1.30_scFv vhCDR3	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 159		
HGNFGDSYVS WFAY		14
SEQ ID NO: 160	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..20	
	note = CD3 High[VL-VH] - [anti-CD3]_L1.47_ H1.30_scFv Linker	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 160		
GKPGSGKPGS GKPGSGKPGS		20
SEQ ID NO: 161	moltype = AA length = 254	
FEATURE	Location/Qualifiers	
REGION	1..254	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..254	
	note = CD3 High-Int #1[VL-VH] - [anti-CD3]_L1.47_ H1.32_scFv scFv	
source	1..254	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 161		
QAVVTQEPSSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV	60	
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVLG KPGSGKPGSG	120	
KPGSGKPGSE VQLVESGGGL VQPGGSLRLS CAASGTFST YAMNWRVQAP KGKLEWVGRI	180	
RSKANNYATY YADSVKGRFT ISRDDSKNTL YLQMNSLRAE DTAVYYCVRH GNFGDSYVSW	240	
PAYWGQTLV TVSS	254	
SEQ ID NO: 162	moltype = AA length = 109	
FEATURE	Location/Qualifiers	
REGION	1..109	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..109	
	note = CD3 High-Int #1[VL-VH] - [anti-CD3]_L1.47_ H1.32_scFv Variable Light (vL) Domain	
source	1..109	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 162		
QAVVTQEPSSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV	60	
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVL	109	
SEQ ID NO: 163	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..14	
	note = CD3 High-Int #1[VL-VH] - [anti-CD3]_L1.47_ H1.32_scFv v1CDR1	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 163		
GSSTGAVTTS NYAN		14
SEQ ID NO: 164	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7	
	note = CD3 High-Int #1[VL-VH] - [anti-CD3]_L1.47_ H1.32_scFv v1CDR2	
source	1..7	

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SEQUENCE: 164	mol_type = protein	
GTNKRAP	organism = synthetic construct	7
SEQ ID NO: 165	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9	
	note = CD3 High-Int #1[VL-VH] - [anti-CD3]_L1.47_	
	H1.32_scFv v1CDR3	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 165		9
ALWYSNHWV		
SEQ ID NO: 166	moltype = AA length = 125	
FEATURE	Location/Qualifiers	
REGION	1..125	
	note = Description of Artificial Sequence: Synthetic	
	polypeptide	
REGION	1..125	
	note = CD3 High-Int #1[VL-VH] - [anti-CD3]_L1.47_	
	H1.32_scFv Variable Heavy (vh) Domain	
source	1..125	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 166		
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKANNYAT	60	
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAWGGQGT	120	
VTVSS	125	
SEQ ID NO: 167	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = CD3 High-Int #1[VL-VH] - [anti-CD3]_L1.47_	
	H1.32_scFv vhCDR1	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 167		5
TYAMN		
SEQ ID NO: 168	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..19	
	note = CD3 High-Int #1[VL-VH] - [anti-CD3]_L1.47_	
	H1.32_scFv vhCDR2	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 168		19
RIRSKANNYA TYYADSVKG		
SEQ ID NO: 169	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..14	
	note = CD3 High-Int #1[VL-VH] - [anti-CD3]_L1.47_	
	H1.32_scFv vhCDR3	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 169		14
HGNFGDSYVS WFAW		
SEQ ID NO: 170	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..20	
	note = CD3 High-Int #1[VL-VH] - [anti-CD3]_L1.47_	

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source          H1.32_scFv Linker
                1..20
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 170
GKPGSGKPGS GKPGSGKPGS                                20

SEQ ID NO: 171      moltype = AA length = 254
FEATURE            Location/Qualifiers
REGION            1..254
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..254
                  note = CD3 High-Int #2 [VL-VH] - [anti-CD3]_L1.47_
                  H1.89_scFv scFv
source            1..254
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 171
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV 60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVLG KPGSGKPGSG 120
KPGSGKPGSE VQLVESGGGL VQPGSLRLS CAASGFTFST YAMNWRVQAP GKGLEWVGRI 180
RSKYNNYATY YADSVKGRFT ISRDDSKNTL YLQMNSLRAE DTAIVYCVRH GNFGDEYVSW 240
FAYWGQGTILV TVSS 254

SEQ ID NO: 172      moltype = AA length = 109
FEATURE            Location/Qualifiers
REGION            1..109
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..109
                  note = CD3 High-Int #2 [VL-VH] - [anti-CD3]_L1.47_
                  H1.89_scFv Variable Light (vl) Domain
source            1..109
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 172
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV 60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVL 109

SEQ ID NO: 173      moltype = AA length = 14
FEATURE            Location/Qualifiers
REGION            1..14
                  note = Description of Artificial Sequence: Synthetic peptide
REGION            1..14
                  note = CD3 High-Int #2 [VL-VH] - [anti-CD3]_L1.47_
                  H1.89_scFv v1CDR1
source            1..14
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 173
GSSTGAVTTS NYAN 14

SEQ ID NO: 174      moltype = AA length = 7
FEATURE            Location/Qualifiers
REGION            1..7
                  note = Description of Artificial Sequence: Synthetic peptide
REGION            1..7
                  note = CD3 High-Int #2 [VL-VH] - [anti-CD3]_L1.47_
                  H1.89_scFv v1CDR2
source            1..7
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 174
GTNKRAP 7

SEQ ID NO: 175      moltype = AA length = 9
FEATURE            Location/Qualifiers
REGION            1..9
                  note = Description of Artificial Sequence: Synthetic peptide
REGION            1..9
                  note = CD3 High-Int #2 [VL-VH] - [anti-CD3]_L1.47_
                  H1.89_scFv v1CDR3
source            1..9
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 175
ALWYSNHWV 9

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SEQ ID NO: 176	moltype = AA length = 125	
FEATURE	Location/Qualifiers	
REGION	1..125	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..125	
	note = CD3 High-Int #2[VL-VH] - [anti-CD3]_L1.47_	
	H1.89_scFv Variable Heavy (vh) Domain	
source	1..125	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 176		
EVQLVESGGG LVQPGGSLRL	SCAASGFTFS TYAMNWRQA	PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT	LYLQMNSLRA EDTAVYYCVR	HGNFGDEYVS WFAYWGGQTL 120
VTVSS		125
SEQ ID NO: 177	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = CD3 High-Int #2[VL-VH] - [anti-CD3]_L1.47_	
	H1.89_scFv vHCDR1	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 177		
TYAMN		5
SEQ ID NO: 178	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..19	
	note = CD3 High-Int #2[VL-VH] - [anti-CD3]_L1.47_	
	H1.89_scFv vHCDR2	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 178		
RIRSKYNNYA TYYADSVKG		19
SEQ ID NO: 179	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..14	
	note = CD3 High-Int #2[VL-VH] - [anti-CD3]_L1.47_	
	H1.89_scFv vHCDR3	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 179		
HGNFGDEYVS WFAV		14
SEQ ID NO: 180	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..20	
	note = CD3 High-Int #2[VL-VH] - [anti-CD3]_L1.47_	
	H1.89_scFv Linker	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 180		
GKPGSGKPGS GKPGSGKPGS		20
SEQ ID NO: 181	moltype = AA length = 254	
FEATURE	Location/Qualifiers	
REGION	1..254	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..254	
	note = CD3 High-Int[VL-VH] - [anti-CD3]_L1.47_ H1.90_scFv	
	scFv	
source	1..254	
	mol_type = protein	
	organism = synthetic construct	

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SEQUENCE: 181
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV 60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVLG KPGSGKPGSG 120
KPGSGKPGSE VQLVESGGGL VQPGGSLRLS CAASGFTFST YAMNWRQAP GKLEWVGRI 180
RSKYNNYATY YADSVKGRFT ISRDDSKNTL YLQMNSLRAE DTAVYYCVRH GNFGDPYVSW 240
FAYWGQGTILV TVSS 254

SEQ ID NO: 182 moltype = AA length = 109
FEATURE Location/Qualifiers
REGION 1..109
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..109
 note = CD3 High-Int [VL-VH] - [anti-CD3]_L1.47_ H1.90_scFv
 Variable Light (vl) Domain
source 1..109
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 182
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV 60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVL 109

SEQ ID NO: 183 moltype = AA length = 14
FEATURE Location/Qualifiers
REGION 1..14
 note = Description of Artificial Sequence: Synthetic peptide
REGION 1..14
 note = CD3 High-Int [VL-VH] - [anti-CD3]_L1.47_ H1.90_scFv
 vLCDR1
source 1..14
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 183
GSSTGAVTTS NYAN 14

SEQ ID NO: 184 moltype = AA length = 7
FEATURE Location/Qualifiers
REGION 1..7
 note = Description of Artificial Sequence: Synthetic peptide
REGION 1..7
 note = CD3 High-Int [VL-VH] - [anti-CD3]_L1.47_ H1.90_scFv
 vLCDR2
source 1..7
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 184
GTNKRAP 7

SEQ ID NO: 185 moltype = AA length = 9
FEATURE Location/Qualifiers
REGION 1..9
 note = Description of Artificial Sequence: Synthetic peptide
REGION 1..9
 note = CD3 High-Int [VL-VH] - [anti-CD3]_L1.47_ H1.90_scFv
 vLCDR3
source 1..9
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 185
ALWYSNHWV 9

SEQ ID NO: 186 moltype = AA length = 125
FEATURE Location/Qualifiers
REGION 1..125
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..125
 note = CD3 High-Int [VL-VH] - [anti-CD3]_L1.47_ H1.90_scFv
 Variable Heavy (vh) Domain
source 1..125
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 186
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNGDPYVS WFAIWGQGTIL 120
VTSS 125

SEQ ID NO: 187 moltype = AA length = 5
FEATURE Location/Qualifiers

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REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = CD3 High-Int [VL-VH] - [anti-CD3]_L1.47_ H1.90_scFv	
	vhCDR1	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 187		
TYAMN		5
SEQ ID NO: 188	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..19	
	note = CD3 High-Int [VL-VH] - [anti-CD3]_L1.47_ H1.90_scFv	
	vhCDR2	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 188		
RIRSKYNNYA TYYADSVKG		19
SEQ ID NO: 189	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..14	
	note = CD3 High-Int [VL-VH] - [anti-CD3]_L1.47_ H1.90_scFv	
	vhCDR3	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 189		
HGNFGDPYVS WFAY		14
SEQ ID NO: 190	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..20	
	note = CD3 High-Int [VL-VH] - [anti-CD3]_L1.47_ H1.90_scFv	
	Linker	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 190		
GKPGSGKPGS GKPGSGKPGS		20
SEQ ID NO: 191	moltype = AA length = 254	
FEATURE	Location/Qualifiers	
REGION	1..254	
	note = Description of Artificial Sequence: Synthetic	
	polypeptide	
REGION	1..254	
	note = Anti-CD3-Intermediate [VL-VH] - [anti-CD3]_L1.47_	
	H1.33_scFv scFv	
source	1..254	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 191		
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV	60	
PARFSGSLLG GKAAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVLG KPGSGKPGSG	120	
KPGSGKPGSE VQLVESGGGL VQPGGSLRLS CAASGFTFST YAMNWVRQAP GKGLEWVGRI	180	
RSKYNNYATY YADSVKGRFT ISRDDSKNTL YLQMNSLRAE DTAVYYCVRH GNFGDSYVSW	240	
FDYWGQGTLLV TVSS	254	
SEQ ID NO: 192	moltype = AA length = 109	
FEATURE	Location/Qualifiers	
REGION	1..109	
	note = Description of Artificial Sequence: Synthetic	
	polypeptide	
REGION	1..109	
	note = Anti-CD3-Intermediate [VL-VH] - [anti-CD3]_L1.47_	
	H1.33_scFv Variable Light (v1) Domain	
source	1..109	
	mol_type = protein	
	organism = synthetic construct	

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source          1..19
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 198
RIRSKYNNYA TYYADSVKG                                     19

SEQ ID NO: 199      moltype = AA length = 14
FEATURE            Location/Qualifiers
REGION             1..14
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..14
                   note = Anti-CD3-Intermediate[VL-VH] - [anti-CD3]_L1.47_
                     H1.33_scFv vhCDR3
source             1..14
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 199
HGNFGDSYVS WFDY                                         14

SEQ ID NO: 200      moltype = AA length = 20
FEATURE            Location/Qualifiers
REGION             1..20
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..20
                   note = Anti-CD3-Intermediate[VL-VH] - [anti-CD3]_L1.47_
                     H1.33_scFv Linker
source             1..20
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 200
GKPGSGKPGS GKPGSGKPGS                                   20

SEQ ID NO: 201      moltype = AA length = 254
FEATURE            Location/Qualifiers
REGION             1..254
                   note = Description of Artificial Sequence: Synthetic
                     polypeptide
REGION             1..254
                   note = CD3 High-Int[VL-VH] - [anti-CD3]_L1.47_ H1.31_scFv
                     scFv
source             1..254
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 201
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV 60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWFV GGGTKLTVLG KPGSGKPGSG 120
KPGSGKPGSE VQLVESGGGL VQPGGSLRLS CAASGFTFST YAMSWVRQAP GKGLEWVGRI 180
RSKYNNYATY YADSVKGRFT ISRDDSKNTL YLQMNSLRAE DTAVYYCVRH GNFGDSYVSW 240
FAYWGQGTILV TVSS                                         254

SEQ ID NO: 202      moltype = AA length = 109
FEATURE            Location/Qualifiers
REGION             1..109
                   note = Description of Artificial Sequence: Synthetic
                     polypeptide
REGION             1..109
                   note = CD3 High-Int[VL-VH] - [anti-CD3]_L1.47_ H1.31_scFv
                     Variable Light (v1) Domain
source             1..109
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 202
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV 60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWFV GGGTKLTVL 109

SEQ ID NO: 203      moltype = AA length = 14
FEATURE            Location/Qualifiers
REGION             1..14
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..14
                   note = CD3 High-Int[VL-VH] - [anti-CD3]_L1.47_ H1.31_scFv
                     v1CDR1
source             1..14
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 203
GSSTGAVTTS NYAN                                         14

SEQ ID NO: 204      moltype = AA length = 7

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-continued

FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7	
	note = CD3 High-Int [VL-VH] - [anti-CD3]_L1.47_ H1.31_scFv	
	vLCDR2	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 204		
GTNKRAP		7
SEQ ID NO: 205	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9	
	note = CD3 High-Int [VL-VH] - [anti-CD3]_L1.47_ H1.31_scFv	
	vLCDR3	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 205		
ALWYSNHWV		9
SEQ ID NO: 206	moltype = AA length = 125	
FEATURE	Location/Qualifiers	
REGION	1..125	
	note = Description of Artificial Sequence: Synthetic	
	polypeptide	
REGION	1..125	
	note = CD3 High-Int [VL-VH] - [anti-CD3]_L1.47_ H1.31_scFv	
	Variable Heavy (vh) Domain	
source	1..125	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 206		
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMSWVRQA PGKGLEWVGR IRSKYNNYAT		60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAIYWGQGTL		120
VTVSS		125
SEQ ID NO: 207	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = CD3 High-Int [VL-VH] - [anti-CD3]_L1.47_ H1.31_scFv	
	vhCDR1	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 207		
TYAMS		5
SEQ ID NO: 208	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..19	
	note = CD3 High-Int [VL-VH] - [anti-CD3]_L1.47_ H1.31_scFv	
	vhCDR2	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 208		
RIRSKYNNYA TYYADSVKG		19
SEQ ID NO: 209	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..14	
	note = CD3 High-Int [VL-VH] - [anti-CD3]_L1.47_ H1.31_scFv	
	vhCDR3	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 209		
HGNFGDSYVS WFAIY		14

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SEQ ID NO: 210      multype = AA  length = 20
FEATURE             Location/Qualifiers
REGION              1..20
                    note = Description of Artificial Sequence: Synthetic peptide
REGION              1..20
                    note = CD3 High-Int[VL-VH] - [anti-CD3]_L1.47_ H1.31_scFv
                        Linker
source              1..20
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 210
GKPGSGKPGS GKPGSGKPGS

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SEQ ID NO: 211	moltype = AA length = 750					
FEATURE	Location/Qualifiers					
REGION	1..750					
	note = sp4609					
source	1..750					
	mol_type = protein					
	organism = Homo sapiens					
SEQUENCE: 211						
MWNLLHETDS	AVATARRPRW	LCAGALVLVLAG	GFFLLGLFLFG	WFIKSSNEAT	NITPKHNMKA	60
FLDELKAENI	KKFLYNFTQI	PHLAGTEQNF	QLAKQIQSQW	KEFGLDSVEL	AHYDVLLSYP	120
NKTHPNYISI	INEDGNEIFN	TSLFEPPPPP	YENVSDIVPP	FSFSPQGMF	EGDLVYVNYA	180
RTDFYFFLEK	DMKINCSCGI	VIARYGKQVFR	GNKVNAQLA	KAGKVILYSD	PADYFAPGVK	240
SYPDGWNLPG	GGVQRGNILN	LNGAGDLPTP	GYPANEYAYR	RGIAEAVGLP	SIPVHPIGYY	300
DAQKLLEKMG	GSAPDDSSWR	GSLSKVPYNVG	PQGTNFSTQI	KVKMHHTSTN	EVTRIYNVIG	360
TLRGAVEBDR	VYILGGHRDS	WVFGGIDPQS	GAAVVHEIVR	SFGTLKKEGW	RRRTILFAS	420
WDAEEFGLLG	STEWAEENSR	LLQERGVAIY	NADSSIEGNY	TLRVDCTPLM	YSLVHNLTKE	480
LKSPDEGFEG	KSLYESWTKK	SPSPSPSGMP	RISLKGSDN	FEVFFQRLGI	ASGRARYTKN	540
WETNKFSGYP	LYHSVYETYE	LVEKFLVDPMF	KYHLTVAQVR	GMGVFELANS	IVLPFDRCRDY	600
AVVLRKYADK	IYSISMKHPQ	EMKTVSVSFD	SLFSAVKNFT	EIASKFSERL	QDFDKSNPIV	660
LRMNMQLMF	LERAFIDPLG	LPDRPPYRHV	IYAPSSHNKY	AGESFPGIYD	ALFDIESKVD	720
PSKAWGEVKR	QIYVAAFTVO	AAAEETISEVA				750

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SEQ ID NO: 212           moltype = AA  length = 707
FEATURE                  Location/Qualifiers
REGION                   1..707
                           note = sp4609 [44]-750
source                   1..707
                           mol_type = protein
                           organism = Homo sapiens

SEQUENCE: 212
KSSNEATNIT PHNMMKAFLD ELKAENIKKF LYNFTQIPLH AGTEQNFQLA KQIQSQWKEF    60
GLDSVELAHY DVLLSYPNKT HPNYSISIME DGNELFNSTL FEPPPPGYEN VSDIVPPFSA    120
FSPQGMPPEGD LVYVINYARTE DFFKLERMDGK INCSEKIVIA RYGKVFVRGNK VKNAQLAGAK    180
GVILYSDDAD YFAPGVKSPY DGNWNLPGGVG QRGNILLNLNG AGDPLTPGYP ANEYAYRRGI    240
AEAVGLRSPIS VHPIGGYDAQ KLEEKMGYVA PDSSWRGSLK KVPYNVGPFGT TGNFSTQKVK    300
MHHISTNEVT RYINWIGTLR GAVEPRDGI    LGGHRDWSVF GGLDDPQSGAA VVHEIVRSFG    360
TLKKEGWRPR RTILFASWDA EEFGLLGSTE WAEENSRLLO ERGVAYINAD SSIEGNYTLR    420
VDCTPLMYSL VHNLTKEKLS PDDEGFEKSL YESWTKKSPS PEPGDMPRIS KLGSGNDFEV    480
FFQRLGIASG RARYTKNWEH NKPSGPLYLH SVYETYLVE KFYDPMFKYH LTVAQVRVGM    540
VFELANSIVL PFDCRDYAVV LRKYADKIYS ISMKHPQEMK TYSVSFDSL F SAVKNFTEIA    600
SKFSERLQDF DKSNNPILVRM MNQMLMFLER AFIDPLGLPD RPFYRHVIA PSSHNNKYAE    660
SFGIYDALF  DIESKVDPSK AWEGVEMKRY VAAFTVQAAA ETLSEVA                    707

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SEQ ID NO: 213           moltype = AA  length = 752
FEATURE                  Location/Qualifiers
REGION                   1..752
                           note = sp5409
source                  1..752
                           mol_type = protein
                           organism = Mus musculus

SEQUENCE: 213
MWNALQDRDS AEVLGHRQRW LRVGTLVLAL TGTFLIGLFL GWFIKPSNEA TGNVSHSGMK 60
KEFLHELKAE NIKKFLYNFT RTPHLAGTQN NFELAKQIHD QWKBFGLDLV ELSHYDVLLS 120
YPNKTHPNYI SIINEDGNEI FKTSLSEQPP PGYENISDVV PPYSAFSPQG TPEGDLVYVN 180
VARTEDYFPL EREMKISCSG KIVIARYGKV FRGNMVKNAQ LAGAKGMILY SDPADYFVPA 240
KSYSPDGWNL PGGGVQQRNV LNLNAGADPL TPGYPANEHA YRHELTNAV LPSIPVHPIG 300
YDDAQKLLEH MGGPAPPDPS WKGGLKVPNY VPGPGAFNFS TQVKMKHIHS YTKVTRIYNV 360
IGTLKGALEP DRYVILGGHR DAWVFGGIDP QSGAAVVHEI VRSFGTLKKK GRRPRRTILF 420
ASWDAAEEFGL LGSTEWAEEH SRLLRQERGVA YINADSSIEG NYTLRVDCTP LMSYSLVYNTL 480
KELQSPDEGF EGKSLYDSWK EKSPSPFEFIF MPRISLKLGS NDFEVFFQRL GIASGRARYT 540
KNWKTNKVSS YPLYHSVYET YELVVKFYDP TPKYHLTVAQ VRGAMVFELA NSIVLPFDCQ 600
SYAVALKKYA DTIYINSMKH PQEMKAYMIS FDSLPSAVNN FTDVASKFNQ RLQELDKSNP 660
ILLRIMNDQL MYLERAIATP LGLPGRPFYR HIIYAPSSHN KYAGESFPPI YDALFDISSK 720
VNASKAMNEV KROISIAATP VOAAAEATLRE VA 752

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SEQ ID NO: 214 moltype = AA length = 708
 FEATURE Location/Qualifiers
 REGION 1..708
 note = sp5409[45]-752
 source 1..708
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 214

KPSNEATGNV	SHSGMKKEFL	HELKAENIKK	FLYNFTRTPH	LAGTQNNFEL	AKQIHDQWKE	60
FGLDLVELSH	YDVLSSYPNK	THPNYISII	EDGNEIFKTS	LSEQPPPGYE	NISDVVPVYS	120
AFSPQGTPEG	DLVYVNYART	EDFFKLEREM	KISCSGKIVI	ARYGKVFRGN	MVKNAQLAGA	180
KGMILYSDPA	DYFVPAVKSY	PDGWNLPGGG	VQRGNVLNLN	GAGDPLTPGY	PANEHAYRHE	240
LTNVAVGLPSI	PVHPIGYDDA	QKLEHMGGP	APDSSWKGG	LKVPYNVPGP	FAGNFSTQKV	300
KMHIHSYTKV	TRIYNVIGTL	KGALEPDRYV	ILGGHRDAWV	FGGIDPQSGA	AVVHEIVRSF	360
GTLKKKGRRP	RRTILFASWD	AEEFGLLGST	EWAEHSRL	QERGVAYINA	DSSIEGNYTL	420
RVDCTPLMYS	LVYNLTKEQ	SPDEGFEKGS	LYDSWKEKSP	SPEFIMPRI	SKLGSGNDFE	480
VFFQRLGIAS	GRARYTKNWK	TNKVSSYPLY	HSVYETYELV	VKFYDPTFKY	HLTVAQVRGA	540
MVFELANSIV	LPPDCQSYAV	ALKKYADTIY	NISMKHPQEM	KAYMISFDSL	FSAVNNFTDV	600
ASKFNQLQLE	LDKSNPILLR	IMNDQLMYLE	RAFIDPLGLP	GRPPYRHIY	APSSHNKYAG	660
ESFPGIYDAL	FDISSKVNAS	KAWNEVKRQI	SIATFTVQAA	AETLREVA		708

SEQ ID NO: 215 moltype = AA length = 750
 FEATURE Location/Qualifiers
 REGION 1..750
 note = trPNF
 source 1..750
 mol_type = protein
 organism = Macaca fascicularis

SEQUENCE: 215

MWNLLETDS	AVATARRPRW	LCAGALVLAG	GFFLLGFLEF	WFIKSSSEAT	NITPKHNMKA	60
FLDELKAENI	KKFLHNFTQI	PHLAGTEQNF	QLAKQIQSQW	KEFGLDSEVL	THYDVLLSYP	120
NKTHPNYISI	INEDGNEIFN	TSLFEPPPPAG	YENVSDIVPP	PSAFSPQGM	EGDLVYVNYA	180
RTEDFFKLER	DMKINCSTGI	VIARYGKVFR	GNKVNAQLA	GATGVILYSD	PDDYFAPGVK	240
SYPDGWNLP	GGVQRGNILN	LNGAGDPLTP	GYPANEYAYR	RGMAEAVGLP	SIPVHPIGYY	300
DAQKLEKMG	GSASPDSSWR	GSLKVPYNVG	PGFTGNFSTQ	KVKMHIHSTS	EVTRIYNVIG	360
TLRGAVEPDR	YVILGGHRDS	WVFGGIDPQS	GAAVVHEIVR	SFGMLKKEGW	RPRRTILFAS	420
WDAEEFGLLG	STEWAEENS	LLQERGVAYI	NADSSIEGNY	TLRVDCTPLM	YSLVYNLTKE	480
LESPDEGFEG	KSLYESWTKK	SPSPEFSGMP	RISKLGSGND	FEVFFQRLGI	ASGRARYTKN	540
WETNKFSYYP	LYHSVYETYE	LVEKFYDPMF	KYHLTVAQVR	GGMVFELENS	VVLPPDCRDY	600
AVVLRKYADK	IYNISMKHPQ	EMKTVSVSFD	SLFSAVKNFT	ELASKFSERL	RDFDKSNPIL	660
LRMNDQLMF	LERAFIDPLG	LPDRPFYRHV	IYAPSSHNKY	AGESFPGIYD	ALFDIESKVD	720
PSQAWGEVKR	QISIATFTVQ	AAEETLSEVA				750

SEQ ID NO: 216 moltype = AA length = 707
 FEATURE Location/Qualifiers
 REGION 1..707
 note = trPNF[44]-750
 source 1..707
 mol_type = protein
 organism = Macaca fascicularis

SEQUENCE: 216

KSSSEATNIT	PKHNMKAFLD	ELKAENIKKF	LHNFTQIPLH	AGTEQNFOLA	KQIQSQWKEF	60
GLDSVELTHY	DVLLSYPNKT	HPNYISIINE	DGNEIFNTSL	FEPPPPAGYEN	VSDIVPPPSA	120
FSPQGMPEGD	LVYVNYARTE	DFPKLERDMK	INCSGKIVIA	RYGKVFRGNK	VKNAQLAGAT	180
GVLISYDDDD	YFAPGVKSY	DGWNLPGGGV	QRGNILNLNG	AGDPLTPGY	ANEYAYRRGM	240
AEAVGLPSIP	VHPIGYYDAQ	KLLEKMGSSA	SPDSSWRGSL	KVPYNVPGPF	TGNFSTQKVK	300
MHIHSTSEVT	RIYNVIGTLR	GAVEPDYRVI	LGGHRDSWVF	GGIDPQSGAA	VVHEIVRSFG	360
MLKKEGWRPR	RTILFASWDA	EEFGLLGST	WAEENSRLQ	ERGVAYINAD	SSIEGNYTLR	420
VDCTPLMYSL	VYNLTKELES	PDEGFEGKSL	YESWTKKSPS	PEFSGMPRIS	KLGSNDGFEV	480
FPQRLGIASG	RARYTKNWET	NKFSSYPLYH	SVYETYELVE	KFYDPMFKYH	LTVAQVRGGM	540
VFELANSVVL	PFDCRDYAVV	LRKYADKIYN	ISMKHPQEMK	TYSVSFDSL	SAVKNFTEIA	600
SKFSERLRDF	DKSNPILLRM	MNDQLMFLE	AFIDPLGLPD	RPFYRHVIYA	PSSHNKYAGE	660
SPPGIYDALF	DIESKVDPSQ	AWGEVKRQIS	IATFTVQAAA	ETLSEVA		707

SEQ ID NO: 217 moltype = AA length = 115
 FEATURE Location/Qualifiers
 REGION 1..115
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..115
 note = PSMA-H_H1L1 Variable heavy (vh) domain
 source 1..115
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 217

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKPFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	LVVSS	115

SEQ ID NO: 218 moltype = AA length = 5

-continued

FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = PSMA-H_H1L1 vhCDR1	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 218		
EYTIH		5
SEQ ID NO: 219	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..17	
	note = PSMA-H_H1L1 vhCDR2	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 219		
NINPNNGGTT YNQKFQG		17
SEQ ID NO: 220	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
REGION	1..6	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..6	
	note = PSMA-H_H1L1 vhCDR3	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 220		
GWNFDY		6
SEQ ID NO: 221	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
REGION	1..107	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..107	
	note = PSMA-H_H1L1 Variable light (vl) domain	
source	1..107	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 221		
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD		60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK		107
SEQ ID NO: 222	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = PSMA-H_H1L1 vlCDR1	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 222		
RASQDVGTAV D		11
SEQ ID NO: 223	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7	
	note = PSMA-H_H1L1 vlCDR2	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 223		
WASTRHT		7
SEQ ID NO: 224	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9	
	note = PSMA-H_H1L1 vlCDR3	

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source                1..9
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 224
QQYNSYPLT                                                    9

SEQ ID NO: 225        moltype = AA  length = 444
FEATURE              Location/Qualifiers
REGION              1..444
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..444
                      note = XENP31858 Chain 1 - PSMA-H_H1_IgG1_PVA_/S267K Heavy
                      Chain
source              1..444
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 225
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EQYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCLVKGFY PSDIAVEWES NGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWQQGNVF 420
SCSVMEALH  NHYTQKSLSL SPGK                                         444

SEQ ID NO: 226        moltype = AA  length = 214
FEATURE              Location/Qualifiers
REGION              1..214
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..214
                      note = XENP31858 Chain 2 - PSMA-H_L1 Light Chain
source              1..214
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 226
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC                               214

SEQ ID NO: 227        moltype = AA  length = 444
FEATURE              Location/Qualifiers
REGION              1..444
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..444
                      note = XENP31604 Chain 1 -
                      PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                      Heavy Chain
source              1..444
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 227
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMEALH  NHYTQKSLSL SPGK                                         444

SEQ ID NO: 228        moltype = AA  length = 214
FEATURE              Location/Qualifiers
REGION              1..214
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..214
                      note = XENP31604 Chain 2 - PSMA-H_L1 Light Chain
source              1..214
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 228
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180

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LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 229      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.1 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 229
DIVMTQSPDS LAVSLGERAT LSCRASNDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 230      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.2 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 230
DIVMTQSPDS LAVSLGERAT LSCRASEDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 231      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.3 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 231
DIVMTQSPDS LAVSLGERAT LSCRASTDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 232      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.4 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 232
DIVMTQSPDS LAVSLGERAT LSCRASSDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 233      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.5 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 233
DIVMTQSPDS LAVSLGERAT LSCRASIDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 234      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.6 Variable Light

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source                1..107
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 234
DIVMTQSPDS LAVSLGERAT LSCRASQDVG SAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 235        moltype = AA length = 107
FEATURE              Location/Qualifiers
REGION               1..107
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION               1..107
                      note = PSMA-H_L1.7 Variable Light
source               1..107
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 235
DIVMTQSPDS LAVSLGERAT LSCRASQDVG NAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 236        moltype = AA length = 107
FEATURE              Location/Qualifiers
REGION               1..107
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION               1..107
                      note = PSMA-H_L1.8 Variable Light
source               1..107
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 236
DIVMTQSPDS LAVSLGERAT LSCRASQDVG EAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 237        moltype = AA length = 107
FEATURE              Location/Qualifiers
REGION               1..107
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION               1..107
                      note = PSMA-H_L1.9 Variable Light
source               1..107
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 237
DIVMTQSPDS LAVSLGERAT LSCRASQDVG IAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 238        moltype = AA length = 107
FEATURE              Location/Qualifiers
REGION               1..107
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION               1..107
                      note = PSMA-H_L1.10 Variable Light
source               1..107
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 238
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TYVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 239        moltype = AA length = 107
FEATURE              Location/Qualifiers
REGION               1..107
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION               1..107
                      note = PSMA-H_L1.11 Variable Light
source               1..107
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 239
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TSVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 240        moltype = AA length = 107
FEATURE              Location/Qualifiers

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REGION 1..107
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..107
 note = PSMA-H_L1.12 Variable Light

source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 240
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TTVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 241 moltype = AA length = 107
 FEATURE Location/Qualifiers

REGION 1..107
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..107
 note = PSMA-H_L1.13 Variable Light

source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 241
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TIVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 242 moltype = AA length = 107
 FEATURE Location/Qualifiers

REGION 1..107
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..107
 note = PSMA-H_L1.14 Variable Light

source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 242
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TGVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 243 moltype = AA length = 107
 FEATURE Location/Qualifiers

REGION 1..107
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..107
 note = PSMA-H_L1.15 Variable Light

source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 243
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TEVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 244 moltype = AA length = 107
 FEATURE Location/Qualifiers

REGION 1..107
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..107
 note = PSMA-H_L1.16 Variable Light

source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 244
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TQVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 245 moltype = AA length = 107
 FEATURE Location/Qualifiers

REGION 1..107
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..107
 note = PSMA-H_L1.17 Variable Light

source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 245

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DIVMTQSPDS LAVSLGERAT LSCRASQDVG TALDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 246      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.18 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 246
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAIDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 247      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.19 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 247
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVAWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 248      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.20 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 248
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 249      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.21 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 249
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 250      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.22 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 250
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIEW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 251      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107

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source          note = PSMA-H_L1.23 Variable Light
                1..107
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 251
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIHW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 252      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.24 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 252
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYF ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 253      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.25 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 253
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYF ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 254      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.26 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 254
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYH ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 255      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.27 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 255
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYQ ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 256      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.28 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 256
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYE ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 257      moltype = AA length = 107

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FEATURE          Location/Qualifiers
REGION           1..107
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION           1..107
                  note = PSMA-H_L1.29 Variable Light
source           1..107
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 257
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYT ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 258      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION             1..107
                    note = PSMA-H_L1.30 Variable Light
source             1..107
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 258
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ISTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 259      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION             1..107
                    note = PSMA-H_L1.31 Variable Light
source             1..107
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 259
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW LSTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 260      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION             1..107
                    note = PSMA-H_L1.32 Variable Light
source             1..107
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 260
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW TSTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 261      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION             1..107
                    note = PSMA-H_L1.33 Variable Light
source             1..107
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 261
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASSRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 262      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION             1..107
                    note = PSMA-H_L1.34 Variable Light
source             1..107
                    mol_type = protein
                    organism = synthetic construct

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SEQUENCE: 262
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASQRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 263      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION            1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION            1..107
                   note = PSMA-H_L1.35 Variable Light
source            1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 263
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASERHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 264      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION            1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION            1..107
                   note = PSMA-H_L1.36 Variable Light
source            1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 264
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASNRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 265      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION            1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION            1..107
                   note = PSMA-H_L1.37 Variable Light
source            1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 265
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASGRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 266      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION            1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION            1..107
                   note = PSMA-H_L1.38 Variable Light
source            1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 266
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRETGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 267      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION            1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION            1..107
                   note = PSMA-H_L1.39 Variable Light
source            1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 267
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHSGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 268      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION            1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide

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REGION 1..107
 note = PSMA-H_L1.40 Variable Light
 source 1..107
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 268
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHEGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107
 SEQ ID NO: 269 moltype = AA length = 107
 FEATURE Location/Qualifiers
 REGION 1..107
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..107
 note = PSMA-H_L1.41 Variable Light
 source 1..107
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 269
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHYGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107
 SEQ ID NO: 270 moltype = AA length = 107
 FEATURE Location/Qualifiers
 REGION 1..107
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..107
 note = PSMA-H_L1.42 Variable Light
 source 1..107
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 270
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCNQ YNSYPLTFGA GTKVEIK 107
 SEQ ID NO: 271 moltype = AA length = 107
 FEATURE Location/Qualifiers
 REGION 1..107
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..107
 note = PSMA-H_L1.43 Variable Light
 source 1..107
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 271
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCEQ YNSYPLTFGA GTKVEIK 107
 SEQ ID NO: 272 moltype = AA length = 107
 FEATURE Location/Qualifiers
 REGION 1..107
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..107
 note = PSMA-H_L1.44 Variable Light
 source 1..107
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 272
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCKQ YNSYPLTFGA GTKVEIK 107
 SEQ ID NO: 273 moltype = AA length = 107
 FEATURE Location/Qualifiers
 REGION 1..107
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..107
 note = PSMA-H_L1.45 Variable Light
 source 1..107
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 273
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCRQ YNSYPLTFGA GTKVEIK 107

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SEQ ID NO: 274      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.46 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 274
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCLQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 275      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.47 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 275
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCHQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 276      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.48 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 276
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ FNSYPLTFGA GTKVEIK 107

SEQ ID NO: 277      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.49 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 277
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ WNSYPLTFGA GTKVEIK 107

SEQ ID NO: 278      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.50 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 278
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ KNSYPLTFGA GTKVEIK 107

SEQ ID NO: 279      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.51 Variable Light
source             1..107
                   mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 279
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ RNSYPLTFGA GTKVEIK 107

SEQ ID NO: 280      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.52 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 280
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ HNSYPLTFGA GTKVEIK 107

SEQ ID NO: 281      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.53 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 281
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ QNSYPLTFGA GTKVEIK 107

SEQ ID NO: 282      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.54 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 282
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ ENSYPLTFGA GTKVEIK 107

SEQ ID NO: 283      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.55 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 283
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ INSYPLTFGA GTKVEIK 107

SEQ ID NO: 284      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.56 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 284
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ LNSYPLTFGA GTKVEIK 107

SEQ ID NO: 285      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic

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REGION                polypeptide
                      1..107
                      note = PSMA-H_L1.57 Variable Light
source                1..107
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 285
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YYSYPLTFGA GTKVEIK 107

SEQ ID NO: 286        moltype = AA length = 107
FEATURE              Location/Qualifiers
REGION              1..107
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..107
                      note = PSMA-H_L1.58 Variable Light
source              1..107
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 286
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YQSYPLTFGA GTKVEIK 107

SEQ ID NO: 287        moltype = AA length = 107
FEATURE              Location/Qualifiers
REGION              1..107
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..107
                      note = PSMA-H_L1.59 Variable Light
source              1..107
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 287
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YESYPLTFGA GTKVEIK 107

SEQ ID NO: 288        moltype = AA length = 107
FEATURE              Location/Qualifiers
REGION              1..107
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..107
                      note = PSMA-H_L1.60 Variable Light
source              1..107
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 288
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YGSYPLTFGA GTKVEIK 107

SEQ ID NO: 289        moltype = AA length = 107
FEATURE              Location/Qualifiers
REGION              1..107
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..107
                      note = PSMA-H_L1.61 Variable Light
source              1..107
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 289
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YTSYPLTFGA GTKVEIK 107

SEQ ID NO: 290        moltype = AA length = 107
FEATURE              Location/Qualifiers
REGION              1..107
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..107
                      note = PSMA-H_L1.62 Variable Light
source              1..107
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 290
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSTPLTFGA GTKVEIK 107

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SEQ ID NO: 291      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.63 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 291
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSFPLTFGA GTKVEIK 107

SEQ ID NO: 292      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.64 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 292
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSWPLTFGA GTKVEIK 107

SEQ ID NO: 293      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.65 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 293
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSHPLTFGA GTKVEIK 107

SEQ ID NO: 294      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.66 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 294
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSQPLTFGA GTKVEIK 107

SEQ ID NO: 295      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.67 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 295
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSEPLTFGA GTKVEIK 107

SEQ ID NO: 296      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.68 Variable Light
source             1..107

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mol_type = protein
organism = synthetic construct

SEQUENCE: 296
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSKPLTFGA GTKVEIK 107

SEQ ID NO: 297      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.69 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 297
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSRPLTFGA GTKVEIK 107

SEQ ID NO: 298      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.70 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 298
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSGPLTFGA GTKVEIK 107

SEQ ID NO: 299      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.71 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 299
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPITFGA GTKVEIK 107

SEQ ID NO: 300      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.72 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 300
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPITFGA GTKVEIK 107

SEQ ID NO: 301      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.73 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 301
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPKTFGA GTKVEIK 107

SEQ ID NO: 302      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107

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note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION      1..107
note = PSMA-H_L1.74 Variable Light
source      1..107
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 302
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPRTFGA GTKVEIK 107

SEQ ID NO: 303      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION      1..107
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION      1..107
note = PSMA-H_L1.75 Variable Light
source      1..107
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 303
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPQTFGA GTKVEIK 107

SEQ ID NO: 304      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION      1..107
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION      1..107
note = PSMA-H_L1.76 Variable Light
source      1..107
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 304
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYY ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPQTFGA GTKVEIK 107

SEQ ID NO: 305      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION      1..107
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION      1..107
note = PSMA-H_L1.77 Variable Light
source      1..107
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 305
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TSVDWYQQKP DQSPKLLIYY ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 306      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION      1..107
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION      1..107
note = PSMA-H_L1.78 Variable Light
source      1..107
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 306
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYY ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ HNSYPLTFGA GTKVEIK 107

SEQ ID NO: 307      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION      1..107
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION      1..107
note = PSMA-H_L1.79 Variable Light
source      1..107
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 307
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYH ASTRHTGVDP 60

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RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPQTFGA GTKVEIK 107

SEQ ID NO: 308      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.80 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 308
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TSVDWYQQKP DQSPKLLIYH ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 309      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.81 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 309
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYH ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ HNSYPLTFGA GTKVEIK 107

SEQ ID NO: 310      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.82 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 310
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYT ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPQTFGA GTKVEIK 107

SEQ ID NO: 311      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.83 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 311
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TSVDWYQQKP DQSPKLLIYT ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIK 107

SEQ ID NO: 312      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = PSMA-H_L1.84 Variable Light
source             1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 312
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYT ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ HNSYPLTFGA GTKVEIK 107

SEQ ID NO: 313      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31618 Chain 1 -

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PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
Heavy Chain
source      1..444
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 313
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDS DGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 314      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31618 Chain 2 - PSMA-H_L1.1 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 314
DIVMTQSPDS LAVSLGERAT LSCRASNDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFI FPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 315      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31619 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 315
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDS DGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 316      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31619 Chain 2 - PSMA-H_L1.2 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 316
DIVMTQSPDS LAVSLGERAT LSCRASEDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFI FPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 317      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31620 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444

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mol_type = protein
organism = synthetic construct

SEQUENCE: 317
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGGTGL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLSDSGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 318      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION              1..214
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION              1..214
                    note = XENP31620 Chain 2 - PSMA-H_L1.3 Light Chain
source              1..214
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 318
DIVMTQSPDS LAVSLGERAT LSCRASTDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIAPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 319      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION              1..444
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION              1..444
                    note = XENP31621 Chain 1 -
                        PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                        Heavy Chain
source              1..444
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 319
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGGTGL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLSDSGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 320      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION              1..214
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION              1..214
                    note = XENP31621 Chain 2 - PSMA-H_L1.4 Light Chain
source              1..214
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 320
DIVMTQSPDS LAVSLGERAT LSCRASTDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIAPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 321      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION              1..444
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION              1..444
                    note = XENP31622 Chain 1 -
                        PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                        Heavy Chain
source              1..444
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 321

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EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDS DGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHAEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 322      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31622 Chain 2 - PSMA-H_L1.5 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 322
DIVMTQSPDS LAVSLGERAT LSCRASIDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFI FPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 323      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31623 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 323
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDS DGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHAEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 324      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31623 Chain 2 - PSMA-H_L1.6 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 324
DIVMTQSPDS LAVSLGERAT LSCRASQDVG SAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFI FPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 325      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31624 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 325
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180

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SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGGS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 326 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP31624 Chain 2 - PSMA-H_L1.7 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 326
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG NAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 327 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..444
 note = XENP31625 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 327
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
 PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGGS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 328 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP31625 Chain 2 - PSMA-H_L1.8 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 328
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG EAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 329 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..444
 note = XENP31626 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 329
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
 PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360

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SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGGS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 330 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP31626 Chain 2 - PSMA-H_L1.9 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 330
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG IAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 331 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..444
 note = XENP31627 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 331
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LTVSSASTKG 120
 PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGGS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 332 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP31627 Chain 2 - PSMA-H_L1.10 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 332
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TYVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 333 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..444
 note = XENP31628 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 333
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LTVSSASTKG 120
 PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGGS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

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SEQ ID NO: 334 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP31628 Chain 2 - PSMA-H_L1.11 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 334
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TSVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 335 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..444
 note = XENP31629 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 335
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LTVSSASTKG 120
 PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDS DGS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 336 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP31629 Chain 2 - PSMA-H_L1.12 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 336
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TTVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 337 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..444
 note = XENP31630 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 337
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LTVSSASTKG 120
 PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDS DGS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 338 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214

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note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..214
note = XENP31630 Chain 2 - PSMA-H_L1.13 Light Chain
source 1..214
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 338
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TIVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 339      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION 1..444
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..444
note = XENP31631 Chain 1 -
      PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
      Heavy Chain
source 1..444
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 339
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LVTSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVPEKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDS DGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 340      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..214
note = XENP31631 Chain 2 - PSMA-H_L1.14 Light Chain
source 1..214
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 340
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TGVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 341      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION 1..444
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..444
note = XENP31632 Chain 1 -
      PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
      Heavy Chain
source 1..444
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 341
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LVTSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVPEKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDS DGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 342      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..214

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source          note = XENP31632 Chain 2 - PSMA-H_L1.15 Light Chain
                1..214
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 342
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TEVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 343      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION            1..444
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..444
                  note = XENP31633 Chain 1 -
                  PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                  Heavy Chain
source            1..444
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 343
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 344      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP31633 Chain 2 - PSMA-H_L1.16 Light Chain
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 344
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TQVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 345      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION            1..444
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..444
                  note = XENP31634 Chain 1 -
                  PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                  Heavy Chain
source            1..444
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 345
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 346      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP31634 Chain 2 - PSMA-H_L1.17 Light Chain
source            1..214
                  mol_type = protein

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organism = synthetic construct

SEQUENCE: 346
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TALDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 347      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31635 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 347
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LVTSSASTKG 120
PSVFFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 348      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31635 Chain 2 - PSMA-H_L1.18 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 348
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAIDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 349      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31636 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 349
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LVTSSASTKG 120
PSVFFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 350      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31636 Chain 2 - PSMA-H_L1.19 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 350
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVAWYQQKP DQSPKLLIYW ASTRHTGVDP 60

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RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 351      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31637 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 351
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTL VTVSSASTKG 120
PSVFFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHAEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 352      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31637 Chain 2 - PSMA-H_L1.20 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 352
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIFW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 353      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31638 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 353
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTL VTVSSASTKG 120
PSVFFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHAEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 354      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31638 Chain 2 - PSMA-H_L1.21 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 354
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIQW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

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SEQ ID NO: 355 moltype = AA length = 444
FEATURE Location/Qualifiers
REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..444
 note = XENP31639 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 355
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGGQTL VTVSSASTKG 120
PSVFFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 356 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..214
 note = XENP31639 Chain 2 - PSMA-H_L1.22 Light Chain
source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 356
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIEW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 357 moltype = AA length = 444
FEATURE Location/Qualifiers
REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..444
 note = XENP31640 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 357
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGGQTL VTVSSASTKG 120
PSVFFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 358 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..214
 note = XENP31640 Chain 2 - PSMA-H_L1.23 Light Chain
source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 358
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIHW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 359 moltype = AA length = 444
FEATURE Location/Qualifiers

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REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..444
 note = XENP31641 Chain 1 - PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S Heavy Chain

source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 359
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LVTSSASTKG 120
 PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 360 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214
 note = XENP31641 Chain 2 - PSMA-H_L1.24 Light Chain

source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 360
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYY ASTRHTGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSPN RGEC 214

SEQ ID NO: 361 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..444
 note = XENP31642 Chain 1 - PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S Heavy Chain

source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 361
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LVTSSASTKG 120
 PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 362 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214
 note = XENP31642 Chain 2 - PSMA-H_L1.25 Light Chain

source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 362
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYF ASTRHTGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSPN RGEC 214

SEQ ID NO: 363 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide

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REGION 1..444
 note = XENP31643 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain

source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 363

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	LVTVSSASTKG	120
PSVFFPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSDTKVD	KKVEPKSCDK	THTCPPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCVV	VDVKHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EEYNSTYRVV	300
SVLTVLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	SREEMTKNQV	360
SLTCDVSGFY	PSDIAVEWES	DGQPENNYKT	TPPVLDSDGS	FFLYSKLTVD	KSRWEQGDVF	420
SCSVMHLEALH	NHYTQKSLSL	SPGK				444

SEQ ID NO: 364 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide

REGION 1..214
 note = XENP31643 Chain 2 - PSMA-H_L1.26 Light Chain

source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 364

DIVMTQSPDS	LAVSLGERAT	LSCRASQDVG	TAVDWYQQKP	DQSPKLLIYH	ASTRHTGVDP	60
RFTGSGSGTD	FTLTISLQA	EDVAVYFCQQ	YNSYPLTFGA	GTKVEIKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYEKHK	VYACEVTHQG	LSSPVTKSPN	RGEC			214

SEQ ID NO: 365 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide

REGION 1..444
 note = XENP31644 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain

source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 365

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	LVTVSSASTKG	120
PSVFFPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSDTKVD	KKVEPKSCDK	THTCPPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCVV	VDVKHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EEYNSTYRVV	300
SVLTVLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	SREEMTKNQV	360
SLTCDVSGFY	PSDIAVEWES	DGQPENNYKT	TPPVLDSDGS	FFLYSKLTVD	KSRWEQGDVF	420
SCSVMHLEALH	NHYTQKSLSL	SPGK				444

SEQ ID NO: 366 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide

REGION 1..214
 note = XENP31644 Chain 2 - PSMA-H_L1.27 Light Chain

source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 366

DIVMTQSPDS	LAVSLGERAT	LSCRASQDVG	TAVDWYQQKP	DQSPKLLIYQ	ASTRHTGVDP	60
RFTGSGSGTD	FTLTISLQA	EDVAVYFCQQ	YNSYPLTFGA	GTKVEIKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYEKHK	VYACEVTHQG	LSSPVTKSPN	RGEC			214

SEQ ID NO: 367 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide

REGION 1..444
 note = XENP31645 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S

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source          Heavy Chain
                1..444
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 367
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTI VTVSSASTKG 120
PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 368      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31645 Chain 2 - PSMA-H_L1.28 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 368
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYE ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 369      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31646 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 369
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTI VTVSSASTKG 120
PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 370      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31646 Chain 2 - PSMA-H_L1.29 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 370
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYT ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 371      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31647 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein

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organism = synthetic construct

SEQUENCE: 371
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NOKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LVTSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 372      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION              1..214
                    note = Description of Artificial Sequence: Synthetic
                        polypeptide
REGION              1..214
                    note = XENP31647 Chain 2 - PSMA-H_L1.30 Light Chain
source              1..214
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 372
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ISTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 373      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION              1..444
                    note = Description of Artificial Sequence: Synthetic
                        polypeptide
REGION              1..444
                    note = XENP31648 Chain 1 -
                        PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                        Heavy Chain
source              1..444
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 373
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NOKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LVTSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 374      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION              1..214
                    note = Description of Artificial Sequence: Synthetic
                        polypeptide
REGION              1..214
                    note = XENP31648 Chain 2 - PSMA-H_L1.31 Light Chain
source              1..214
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 374
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW LSTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 375      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION              1..444
                    note = Description of Artificial Sequence: Synthetic
                        polypeptide
REGION              1..444
                    note = XENP31649 Chain 1 -
                        PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                        Heavy Chain
source              1..444
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 375
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60

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NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTI VTVSSASTKG 120
 PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 376 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP31649 Chain 2 - PSMA-H_L1.32 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 376
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW TSTRHTGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 377 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..444
 note = XENP31650 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 377
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTI VTVSSASTKG 120
 PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 378 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP31650 Chain 2 - PSMA-H_L1.33 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 378
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASSRHTGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 379 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..444
 note = XENP31651 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 379
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTI VTVSSASTKG 120
 PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240

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PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 380 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP31651 Chain 2 - PSMA-H_L1.34 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 380
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASQRHTGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 381 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..444
 note = XENP31652 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 381
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LTVSSASTKG 120
 PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 382 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP31652 Chain 2 - PSMA-H_L1.35 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 382
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASERHTGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 383 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..444
 note = XENP31653 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 383
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LTVSSASTKG 120
 PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420

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SCSVMEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 384 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP31653 Chain 2 - PSMA-H_L1.36 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 384
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASNRHTGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSLSTLT 180
 LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 385 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..444
 note = XENP31654 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 385
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTL VTVSSASTKG 120
 PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 386 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP31654 Chain 2 - PSMA-H_L1.37 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 386
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASGRHTGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSLSTLT 180
 LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 387 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..444
 note = XENP31655 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 387
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTL VTVSSASTKG 120
 PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 388 moltype = AA length = 214

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FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP31655 Chain 2 - PSMA-H_L1.38 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 388
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRETGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

 SEQ ID NO: 389 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..444
 note = XENP31656 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 389
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGGQTL VTVSSASTKG 120
 PSVFLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSDGS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

 SEQ ID NO: 390 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP31656 Chain 2 - PSMA-H_L1.39 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 390
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHSGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

 SEQ ID NO: 391 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..444
 note = XENP31657 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 391
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGGQTL VTVSSASTKG 120
 PSVFLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSDGS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

 SEQ ID NO: 392 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic

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REGION          polypeptide
                  1..214
note = XENP31657 Chain 2 - PSMA-H_L1.40 Light Chain
source          1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 392
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHEGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEHKK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 393    moltype = AA length = 444
FEATURE          Location/Qualifiers
REGION          1..444
note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION          1..444
note = XENP31658 Chain 1 -
                  PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                  Heavy Chain
source          1..444
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 393
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSDGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMEALH NHYTQKLSL SPKG 444

SEQ ID NO: 394    moltype = AA length = 214
FEATURE          Location/Qualifiers
REGION          1..214
note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION          1..214
note = XENP31658 Chain 2 - PSMA-H_L1.41 Light Chain
source          1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 394
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHYGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEHKK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 395    moltype = AA length = 444
FEATURE          Location/Qualifiers
REGION          1..444
note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION          1..444
note = XENP31659 Chain 1 -
                  PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                  Heavy Chain
source          1..444
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 395
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSDGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMEALH NHYTQKLSL SPEK 444

SEQ ID NO: 396    moltype = AA length = 214
FEATURE          Location/Qualifiers
REGION          1..214
note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION          1..214
note = XENP31659 Chain 2 - PSMA-H_L1.42 Light Chain

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source          1..214
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 396
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCNQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 397      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION            1..444
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..444
                  note = XENP31660 Chain 1 -
                  PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                  Heavy Chain
source            1..444
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 397
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LVTSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGPENNYKT TPPVLDSGDS PFLYSKLTVD KSRWEQGDVF 420
SCSVMEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 398      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP31660 Chain 2 - PSMA-H_L1.43 Light Chain
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 398
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCEQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 399      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION            1..444
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..444
                  note = XENP31661 Chain 1 -
                  PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                  Heavy Chain
source            1..444
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 399
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LVTSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGPENNYKT TPPVLDSGDS PFLYSKLTVD KSRWEQGDVF 420
SCSVMEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 400      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP31661 Chain 2 - PSMA-H_L1.44 Light Chain
source            1..214
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 400
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCRQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 401      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31662 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 401
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLWGMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DQQPENNYKT TPPVLDSDGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 402      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31662 Chain 2 - PSMA-H_L1.45 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 402
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCRQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 403      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31663 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 403
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLWGMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DQQPENNYKT TPPVLDSDGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 404      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31663 Chain 2 - PSMA-H_L1.46 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 404
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCLQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120

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SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 405      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31664 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 405
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 406      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31664 Chain 2 - PSMA-H_L1.47 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 406
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCHQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 407      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31665 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 407
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 408      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31665 Chain 2 - PSMA-H_L1.48 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 408
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ FNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

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SEQ ID NO: 409      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31666 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 409
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 410      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31666 Chain 2 - PSMA-H_L1.49 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 410
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ WNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 411      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31667 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 411
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 412      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31667 Chain 2 - PSMA-H_L1.50 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 412
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ KNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 413      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION             1..444

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note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..444
note = XENP31668 Chain 1 -
      PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
      Heavy Chain
source 1..444
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 413
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 414      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..214
note = XENP31668 Chain 2 - PSMA-H_L1.51 Light Chain
source 1..214
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 414
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ RNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSK STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 415      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION 1..444
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..444
note = XENP31669 Chain 1 -
      PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
      Heavy Chain
source 1..444
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 415
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 416      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..214
note = XENP31669 Chain 2 - PSMA-H_L1.52 Light Chain
source 1..214
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 416
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ HNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSK STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 417      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION 1..444
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..444

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note = XENP31670 Chain 1 -
      PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
      Heavy Chain
source      1..444
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 417
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 418      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP31670 Chain 2 - PSMA-H_L1.53 Light Chain
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 418
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ QNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 419      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION            1..444
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..444
                  note = XENP31671 Chain 1 -
                  PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                  Heavy Chain
source            1..444
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 419
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 420      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP31671 Chain 2 - PSMA-H_L1.54 Light Chain
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 420
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ ENSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 421      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION            1..444
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..444
                  note = XENP31672 Chain 1 -
                  PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                  Heavy Chain

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source          1..444
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 421
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFPQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 422      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31672 Chain 2 - PSMA-H_L1.55 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 422
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ INSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 423      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31673 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 423
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFPQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 424      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31673 Chain 2 - PSMA-H_L1.56 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 424
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ LNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 425      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31674 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein
                   organism = synthetic construct

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SEQUENCE: 425
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFPQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 426      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP31674 Chain 2 - PSMA-H_L1.57 Light Chain
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 426
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YYSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 427      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION            1..444
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..444
                  note = XENP31675 Chain 1 -
                  PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                  Heavy Chain
source            1..444
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 427
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFPQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 428      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP31675 Chain 2 - PSMA-H_L1.58 Light Chain
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 428
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YQSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 429      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION            1..444
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..444
                  note = XENP31676 Chain 1 -
                  PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                  Heavy Chain
source            1..444
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 429
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFPQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120

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PSVFLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVTVTPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 430 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..214
 note = XENP31676 Chain 2 - PSMA-H_L1.59 Light Chain
source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 430
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YESYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 431 moltype = AA length = 444
FEATURE Location/Qualifiers
REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..444
 note = XENP31677 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 431
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVTVTPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 432 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..214
 note = XENP31677 Chain 2 - PSMA-H_L1.60 Light Chain
source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 432
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YGSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 433 moltype = AA length = 444
FEATURE Location/Qualifiers
REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..444
 note = XENP31678 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 433
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVTVTPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300

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SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHREALH NHYTQKSLSL SPGK 444

SEQ ID NO: 434      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31678 Chain 2 - PSMA-H_L1.61 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 434
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YTSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 435      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31679 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 435
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLLEWMGN INPNNGGTTY 60
NQKFPQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGGQTL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHREALH NHYTQKSLSL SPGK 444

SEQ ID NO: 436      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP31679 Chain 2 - PSMA-H_L1.62 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 436
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSTPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 437      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP31680 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                   Heavy Chain
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 437
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLLEWMGN INPNNGGTTY 60
NQKFPQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGGQTL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHREALH NHYTQKSLSL SPGK 444

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SEQ ID NO: 438 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP31680 Chain 2 - PSMA-H_L1.63 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 438
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSFPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 439 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..444
 note = XENP31681 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 439
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
 PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSDGS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 440 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP31681 Chain 2 - PSMA-H_L1.64 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 440
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSWPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 441 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..444
 note = XENP31682 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
 Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 441
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
 PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSDGS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 442 moltype = AA length = 214
 FEATURE Location/Qualifiers

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REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214
note = XENP31682 Chain 2 - PSMA-H_L1.65 Light Chain

source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 442
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSHPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 443 moltype = AA length = 444
FEATURE Location/Qualifiers
REGION 1..444
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..444
note = XENP31683 Chain 1 -
PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S Heavy Chain

source 1..444
mol_type = protein
organism = synthetic construct

SEQUENCE: 443
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVPEKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 444 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214
note = XENP31683 Chain 2 - PSMA-H_L1.66 Light Chain

source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 444
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSQPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 445 moltype = AA length = 444
FEATURE Location/Qualifiers
REGION 1..444
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..444
note = XENP31684 Chain 1 -
PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S Heavy Chain

source 1..444
mol_type = protein
organism = synthetic construct

SEQUENCE: 445
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVPEKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 446 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide

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REGION 1..214
 note = XENP31684 Chain 2 - PSMA-H_L1.67 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 446
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSEPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 447 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..444
 note = XENP31685 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 447
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGGQTL VTVSSASTKG 120
 PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 448 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP31685 Chain 2 - PSMA-H_L1.68 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 448
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSKPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 449 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..444
 note = XENP31686 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 449
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGGQTL VTVSSASTKG 120
 PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 450 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP31686 Chain 2 - PSMA-H_L1.69 Light Chain
 source 1..214

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mol_type = protein
organism = synthetic construct

SEQUENCE: 450
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSRPLTPGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 451      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION              1..444
                    note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..444
                    note = XENP31687 Chain 1 -
                      PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                      Heavy Chain
source              1..444
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 451
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NOKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGGQTL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 452      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION              1..214
                    note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..214
                    note = XENP31687 Chain 2 - PSMA-H_L1.70 Light Chain
source              1..214
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 452
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSGPLTPGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 453      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION              1..444
                    note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..444
                    note = XENP31688 Chain 1 -
                      PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S
                      Heavy Chain
source              1..444
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 453
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NOKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGGQTL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 454      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION              1..214
                    note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..214
                    note = XENP31688 Chain 2 - PSMA-H_L1.71 Light Chain
source              1..214
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 454

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DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPITFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 455 moltype = AA length = 444
FEATURE Location/Qualifiers
REGION 1..444
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..444
note = XENP31689 Chain 1 -
PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S Heavy Chain
source 1..444
mol_type = protein
organism = synthetic construct

SEQUENCE: 455
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 456 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..214
note = XENP31689 Chain 2 - PSMA-H_L1.72 Light Chain
source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 456
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPITFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 457 moltype = AA length = 444
FEATURE Location/Qualifiers
REGION 1..444
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..444
note = XENP31690 Chain 1 -
PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S Heavy Chain
source 1..444
mol_type = protein
organism = synthetic construct

SEQUENCE: 457
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 458 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..214
note = XENP31690 Chain 2 - PSMA-H_L1.73 Light Chain
source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 458
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPITFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180

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LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 459 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..444
 note = XENP31691 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 459
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
 PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNATKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 460 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP31691 Chain 2 - PSMA-H_L1.74 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 460
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPRTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 461 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..444
 note = XENP31692 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_PVA_/S267K /L368D/K370S Heavy Chain
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 461
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
 PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNATKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 462 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP31692 Chain 2 - PSMA-H_L1.75 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 462
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPQTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 463 moltype = AA length = 444

-continued

FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..444
 note = XENP14484 Chain 1 -
 PSMA-H_H1_IgG1_PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S

source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 463

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	VTVSSASTKG	120
PSVFLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSDTKVD	KKVEPKSCDK	THTCPPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCVV	VDVKHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EEYNSTYRVV	300
SVLTVLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	SREEMTKNQV	360
SLTCDVSGFY	PSDIAVEWES	DGQPENNYKT	TPPVLDSDGS	FFLYSKLTV	KSRWEQGDVF	420
SCSVMHLEALH	NHYTQKSLSL	SPGK				444

SEQ ID NO: 464 moltype = AA length = 485

FEATURE Location/Qualifiers
 REGION 1..485
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..485
 note = XENP14484 Chain 2 -
 [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_C220S/S364K/E357Q

source 1..485
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 464

EVQLVESGGG	LVQPGGSLRL	SCAASGFTFS	TYAMNWVRQA	PGKGLEWVGR	IRSKYNNYAT	60
YYADSVKGRF	TISRDDSKNT	LYLQMNSLRA	EDTAVYYCVR	HGNFGDSYVS	WFAYWGQGT	120
VTVSSGKPGS	GKPGSGKPGS	GKPGSQAVVT	QEPSLTVSPG	GTVTLTGCGS	TGAVTTSNYA	180
NWVQKPGKGS	PRGLIGGTNK	RAPGVPARFS	GSLLGKKAAL	TISGAQPEDE	ADYYCALWYS	240
NHWVFGGGTK	LTVLEPKSSD	KTHTCPPCPA	PPVAGPSVFL	PPKPKDTLM	ISRTPEVTCV	300
VVDVKHEDPE	VKFNWYVDGV	EVHNAKTKPR	EEQYNSTYRV	VSVLTVLHQD	WLNKEYKCK	360
VSNKALPAPI	EKTISKAKGQ	PREPQVYTL	PSREQMTKNQ	VKLTCLVKGF	YPSDIAVEWE	420
SNGQPENNYK	TTPPVLDSDG	SFFLYSKLTV	DKSRWQQGNV	FSCSVMHLEAL	HNHYTQKSL	480
LSPGK						485

SEQ ID NO: 465 moltype = AA length = 214

FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214
 note = XENP14484 Chain 3 - PSMA-H_L1 Light Chain

source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 465

DIVMTQSPDS	LAVSLGERAT	LSCRASQDVG	TAVDWYQQKP	DQSPKLLIYW	ASTRHTGVPD	60
RFTGSGSGTD	FTLTISLQA	EDVAVYFCQQ	YNSYPLTFGA	GTKVEIKRTV	AAPSVFIAPP	120
SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYKHKH	VYACEVTHQG	LSSPVTKSFN	RGEC			214

SEQ ID NO: 466 moltype = AA length = 444

FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..444
 note = XENP33755 Chain 1 -
 PSMA-H_H1_IgG1_PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S

source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 466

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	VTVSSASTKG	120
PSVFLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSDTKVD	KKVEPKSCDK	THTCPPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCVV	VDVKHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EEYNSTYRVV	300
SVLTVLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	SREEMTKNQV	360
SLTCDVSGFY	PSDIAVEWES	DGQPENNYKT	TPPVLDSDGS	FFLYSKLTV	KSRWEQGDVF	420
SCSVMHLEALH	NHYTQKSLSL	SPGK				444

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SEQ ID NO: 467      moltype = AA  length = 485
FEATURE            Location/Qualifiers
REGION             1..485
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..485
                   note = XENP33755 Chain 2 -
                   [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1
                   _PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/
                   E357Q
source             1..485
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 467
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWVRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGTL 120
VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSLTVSPG GTVTLTCGSS TGAVTTSNYA 180
NWVQKPGKS PRGLIGGTNK RAPGVPARFS GSLLGKKAAL TISGAQPEDE ADYYCALWYS 240
NHWVFGGGTK LTVLEPKSSD KTHTCPPCPA PPVAGPSVFL PPPKPKDTLM ISRTPEVTCV 300
VVDVKHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV VSVLTVLHQD WLNKEYKCK 360
VSNKALPAPI EKTISKAKGQ PREPQVYTL PIREQMTKNQ VKLTCLVKG FYPDI AVEWE 420
SNGQPENNYK TTPPVLDSDG SFFLYSKLTV DKSRWQQGNV FSCSVMHEAL HNHYTQKSL 480
LSPGK 485

SEQ ID NO: 468      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP33755 Chain 3 - PSMA-H_L1.58 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 468
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YQSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 469      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP33756 Chain 1 -
                   PSMA-H_H1_IgG1_PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 469
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NKKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVTVTPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNKKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGPENNYKT TPPVLDSDGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 470      moltype = AA  length = 485
FEATURE            Location/Qualifiers
REGION             1..485
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..485
                   note = XENP33756 Chain 2 -
                   [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1
                   _PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/
                   E357Q
source             1..485
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 470
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWVRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGTL 120
VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSLTVSPG GTVTLTCGSS TGAVTTSNYA 180

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NWVQKPGKGS PRGLIGGTNK RAPGVPARFS GSSLGGKAAL TISGAQPEDE ADYYCALWYS 240
NHWFPGGGTK LTVLEPKSSD KTHTCPPCPA PPVAGPSVFL FPPKPKDTLM ISRTPEVTCV 300
VVDVKHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV VSVLTVLHQD WLNKEYKCK 360
VSNKALPAPI EKTISKAKGQ PREPQVYTL PPSREQMTKNQ VKLTCLVKGF YPSDIAVEWE 420
SNGQPENNYK TPPVLDSGD SFPLYSKLTV DKSRWQQGNV FSCVMHEAL HNHYTQKSL 480
LSPGK 485

SEQ ID NO: 471      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP33756 Chain 3 - PSMA-H_L1.11 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 471
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TSVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTSGSGSDT FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 472      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP33757 Chain 1 -
                   PSMA-H_H1_IgG1_PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 472
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NOKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LVTSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVTVTPSSS LGTQTYICNV NHKPSDTKVD KKEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCV VDVKHEDPEV KFNWYVDGVE VHNNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DQQPENNYKT TPPVLDSGD FFLYSKLTV DKSWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 473      moltype = AA length = 485
FEATURE            Location/Qualifiers
REGION             1..485
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..485
                   note = XENP33757 Chain 2 -
                   [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1
                   _PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/
                   E357Q
source             1..485
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 473
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGT L 120
VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSLTVSPG GTVTLTCGSS TGAVTTSNYA 180
NWVQKPGKGS PRGLIGGTNK RAPGVPARFS GSSLGGKAAL TISGAQPEDE ADYYCALWYS 240
NHWFPGGGTK LTVLEPKSSD KTHTCPPCPA PPVAGPSVFL FPPKPKDTLM ISRTPEVTCV 300
VVDVKHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV VSVLTVLHQD WLNKEYKCK 360
VSNKALPAPI EKTISKAKGQ PREPQVYTL PPSREQMTKNQ VKLTCLVKGF YPSDIAVEWE 420
SNGQPENNYK TPPVLDSGD SFPLYSKLTV DKSRWQQGNV FSCVMHEAL HNHYTQKSL 480
LSPGK 485

SEQ ID NO: 474      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP33757 Chain 3 - PSMA-H_L1.24 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 474

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DIVMTQSPDS	LAVSLGERAT	LSCRASQDVG	TAVDWYQQKP	DQSPKLLIY	ASTRHTGVPD	60
RFTGSGSGTD	FTLTISLQA	EDVAVYFCQQ	YNSYPLTFGA	GTKVEIKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLLNNFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYEKKH	VYACEVTHQG	LSSPVTKSFN	RGEC			214

SEQ ID NO: 475 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..444
 note = XENP33758 Chain 1 -
 PSMA-H_H1_IgG1_PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 475

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGTL	VTVSSASTKG	120
PSVFFPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSDTKVD	KKVEPKSCDK	THTCPPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCVV	VDVKHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EEYNSTYRVV	300
SVLTVLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	SREEMTKNQV	360
SLTCDVSGFY	PSDIAVEWES	DGQPENNYKT	TPPVLDSDGS	FFLYSKLTV	KSRWEQGDVF	420
SCSVMHEALH	NHYTQKSLSL	SPGK				444

SEQ ID NO: 476 moltype = AA length = 485
 FEATURE Location/Qualifiers
 REGION 1..485
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..485
 note = XENP33758 Chain 2 -
 [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1
 PVA/S267K_pI(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/
 E357Q
 source 1..485
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 476

EVQLVESGGG	LVQPGGSLRL	SCAASGFTFS	TYAMNWRQA	PGKGLEWVGR	IRSKYNNYAT	60
YYADSVKGRF	TISRDDSKNT	LYLQMNLSLR	EDTAVYVCVR	HGNFGDSYVS	WFAYWGQGTL	120
VTVSSGKPGS	GKPGSGKPGS	GKPGSQAVVT	QEPSTVSPG	GTVTLTCGSS	TGAVTTSNYA	180
NWVQQKPGKS	PRGLIGGTNK	RAPGVPARFS	GSLGGKAAL	TISGAQPEDE	ADYYCALWYS	240
NHWVFGGGTK	LTVLEPKSSD	KTHTCPPCPA	PPVAGPSVFL	FPPKPKDTLM	ISRTPEVTCV	300
VVDVKHEDPE	VKFNWYVDGV	EVHNAKTKPR	EEQYNSTYRV	VSVLTVLHQD	WLNGKEYKCK	360
VSNKALPAPI	EKTISKAKGQ	PREPQVYTL	PSREQMTKNQ	VKLTCVLKGF	YPSDIAVEWE	420
SNGQPENNYK	TPPVLDSDG	SFFLYSKLTV	DKSRWQQGNV	FSCSVMHEAL	HNHYTQKSLS	480
LSPGK						485

SEQ ID NO: 477 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP33758 Chain 3 - PSMA-H_L1.26 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 477

DIVMTQSPDS	LAVSLGERAT	LSCRASQDVG	TAVDWYQQKP	DQSPKLLIYH	ASTRHTGVPD	60
RFTGSGSGTD	FTLTISLQA	EDVAVYFCQQ	YNSYPLTFGA	GTKVEIKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLLNNFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYEKKH	VYACEVTHQG	LSSPVTKSFN	RGEC			214

SEQ ID NO: 478 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..444
 note = XENP33759 Chain 1 -
 PSMA-H_H1_IgG1_PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 478

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
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NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
 PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGGS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 479 moltype = AA length = 485
 FEATURE Location/Qualifiers
 REGION 1..485
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..485
 note = XENP33759 Chain 2 -
 [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1
 PVA/S267K_pI(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/
 E357Q
 source 1..485
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 479
 EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT 60
 YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGT L 120
 VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSLTVSPG GTVTLTCGSS TGAVTTSNYA 180
 NWVQQKPGKS PRGLIGGTNK RAPGVPARFS GSLGGKAAL TISGAQPEDE ADYYCALWYS 240
 NHWVFGGGTK LTVLEPKSSD KTHTCPPCPA PPVAGPSVFL FPPKPKDTLM ISRTPEVTCV 300
 VVDVKHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV VSVLTVLHQD WLNGKEYKCK 360
 VSNKALPAPI EKTISKAKGQ PREPQVYTL PPSREQMTKNQ VKLTCLVKGF YPSDIAVEWE 420
 SNGQPENNYK TTPVLDSGDS SFPLYSKLTV DKSRWQQGNV FSCVMHEAL HHYTQKSLSL 480
 LSPGK 485

SEQ ID NO: 480 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP33759 Chain 3 - PSMA-H_L1.75 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 480
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
 RFTGSGSGTD FTLTISLQA EDVAVYFCQQ YNSYPQTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNPFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSLSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 481 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..444
 note = XENP33760 Chain 1 -
 PSMA-H_H1_IgG1_PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 481
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
 PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGGS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 482 moltype = AA length = 485
 FEATURE Location/Qualifiers
 REGION 1..485
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..485
 note = XENP33760 Chain 2 -
 [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1
 PVA/S267K_pI(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/
 E357Q

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source          1..485
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 482
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGTL 120
VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSTLVSPG GTVTLTCGSS TGAVTTSNYA 180
NWVQKPKGKS PRGLIGGTNK RAPGVPARFS GSLLGKKAAL TISGAQPEDE ADYYCALWYS 240
NHWVFGGGTK LTVLEPKSSD KTHTCPPCPA PPVAGPSVFL FPPKPKDTLM ISRTPEVTCV 300
VVDVKHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV VSVLTVLHQD WLNKEYKCK 360
VSNKALPAPI EKTISKAKGQ PREPQVYTL PPREQMTKNQ VKLTCLVKGF YPSDIAVEWE 420
SNGQPENNYK TTPPVLDSDG SFFLYSKLTV DKSRWQQGNV FSCSVMEAL HNHYTQKSLS 480
LSPGK                                             485

SEQ ID NO: 483      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP33760 Chain 3 - PSMA-H_L1.68 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 483
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISLQA EDVAVYFCQQ YNSKPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSLSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC                                             214

SEQ ID NO: 484      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP33761 Chain 1 -
                   PSMA-H_H1_IgG1_PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S
source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 484
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLWGMN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVLPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNKATKPRE EEYNSTYRVV 300
SVLTVLHQDW LNKKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DQQPENNYKT TPPVLDSDGS FFLYSLKLTVD KSRWEQGDVF 420
SCSVMEALH HNHYTQKSLS SPGK                                             444

SEQ ID NO: 485      moltype = AA length = 485
FEATURE            Location/Qualifiers
REGION             1..485
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..485
                   note = XENP33761 Chain 2 -
                   [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1
                   PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/
                   E357Q
source             1..485
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 485
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGTL 120
VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSTLVSPG GTVTLTCGSS TGAVTTSNYA 180
NWVQKPKGKS PRGLIGGTNK RAPGVPARFS GSLLGKKAAL TISGAQPEDE ADYYCALWYS 240
NHWVFGGGTK LTVLEPKSSD KTHTCPPCPA PPVAGPSVFL FPPKPKDTLM ISRTPEVTCV 300
VVDVKHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV VSVLTVLHQD WLNKEYKCK 360
VSNKALPAPI EKTISKAKGQ PREPQVYTL PPREQMTKNQ VKLTCLVKGF YPSDIAVEWE 420
SNGQPENNYK TTPPVLDSDG SFFLYSKLTV DKSRWQQGNV FSCSVMEAL HNHYTQKSLS 480
LSPGK                                             485

SEQ ID NO: 486      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic

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REGION          polypeptide
                1..214
note = XENP33761 Chain 3 - PSMA-H_L1.29 Light Chain
source          1..214
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 486
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYT ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 487      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION            1..444
note = Description of Artificial Sequence: Synthetic
                polypeptide
REGION            1..444
note = XENP33762 Chain 1 -
                PSMA-H_H1_IgG1_PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S
source            1..444
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 487
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFPQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGPENNYKT TPPVLDSGGS FFLYSKLTV D KSRWEQGDVF 420
SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 488      moltype = AA length = 485
FEATURE            Location/Qualifiers
REGION            1..485
note = Description of Artificial Sequence: Synthetic
                polypeptide
REGION            1..485
note = XENP33762 Chain 2 -
                [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1
                _PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/
                E357Q
source            1..485
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 488
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWVRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGT L 120
VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSTLTVSPG GTVTLTCGSS TGAVTTSNYA 180
NHWVQKPGKS PRGLIGGTNK RAPGVPARFS GSLLGKKAAL TISGAQPEDE ADYYCALWYS 240
NHWVFGGGTK LTVLEPKSSD KTHTCPPCPA PPVAGPSVFL FPPKPKDTLM ISRTPEVTCV 300
VVDVKHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV VSVLTVLHQD WLNKEYKCK 360
VSNKALPAPI EKTISKAKGQ PREPQVYTL P SREQMTKNQ VKLTCLVKGF YPSDIAVEWE 420
SNGQPENNYK TPPVLDSG SFFLYSKLTV D KSRWQQGNV FSCSVMHEAL HNHYTQKSLS 480
LSPGK 485

SEQ ID NO: 489      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
note = Description of Artificial Sequence: Synthetic
                polypeptide
REGION            1..214
note = XENP33762 Chain 3 - PSMA-H_L1.52 Light Chain
source            1..214
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 489
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ HNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 490      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION            1..444
note = Description of Artificial Sequence: Synthetic
                polypeptide
REGION            1..444

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note = XENP34234 Chain 1 -
PSMA-H_H1_IgG1_PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S

source 1..444
mol_type = protein
organism = synthetic construct

SEQUENCE: 490

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	LVSSASTKG	120
PSVFPPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSDTKVD	KKVEPKSCDK	THTCPPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCVV	VDVKHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EEYNSTYRVV	300
SVLTVLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	SREEMTKNQV	360
SLTCDVSGFY	PSDIAVEWES	DGQPENNYKT	TPPVLDSDGS	FFLYSKLTVD	KSRWEQGDVF	420
SCSVMHEALH	NHYTQKSLSL	SPGK				444

SEQ ID NO: 491 moltype = AA length = 485
FEATURE Location/Qualifiers
REGION 1..485
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..485
note = XENP34234 Chain 2 -
[CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/E357Q

source 1..485
mol_type = protein
organism = synthetic construct

SEQUENCE: 491

EVQLVESGGG	LVQPGGSLRL	SCAASGFTFS	TYAMNWRQA	PGKGLEWVGR	IRSKYNNYAT	60
YYADSVKGRF	TISRDDSKNT	LYLQMNSLRA	EDTAVYYCVR	HGNFGDSYVS	WFAYWGQGT	120
VTVSSGKPGS	GKPGSGKPGS	GKPGSQAVVT	QEPSTLTVSPG	GTVTLTGCGS	TGAVTTSNYA	180
NWVQQKPGKS	PRGLIGGTNK	RAPGVPARFS	GSLLGKKAAL	TISGAQPEDE	ADYYCALWYS	240
NHWVFGGGTK	LTVLEPKSSD	KHTTCPPCPA	PPVAGPSVFL	FPPKPKDTLM	ISRTPEVTCV	300
VVDVKHEDPE	VKFNWYVDGV	EVHNAKTKPR	EEQYNSTYRV	VSVLTVLHQD	WLNGKEYKCK	360
VSNKALPAPI	EKTISKAKGQ	PREPQVYTL	PSREQMTKNQ	VKLTCLVKGF	YPSDIAVEWE	420
SNQCPENNYK	TPPVLDSDG	SFFLYSKLTV	DKSRWQQGNV	FSCSVMHEAL	HNHYTQKSLS	480
LSPGK						485

SEQ ID NO: 492 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214
note = XENP34234 Chain 3 - PSMA-H_L1.78 Light Chain

source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 492

DIVMTQSPDS	LAVSLGERAT	LSCRASQDVG	TAVDWYQQKP	DQSPKLLIYY	ASTRHTGVDP	60
RFTGSGSGTD	FTLTISLQA	EDVAVYFCQQ	HNSYPLTFGA	GTKVEIKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSLSTLT	180
LSKADYEKKH	VYACEVTHQG	LSSPVTKSFN	RGEC			214

SEQ ID NO: 493 moltype = AA length = 444
FEATURE Location/Qualifiers
REGION 1..444
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..444
note = XENP34235 Chain 1 -
PSMA-H_H1_IgG1_PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S

source 1..444
mol_type = protein
organism = synthetic construct

SEQUENCE: 493

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	LVSSASTKG	120
PSVFPPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSDTKVD	KKVEPKSCDK	THTCPPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCVV	VDVKHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EEYNSTYRVV	300
SVLTVLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	SREEMTKNQV	360
SLTCDVSGFY	PSDIAVEWES	DGQPENNYKT	TPPVLDSDGS	FFLYSKLTVD	KSRWEQGDVF	420
SCSVMHEALH	NHYTQKSLSL	SPGK				444

SEQ ID NO: 494 moltype = AA length = 485
FEATURE Location/Qualifiers
REGION 1..485

-continued

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note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..485
note = XENP34235 Chain 2 -
      [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1
      _PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/
      E357Q
source 1..485
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 494
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGTL 120
VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSLTVSPG GTVTLTCGSS TGAVTTSNYA 180
NWVQQKPGKS PRGLIGGTNK RAPGVPARFS GSLLGKKAAL TISGAQPEDE ADYYCALWYS 240
NHWVFGGGTK LTVLEPKSSD KTHTCPPCPA PPVAGPSVFL FPPKPKDTLM ISRTPEVTCV 300
VVDVKHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV VSVLTVLHQD WLNKEYKCK 360
VSNKALPAPI EKTISKAKGQ PREPQVYTL PPREQMTKNQ VKLTCLVKGF YPSDIAVEWE 420
SNGQPENNYK TTPVLDSDG SFPLYSKLTV DKSRWQQGNV FSCSVMEAL HNHYTQKSLS 480
LSPGK 485

SEQ ID NO: 495      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..214
note = XENP34235 Chain 3 - PSMA-H_L1.81 Light Chain
source 1..214
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 495
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYH ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ HNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 496      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION 1..444
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..444
note = XENP34236 Chain 1 -
      PSMA-H_H1_IgG1_PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S
source 1..444
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 496
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFKQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTL VTVSSASTKG 120
PSVFLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWSNGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCV VDKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNKKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSDGS FFLYSLKLTVD KSRWEQGDVF 420
SCSVMEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 497      moltype = AA length = 485
FEATURE            Location/Qualifiers
REGION 1..485
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..485
note = XENP34236 Chain 2 -
      [CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1
      _PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S-IgG1_C220S/S364K/
      E357Q
source 1..485
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 497
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGTL 120
VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSLTVSPG GTVTLTCGSS TGAVTTSNYA 180
NWVQQKPGKS PRGLIGGTNK RAPGVPARFS GSLLGKKAAL TISGAQPEDE ADYYCALWYS 240
NHWVFGGGTK LTVLEPKSSD KTHTCPPCPA PPVAGPSVFL FPPKPKDTLM ISRTPEVTCV 300
VVDVKHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV VSVLTVLHQD WLNKEYKCK 360
VSNKALPAPI EKTISKAKGQ PREPQVYTL PPREQMTKNQ VKLTCLVKGF YPSDIAVEWE 420

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SNGQPENNYK TTPVLDSDG SFLYSLKLTV DKSRWQQGNV FSCSVMEAL HNHYTQKSLS 480
LSPGK 485

SEQ ID NO: 498 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..214
 note = XENP34236 Chain 3 - PSMA-H_L1.84 Light Chain
source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 498
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYT ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ HNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 499 moltype = AA length = 444
FEATURE Location/Qualifiers
REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..444
 note = XENP16873 Chain 1 -
 PSMA-H_H1_IgG1_PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S
source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 499
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NOKFQGRVTI TVDKSTSTAY MELSSSLRSED TAVYYCAAGW NFDYWGQGT LVTSSASTKG 120
PSVFLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSDGS FFLYSLKLTVD KSRWEQGDVF 420
SCSVMEALH HNHYTQKSLS SPGK 444

SEQ ID NO: 500 moltype = AA length = 485
FEATURE Location/Qualifiers
REGION 1..485
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..485
 note = XENP16873 Chain 2 -
 [CD3]_H1.33_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_
 C220S/S364K/E357Q
source 1..485
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 500
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFDYWGQGT L 120
VTSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSLTVSPG GTVTLTCGSS TGAVTTSNYA 180
NHWVQKPGKS PRGLIGGTNK RAPGVPARFS GSLLGKKAAL TISGAQPEDE ADYICALWYS 240
NHWVFGGGTK LTVLEPKSSD KTHTCPPCPA PPVAGPSVFL FPPKPKDTLM ISRTPEVTCV 300
VVDVKHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV VSVLTVLHQD WLNKEYKCK 360
VSNKALPAPI EKTISKAKGQ PREPQVYTLPP PSREMTKNQ VKLTCLVKGF YPSDIAVEWE 420
SNGQPENNYK TTPVLDSDG SFLYSLKLTV DKSRWQQGNV FSCSVMEAL HNHYTQKSLS 480
LSPGK 485

SEQ ID NO: 501 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..214
 note = XENP16873 Chain 3 - PSMA-H_L1 Light Chain
source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 501
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

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SEQ ID NO: 502 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..444
 note = XENP16874 Chain 1 -
 PSMA-H_H1_IgG1_PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 502

EVQLVQSGAE	VKKPGASVKV	CKTS	SGYTFT	EYTIHWVRQA	PGQ	SLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSS	LRSED	TAVYYCAAGW	NFDY	WGQGT	LVTVSSASTKG	120
PSVFPPLAPSS	KSTSGGTAAL	GCLVKD	YFPE	PVTVSWNSGA	LTSG	VHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSD	TKVD	KKVEPKSCDK	THTC	PPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCVV	VDVKHED	PEV	KFNWYVDGVE	VHNA	KTTPRE	EEYNSTYRVV	300
SVLTVLHQDW	LNGKEYKCKV	SNKALP	APIE	KTISKAKGQP	REPQ	VYTLPP	SREEMTKNQV	360
SLTCDVSGFY	PSDIAVEWES	DGQPE	NNYKT	TPPVLDSDGS	FPLY	SKLTV	KSRWEQGDVF	420
SCSVMHEALH	NHYTQKSLSL	SPGK						444

SEQ ID NO: 503 moltype = AA length = 485
 FEATURE Location/Qualifiers
 REGION 1..485
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..485
 note = XENP16874 Chain 2 -
 [CD3]_H1.31_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_
 C220S/S364K/E357Q
 source 1..485
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 503

EVQLVESGGG	LVQPGGSLRL	SCAASG	FTFS	TYAMSWVRQA	PGK	GLEWVGR	IRSKYNNYAT	60
YYADSVKGRF	TISRDDSKNT	LYLQM	NLSLRA	EDTAVYYCVR	HGN	FGDSYVS	WFAYWGQGT	120
VTVSSGKPGS	GKPGSGKPGS	GKPGSQ	AVVT	QEP	SLTVSPG	GT	VTLTGSS	180
NWVQKPKGKS	PRGLIGGTNK	RAPGV	PARFS	GSL	LGGKAAL	TISGA	QPEDE	240
NHWVGGG	GTK	LTVLE	PKSSD	KTH	TCPPCPA	PPV	AGPSVFL	300
VVDVKHEDPE	VKFNWYVDGV	EVHNA	KTTPRE	EEQ	YNSTYRV	VSV	LTVLHQD	360
VSNKALPAPI	EKTISKAKGQ	PREPQ	VYTLPP	PSR	EQMTKNQ	VKL	TCLVKGF	420
SNGQPENNYK	TTPPVLDSDG	SFPLY	SKLTV	DKSR	WQGNV	FSC	SVMHEAL	480
LSPGK								485

SEQ ID NO: 504 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP16874 Chain 3 - PSMA-H_L1 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 504

DIVMTQSPDS	LAVSLGERAT	LSCRASQ	DVG	TAVDWYQQKP	DQSP	KLLIYW	ASTRHTGVPD	60
RFTGSGSGTD	FTLTISLQA	EDVAVY	FCQQ	YNSYPLT	FGA	GTKVEIK	RTV	120
SDEQLKSGTA	SVVCLNNFY	PREAKV	QWKV	DNALQ	SGNSQ	ESVTE	QDSKD	180
LSKADYEKHK	VYACEVTHQG	LSSPV	TKSFN	RGEC				214

SEQ ID NO: 505 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..444
 note = XENP19722 Chain 1 -
 PSMA-H_H1_IgG1_PVA_/S267K_pI(-)_Isosteric_A_L368D/K370S
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 505

EVQLVQSGAE	VKKPGASVKV	CKTS	SGYTFT	EYTIHWVRQA	PGQ	SLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSS	LRSED	TAVYYCAAGW	NFDY	WGQGT	LVTVSSASTKG	120
PSVFPPLAPSS	KSTSGGTAAL	GCLVKD	YFPE	PVTVSWNSGA	LTSG	VHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSD	TKVD	KKVEPKSCDK	THTC	PPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCVV	VDVKHED	PEV	KFNWYVDGVE	VHNA	KTTPRE	EEYNSTYRVV	300
SVLTVLHQDW	LNGKEYKCKV	SNKALP	APIE	KTISKAKGQP	REPQ	VYTLPP	SREEMTKNQV	360
SLTCDVSGFY	PSDIAVEWES	DGQPE	NNYKT	TPPVLDSDGS	FPLY	SKLTV	KSRWEQGDVF	420

-continued

SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 506 moltype = AA length = 485
 FEATURE Location/Qualifiers
 REGION 1..485
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..485
 note = XENP19722 Chain 2 -
 [CD3]_H1.32_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_ C220S/S364K/E357Q
 source 1..485
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 506
 EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWVRQA PGKGLEWVGR IRSKANNYAT 60
 YYADSVKGRF TISRDDSNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGT 120
 VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSLTVSPG GTVTLTCGSS TGAVTTSNYA 180
 NWVQKPGKS PRGLIGGTNK RAPGVPARFS GSLLGGKAL TISGAQPEDE ADYICALWYS 240
 NHWVFGGGTK LTVLEPKSSD KTHTCPPCPA PPVAGPSVFL PPPKPKDTLM ISRTPEVTCV 300
 VVDVKHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV VSVLTVLHQD WLNKEYKCK 360
 VSNKALPAPI EKTISKAKGQ PREPQVYTL PAREQMTKNQ VKLTCLVKG FYPDI AVEWE 420
 SNGQPENNYK TTPPVLDSDG SFFLYSKLTV DKSRWQGGNV FSCSVMHLEAL HHYTQKSLSL 480
 LSPGK 485

SEQ ID NO: 507 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP19722 Chain 3 - PSMA-H_L1 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 507
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTPGA GTKVEIKRTV AAPSVPFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 508 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..444
 note = XENP31602 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 508
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT VTVSSASTKG 120
 PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKEPKKCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNKKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGPENNYKT TPPVLDSDGS FFLYSKLTV DKSREWGQDVF 420
 SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 509 moltype = AA length = 717
 FEATURE Location/Qualifiers
 REGION 1..717
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..717
 note = XENP31602 Chain 2 -
 PSMA-H_H1_CH1_(G4S)2_[CD3]_H1.30_L1.47_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q
 source 1..717
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 509
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT VTVSSASTKG 120
 PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSNTKVD KKEPKKSCGG GSGGGGSEV QLVESGGGLV 240

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QPGGSLRLSC AASGFTFSTY AMNWRQAPG KGLEWVGRIR SKYNNYATYY ADSVKGRFTI 300
SRDDSKNTLY LQMNSLRAED TAVYYCVRHG NFGDSYVSWF AYWGQGTLLVT VSSGKPGSGK 360
PGSGKPGSGK PGSAVVTQE PSLTVSPGGT VTLTCGSSTG AVTTSNYANW VQKPGKSPR 420
GLIGGTNKRRA PGVPARFSGS LLGKKAALTI SGAQPEDEAD YYCALWYSNH WVFGGGTKLT 480
VLGGGGSGGG GSKTHTCPPC PAPPVAGPSV FLFPPKPKDT LMISRTPEVT CVVVDVKHED 540
PEVKFNWYVD GVEVHNAKTK PREEQYNSTY RVVSVLTVLH QDWLNGKEYK CKVSNKALPA 600
PIEKTISKAK GQPREPQVYT LPPSRQMTK NQVKLTCVLK GFYPSDIAVE WESNGQPENN 660
YKTTTPPVLD DGSFFLYSKL TVDKSRWQQG NVFSCSVME ALHNHYTQKS LSLSPGK 717

SEQ ID NO: 510 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..214
 note = XENP31602 Chain 3 - PSMA-H_L1 Light Chain
source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 510
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 511 moltype = AA length = 444
FEATURE Location/Qualifiers
REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..444
 note = XENP31603 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 511
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMEALH NHYTKSLSL SPGK 444

SEQ ID NO: 512 moltype = AA length = 717
FEATURE Location/Qualifiers
REGION 1..717
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..717
 note = XENP31603 Chain 2 -
 PSMA-H_H1_CH1_(G4S)2_[CD3]_H1.32_L1.47_scFv
 (KPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q
source 1..717
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 512
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCG GSGGGGSEV QLVESGGGLV 240
QPGGSLRLSC AASGFTFSTY AMNWRQAPG KGLEWVGRIR SKANNYATYY ADSVKGRFTI 300
SRDDSKNTLY LQMNSLRAED TAVYYCVRHG NFGDSYVSWF AYWGQGTLLVT VSSGKPGSGK 360
PGSGKPGSGK PGSAVVTQE PSLTVSPGGT VTLTCGSSTG AVTTSNYANW VQKPGKSPR 420
GLIGGTNKRRA PGVPARFSGS LLGKKAALTI SGAQPEDEAD YYCALWYSNH WVFGGGTKLT 480
VLGGGGSGGG GSKTHTCPPC PAPPVAGPSV FLFPPKPKDT LMISRTPEVT CVVVDVKHED 540
PEVKFNWYVD GVEVHNAKTK PREEQYNSTY RVVSVLTVLH QDWLNGKEYK CKVSNKALPA 600
PIEKTISKAK GQPREPQVYT LPPSRQMTK NQVKLTCVLK GFYPSDIAVE WESNGQPENN 660
YKTTTPPVLD DGSFFLYSKL TVDKSRWQQG NVFSCSVME ALHNHYTQKS LSLSPGK 717

SEQ ID NO: 513 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..214
 note = XENP31603 Chain 3 - PSMA-H_L1 Light Chain

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source          1..214
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 513
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSLSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSPN RGEC 214

SEQ ID NO: 514      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION            1..444
                  note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION            1..444
                  note = XENP31855 Chain 1 -
                    PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
source            1..444
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 514
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 515      moltype = AA length = 717
FEATURE            Location/Qualifiers
REGION            1..717
                  note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION            1..717
                  note = XENP31855 Chain 2 -
                    PSMA-H_H1_CH1_(G4S)2_Fc(222)_IgG1_PVA_/S267K /S364K/E357Q
source            1..717
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 515
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GSGGGGSQA VVTQEPSLTV 240
SPGGTVTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFSGSLLGGK 300
AALTISGAQP EDEADYYCAL WYNNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGGSRLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KANNYATYYA 420
DSVKGRTTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDSYVSWFA YWQGT LVTV 480
SGGGGGSGGG GSKTHTCPPC PAPPVAGPSV FLFPPKPKDT LMISRTPEVT CVVVDVKHED 540
PEVKFNWYVD GVEVHNAKTK PREEQYNSTY RVSVLTVLH QDWLNGKEYK CKVSNKALPA 600
PIEKTISKAK GQPREPQVYT LPPSREQMTK NQVKLTCLVK GFYPSDIAVE WESNGQPENN 660
YKTTTPVLDS DGSFFLYSKL TVDKSRWQQG NVFSCSVME ALHNHYTQKS LSLSPGK 717

SEQ ID NO: 516      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION            1..214
                  note = XENP31855 Chain 3 - PSMA-H_L1 Light Chain
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 516
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSLSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSPN RGEC 214

SEQ ID NO: 517      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION            1..444
                  note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION            1..444
                  note = XENP32218 Chain 1 -
                    PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S

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source          1..444
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 517
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFPQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKKCV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMEALHL NHYTQKSLSL SPGK 444

SEQ ID NO: 518      moltype = AA length = 717
FEATURE            Location/Qualifiers
REGION             1..717
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..717
                   note = XENP32218 Chain 2 -
                   PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv
                   (GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q

source             1..717
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 518
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFPQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GSGGGGGSQA VVTQEPSLTV 240
SPGGTVTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFGSGLLGK 300
AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGGSLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KANNYATYYA 420
DSVKGRTTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDSYVSWFA YWGQGLTVTV 480
SSGGGGSGGG GSKTHTCPPC PAPPVAGPSV FLFPPKPKDT LMISRTPEVT CVVVDVKHED 540
PEVKFNWYVD GVEVHNAKTK PREEQYNSTY RVSVTLTVLH QDWLNGKEYK CKVSNKALPA 600
PIEKTISKAK GQPREPQVYT LPPSREQMTK NQVKLTCLVK GFYPSDIAVE WESNGQPENN 660
YKTPPVVLDS DGSFFLYSKL TVDKSRWQQG NVFSCSVME ALHNHYTQKS LSLSPGK 717

SEQ ID NO: 519      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP32218 Chain 3 - PSMA-H_L1.58 Light Chain

source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 519
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YQSYPLTFGA GTKVEIKRTV AAPSVFIAPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSLSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 520      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP32219 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S

source             1..444
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 520
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFPQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKKCV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMEALHL NHYTQKSLSL SPGK 444

SEQ ID NO: 521      moltype = AA length = 717
FEATURE            Location/Qualifiers
REGION             1..717

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note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..717

note = XENP32219 Chain 2 -
PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv
(GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q

source 1..717
mol_type = protein
organism = synthetic construct

SEQUENCE: 521

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKPFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	VTVSSASTKG	120
PSVFLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSNTKVD	KKVEPKSCGG	GGSGGGGSQA	VVTQEPSLTV	240
SPGGTVTLTC	GSSTGAVTTS	NYANWVQQKP	GKSPRGLIGG	TNKRAPGVPA	RFGSGLLGK	300
AALTISGAQP	EDEADYYCAL	WYSNHWVFGG	GTKLTVLGKP	GSQKPGSGKP	GSQKPGSEVQ	360
LVESGGGLVQ	PGGSLRLSCA	ASGFTFSTYA	MNWRQAPGK	GLEWVGRIRS	KANNYATYYA	420
DSVKGRFTIS	RDDSKNTLYL	QMNSLRAEDT	AVYYCVRHGN	FGDSYVSWFA	YWGQGT	480
SSGGGGSGGG	GSKTHTCPPC	PAPPVAGPSV	FLFPPKPKDT	LMISRTPEVT	CVVVDVKHED	540
PEVKFNWYVD	GVEVHNAKTK	PREEQYNSTY	RVVSVLT	VLH QDWLNGKEYK	CKVSNKALPA	600
PIEKTISKAK	GQPREPQVYT	LPPSR	REQMTK	NQV	KLTC	660
YKTT	PPVLDS	DGSF	FLYSKL	TVDKSRWQQG	NV	717

SEQ ID NO: 522 moltype = AA length = 214

FEATURE Location/Qualifiers

REGION 1..214

note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214

note = XENP32219 Chain 3 - PSMA-H_L1.11 Light Chain

source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 522

DIVMTQSPDS	LAVSLGERAT	LSCRASQDVG	TSVDWYQQKP	DQSPKLLIYW	ASTRHTGVPD	60
RFTGSGSGTD	FTLTISLQA	EDVAVYFCQQ	YNSYPLTFGA	GTKVEIKRTV	AAPS	120
SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYEKHK	VYACEVTHQG	LSSPVT	KSPN	RGEC		214

SEQ ID NO: 523 moltype = AA length = 444

FEATURE Location/Qualifiers

REGION 1..444

note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..444

note = XENP32220 Chain 1 -
PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S

source 1..444
mol_type = protein
organism = synthetic construct

SEQUENCE: 523

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKPFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	VTVSSASTKG	120
PSVFLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSDTKVD	KKVEPKSCDK	THTCPPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCVV	VDVKHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EEYNSTYRVV	300
SVLT	TVLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	360
SLTCDVSGFY	PSDIAVEVES	DGQ	PENNYKT	TPPVLDSDGS	FFLYSKLTVD	420
SCSCVMHEALH	NHYTQKSLSL	SPGK				444

SEQ ID NO: 524 moltype = AA length = 717

FEATURE Location/Qualifiers

REGION 1..717

note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..717

note = XENP32220 Chain 2 -
PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv
(GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q

source 1..717
mol_type = protein
organism = synthetic construct

SEQUENCE: 524

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKPFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	VTVSSASTKG	120
PSVFLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSNTKVD	KKVEPKSCGG	GGSGGGGSQA	VVTQEPSLTV	240
SPGGTVTLTC	GSSTGAVTTS	NYANWVQQKP	GKSPRGLIGG	TNKRAPGVPA	RFGSGLLGK	300
AALTISGAQP	EDEADYYCAL	WYSNHWVFGG	GTKLTVLGKP	GSQKPGSGKP	GSQKPGSEVQ	360

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LVESGGGLVQ  PGGSLRLSCA  ASGFTFSTYA  MNWVRQAPGK  GLEWVGRIRS  KANNYATYYA  420
DSVKGGRFTIS RDDSKNTLYL  QMNSLRAEDT  AVYVCVRHGN  FGDSYVSWFA  YWQGGLVTV  480
SSGGGSGGGG  GSKTHTCPPC  PAPPVAGPSV  FLFPKPKD  LMISRTPEVT  CVVVDVKHED  540
PEVKFNWYVD  GVEVHNAKTK  PREEQYNSTY  RVVSVLTVLH  QDWLNGKEYK  CKVSNKALPA  600
PIEKTISKAK  GQPREPQVYT  LPPSRQMTK  NQVKLTCLVK  GFYPSDIAVE  WESNGQPENN  660
YKTTTPVLDS  DGSFFLYSKL  TVDKSRWQQG  NVFSCVMHE  ALHNHYTQKS  LSLSPGK  717

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SEQ ID NO: 525      moltype = AA  length = 214
FEATURE             Location/Qualifiers
REGION              1..214
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION              1..214
                    note = XENP32220 Chain 3 - PSMA-H_L1.24 Light Chain
source              1..214
                    mol_type = protein
                    organism = synthetic construct

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SEQUENCE: 525
DIVMTQSPDS  LAVSLGERAT  LSCRASQDVG  TAVDWYQQKP  DQSPKLLIYY  ASTRHTGVPD  60
RFTGSGSGTD  FTLTISSLQA  EDVAVYFCQQ  YNSYPLTFGA  GTKVEIKRTV  AAPSVFIFPP  120
SDEQLKSGTA  SVVCLLNIFY  PREAKVQWKV  DNALQSGNSQ  ESVTEQDSKD  STYLSSTLT  180
LSKADYEKKH  VYACEVTHQG  LSSPVTKSFN  RGEC  214

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SEQ ID NO: 526      moltype = AA  length = 444
FEATURE             Location/Qualifiers
REGION              1..444
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION              1..444
                    note = XENP32221 Chain 1 -
                    PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
source              1..444
                    mol_type = protein
                    organism = synthetic construct

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SEQUENCE: 526
EVQLVQSGAE  VKKPGASVKV  SCKTSGYTFT  EYTIHWVRQA  PGQSLEWMGN  INPNNGGTTY  60
NQKFQGRVTI  TVDKSTSTAY  MELSSLRSED  TAVYYCAAGW  NFDYWGQGT  LTVSSASTKG  120
PSVFPLAPSS  KSTSGGTAAL  GCLVKDYFPE  PVTVSWNSGA  LTSGVHTFPA  VLQSSGLYSL  180
SSVTVTPSSS  LGTQTYICNV  NHKPSDTKVD  KKVEPKSCDK  THTCPPCPAP  PVAGPSVFLF  240
PKPKDITLMI  SRTPEVTCV  VDVKHEDPEV  KFNWYVDGVE  VHNAKTKPRE  EEYNSTYRVV  300
SVLTVLHQDW  LNGKEYKCKV  SNKALPAPIE  KTISKAKGQP  REPQVYTLPP  SREEMTKNQV  360
SLTCDVSGFY  PSDIAVEWES  DQQPENNYKT  TPPVLDSGDS  FFLYSKLTVD  KSRWEQGDVF  420
SCSVMHLEALH  NHYTQKSLSL  SPGK  444

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SEQ ID NO: 527      moltype = AA  length = 717
FEATURE             Location/Qualifiers
REGION              1..717
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION              1..717
                    note = XENP32221 Chain 2 -
                    PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv
                    (GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q
source              1..717
                    mol_type = protein
                    organism = synthetic construct

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SEQUENCE: 527
EVQLVQSGAE  VKKPGASVKV  SCKTSGYTFT  EYTIHWVRQA  PGQSLEWMGN  INPNNGGTTY  60
NQKFQGRVTI  TVDKSTSTAY  MELSSLRSED  TAVYYCAAGW  NFDYWGQGT  LTVSSASTKG  120
PSVFPLAPSS  KSTSGGTAAL  GCLVKDYFPE  PVTVSWNSGA  LTSGVHTFPA  VLQSSGLYSL  180
SSVTVTPSSS  LGTQTYICNV  NHKPSNTKVD  KKVEPKSCGG  GSGGGGSQA  VVTQEPSLTV  240
SPGGTVTLTC  GSSTGAVTTS  NYANWVQQKP  GKSPRGLIGG  TNKRAPGVPA  RFGSGLLGK  300
AALTISGAQP  EDEADYYCAL  WYSNHWVFGG  GTKLTVLGKP  GSGKPGSGKP  GSGKPGSEVQ  360
LVESGGGLVQ  PGGSLRLSCA  ASGFTFSTYA  MNWVRQAPGK  GLEWVGRIRS  KANNYATYYA  420
DSVKGGRFTIS RDDSKNTLYL  QMNSLRAEDT  AVYVCVRHGN  FGDSYVSWFA  YWQGGLVTV  480
SSGGGSGGGG  GSKTHTCPPC  PAPPVAGPSV  FLFPKPKD  LMISRTPEVT  CVVVDVKHED  540
PEVKFNWYVD  GVEVHNAKTK  PREEQYNSTY  RVVSVLTVLH  QDWLNGKEYK  CKVSNKALPA  600
PIEKTISKAK  GQPREPQVYT  LPPSRQMTK  NQVKLTCLVK  GFYPSDIAVE  WESNGQPENN  660
YKTTTPVLDS  DGSFFLYSKL  TVDKSRWQQG  NVFSCVMHE  ALHNHYTQKS  LSLSPGK  717

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SEQ ID NO: 528      moltype = AA  length = 214
FEATURE             Location/Qualifiers
REGION              1..214
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION              1..214
                    note = XENP32221 Chain 3 - PSMA-H_L1.26 Light Chain
source              1..214
                    mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 528
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYH ASTRHTGVDP 60
RFTGSGSGTD PTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 529      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION            1..444
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..444
                  note = XENP32222 Chain 1 -
                  PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
source            1..444
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 529
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSDGS PFLYSKLTVD KSRWEQGDVF 420
SCSVMHEALH NHYQKSLSL SPGK 444

SEQ ID NO: 530      moltype = AA  length = 717
FEATURE            Location/Qualifiers
REGION            1..717
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..717
                  note = XENP32222 Chain 2 -
                  PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv
                  (GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q
source            1..717
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 530
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GSGGGGGSQA VVTQEPSLTV 240
SPGGTVTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFGSGLLGK 300
AALTISGAQP EDEADYYCAL WYSNHWFVGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGGSRLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KANNYATYYA 420
DSVKGGRFTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDSYVSWFA YWGQGLVTV 480
SSGGGGSGGG GSKTHTCPPC PAPPVAGPSV FLFPPKPKDT LMISRTPEVT CVVVDVKHED 540
PEVKFNWYVD GVEVHNAKTK PREEQYNSTY RVVSVLTVLH QDWLNGKEYK CKVSNKALPA 600
PIEKTISKAK QPREPQVYT LPSPREQMTK NQVKLTCLVK GFYPDSIAVE WESNGQPENN 660
YKTPPVVLD SGSFFLYSKL TVDKSRWQQG NVFSCSVME ALHNHYTQKS LSLSPGK 717

SEQ ID NO: 531      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP32222 Chain 3 - PSMA-H_L1.75 Light Chain
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 531
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD PTLTISSLQA EDVAVYFCQQ YNSYPTTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 532      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION            1..444
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..444
                  note = XENP32223 Chain 1 -
                  PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
source            1..444

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mol_type = protein
organism = synthetic construct

SEQUENCE: 532

EVQLVQSGAE	VKKPGASVKV	CKTSGYTFT	EYTIHWVRQA	PGQSLWWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGGQTL	VTVSSASTKG	120
PSVFPPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSDTKVD	KKVEPKSCDK	THTCPPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCVV	VDVKHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EEYNSTYRVV	300
SVLTVLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	SREEMTKNQV	360
SLTCDVSGFY	PSDIAVEWES	DGQPENNYKT	TPPVLDSDGS	FFLYSKLTVD	KSRWEQGDVF	420
SCSVMHAEALH	NHYTQKSLSL	SPGK				444

SEQ ID NO: 533

FEATURE

REGION

1..717

note = Description of Artificial Sequence: Synthetic polypeptide

REGION

1..717

note = XENP32223 Chain 2 -
PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv
(GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q

source

1..717

mol_type = protein
organism = synthetic construct

SEQUENCE: 533

EVQLVQSGAE	VKKPGASVKV	CKTSGYTFT	EYTIHWVRQA	PGQSLWWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGGQTL	VTVSSASTKG	120
PSVFPPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSNTKVD	KKVEPKSCGG	GGSGGGGSQA	VVTQEPSLTV	240
SPGGTVTLTC	GSSTGAVTTS	NYANWVQQKP	GKSPRGLIGG	TNKRAPGVPA	RFGSGLLGK	300
AALTISGAQP	EDEADYYCAL	WYSNHWVFGG	GTKLTVLGKP	GSQKPGSGKP	GSQKPGSEVQ	360
LVESGGGLVQ	PGGSLRLSCA	ASGFTFSTYA	MNWVRQAPGK	GLEWVGRISS	KANNYATYYA	420
DSVKGRFTIS	RDDSKNTLYL	QMNSLRAEDT	AVYYCVRHGN	FGDSYVSWFA	YWGQGLTVTV	480
SSGGGSGGGG	GSKTHTCPPC	PAPPVAGPSV	FLFPPKPKDT	LMISRTPEVT	CVVVDVKHED	540
PEVKFNWYVD	GVEVHNAKTK	PREEQYNSTY	RVVSVLTVLH	QDWLNGKEYK	CKVSNKALPA	600
PIEKTISKAK	GQPREPQVYT	LPPSRQMTK	NQVKLTCLVK	GFYPSDIAVE	WESNGQPENN	660
YKTTTPVLDS	DGSFFLYSKL	TVDKSRWQQG	NVPSCSVMHE	ALHNHYTQKS	LSLSPGK	717

SEQ ID NO: 534

FEATURE

REGION

1..214

note = Description of Artificial Sequence: Synthetic polypeptide

REGION

1..214

note = XENP32223 Chain 3 - PSMA-H_L1.68 Light Chain

source

1..214

mol_type = protein
organism = synthetic construct

SEQUENCE: 534

DIVMTQSPDS	LAVSLGERAT	LSCRASQDVG	TAVDWYQQKP	DQSPKLLIYW	ASTRHTGVDP	60
RFTGSGSGTD	FTLTISLQA	EDVAVYFCQQ	YNSKPLTFGA	GTKVEIKRTV	AAPSVPFIPP	120
SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYEKHK	VYACEVTHQG	LSSPVTKSFN	RGEC			214

SEQ ID NO: 535

FEATURE

REGION

1..444

note = Description of Artificial Sequence: Synthetic polypeptide

REGION

1..444

note = XENP32224 Chain 1 -
PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K/L368D/K370S

source

1..444

mol_type = protein
organism = synthetic construct

SEQUENCE: 535

EVQLVQSGAE	VKKPGASVKV	CKTSGYTFT	EYTIHWVRQA	PGQSLWWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGGQTL	VTVSSASTKG	120
PSVFPPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSDTKVD	KKVEPKSCDK	THTCPPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCVV	VDVKHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EEYNSTYRVV	300
SVLTVLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	SREEMTKNQV	360
SLTCDVSGFY	PSDIAVEWES	DGQPENNYKT	TPPVLDSDGS	FFLYSKLTVD	KSRWEQGDVF	420
SCSVMHAEALH	NHYTQKSLSL	SPGK				444

SEQ ID NO: 536

FEATURE

REGION

1..717

note = Description of Artificial Sequence: Synthetic

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REGION          polypeptide
                  1..717
note = XENP32224 Chain 2 -
      PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv
      (GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q
source          1..717
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 536
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFGQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GSGGGGSQA VVTQEPSLTV 240
SPGGTVTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFGSGLLGK 300
AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGGSLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KANNYATYYA 420
DSVKGRFTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDSYVSWFA YWGQGT LVTV 480
SSGGGSGGG GSKTHTCPPC PAPPVAGPSV FLFPPKPKDT LMISRTPEVT CVVVDVKHED 540
PEVKFNWYVD GVEVHNAKTK PREEQYNSTY RVSVLTVLH QDWLNGKEYK CKVSNKALPA 600
PIEKTISKAK GQPREPQVYT LPPSREQMTK NQVKLTCLVK GFYPSDIAVE WESNGQPENN 660
YKTTTPVLDL DGSFPLYSKL TVDKSRWQQG NVFSCSVMHE ALHNHYTQKS LSLSPGK 717

SEQ ID NO: 537      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION             1..214
note = XENP32224 Chain 3 - PSMA-H_L1.29 Light Chain
source             1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 537
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYT ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTPGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 538      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION             1..444
note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION             1..444
note = XENP32225 Chain 1 -
      PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
source             1..444
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 538
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFGQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PKPKDITLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNATKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYCKCV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHAEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 539      moltype = AA length = 717
FEATURE            Location/Qualifiers
REGION             1..717
note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION             1..717
note = XENP32225 Chain 2 -
      PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv
      (GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q
source             1..717
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 539
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFGQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GSGGGGSQA VVTQEPSLTV 240
SPGGTVTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFGSGLLGK 300
AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGGSLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KANNYATYYA 420

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SEQUENCE: 543
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TIVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
 RFTGSGSGTD FTLTISLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY BREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSLSTLT 180
 LSKADYKHKH VYACEVTHQG LSSPVTKSPN RGEC 214

SEQ ID NO: 544 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..444
 note = XENP34237 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 544
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTL VTVSSASTKG 120
 PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEVES DGQPENNYKT TPPVLDSGGS FFLYSLKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 545 moltype = AA length = 717
 FEATURE Location/Qualifiers
 REGION 1..717
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..717
 note = XENP34237 Chain 2 -
 PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv
 (GKPGS)4_(G4S)2_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q
 source 1..717
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 545
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGTL VTVSSASTKG 120
 PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GSGGGGGSQA VVTQEPSLTV 240
 SPGGVTTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIG TNKRAPGVPA RFGSGLLGK 300
 AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
 LVESGGGLVQ PGGSRLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGIRIS KANNYATYYA 420
 DSVKGRFTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDSYVSWFA YWGQGTLLTV 480
 SSGGGGSGGG GSKTHTCPPC PAPPVAGPSV FLFPPKPKDT LMISRTPEVT CVVVDVKHED 540
 PEVKFNWYVD GVEVHNAKTK PREEQYNSTY RVVSVLTVLH QDWLNGKEYK CKVSNKALPA 600
 PIEKTISKAK GQPREPQVYT LPPSREQMTK NQVKLTCLVK GFYPSDIAVE WESNGQPENN 660
 YKTTTPVLDS DGSFFLYSKL TVDKSRWQQG NVFSCSVMHE ALHNHYTQKS LSLSPGK 717

SEQ ID NO: 546 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP34237 Chain 3 - PSMA-H_L1.78 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 546
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYY ASTRHTGVPD 60
 RFTGSGSGTD FTLTISLQA EDVAVYFCQQ HNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY BREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSLSTLT 180
 LSKADYKHKH VYACEVTHQG LSSPVTKSPN RGEC 214

SEQ ID NO: 547 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..444
 note = XENP34238 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
 source 1..444
 mol_type = protein

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organism = synthetic construct

SEQUENCE: 547
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYS L 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDS DGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 548      moltype = AA length = 717
FEATURE            Location/Qualifiers
REGION              1..717
                    note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..717
                    note = XENP34238 Chain 2 -
                      PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv
                      (GKPGS)4_(G4S)2_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q
source              1..717
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 548
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYS L 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCG GSGGGGGSQA VVTOEPSLT V 240
SPGGTVTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFSGSLLGGK 300
AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGGSRLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KANNYATYYA 420
DSVKGRTTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDSYVSWFA YWGQGT L VTV 480
SSGGGGSGGG GSKTHTCPPC PAPPVAGPSV FLFPPKPKDT LMI SRTPEVT CVVVDVKHED 540
PEVKFNWYVD GVEVHNAKTK PREEQYNSTY RVVSVLTVLH QDWLNGKEYK CKVSNKALPA 600
PIEKTISKAK GQPREPQVYT LPPSREQMTK NQVKLTCLVK GFYPSDIAVE WESNGQPENN 660
YKTTTPVLDS DGSFFLYSKL TVDKSRWQQG NVFSCSVMHE ALHNHYTQKS LSLSPGK 717

SEQ ID NO: 549      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION              1..214
                    note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..214
                    note = XENP34238 Chain 3 - PSMA-H_L1.81 Light Chain
source              1..214
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 549
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYH ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ HNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSK STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 550      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION              1..444
                    note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..444
                    note = XENP34239 Chain 1 -
                      PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
source              1..444
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 550
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYS L 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDS DGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 551      moltype = AA length = 717
FEATURE            Location/Qualifiers
REGION              1..717
                    note = Description of Artificial Sequence: Synthetic
                      polypeptide

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REGION 1..717
note = XENP34239 Chain 2 -
PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv
(GKPGS)4_(G4S)2_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q

source 1..717
mol_type = protein
organism = synthetic construct

SEQUENCE: 551

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	LVTVSSASTKG	120
PSVFPPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSNTKVD	KKVEPKSCGG	GGSGGGGSQA	VVTQEPSLTV	240
SPGGTVTLTC	GSSTGAVTTS	NYANWVQQKP	GKSPRGLIGG	TNKRAPGVPA	RFGSLLGGK	300
AALTISGAQP	EDEADYYCAL	WYSNHWVFGG	GTKLTVLGKP	GSQKPGSGKP	GSQKPGSEVQ	360
LVESGGGLVQ	PGGSLRLSCA	ASGFTFSTYA	MNWRQAPGK	GLEWVGRIRS	KANNYATYYA	420
DSVKGRTTIS	RDDSKNTLYL	QMNSLRAEDT	AVYYCVRHGN	FGDSYVSWFA	YWGQGT	480
SSGGGGSGGG	GSKTHTCPPC	PAPPVAGPSV	FLFPPKPKDT	LMISRTPEVT	CVVVDVKHED	540
PEVKFNWYVD	GVEVHNAKTK	PREEQYNSTY	RVSVLT	VLH QDWLNGKEYK	CKVSNKALPA	600
PIEKTISKAK	GQPREPQVYT	LPPSRQMTK	NQVLTCLVK	GFYPSDIAVE	WESNGQPENN	660
YKTTPPVLD	DGSFFLYSKL	TVDKSRWQQG	NVFSCSVME	ALHNHYTQKS	LSLSPGK	717

SEQ ID NO: 552 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214
note = XENP34239 Chain 3 - PSMA-H_L1.84 Light Chain

source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 552

DIVMTQSPDS	LAVSLGERAT	LSCRASQDVG	TAVDWYQQKP	DQSPKLLIYT	ASTRHTGVDP	60
RFTGSGSGTD	FTLTISLQA	EDVAVYFCQQ	HNSYPLTFGA	GTKVEIKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYEKHK	VYACEVTHQG	LSSPVTKSFN	RGEC			214

SEQ ID NO: 553 moltype = AA length = 444
FEATURE Location/Qualifiers
REGION 1..444
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..444
note = XENP34625 Chain 1 -
PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K
/L368D/K370S/M428L/N434S

source 1..444
mol_type = protein
organism = synthetic construct

SEQUENCE: 553

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	LVTVSSASTKG	120
PSVFPPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSDTKVD	KKVEPKSCDK	THTCPPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCV	VDVKHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EEYNSTYRVV	300
SVLTVLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	SREEMTKNQV	360
SLTCDVSGFY	PSDIAVEWES	DGQPENNYKT	TPPVLDSDGS	FFLYSKLTVD	KSRWEQGDVF	420
SCSVLHEALH	SHYTQKSLSL	SPGK				444

SEQ ID NO: 554 moltype = AA length = 717
FEATURE Location/Qualifiers
REGION 1..717
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..717
note = XENP34625 Chain 2 -
PSMA-H_H1L1_Fab_(G4S)_[CD3]_L1.47_H1.32_scFv
(GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q/M428L/N434S

source 1..717
mol_type = protein
organism = synthetic construct

SEQUENCE: 554

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	LVTVSSASTKG	120
PSVFPPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSNTKVD	KKVEPKSCGG	GGSGGGGSQA	VVTQEPSLTV	240
SPGGTVTLTC	GSSTGAVTTS	NYANWVQQKP	GKSPRGLIGG	TNKRAPGVPA	RFGSLLGGK	300
AALTISGAQP	EDEADYYCAL	WYSNHWVFGG	GTKLTVLGKP	GSQKPGSGKP	GSQKPGSEVQ	360

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LVESGGGLVQ PGGSLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KANNYATYYA 420
 DSVKGRFTIS RDDSKNTLYL QMNSLRAEDT AVYVCVRHGN FGDSYVSWFA YWQGGLTVTV 480
 SSGGGGSGGG GSKTHTCPPC PAPPVAGPSV FLFPKPKDT LMISRTPEVT CVVVDVKHED 540
 PEVKFNWYVD GVEVHNAKTK PREEQYNSTY RVSVLTVLH QDWLNGKEYK CKVSNKALPA 600
 PIEKTISKAK GQPREPQVYT LPPSRQMTK NQVKLTCLVK GFYPSDIAVE WESNGQPENN 660
 YKTTTPVLDS DGSFFLYSKL TVDKSRWQQG NVFSCSVLHE ALHSHYTQKS LSLSPGK 717

SEQ ID NO: 555 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP34625 Chain 3 - PSMA-H_L1 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 555
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
 RFTSGSGSDT FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSK STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 556 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..444
 note = XENP34626 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K
 /L368D/K370S/M428L/N434S
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 556
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWQGQTL VTVSSASTKG 120
 PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVES DGQPENNYKT TPPVLDSGGS FFLYSKLTVD KSRWEQGDVF 420
 SCSVLHEALH SHYTQKSLSL SPGK 444

SEQ ID NO: 557 moltype = AA length = 717
 FEATURE Location/Qualifiers
 REGION 1..717
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..717
 note = XENP34626 Chain 2 -
 PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv
 (GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q/M428L/N434S
 source 1..717
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 557
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWQGQTL VTVSSASTKG 120
 PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GSGGGGGSQA VVTQEPSTLV 240
 SPGGTVTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFGSGLLGK 300
 AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
 LVESGGGLVQ PGGSLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KANNYATYYA 420
 DSVKGRFTIS RDDSKNTLYL QMNSLRAEDT AVYVCVRHGN FGDSYVSWFA YWQGGLTVTV 480
 SSGGGGSGGG GSKTHTCPPC PAPPVAGPSV FLFPKPKDT LMISRTPEVT CVVVDVKHED 540
 PEVKFNWYVD GVEVHNAKTK PREEQYNSTY RVSVLTVLH QDWLNGKEYK CKVSNKALPA 600
 PIEKTISKAK GQPREPQVYT LPPSRQMTK NQVKLTCLVK GFYPSDIAVE WESNGQPENN 660
 YKTTTPVLDS DGSFFLYSKL TVDKSRWQQG NVFSCSVLHE ALHSHYTQKS LSLSPGK 717

SEQ ID NO: 558 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP34626 Chain 3 - PSMA-H_L1.58 Light Chain

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source          1..214
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 558
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YQSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSPN RGEC 214

SEQ ID NO: 559      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION            1..444
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..444
                  note = XENP34627 Chain 1 -
                  PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K
                  /L368D/K370S/M428L/N434S

source          1..444
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 559
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGGQTL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DQQPENNYKT TPPVLDSGDS PFLYSKLTVD KSRWEQGDVF 420
SCSVLHEALH SHYQKSLSL SPGK 444

SEQ ID NO: 560      moltype = AA length = 717
FEATURE            Location/Qualifiers
REGION            1..717
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..717
                  note = XENP34627 Chain 2 -
                  PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.32_scFv
                  (GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q/M428L/N4
                  34S

source          1..717
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 560
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGGQTL VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGQ GSGGGGGSQA VVTQEPSLTV 240
SPGGTVTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFSGSLLGGK 300
AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGGSRLRLCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KANNYATYYA 420
DSVKGRTTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDSYVSWFA YWGQGLVTV 480
SSGGGGSGGG GSKTHTCPPC PAPPVAGPSV FLFPPKPKDT LMISRTPEVT CVVVDVKHED 540
PEVKFNWYVD GVEVHNAKTK PREEQYNSTY RVVSVLTVLH QDWLNGKEYK CKVSNKALPA 600
PIEKTISKAK GQPREPQVYT LPPSRQMTK NQVKLTCLVK GFYPSDIAVE WESNGQPENN 660
YKTTTPVLDL DGSFFLYSKL TVDKSRWQQG NVFSCSVLHE ALHSHYTQKS LSLSPGK 717

SEQ ID NO: 561      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP34627 Chain 3 - PSMA-H_L1.24 Light Chain

source          1..214
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 561
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSPN RGEC 214

SEQ ID NO: 562      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION            1..444
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide

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REGION 1..444
 note = XENP34628 Chain 1 -
 PSMA-H H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K
 /L368D/K370S/M428L/N434S

source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 562

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	LVTVSSASTKG	120
PSVFFPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSDTKVD	KKVEPKSCDK	THTCPPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCVV	VDVKHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EEYNSTYRVV	300
SVLTVLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	SREEMTKNQV	360
SLTCDVSGFY	PSDIAVEWES	DGQPENNYKT	TPPVLDSDGS	FFLYSKLTVD	KSRWEQGDVF	420
SCSVLHEALH	SHYTQKSLSL	SPGK				444

SEQ ID NO: 563 moltype = AA length = 717
 FEATURE Location/Qualifiers
 REGION 1..717
 note = Description of Artificial Sequence: Synthetic
 polypeptide

REGION 1..717
 note = XENP34628 Chain 2 -
 PSMA-H H1_(G4S)_[CD3]_L1.47_H1.32_scFv
 (GKPGS)4_Fc(222)_IgG1_C220S/PVA_/S267K/S364K/E357Q/M428L/N4
 34S

source 1..717
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 563

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	LVTVSSASTKG	120
PSVFFPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSNTKVD	KKVEPKSCGG	GGSGGGGSQA	VVTQEPSLTV	240
SPGGTVTLTC	GSSTGAVTTS	NYANWVQQKP	GKSPRGLIGG	TNKRAPGVPA	RFGSGLLGK	300
AALTISGAQP	EDEADYYCAL	WYSNHWVFGG	GTKLTVLGKP	GSGKPGSGKP	GSGKPGSEVQ	360
LVESGGGLVQ	PGGSLRLSCA	ASGFTFSTYA	MNWVRQAPGK	GLEWVGIRIS	KANNYATYYA	420
DSVKGGRFTIS	RDDSKNTLYL	QMNSLRAEDT	AVYYCVRHGN	FGDSYVSWFA	YWGQGTLVTV	480
SSGGGSGGGG	GSKTHTCPPC	PAPPVAGPSV	FLFPPKPKDT	LMISRTPEVT	CVVVDVKHED	540
PEVKFNWYVD	GVEVHNAKTK	PREEQYNSTY	RVVSVLTVLH	QDWLNGKEYK	CKVSNKALPA	600
PIEKTISKAK	GQPREPQVYT	LPPSRQMTK	NQVKLTCVVK	GFYPSDIAVE	WESNGQPENN	660
YKTTTPVLDS	DGSFFLYSKL	TVDKSRWQQG	NVPSCSVLHE	ALHSHYTQKS	LSLSPGK	717

SEQ ID NO: 564 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide

REGION 1..214
 note = XENP34628 Chain 3 - PSMA-H L1.29 Light Chain

source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 564

DIVMTQSPDS	LAVSLGERAT	LSCRASQDVG	TAVDWYQQKP	DQSPKLLIYT	ASTRHTGVDP	60
RFTGSGSGTD	FTLTISLQA	EDVAVYFCQQ	YNSYPLTFGA	GTKVEIKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYEKHK	VYACEVTHQG	LSSPVTKSFN	RGEC			214

SEQ ID NO: 565 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide

REGION 1..444
 note = XENP31853 Chain 1 -
 PSMA-H H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S

source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 565

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	LVTVSSASTKG	120
PSVFFPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSDTKVD	KKVEPKSCDK	THTCPPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCVV	VDVKHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EEYNSTYRVV	300
SVLTVLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	SREEMTKNQV	360
SLTCDVSGFY	PSDIAVEWES	DGQPENNYKT	TPPVLDSDGS	FFLYSKLTVD	KSRWEQGDVF	420

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SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 566 moltype = AA length = 717
 FEATURE Location/Qualifiers
 REGION 1..717
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..717
 note = XENP31853 Chain 2 -
 PSMA-H_H1_CH1_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q
 source 1..717
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 566

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	VTVSSASTKG	120
PSVFPPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSNTKVD	KKVEPKSCGG	GGSGGGGSEV	QLVESGGGLV	240
QPGGSLRLSC	AASGFTFSTY	AMNWRQAPG	KGLEWVGRIR	SKYNNYATYY	ADSVKGRFTI	300
SRDDSKNTLY	LQMNSLRAED	TAVYYCVRHG	NFGDEYVSWF	AYWGQGT	LVTVSSGKPGSGK	360
PGSGKPGSGK	PGSQAVVTQE	PSLTVSPGGT	VTLTCSSTG	AVTTSNYANW	VQOKPGKSPR	420
GLIGGTNKR	PGVPARFSGS	LLGKKAALTI	SGAQPEDEAD	YYCALWYSNH	WVFGGGLT	480
VLGGGGSGGG	GSKTHTCPPC	PAPPVAGPSV	FLFPPKPKDT	LMISRTPEVT	CVVVDVKHED	540
PEVKFNWYVD	GVEVHNAKTK	PREEQYNSTY	RVSVLT	VLH QDWLNGKEYK	CKVSNKALPA	600
PIEKTISKAK	GQPREPQVYT	LPSPREQMTK	NQVLTCLVK	GFYPSDIAVE	WESNGQPENN	660
YKTTPPVLD	DGSFFLYSKL	TVDKSRWQQG	NVFS	SCVMHE	ALHNHYTQKS	717
					LSLSPGK	

SEQ ID NO: 567 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP31853 Chain 3 - PSMA-H_L1 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 567

DIVMTQSPDS	LAVSLGERAT	LSCRASQDVG	TAVDWYQQKP	DQSPKLLIYW	ASTRHTGVDP	60
RFTGSGSGTD	FTLTISSLQA	EDVAVYFCQQ	YNSYPLTFGA	GTKVEIKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYEKKH	VYACEVTHQG	LSSPVT	KSFN RGE	C		214

SEQ ID NO: 568 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..444
 note = XENP31856 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K
 /L368D/K370S-Fc(216)_IgG1_PVA_/S267K/S364K/E357Q
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 568

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	VTVSSASTKG	120
PSVFPPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSDTKVD	KKVEPKSCDK	THTCPPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCVV	VDVKHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EEYNSTYRVV	300
SVLT	VLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	360
SLTCDVSGFY	PSDIAVEWES	DGQPENNYKT	TPPVLDSDGS	FFLYSKLTVD	KSRWEQGDVF	420
SCSVMHLEALH	NHYTQKSLSL	SPGK				444

SEQ ID NO: 569 moltype = AA length = 713
 FEATURE Location/Qualifiers
 REGION 1..713
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..713
 note = XENP31856 Chain 2 -
 PSMA-H_H1_CH1_(G4S)2_[CD3]_L1.47_H1.89_scFv(
 GKPGS)4-IgG1_pI(-)_Isosteric_A_/PVA_/S267K/L368D/K370S-Fc(2
 16)_IgG1_PVA_/S267K/S364K/E357Q
 source 1..713
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 569

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EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GSGGGGSQA VVTQEPSLTV 240
SPGGTVTLTC GSSTGAVTTS NYANWVQKPK GKSPRGLIGG TNKRAPGVPA RFSGSLGK 300
AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGSSLRSLCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KYNNYATYYA 420
DSVKGRTTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDEYVSWFA YWGQGT LVTV 480
SSEPKSSDKT HTPPCPAPP VAGPSVFLFP PKPKDTLMIS RTPEVTCVVV DVKHEDPEVK 540
FNWYVDGVEV HNAKTKPREE QYNSTYRVVS VLTVLHQDWL NGKEYCKKVS NKALPAPIEK 600
TISKAKGQPR EPQVYTLPPS REQMTKNQVK LTCLVKGFPY SDIAVEWESN GQPENNYKTT 660
PPVLDSGDSF FLYSKLTVDK SRWQQGNVFS CSVMHEALHN HYTKSLSL PGK 713

SEQ ID NO: 570      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION              1..214
                    note = Description of Artificial Sequence: Synthetic
                        polypeptide
REGION              1..214
                    note = XENP31856 Chain 3 - PSMA-H_L1 Light Chain
source              1..214
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 570
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQKPK DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 571      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION              1..444
                    note = Description of Artificial Sequence: Synthetic
                        polypeptide
REGION              1..444
                    note = XENP33063 Chain 1 -
                        PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
source              1..444
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 571
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PKPKDTLMIS SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYCKKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 572      moltype = AA  length = 713
FEATURE            Location/Qualifiers
REGION              1..713
                    note = Description of Artificial Sequence: Synthetic
                        polypeptide
REGION              1..713
                    note = XENP33063 Chain 2 -
                        PSMA-H_H1_(G4S)_ [CD3]_L1.47_H1.89_scFv
                        (GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q
source              1..713
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 572
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GSGGGGSQA VVTQEPSLTV 240
SPGGTVTLTC GSSTGAVTTS NYANWVQKPK GKSPRGLIGG TNKRAPGVPA RFSGSLGK 300
AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGSSLRSLCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KYNNYATYYA 420
DSVKGRTTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDEYVSWFA YWGQGT LVTV 480
SSEPKSSDKT HTPPCPAPP VAGPSVFLFP PKPKDTLMIS RTPEVTCVVV DVKHEDPEVK 540
FNWYVDGVEV HNAKTKPREE QYNSTYRVVS VLTVLHQDWL NGKEYCKKVS NKALPAPIEK 600
TISKAKGQPR EPQVYTLPPS REQMTKNQVK LTCLVKGFPY SDIAVEWESN GQPENNYKTT 660
PPVLDSGDSF FLYSKLTVDK SRWQQGNVFS CSVMHEALHN HYTKSLSL PGK 713

SEQ ID NO: 573      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION              1..214

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note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..214
note = XENP33063 Chain 3 - PSMA-H_L1.58 Light Chain
source 1..214
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 573
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YQSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 574      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION 1..444
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..444
note = XENP33064 Chain 1 -
      PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
source 1..444
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 574
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFGQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS PFLYSKLTVD KSRWEQGDVF 420
SCSVMHAEALH NHYQKSLSL SPGK 444

SEQ ID NO: 575      moltype = AA length = 713
FEATURE            Location/Qualifiers
REGION 1..713
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..713
note = XENP33064 Chain 2 -
      PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv
      (GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q
source 1..713
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 575
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFGQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GSGGGGGSQA VVTQEPSLTV 240
SPGGTVTLTC GSSTGAVTTS NYANWVQOKP GKSPRGLIGG TNKRAPGVPA RFSGSLGGK 300
AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGGSLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KYNNYATYYA 420
DSVKGGRFTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDEYVSWFA YWGQGLVTV 480
SSEPKSSDKT HTPPCPPAPP VAGPSVFLFP PKPKDTLMIS RTPEVTCVVV DVKHEDPEVK 540
FNWYVDGVEV HNAKTKPREE QYNSTYRVVS VLTVLHQDWL NGKEYKCKVS NKALPAPIEK 600
TISKAKGQPR EPQVYTLPPS REQMTKNQVK LTCLVKGFYP SDIAVEWESN GQPENNYKTT 660
PPVLDSGGSF FLYSKLTVDK SRWQQGNVFS CSMVHEALHN HYTKSLSL PGK 713

SEQ ID NO: 576      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..214
note = XENP33064 Chain 3 - PSMA-H_L1.11 Light Chain
source 1..214
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 576
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TSVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 577      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION 1..444

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note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..444
note = XENP33065 Chain 1 -
      PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
source 1..444
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 577
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYCKKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 578      moltype = AA length = 713
FEATURE            Location/Qualifiers
REGION 1..713
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..713
note = XENP33065 Chain 2 -
      PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv
      (GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q
source 1..713
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 578
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GGSGGGGSQA VVTQEPSLTV 240
SPGGVTTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFGSGLLGGK 300
AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGGSRLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KYNNYATYYA 420
DSVKGGRFTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDEYVSWFA YWGQGT L VTV 480
SSEPSSDKT HTPCPPCPAP VAGPSVFLFP PKPKDTLMIS RTPEVTCVVV DVKHEDPEVK 540
FNWYVDGVEV HNAKTKPREE QYNSTYRVVS VLTVLHQDWL NGKEYCKKVS NKALPAPIEK 600
TISKAKGQPR EPQVYTLPPS REQMTKNQVK LTCLVKGFYP SDIAVEWESN GQPENNYKT 660
PPVLDSGDSF FLYSKLTVDK SRWQQGNVFS CSVMHEALHN HYTKSLSLSL PGK 713

SEQ ID NO: 579      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..214
note = XENP33065 Chain 3 - PSMA-H_L1.24 Light Chain
source 1..214
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 579
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYY ASTRHTGVDP 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTPGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSLSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 580      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION 1..444
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..444
note = XENP33066 Chain 1 -
      PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
source 1..444
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 580
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYCKKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGGS FFLYSKLTVD KSRWEQGDVF 420

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SCSVMEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 581 moltype = AA length = 713
 FEATURE Location/Qualifiers
 REGION 1..713
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..713
 note = XENP33066 Chain 2 -
 PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv
 (GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q
 source 1..713
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 581

EVQLVQSGAE VKKPGASVKV	SCKTSGYTFT EYTIHWVRQA	PGQSLEWMGN INPNNGGTTY	60
NQKFQGRVTI TVDKSTSTAY	MELSSLRSED TAVYYCAAGW	NFDYWGQGT L VTVSSASTKG	120
PSVFPPLAPSS KSTSGGTAAL	GCLVKDYFPE PVTVSWNSGA	LTSGVHTFPA VLQSSGLYSL	180
SSVVTVPSSS LGTQTYICNV	NHKPSNTKVD KKVEPKSCGG	GGSGGGGSQA VVTQEPSLTV	240
SPGGTVTLTC GSSTGAVTTS	NYANWVQQKP GKSPRGLIGG	TNKRAPGVPA RFGSGLLGK	300
AALTISGAQP EDEADYYCAL	WYSNHWVFGG GTKLTVLGKP	GSQKPGSGKP GSGKPGSEVQ	360
LVESGGGLVQ PGGSRLRLSCA	ASGFTFTSYA MNWVRQAPGK	GLEWVGRIRS KYNNTATYYA	420
DSVKGRTTIS RDDSKNTLYL	QMNSLRAEDT AVYYCVRHGN	FGDEYVSWFA YWGQGT LVTV	480
SSEPKSSDKT HTCPCPCAPP	VAGPSVFLFP PKPKDTLMIS	RTPEVTCVVV DVKHEDPEVK	540
FNWYVDGVEV HNAKTKPREE	QYNSTYRVVS VLTVLHQDWL	NGKEYKCKVS NKALPAPIEK	600
TISKAKGQPR EPQVYTLPPS	REQMTKNQVK LTCLVKGFYP	SDIAVEWESN GQPENNYKTT	660
PPVLDSGGSF FLYSKLTVDK	SRWQQGNVFS CSMVMEALHN	HYTQKSLSLS PGK	713

SEQ ID NO: 582 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP33066 Chain 3 - PSMA-H_L1.26 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 582

DIVMTQSPDS LAVSLGERAT	LSCRASQDVG TAVDWYQQKP	DQSPKLLIYH ASTRHTGVPD	60
RFTGSGSGTD FTLTISSLQA	EDVAVYFCQQ YNSYPLTFGA	GTKVEIKRTV AAPS VFIFPP	120
SDEQLKSGTA SVVCLLNIFY	PREAKVQWKV DNALQSGNSQ	ESVTEQDSK STYLSSTLT	180
LSKADYKHKH VYACEVTHQG	LSSPVTKSFN RGEK		214

SEQ ID NO: 583 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..444
 note = XENP33067 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 583

EVQLVQSGAE VKKPGASVKV	SCKTSGYTFT EYTIHWVRQA	PGQSLEWMGN INPNNGGTTY	60
NQKFQGRVTI TVDKSTSTAY	MELSSLRSED TAVYYCAAGW	NFDYWGQGT L VTVSSASTKG	120
PSVFPPLAPSS KSTSGGTAAL	GCLVKDYFPE PVTVSWNSGA	LTSGVHTFPA VLQSSGLYSL	180
SSVVTVPSSS LGTQTYICNV	NHKPSDTKVD KKVEPKSCDK	THTCPPCPAP PVAGPSVFLF	240
PPKPKDTLMI SRTPEVTCVV	VDVKHEDPEV KFNWYVDGVE	VHNAKTKPRE EEYNSTYRVV	300
SVLTVLHQDW LNGKEYKCKV	SNKALPAPIE KTISKAKGQP	REPQVYTLPP SREEMTKNQV	360
SLTCDVSGFY PSDIAVEWES	DGQPENNYKT TPPVLDSGGS	FPLYSKLTVD KSRWEQGDVF	420
SCSVMEALH NHYTQKSLSL	SPGK		444

SEQ ID NO: 584 moltype = AA length = 713
 FEATURE Location/Qualifiers
 REGION 1..713
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..713
 note = XENP33067 Chain 2 -
 PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv
 (GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q
 source 1..713
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 584

EVQLVQSGAE VKKPGASVKV	SCKTSGYTFT EYTIHWVRQA	PGQSLEWMGN INPNNGGTTY	60
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NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GSGGGGGSQA VVTQEPSLTV 240
SPGGTVTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFGSGLLGK 300
AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGGSLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KYNNYATYYA 420
DSVKGRFTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDEYVSWFA YWGQGLVTV 480
SSEPKSSDKT HTPPCPAPP VAGPSVFLFP PKPKDTLMIS RTPEVTCVVV DVKHEDPEVK 540
FNWYVDGVEV HNAKTKPREE QYNSTYRVVS VLTVLHQDWL NGKEYKCKVS NKALPAPIEK 600
TISKAKGQPR EPQVYTLPPS REQMTKNQVK LTCLVKGFYP SDIAVEWESN GQPENNYKTT 660
PPVLDSGDSF FLYSKLTVDK SRWQQGNVFS CSVMHEALHN HYTKSLSLSG PGK 713

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SEQ ID NO: 585      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP33067 Chain 3 - PSMA-H_L1.75 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

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SEQUENCE: 585
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD PTLTISSLQA EDVAVYFCQQ YNSYPQTEGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

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SEQ ID NO: 586      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..444
                   note = XENP33068 Chain 1 -
                   PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
source             1..444
                   mol_type = protein
                   organism = synthetic construct

```

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SEQUENCE: 586
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV DVKHEDPEV KFNWYVDGVE VHNKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGPY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSMHEALH NHYTQKSLSL SPGK 444

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SEQ ID NO: 587      moltype = AA  length = 713
FEATURE            Location/Qualifiers
REGION             1..713
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..713
                   note = XENP33068 Chain 2 -
                   PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv
                   (GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q
source             1..713
                   mol_type = protein
                   organism = synthetic construct

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SEQUENCE: 587
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GSGGGGGSQA VVTQEPSLTV 240
SPGGTVTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFGSGLLGK 300
AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGGSLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KYNNYATYYA 420
DSVKGRFTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDEYVSWFA YWGQGLVTV 480
SSEPKSSDKT HTPPCPAPP VAGPSVFLFP PKPKDTLMIS RTPEVTCVVV DVKHEDPEVK 540
FNWYVDGVEV HNAKTKPREE QYNSTYRVVS VLTVLHQDWL NGKEYKCKVS NKALPAPIEK 600
TISKAKGQPR EPQVYTLPPS REQMTKNQVK LTCLVKGFYP SDIAVEWESN GQPENNYKTT 660
PPVLDSGDSF FLYSKLTVDK SRWQQGNVFS CSVMHEALHN HYTKSLSLSG PGK 713

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SEQ ID NO: 588      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic

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REGION polypeptide
1..214
note = XENP33068 Chain 3 - PSMA-H_L1.68 Light Chain
source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 588
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSKPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 589 moltype = AA length = 444
FEATURE Location/Qualifiers
REGION 1..444
note = Description of Artificial Sequence: Synthetic
polypeptide
REGION 1..444
note = XENP33069 Chain 1 -
PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
source 1..444
mol_type = protein
organism = synthetic construct

SEQUENCE: 589
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFPQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGPENNYKT TPPVLDSGGS FFLYSLKLTVD KSRWEQGDVF 420
SCSVMEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 590 moltype = AA length = 713
FEATURE Location/Qualifiers
REGION 1..713
note = Description of Artificial Sequence: Synthetic
polypeptide
REGION 1..713
note = XENP33069 Chain 2 -
PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv
(GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q
source 1..713
mol_type = protein
organism = synthetic construct

SEQUENCE: 590
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFPQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GSGGGGGSQA VVTQEPSLTV 240
SPGGTVTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFGSGLLGGK 300
AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGGSRLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGIRIS KYNNYATYYA 420
DSVKGRTFIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDEYVSWFA YWQGT L VTV 480
SSEP KSSDKT HTCPCPAPP VAGPSVFLFP PKPKDTLMIS RTPEVTCVVV DVKHEDPEVK 540
FNWYVDGVEV HNAKTKPREE QYNSTYRVVS VLTVLHQDWL NGKEYKCKVS NKALPAPIEK 600
TISKAKGQPR EPQVYTLPPS REQMTKNQVK LTCLVKGFYP SDIAVEWESN GQPENNYKT 660
PPVLDSGDSF FLYSKLTVDK SRWQQGNVFS CSVMHEALHN HYTKSLSLSL PGK 713

SEQ ID NO: 591 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic
polypeptide
REGION 1..214
note = XENP33069 Chain 3 - PSMA-H_L1.29 Light Chain
source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 591
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYT ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 592 moltype = AA length = 444
FEATURE Location/Qualifiers
REGION 1..444
note = Description of Artificial Sequence: Synthetic

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REGION          polypeptide
                  1..444
note = XENP33070 Chain 1 -
source          PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
                  1..444
mol_type = protein
organism = synthetic construct

SEQUENCE: 592
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LVTSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDS DGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 593      moltype = AA length = 713
FEATURE            Location/Qualifiers
REGION            1..713
note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..713
note = XENP33070 Chain 2 -
                  PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv
                  (GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q
source            1..713
mol_type = protein
organism = synthetic construct

SEQUENCE: 593
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LVTSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCG GSGGGGGSQA VVTQEPSLTV 240
SPGGTVTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFGSLLGGK 300
AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGGSRLRLSCA ASGFTFTSYA MNWVRQAPGK GLEWVGRIRS KYNMYATYYA 420
DSVKGRTTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDEYVSWFA YWGQGT LVTV 480
SSEPKSSDKT HTPPCPAPP VAGPSVFLFP PKPKDTLMIS RTPEVTCVVV DVKHEDPEVK 540
FNWYVDGVEV HNAKTKPRE QYNSTYRVVS VLTVLHQDWL NGKEYKCKVS NKALPAPIEK 600
TISKAKGQPR EPQVYTLPPS REQMTKNQVK LTCLVKGFYP SDIAVEWESN GQPENNYKTT 660
PPVLDS DGSF FLYSKLTVDK SRWQQGNVFS CSMHLEALHN HYTKSLSL PGK 713

SEQ ID NO: 594      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
note = XENP33070 Chain 3 - PSMA-H_L1.52 Light Chain
source            1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 594
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ HNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSK STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 595      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION            1..444
note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..444
note = XENP33071 Chain 1 -
                  PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
source            1..444
mol_type = protein
organism = synthetic construct

SEQUENCE: 595
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LVTSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDS DGS FFLYSKLTVD KSRWEQGDVF 420
SCSVMHLEALH NHYTQKSLSL SPGK 444

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SEQ ID NO: 596 moltype = AA length = 713
FEATURE Location/Qualifiers
REGION 1..713
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..713
 note = XENP33071 Chain 2 -
 PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv
 (GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q
source 1..713
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 596
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFP LAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GSGGGGGSQA VVTQEPSLTV 240
SPGGTVTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFSGSLGGK 300
AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGGSLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KYNNYATYYA 420
DSVKGRTTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDEYVSWFA YWGQGLVTV 480
SSEPKSSDKT HTCPCPAPP VAGPSVFLFP PKPKDTLMIS RTPEVTCVVV DVKHEDPEVK 540
FNNWYVDGVE HNAKTKPREE QYNSTYRVVS VLTVLHQDWL NGKEYKCKVS NKALPAPIEK 600
TISKAKGQPR EPQVYTLPPS REQMTKNQVK LTCLVKGFYP SDIAVEWESN GQPENNYKTT 660
PPVLDSGGSF FLYSKLTVDK SRWQQGNVFS CSMVHEALHN HYTQKSLSLS PGK 713

SEQ ID NO: 597 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..214
 note = XENP33071 Chain 3 - PSMA-H_L1.13 Light Chain
source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 597
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TIVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 598 moltype = AA length = 444
FEATURE Location/Qualifiers
REGION 1..444
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..444
 note = XENP34240 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 598
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
PSVFP LAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV FNNWYVDGVE VHNKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGGS FFLYSKLTVD KSRWEQGDVF 420
SCSMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 599 moltype = AA length = 713
FEATURE Location/Qualifiers
REGION 1..713
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..713
 note = XENP34240 Chain 2 -
 PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv
 (GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q
source 1..713
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 599
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120

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PSVFLPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GGSGGGGSQA VVTQEPSLTV 240
SPGGTVTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFGSGLLGGK 300
AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGGSRLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KYNNYATYYA 420
DSVKGGRFTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDEYVSWFA YWGQGTLLVTV 480
SSEPKSSDKT HTPPCPPAPP VAGPSVFLFP PKPKDTLMIS RTPEVTCVVV DVKHEDPEVK 540
FNWYVDGVEV HNAKTKPREE QYNSTYRVVS VLTVLHQDWL NGKEYKCKVS NKALPAPIEK 600
TISKAKGQPR EPQVYTLPPS REQMTKNQVK LTCLVKGFYP SDIAVEWESN GQPENNYKTT 660
PPVLDSDSGS FLYSKLTVDK SRWQQGNVFS CSVMEALHN HTYQKSLSLS PGK 713

SEQ ID NO: 600      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION              1..214
                    note = Description of Artificial Sequence: Synthetic
                        polypeptide
REGION              1..214
                    note = XENP34240 Chain 3 - PSMA-H_L1.78 Light Chain
source              1..214
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 600
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYY ASTRHTGVPD 60
RFTSGSGSGTD FTLTISLQA EDVAVYFCQQ HNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSLSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 601      moltype = AA length = 444
FEATURE            Location/Qualifiers
REGION              1..444
                    note = Description of Artificial Sequence: Synthetic
                        polypeptide
REGION              1..444
                    note = XENP34241 Chain 1 -
                        PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
source              1..444
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 601
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFPQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGGQTL VTVSSASTKG 120
PSVFLPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DQPENNYKT TPPVLDSGS FFLYSKLTV D KSRWEQGDVF 420
SCSVMEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 602      moltype = AA length = 713
FEATURE            Location/Qualifiers
REGION              1..713
                    note = Description of Artificial Sequence: Synthetic
                        polypeptide
REGION              1..713
                    note = XENP34241 Chain 2 -
                        PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv
                        (GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q
source              1..713
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 602
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFPQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGGQTL VTVSSASTKG 120
PSVFLPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GGSGGGGSQA VVTQEPSLTV 240
SPGGTVTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFGSGLLGGK 300
AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGGSRLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KYNNYATYYA 420
DSVKGGRFTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDEYVSWFA YWGQGTLLVTV 480
SSEPKSSDKT HTPPCPPAPP VAGPSVFLFP PKPKDTLMIS RTPEVTCVVV DVKHEDPEVK 540
FNWYVDGVEV HNAKTKPREE QYNSTYRVVS VLTVLHQDWL NGKEYKCKVS NKALPAPIEK 600
TISKAKGQPR EPQVYTLPPS REQMTKNQVK LTCLVKGFYP SDIAVEWESN GQPENNYKTT 660
PPVLDSDSGS FLYSKLTVDK SRWQQGNVFS CSVMEALHN HTYQKSLSLS PGK 713

SEQ ID NO: 603      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION              1..214
                    note = Description of Artificial Sequence: Synthetic
                        polypeptide

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REGION 1..214
 note = XENP34241 Chain 3 - PSMA-H_L1.81 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 603
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYH ASTRHTGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ HNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 604 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..444
 note = XENP34242 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S

source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 604
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGGTGL VTVSSASTKG 120
 PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSDTKVD KKEPKKSCDK THTCPPCPAP PVAGPSVFLF 240
 PKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 605 moltype = AA length = 713
 FEATURE Location/Qualifiers
 REGION 1..713
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..713
 note = XENP34242 Chain 2 -
 PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv
 (GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q

source 1..713
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 605
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
 NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGGTGL VTVSSASTKG 120
 PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHKPSNTKVD KKEPKKSCGG GSGGGGGSQA VVTQEPSLTV 240
 SPGGTVTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFSGSLLGGK 300
 AALTISGAQP EDEADYYCAL WYNSHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
 LVESGGGLVQ PGGSRLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGRIRS KYNNYATYYA 420
 DSVKGRFTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDEYVSWFA YWQGTLLVTV 480
 SSEPKSSDKT HTPPCPAPP VAGPSVFLFP PKPKDTLMIS RTPEVTCVVV DVKHEDPEVK 540
 FNWYVDGVEV HNAKTKPREE QYNSTYRVVS VLTVLHQDWL NGKEYKCKVS NKALPAPIEK 600
 TISKAKGQPR EPQVYTLPPS REQMTKNQVK LTCLVKGFPY SDIAVEWESN GQPENNYKTT 660
 PPVLDSGGSF FLYSKLTVDK SRWQQGNVFS CSMHEALHN HYTQKSLSL PGK 713

SEQ ID NO: 606 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214
 note = XENP34242 Chain 3 - PSMA-H_L1.84 Light Chain

source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 606
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYT ASTRHTGVPD 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ HNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 607 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide

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REGION 1..444
note = XENP34629 Chain 1 -
PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K
/L368D/K370S/M428L/N434S

source 1..444
mol_type = protein
organism = synthetic construct

SEQUENCE: 607

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	LVTVSSASTKG	120
PSVFLAPASS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSDTKVD	KKVEPKSCDK	THTCPPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCVV	VDVKHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EEYNSTYRVV	300
SVLTVLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	SREEMTKNQV	360
SLTCDVSGFY	PSDIAVEWES	DGQPENNYKT	TPPVLDSDGS	FFLYSKLTVD	KSRWEQGDVF	420
SCSVLHEALH	SHYTQKSLSL	SPGK				444

SEQ ID NO: 608 moltype = AA length = 713
FEATURE Location/Qualifiers
REGION 1..713
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..713
note = XENP34629 Chain 2
-PSMA-H_H1L1_Fab_(G4S)_[CD3]_L1.47_H1.89_scFv(
GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q/M428L/N43
4S

source 1..713
mol_type = protein
organism = synthetic construct

SEQUENCE: 608

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	LVTVSSASTKG	120
PSVFLAPASS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSNTKVD	KKVEPKSCGG	GGSGGGGSQA	VVTQEPSLTV	240
SPGGVTTLTC	GSSTGAVTTS	NYANWVQQKP	GKSPRGLIGG	TNKRAPGVPA	RFGSGLGGK	300
AALTIISGAQP	EDEADYYCAL	WYSNHWVFGG	GTKLTVLGKP	GSGKPGSGKP	GSGKPGSEVQ	360
LVESGGGLVQ	PGGSLRLSCA	ASGFTFSTYA	MNWVRQAPGK	GLEWVGIRIS	KYNINYATYYA	420
DSVKGGRFTIS	RDDSKNTLYL	QMNSLRAEDT	AVYYCVRHGN	FGDEYVSWFA	YWGQGT	480
SSEPKSSDKT	HTCPCPCAPP	VAGPSVFLFP	PKPKDTLMIS	RTPEVTCVVV	VDVKHEDPEVK	540
FNWYVDGVEV	HNAKTKPREE	QYNSTYRVVS	VLTVLHQDWL	NGKEYKCKVS	NKALPAPIEK	600
TSKAKGQPR	EPQVYTLPPS	REQMTKNQVK	LTCLVKGFYP	SDIAVEWESN	GQPENNYKT	660
PPVLDSDGSF	FLYSKLTVDK	SRWQQGNVFS	CSVLHEALHS	HYTQKSLSL	PGK	713

SEQ ID NO: 609 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214
note = XENP34629 Chain 3 - PSMA-H_L1 Light Chain

source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 609

DIVMTQSPDS	LAVSLGERAT	LSCRASQDVG	TAVDWYQQKP	DQSPKLLIYW	ASTRHTGVPD	60
RFTGSGSGTD	FTLTISLQA	EDVAVYFCQQ	YNSYPLTFGA	GTKVEIKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYEKHK	VYACEVTHQG	LSSPVTKSFN	RGEC			214

SEQ ID NO: 610 moltype = AA length = 444
FEATURE Location/Qualifiers
REGION 1..444
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..444
note = XENP34630 Chain 1 -
PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K
/L368D/K370S/M428L/N434S

source 1..444
mol_type = protein
organism = synthetic construct

SEQUENCE: 610

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLEWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGQGT	LVTVSSASTKG	120
PSVFLAPASS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSDTKVD	KKVEPKSCDK	THTCPPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCVV	VDVKHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EEYNSTYRVV	300
SVLTVLHQDW	LNGKEYKCKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	SREEMTKNQV	360

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SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGGS FFLYSKLTVD KSRWEQGDVF 420
SCSVLHEALH SHYTQKSLSL SPGK 444
```

```
SEQ ID NO: 611      moltype = AA  length = 713
FEATURE            Location/Qualifiers
REGION             1..713
                  note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION             1..713
                  note = XENP34630 Chain 2 -
                    PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv
                    (GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q/M428L/N4
                    34S
source             1..713
                  mol_type = protein
                  organism = synthetic construct
```

```
SEQUENCE: 611
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFPQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LVTSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GSGGGGGSQA VVTQEPSLTV 240
SPGGTVTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFGSLLGGK 300
AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGGSRLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGRISS KYNNTATYYA 420
DSVKGRTTIS RDDSKNTLYL QMNSLRAEDT AVYYCVRHGN FGDEYVSWFA YWGQGLVTV 480
SSEPKSSDKT HTCPCPCAPP VAGPSVFLFP PKPKDTLMIS RTPEVTCVVV DVKHEDPEVK 540
FNWYVDGVEV HNAKTKPREE QYNSTYRVVS VLTVLHQDWL NGKEYKCKVS NKALPAPIEK 600
TISKAKGQPR EPQVYTLPPS REQMTKNQVK LTCLVKGFYP SDIAVEWESN GQPENNYKTT 660
PPVLDSGGSF FLYSKLTVDK SRWQQGNVFS CCSVLHEALHS HYTQKSLSL PGK 713
```

```
SEQ ID NO: 612      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                  note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION             1..214
                  note = XENP34630 Chain 3 - PSMA-H_L1.58 Light Chain
source             1..214
                  mol_type = protein
                  organism = synthetic construct
```

```
SEQUENCE: 612
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
RFTGSGSGTD FTLTISLQA EDVAVYFCQQ YQSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214
```

```
SEQ ID NO: 613      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION             1..444
                  note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION             1..444
                  note = XENP34631 Chain 1 -
                    PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K
                    /L368D/K370S/M428L/N434S
source             1..444
                  mol_type = protein
                  organism = synthetic construct
```

```
SEQUENCE: 613
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFPQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT LVTSSASTKG 120
PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGGS FFLYSKLTVD KSRWEQGDVF 420
SCSVLHEALH SHYTQKSLSL SPGK 444
```

```
SEQ ID NO: 614      moltype = AA  length = 713
FEATURE            Location/Qualifiers
REGION             1..713
                  note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION             1..713
                  note = XENP34631 Chain 2 -
                    PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv
                    (GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q/M428L/N4
                    34S
source             1..713
```

-continued

mol_type = protein
organism = synthetic construct

SEQUENCE: 614

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLWWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGGQTL	VTVSSASTKG	120
PSVFPPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSNTKVD	KKVEPKSCGG	GGSGGGGSQA	VVTQEPSLTV	240
SPGGTVTLTC	GSSTGAVTTS	NYANWVQQKP	GKSPRGLIGG	TNKRAPGVPA	RFSGSLGGK	300
AALTISGAQP	EDEADYYCAL	WYSNHWVFGG	GTKLTVLGKP	GSQKPGSGKP	GSQKPGSEVQ	360
LVESGGGLVQ	PGGSLRLSCA	ASGFTFSTYA	MNWVRQAPGK	GLEWVGRIRS	KYNNYATYYA	420
DSVKGRTTIS	RDDSKNTLYL	QMNSLRAEDT	AVYYCVRHGN	FGDEYVSWFA	YWGGQTLVTV	480
SSEPKSSDKT	HTCPPCPAPP	VAGPSVFLFP	PKPKDTLMIS	RTPEVTCVVV	DVKHEDPEVK	540
FNWYVDGVEV	HNAKTKPREE	QYNSTYRVVS	VLTVLHQDWL	NGKEYCKKVS	NKALPAPIEK	600
TISKAKGQPR	EPQVYTLPPS	REQMTKNQVK	LTCLVKGFYP	SDIAVEWESN	GQPENNYKTT	660
PPVLDSDGGSF	FLYSKLTVDK	SRWQQGNVFS	CSVLHEALHS	HYTKQSLSL	PGK	713

SEQ ID NO: 615

moltype = AA length = 214

FEATURE Location/Qualifiers

REGION 1..214

note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214

note = XENP34631 Chain 3 - PSMA-H_L1.24 Light Chain

source 1..214

mol_type = protein

organism = synthetic construct

SEQUENCE: 615

DIVMTQSPDS	LAVSLGERAT	LSCRASQDVG	TAVDWYQQKP	DQSPKLLIYY	ASTRHTGVPD	60
RFTGSGSGTD	FTLTISLQA	EDVAVYFCQQ	YNSYPLTFGA	GTKVEIKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYKHKH	VYACEVTHQG	LSSPVTKSFN	RGEC			214

SEQ ID NO: 616

moltype = AA length = 444

FEATURE Location/Qualifiers

REGION 1..444

note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..444

note = XENP34632 Chain 1 - PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K/L368D/K370S/M428L/N434S

source 1..444

mol_type = protein

organism = synthetic construct

SEQUENCE: 616

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLWWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGGQTL	VTVSSASTKG	120
PSVFPPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSDTKVD	KKVEPKSCDK	THTCPPCPAP	PVAGPSVFLF	240
PPKPKDTLMI	SRTPEVTCVV	VDVKHEDPEV	KFNWYVDGVE	VHNAKTKPRE	EEYNSTYRVV	300
SVLTVLHQDW	LNGKEYCKKV	SNKALPAPIE	KTISKAKGQP	REPQVYTLPP	SREEMTKNQV	360
SLTCDVSGFY	PSDIAVEWES	DGQPENNYKT	TPPVLDSDGS	FFLYSKLTVD	KSRWEQGDVF	420
SCSVLHEALH	SHYTQKSLSL	SPGK				444

SEQ ID NO: 617

moltype = AA length = 713

FEATURE Location/Qualifiers

REGION 1..713

note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..713

note = XENP34632 Chain 2 - PSMA-H_H1_(G4S)_[CD3]_L1.47_H1.89_scFv (GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q/M428L/N434S

source 1..713

mol_type = protein

organism = synthetic construct

SEQUENCE: 617

EVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTIHWVRQA	PGQSLWWMGN	INPNNGGTTY	60
NQKFQGRVTI	TVDKSTSTAY	MELSSLRSED	TAVYYCAAGW	NFDYWGGQTL	VTVSSASTKG	120
PSVFPPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA	LTSGVHTFPA	VLQSSGLYSL	180
SSVVTVPSSS	LGTQTYICNV	NHKPSNTKVD	KKVEPKSCGG	GGSGGGGSQA	VVTQEPSLTV	240
SPGGTVTLTC	GSSTGAVTTS	NYANWVQQKP	GKSPRGLIGG	TNKRAPGVPA	RFSGSLGGK	300
AALTISGAQP	EDEADYYCAL	WYSNHWVFGG	GTKLTVLGKP	GSQKPGSGKP	GSQKPGSEVQ	360
LVESGGGLVQ	PGGSLRLSCA	ASGFTFSTYA	MNWVRQAPGK	GLEWVGRIRS	KYNNYATYYA	420
DSVKGRTTIS	RDDSKNTLYL	QMNSLRAEDT	AVYYCVRHGN	FGDEYVSWFA	YWGGQTLVTV	480
SSEPKSSDKT	HTCPPCPAPP	VAGPSVFLFP	PKPKDTLMIS	RTPEVTCVVV	DVKHEDPEVK	540
FNWYVDGVEV	HNAKTKPREE	QYNSTYRVVS	VLTVLHQDWL	NGKEYCKKVS	NKALPAPIEK	600
TISKAKGQPR	EPQVYTLPPS	REQMTKNQVK	LTCLVKGFYP	SDIAVEWESN	GQPENNYKTT	660

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PPVLDSGGSF FLYSKLTVDK SRWQQGNVFS CSVLHEALHS HYTQKSLSLSG PGK 713

SEQ ID NO: 618 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP34632 Chain 3 - PSMA-H_L1.29 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 618
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYT ASTRHTGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 619 moltype = AA length = 444
 FEATURE Location/Qualifiers
 REGION 1..444
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..444
 note = XENP31854 Chain 1 -
 PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
 source 1..444
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 619
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLLEWMGN INPNNGGTTY 60
 NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
 PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
 PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
 SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
 SLTCDVSGFY PSDIAVEWES DGPENNYKT TPPVLDSGGS FFLYSLKLTVD KSRWEQGDVF 420
 SCSVMHEALH NHYTQKSLSL SPGK 444

SEQ ID NO: 620 moltype = AA length = 717
 FEATURE Location/Qualifiers
 REGION 1..717
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..717
 note = XENP31854 Chain 2 -
 PSMA-H_H1_CH1_(G4S)2_Fc(222)_IgG1_PVA_/S267K /S364K/E357Q
 source 1..717
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 620
 EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLLEWMGN INPNNGGTTY 60
 NQKPFQGRVTI TVDKSTSTAY MELSSLRSED TAVYYCAAGW NFDYWGQGT L VTVSSASTKG 120
 PSVFPPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GSGGGGSEV QLVESSGGLV 240
 QPGGSLRLSC AASGFTFSTY AMNWVRQAPG KGLEWVGRIR SKYNNYATYY ADSVKGRFTI 300
 SRDDSKNTLY LQMNSLRAED TAVYYCVRHG NFGDSYVSWF DYWGQGT LVT VSSGKPGSGK 360
 PGSGKPGSGK PGSQAVVTQE PSLTVSPGGT VTLTCGSGTG AVTTSNYANW VQKPKGKSPR 420
 GLIGGTNKRA PGVPARFSGS LLGGKAALTI SGAQPEDEAD YYCALWYSNH WVFGGGTKLT 480
 VLGGGGSGGG GSKTHTCPPC PAPPVAGPSV FLFPPKPKDT LMI SRTPEVT CVVVDVKHED 540
 PEVKFNWYVD GVEVHNAKTK PREEQYNSTY RVVSVLTVLH QDWLNGKEYK CKVSNKALPA 600
 PIEKTISKAK GQPREPQVYT LPPSREQMTK NQVKTCLVK GFYPSDIAVE WESNGQPENN 660
 YKTTTPVLDS DGSFFLYSKL TVDKSRWQQG NVFSCSVMHE ALHNHYTQKS LSLSPGK 717

SEQ ID NO: 621 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP31854 Chain 3 - PSMA-H_L1 Light Chain
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 621
 DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVDP 60
 RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

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SEQ ID NO: 622      moltype = AA  length = 444
FEATURE            Location/Qualifiers
REGION            1..444
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..444
                  note = XENP31857 Chain 1 -
                  PSMA-H_H1_IgG1_pI(-)_Isosteric_A_/PVA_/S267K /L368D/K370S
source            1..444
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 622
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFGQGRVTI TVDKSTSTAY MELSSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSDTKVD KKVEPKSCDK THTCPPCPAP PVAGPSVFLF 240
PPKPKDTLMI SRTPEVTCVV VDVKHEDPEV KFNWYVDGVE VHNAKTKPRE EEYNSTYRVV 300
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP SREEMTKNQV 360
SLTCDVSGFY PSDIAVEWES DGQPENNYKT TPPVLDSGDS FFLYSKLTVD KSRWEQGDVF 420
SCSVMEALHL NHYQKSLSL SPGK 444

SEQ ID NO: 623      moltype = AA  length = 717
FEATURE            Location/Qualifiers
REGION            1..717
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..717
                  note = XENP31857 Chain 2 -
                  PSMA-H_H1_CH1_(G4S)2_Fc(222)_IgG1_PVA_/S267K /S364K/E357Q
source            1..717
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 623
EVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTIHWVRQA PGQSLEWMGN INPNNGGTTY 60
NQKFGQGRVTI TVDKSTSTAY MELSSSLRSED TAVYYCAAGW NFDYWGQGTLL VTVSSASTKG 120
PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYICNV NHKPSNTKVD KKVEPKSCGG GSGGGGGSQA VVTQEPSLTV 240
SPGGVTTLTC GSSTGAVTTS NYANWVQQKP GKSPRGLIGG TNKRAPGVPA RFGSGLLGK 300
AALTISGAQP EDEADYYCAL WYSNHWVFGG GTKLTVLGKP GSGKPGSGKP GSGKPGSEVQ 360
LVESGGGLVQ PGGSLRLSCA ASGFTFSTYA MNWVRQAPGK GLEWVGIRIS KYNNYATYYA 420
DSVKGRTFTS RDDSKNTLYL QMNSLRAEDT AVYCYVRHGN FGDSYVSWFD YWGQGTLLTV 480
SGGGGSGGGG GSKTHTCPPC PAPPVAGPSV FLFPPKPKDT LMISRTPEVT CVVVDVKHED 540
PEVKFNWYVD GVEVHNAKTK PREEQYNSTY RVVSVLTVLH QDWLNGKEYK CKVSNKALPA 600
PIEKTISKAK QQPREPQVYT LPPSRQMTK NQVKLTCLVK GFYPSPDIAVE WESNGQPENN 660
YKTTTPVLDS DGSFFLYSKL TVDKSRWQQG NVFSCSVME ALHNHYTQKS LSLSPGK 717

SEQ ID NO: 624      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP31857 Chain 3 - PSMA-H_L1 Light Chain
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 624
DIVMTQSPDS LAVSLGERAT LSCRASQDVG TAVDWYQQKP DQSPKLLIYW ASTRHTGVPD 60
RFTGSGSGTD FTLTISSLQA EDVAVYFCQQ YNSYPLTFGA GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNFFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 625      moltype = AA  length = 499
FEATURE            Location/Qualifiers
REGION            1..499
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..499
                  note = XENP34282
source            1..499
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 625
EVQLVESGGG LVQPGGSLTL SCAASRFMIS EYHMHWRQA PGKGLEWVST INPAGTTDYA 60
ESVKGRTFTS RDNAKNTLYL QMNSLKPEDT AVYYCDSYGY RGQGTQVTVS SGGGGSGGGS 120
EVQLVESGGG LVQPGNSLRL SCAASGFTFS KFGMSWVRQA PGKGLEWVSS ISGSGRDTLY 180
ADSVKGRFTI SRDNAKNTLY LQMNSLRPED TAVYYCTIGG SLVSSQGTLL VTVSSGGGGS 240
GGGSEVQLVE SGGGLVQPGG SLKLSCAASG FTFNKYAINW VRQAPGKGLE WVARIRSKYN 300

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NYATYYADQV KDRFTISRDD SKNTAYLQMN NLKTEDTAVY YCVRHANFGN SYISYWAYWG 360
QGTLTVTSSG GGGSGGGGSG GGGSQTVVTQ EPSLTVSPGG TVTLTCASST GAVTSGNYPN 420
WVQKPGQAP RGLIGGTKFL VPGTPARFSG SLLGGKAALT LSGVQPEDEA EYYCTLWYSN 480
RWVFGGGTKL TVLHHHHHH 499

SEQ ID NO: 626      moltype = AA length = 504
FEATURE            Location/Qualifiers
REGION             1..504
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..504
                   note = XENP34283
source             1..504
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 626
QVQLVESGGG LVKPGESLRL SCAASGFTFS DYYMYWVRQA PGKGLEWVAI ISDGGYYTTY 60
SDIIKGRFTI SRDNAKNSLY LQMNSLKAED TAVYYCARGF PLLRHGAMDY WGQGTLVTVS 120
SGGGSGGGG GGGGSDIQM TQSPSSLAS VGDRVITICK ASQNVDTNVA WYQKPGQAP 180
KSLIYSASYR YSDVPSRFSG SASGTDFTLT ISSVQSEDFA TYYCQYDSY PYTFGGGTKL 240
EIKSGGGGSE VQLVESGGGL VQPGGSLKLS CAASGFTFNK YAMNWVRQAP GKGLEWVARI 300
RSKYNMYATY YADSVKDRFT ISRDDSKNTA YLQMNNLKTE DTAVYYCVRH GNFGNSYISY 360
WAYWGQGTIV TVSSGGGGSG GGGSGGGGSG TVVTQEPSTL VSPGGTVTLT CGSSTGAVTS 420
GNYPNWWQK PGQAPRGLIG GTKFLAPGTP ARFSGSLLGG KAALTLSGVQ PEDEAEYYCV 480
LWYSNRWVFG GGTKLTVLHH HHHH 504

SEQ ID NO: 627      moltype = AA length = 5
FEATURE            Location/Qualifiers
REGION             1..5
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..5
                   note = See specification as filed for detailed description
                   of substitutions and preferred embodiments
source             1..5
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 627
GGGGS

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What is claimed is:

1. A composition comprising a Prostate Specific Membrane Antigen (PSMA) binding domain comprising:

a) a variable heavy domain comprising the variable heavy complementarity determining regions vhCDR1, vhCDR2, and vhCDR3 of a variable heavy domain having an amino acid sequence of SEQ ID NO: 217; and

b) a variable light domain comprising the variable light complementarity determining regions vlCDR1, vlCDR2, and vlCDR3 of a variable light domain having an amino acid sequence of one of the following: SEQ ID NO: 252, SEQ ID NO: 257, and SEQ ID NO: 286.

2. A composition comprising a Prostate Specific Membrane Antigen (PSMA) binding domain comprising:

a) a variable heavy domain having an amino acid sequence of SEQ ID NO: 217; and

b) a variable light domain having an amino acid sequence of one of the following: SEQ ID NO: 252, SEQ ID NO: 257, and SEQ ID NO: 286.

3. A nucleic acid composition comprising:

a) a first nucleic acid encoding the variable heavy domain according to claim 2; and

b) a second nucleic acid encoding the variable light domain according to claim 2.

4. An expression vector composition comprising:

a) a first expression vector comprising the first nucleic acid of claim 3; and

b) a second expression vector comprising the second nucleic acid of claim 3.

5. A host cell comprising the expression vector composition according to claim 4.

6. A method of making a Prostate Specific Membrane Antigen (PSMA) binding domain comprising culturing the host cell according to claim 5 under conditions wherein the PSMA binding domain is expressed, and recovering the PSMA binding domain.

* * * * *